GRUT

Geometric Relational Unified Topology Theory

Derivations from the seed $h = r$, the C_3 triangle and the force axiom

Version 1.23.2

Jose Mancilla

18 October 2026

Abstract

GRUT derives gravity, horizon thermodynamics, cosmology and the lepton sector from the speed of light and three geometric primitives: the seed $h = r$ (the null cone), three equivalent generators (the equilateral C_3 triangle, total deficit π, $\pi/3$ each), and the force axiom $FG = c^4$, whose quarter $c^4/4G$ is the force carried by one cone. Every equation is derived, every constant is calibrated once in a ledger, and every result carries a status tag, so that what the theory adds is visible line by line.

The axiom turns length into mass, $M_{\rm H} = Fr/2c^2$: the horizon is the Schwarzschild radius of its own contents, which is flatness. From it follow $G = c^4/F$, $\hbar = 2\pi Fr^2/Nc$, the entropy $S = N/2$, the temperature $k_B T = \hbar c/2\pi r$ and the exact exchange law $E_{\rm H} = k_B T\, S$. Every horizon quantity $X \propto r^k$ flows at $\dot{X}/X = k(1+q)H$, and the horizon's energy grows at $3(1+w_{\rm eff})\, c^5/4G$: four, three, two, one and zero times the maximum power through radiation, matter, strings ($q = 0$), walls ($\Omega_m = \frac{1}{3}$, $q = -\frac{1}{2}$) and vacuum. The present is a resonance: the horizon's C_3 mode reaches the final expansion rate when $\Omega_\Lambda = 27/4\pi^2 = 0.684$ (Planck 0.685 ± 0.007). With the derived baryon ratio and the derived neutrino masses, a Boltzmann fit to the acoustic angle gives $H_0 = 67.35$, $\omega_b = 0.02248$, $\omega_c = 0.12022$ and $\Omega_m = 0.316$, within 1σ of Planck's densities. Dark energy is the Planck density shared over the final horizon's cells, $\rho_\Lambda = \frac{3}{4}\rho_P/N_\infty = 1.13 \times 10^{-123}\rho_P$.

The horizon's generating arcs spin at $\omega = H_\infty$ with rims at light speed. The spin fixes $\Lambda = 3\omega^2/c^2$, adds $8\pi\ell_{\rm P}^2$ of area per quantum to any horizon, predicts the CMB's birefringence, $\beta = (2\alpha/\pi)\,\mathrm{artanh}(3\sqrt{3}/2\pi) = 0.314^\circ$ (measured $0.277^\circ \pm 0.057^\circ$), and the galactic acceleration scale, $a_0 = (2/3\pi)\, c^2\sqrt{\Lambda/3} = (\sqrt{3}/\pi^2)\, cH_0 = 1.148 \times 10^{-10}\ \mathrm{m\,s^{-2}}$. Across 153 SPARC rotation curves the threshold law fits with tail orders $1.62 \le n \le 1.91$, the predicted scale 1.0σ from the likelihood value. The extra gravity is carried by the universe's rotation, a phase held at its rate that dilutes as cold dust; a system holds it only where its escape speed exceeds the motion of the field around it, so the Solar System and wide binaries stay Newtonian.

The dark sector is one geometry: dark energy is spacetime unexcited, what is called dark matter is the same spacetime excited, and the whole has $w = -1 + \Theta$. What lensing infers as dark matter is interpreted by GRUT as a dilation $\mathcal{S} = 1/\cos_n\theta$, $\tan\theta = a_0/g$: one at the centre, growing toward the edge and varying with capture and relaxation. In the Bullet Cluster, lensing-inferred mass peaks are offset from the X-ray-bright gas and lie closer to the galaxies; this is consistent with a collisionless gravitating component under standard lensing/GR interpretations, but does not by itself identify that component as GRUT's phase sector or establish a cosmological reset. On the unbound web the horizon dilates clocks and rulers together from the onset of acceleration, above 68 Mpc, so matter feels it and light does not: S_8, cosmic shear and CMB lensing keep the background's values, the late Sachs–Wolfe signal falls, and the bulk flows Cosmicflows-4 measures (below 0.03% in ΛCDM) lie at the 78th–90th percentiles. The framework has one expansion rate, 67.35; the dilation does not move it, and the supernova Hubble diagram rules out a local void, so its prediction is that distance ladders calibrated alike converge on 67.3 (the 2026 network lies 6.1σ away, the JWST red-giant ladder 1.5σ).

The present is also fixed in Planck cells by the electric coupling, $r = 8\pi\ell_{\rm P}\, e^{1/\alpha + 3\pi\alpha/2}$: with no cosmological input it gives $H_0 = 67.377$, 0.4σ from the CMB fit, and $\Lambda\ell_{\rm P}^2 = (81/256\pi^4)\, e^{-2/\alpha - 3\pi\alpha}$. The four interactions are then four features of one shape—the cone, the generators' phase, the conjugate pair and the triangle—written as one action with one gauge group, $(SU(3) \times SU(2) \times U(1))/\mathbb{Z}_6$, whose content fixes the running ($b_3 = -7$). The horizon and the nucleon are the transmutations of the electric and the strong coupling from one Planck cell, gravity's weakness at the nucleon is $e^{-2/3\alpha}$, and there is no unification scale and so no gauge-mediated proton decay.

In generation space the seed cone is Koide's relation, with the lepton cone at the axiom's vacuum: $m_\tau = 1776.969$ MeV, $\sum m_\nu = 59.34$ meV (normal) or 102.0 meV (inverted), $\sin^2\theta_{12} = 0.3180$. The fine-structure constant follows from $S^1(2\pi) \times S^3(1)$, $1/\alpha = 4\pi^3 + \pi^2 + \pi - \frac{\alpha}{24}(1 + e^{-2\pi}) = 137.035\,999\,152$, 1.2σ from CODATA 2022. The primordial tilt measures the reset carrier's mass, with negligible running and zero tensor tilt; the 21-cm onset of cosmic dawn is set by the CMB's twist, $42.7^{+5.6}_{-6.0}$ MHz; and the Schwarzschild radii of everything inside the horizon add up to it, $\Omega_{\rm tot} = \chi/2 = 1$. Auxoviscausality identifies the phase-flow

integral of proper-volume expansion as retained geometric strain, with its local Ricci-load/shear response fixed by the existing Raychaudhuri/Einstein system. At the conformal junction, the raw strain diverges rather than matching continuously; the finite conformal-volume remainder matches under the stated smooth-metric and common-flow-tube assumptions. This does not preserve the entire prior FLRW expansion integral as a separate state. Every prediction is listed with its dependencies and falsification condition (Appendices C and E).

What this work contributes

This framework is original work. Its contributions are:

1. The four aspects of the geometry kept distinct: reality, the spin; actuality, the swept sphere; occurrence, the crossings; representation, the boundary. The present is the occurrence at which actuality falls into step with reality (the opening principle).
2. The two-C/S geometry of the horizon and its C_3 structure: one closed shape that organizes the cosmos, galaxies and the nucleon (Sections 23 and 50).
3. The spinning-arc principle, $\omega R = c$: the arcs that sweep each sphere spin at a fixed rate with rims at light speed; one spin sets the final horizon, the cosmological constant and the present, is measured by the CMB's birefringence, and gives each particle its Compton radius (§36.9). For the final horizon it is derived: its thermal geometry is the spinning arcs, with the observer on the rim; the charge of the spin is the horizon's angular momentum and its entropy over 2π; and the quantum of the spin is the area step $8\pi\ell_{\mathrm{P}}^2$ of every horizon (§36.10–36.12).
4. The threefold count as a beat: the fold, turning twice for each turn of the arc, beats against the conjugate arc three times per turn and keeps exactly $\mathbb{Z}_3$, as the conjugate pair keeps $\mathbb{Z}_2$ (§1.11).
5. The birefringence angle in closed form, $\beta = (2\alpha/\pi)\,\mathrm{artanh}(3\sqrt{3}/2\pi)$, and the acceleration scale from the cosmological constant, $a_0 = (2/3\pi)\,c^2\sqrt{\Lambda/3}$ (§36.13).
6. The quantum of the universe's rotation as the quantum of the horizon's spin, $\Omega_0 = jH_\infty$ with $j = 1$ measured (§36.14).
7. The dark ratio from the resonance, 2.580, in place of an identification of densities with powers (§25.11).
8. Koide's relation as the vacuum of the axiom's potential, seen from the generators under the seed's right angle (§42.8).
9. First starlight counted in cycles: the 21-cm onset fixed by the CMB's birefringence, $\nu_\star/\nu_{21} = [\sqrt{\Omega_m/\Omega_\Lambda}\,\sinh(\pi\beta/2)]^{2/3}$, the fine-structure constant, the clock and the Hubble constant cancelling (§48.8).
10. The dark sector as one geometry: dark energy as spacetime unexcited and the cosmic share as spacetime excited—the spin's quanta held by the horizon and carried by the rotation—with the equation of state $w = -1+\Theta$ of the whole, and what lensing infers as dark matter as the dilation $\mathcal{S} = 1/\cos_n\theta$, $\tan\theta = a_0/g$; the dilation field as the index of the optical metric, the switch as the capture fraction, and the middle scales as geometric means of the outer ones; the background unmoved by the dilation, and the web dilated as a clock is slowed—from the onset of acceleration, above 68 Mpc, on clocks and not on light—carrying the measured bulk flows with S_8 unchanged (§53).
11. The closure law in 3+1 dimensions: energy as defect on the cosmological horizon, $\sum r_s = (\chi/2)r$ and $\Omega_{\mathrm{tot}} = \chi/2$, so that flatness is topology (§50.1).
12. The scale ladder as the horizon's symmetric cuts, ending at one Planck cell whose energy is the framework's energy per bit (§50.2).
13. The MOND acceleration from the horizon's C_3 eigenvalue, $a_0 = (\sqrt{3}/\pi^2)\,cH_0$ (§22).
14. Dark matter as the internal rotation of the horizon's phase: one rotation that sets a_0, carries the cosmic cold share and needs no particle (§23).
15. The birefringence–acceleration relation $\beta = \frac{9}{2}\alpha\,a_0t_0/c$: galaxy rotation curves predict the rotation of the CMB's polarization (§49.13).
16. The derived baryon ratio $\Omega_b/\Omega_c = 0.18697$, which passes the full CMB spectrum (§25.9).
17. The footprint balance, which places the two-thirds point with its coefficient fixed and reduces the Eddington–Weinberg coincidence to the framework's scale data (§50.8).

18. The nucleon as the horizon's closed C_3 at the strong scale, with integer charge from the $\mathbb{Z}_2 \times \mathbb{Z}_3$ closure (§50.10).

19. The electroweak-volume reading of α, the lepton cone with m_τ, and the closure of $N_c = 3$ by C_3, each labelled derived or identified in the text.

20. Auxoviscausality: logarithmic proper-volume strain follows the phase-flow integral and its Ricci-load/shear response follows from Raychaudhuri/Einstein. At the conformal junction the raw strain is not continuous; the regularized conformal-volume remainder matches under explicit geometric regularity assumptions, without adding a field or claiming microscopic irreversibility.

Earlier results are cited where this work meets them. Each citation marks a point of agreement with established physics, not a source of the framework.

Results at a glance

Result	Status	Data	§
Closure law: $\sum r_s = (\chi/2)r$, $\Omega_{\rm tot} = 1$	derived	1.0007 ± 0.0019, 0.4σ	50.1
MOND scale $a_0 = (\sqrt{3}/\pi^2)\, cH_0$	derived	-0.7σ	22
Birefringence from galaxies, $\beta = \frac{9}{2}\alpha\, a_0 t_0/c$	derived	$0.33^\circ \pm 0.07^\circ$ vs $0.277^\circ \pm 0.057^\circ$	49.13
Dark sector as the internal rotation	derived	dust at linear order; wide binaries Newtonian	23
Baryon ratio $\Omega_b/\Omega_c = 0.18697$	derived	CMB spectra, $\Delta\chi^2 = 0.2$	25.9
Dark-energy ratio $\Omega_\Lambda/\Omega_c = 5/2$	withdrawn	disfavoured, 4.6σ; data 2.61 ± 0.02	25.9, 52.6
Dark ratio from the resonance, 2.580	from H3	2.611 ± 0.024, -1.3σ	25.11
$1/\alpha = 137.035999152$	identified	-1.2σ	43F
$m_\tau = 1776.97$ MeV from the lepton cone	identified	0.4σ	42.7
Two-thirds point, $m_*^3 = \hbar^2 H/2Gc$	derived	matter at 0.01–20 m_*	50.8
Nucleon as a closed C_3; integer charge	derived	lattice QCD junction at 120°	50.10
Now as a resonance, $\Omega_\Lambda = 27/4\pi^2$	identified	0.6847 ± 0.0073, 0.1σ	36.7
Spinning arcs of the final horizon, $\omega R = c$, $\Lambda = 3\omega^2/c^2$	derived	spin from birefringence $(0.73 \pm 0.15)H_0$, 0.6σ	36.9–36.10
Charge of the spin, $S = 2\pi L/\hbar$; area quantum $8\pi\ell_{\rm P}^2$	derived	LVK ringdowns $\alpha = 15.6^{+20.5}_{-13.3}$ vs 8π, 0.5σ	36.11–36.12
Birefringence in closed form, $\beta = 0.3137^\circ$	predicted	$0.277^\circ \pm 0.057^\circ$, 0.6σ	36.13
Quantum of the rotation, $\Omega_0 = jH_\infty$, $j = 1$	derived	j measured: 0.88 ± 0.18; $j = 2$ out at 6.2σ	36.14
Acceleration scale tracks the horizon, $a_0(z) = a_0 E(z)$	derived	MUSE-DARK: rising, constant excluded	36.15
Koide's cone as the vacuum, $Q = (\Xi + 2)/3\Xi$, given I6	derived	$\Xi_\ell = 2.000013 \pm 0.000031$	42.8
Cosmic dawn from the twist, $H_\infty t_\star = (\pi/3)\beta$	hypothesis	from β: $42.7^{+5.6}_{-6.0}$ MHz; closed form 46.4 MHz	48.8
Dark sector as one geometry, $w = -1 + \Theta$	derived	equality at $z = 0.372$; background unchanged	53.1–53.3
Dark matter as the dilation, $\mathcal{S} = 1/\cos_n\theta$, $\tan\theta = a_0/g$	derived	1 at the centre, r/ℓ toward the edge; varies	53.4
Dilation field $S = e^{-\psi}$; switch $= \Gamma$; $\ell^2 = r_s R_{a_0}$	derived	no screening inside R_∞; Sun $\Gamma = 7 \times 10^{-9}$	53.7–53.8
Web dilation on clocks, not light, above 68 Mpc	hypothesis	bulk flows at the 78–90th percentile; S_8 +0.16%; H_0 unmoved	53.9
One expansion rate, $H_0 = 67.35 \pm 0.06$; ladders calibrated alike converge on 67.3	predicted	network 6.1σ (local scatter included); red-giant ladder 1.5σ	26.7, 53.9
The present in Planck cells, $r = 8\pi\ell_{\rm P} e^{1/\alpha + 3\pi\alpha/2}$	hypothesis (H4)	$H_0 = 67.3765$ against the CMB fit 67.35 ± 0.06, 0.4σ	48.12
Four interactions from one shape, $(SU(3) \times SU(2) \times U(1))/\mathbb{Z}_6$; $Gm_p^2/\hbar c = K^2(16\pi)^{-2/3} e^{-2/3\alpha - \pi\alpha}$	derived (given H4)	5.906×10^{-39}; no proton decay, $> 2.4 \times 10^{34}$ yr	54
Bullet Cluster: $M_{\rm tot}/M_b = 6.35$, offset with the galaxies	derived	8σ offset; $\sigma/m < 0.63$ cm^2 g^{-1}	53.6
Mergers: excess stays on the galaxies, drift $f = 0.5\%$ at the Bullet, $\leq 6\%$ at 2 Gyr	derived	synthetic maps separate GRUT and CDM from baryon-following ($T = 98$), not from each other	47.9
Threefold count as the beat of fold and conjugate arc	derived	the $\mathbb{Z}_3$ harmonic of §45.1	1.11

Material tensions at the resonance budget (§25.11, Appendix D): DESI DR2's Ω_m at $+2.2\sigma$ and DES-Y5's at -2.1σ, bracketing the budget from opposite sides as ΛCDM's own CMB fit does; and the distance ladder, 7.6σ from the 2026 network (6.1σ with the local scatter the dilated flows add). The framework has one expansion rate and predicts that ladders calibrated alike converge on 67.3 (§53.9).

Contents

The opening principle: reality, actuality, occurrence and representation 5

0 Method, anchors and calibration 6

1 The Geometry 8

2 The Method 9

3 The Seed and the Cone 10

4 The Defect and the Fold 10

5 The Cycle C_3 11

6 Probability and the Born Rule 11

7 Effective Area and Cell Count 11

8 The Axiom 12

9 The Zero Equation 12

10 Derivation of r 13

11 Derivation of N 13

12 Derivation of G 13

13 Derivation of $\hbar$ 13

14 Derivation of F 14

15 The Fine-Structure Constant α 14

16 Derivation of H_0 14

17 Energy 15

18 Mass 15

19 Momentum 15

20 Information 16

21 Angular Momentum 16

22 Critical Acceleration and a_0 16

23 The Rotation of the Universe 21

24 The Three Currencies 23

25 The Cosmic Budget 23

26 The Arrow of Time: Three of Space, One of Time **26**

27 The Light Cone and the Speed of Light **28**

28 The Lorentz Structure **29**

29 Newtonian Gravity **29**

30 General Relativity **29**

31 Electromagnetism **30**

32 Quantum Mechanics **31**

33 The Schrödinger Equation **31**

34 Thermodynamics **32**

35 Statistical Mechanics **32**

36 Cosmology **33**

37 Dark Matter **38**

38 Dark Energy **40**

39 The CMB **41**

40 Black Holes and Information **42**

41 Gravity and the Scalar Action **42**

42 Three Generations on the Seed Cone **43**

43 Causal Time and the Electric Coupling **47**

44 TRSD and GRUT: the formal bridge **52**

45 Primordial Spectrum from the Conformal Reset **55**

46 The Complete Gravity Equation: Dark-Matter Boosting **59**

47 Kinematic Evolution and Gravitational Condensation of the Topological Field **61**

48 Limits and the Finite Clock **67**

49 The Local Baryonic Coupling **73**

50 The closure law and the scale ladder **78**

51 Status of the Framework **81**

52 Superseded and Excluded Results **82**

53 The Dark Sector as One Geometry **85**

54 The Four Interactions from One Shape **92**

55 Collision Lag, Geometric Recoil, and CHIRP **94**

Update supplement: derivations and the status of the open equations **102**

Junction closure: phase current, curvature perturbation, and conformal information **105**

A Numerical values **108**

B From the draft to the derived form **109**

C Prediction register **111**

D Variance analysis: prediction against observation **116**

E Dependencies of the predictions **119**

F Next Derivations: the List and Where Each Was Carried Out **120**

G Version history and verification **122**

The opening principle: reality, actuality, occurrence and representation

The framework distinguishes four aspects of one geometry and keeps them apart throughout.

Reality is the spin: the C-arcs that generate the horizon and their fixed rate ω, set once and unchanged, since nothing acts on them (§36.9). What is real does not change.

Actuality is the swept sphere: what the spin has made actual so far, the horizon at this moment, its radius and its cells. Reality is the rate; actuality is how much of the sphere the rate has swept. Actuality grows toward the final horizon $R_\infty = c/\omega$, where the arcs' rims reach light speed.

Occurrence is a crossing. Something happens only where lines or rates cross: the arcs crossing at the three C_3 nodes, which are the defects; the wobble falling into step with the spin, which is the present (§36.7); the reset, where one aeon's future meets the next one's past; a horizon, where time and radius exchange; and each activation of the threshold law (§22). Between crossings the motion continues and nothing occurs.

Representation is the boundary. Every energy inside the horizon appears on it as a defect, its share of the horizon's cells (§50.2), so the sphere carries an image of everything it contains: the universe represents itself on its own surface. Measurements read that image; the CMB's birefringence reads the spin (§36.9). The framework's equations, and the picture of the arcs itself, are representations of the same reality.

Physics has often merged these. The block universe of relativity treats reality as if it were all actuality, as though the whole sphere were already swept; quantum theory blurs representation, the wavefunction, with reality. Kept apart, they define the present. The wobble belongs to actuality, its frequency falling as the swept sphere grows; the spin belongs to reality, and is fixed. *The present is the occurrence at which actuality falls into step with reality.*

0 Method, anchors and calibration

0.1 Equations are derived; values are calibrated

The form of each law is derived and its constants are measured once: each observation calibrates one constant (the bootstrap of §2.7), and every number downstream is a prediction.

Theorem 1 (Nothing dimensionful follows from c alone)**.** DERIVED *Any quantity computed from c and dimensionless numbers has dimension $(\mathrm{L}\,\mathrm{T}^{-1})^k$, so no length, time or mass follows from c alone; three dimensionful anchors are conventions and carry no predictive content.*

Proof. Dimensions multiply, transcendental functions need dimensionless arguments and summed terms share one dimension, so anything built from c is c^k times a pure number, of dimension $\mathrm{L}^k\mathrm{T}^{-k}$: a length needs $k = 1$ and $k = 0$, a time $k = 0$ and $k = -1$, and no k gives mass. L, T and M thus need three independent anchors, whose values a change of units rescales at will; only their dimensionless combinations are predictive. □

0.2 Anchors and dimensional closure

Four anchors carry the dimensions: c converts time to length (the seed $h = r$); r, the seed's scale, flows with the expansion ($r_0 = c/H_0$ today); F, the force of the axiom $FG = c^4$, converts length to mass; ℓ_{P}, the horizon's cell size, converts area to count. Hence

$$[\text{length}] = r, \quad [\text{time}] = \frac{r}{c}, \quad [\text{mass}] = \frac{Fr}{c^2}, \quad [\text{energy}] = Fr, \quad [\text{action}] = \frac{Fr^2}{c}, \quad [\text{count}] = \frac{r^2}{\ell_{\mathrm{P}}^2}. \tag{1}$$

c, r, F fix the units; ℓ_{P} enters only through the pure number $N = 2\pi r^2/\ell_{\mathrm{P}}^2$, the horizon cell count (§7), so $\hbar = (2\pi/N)\,Fr^2/c$ and $G = c^4/F$. Every quantity not calibrated in §0.3 is a geometric constant—built from π, $\sqrt{2}$, $\sqrt{3}$, the deficits and the C_3 eigenvalues—times a monomial in (1), possibly times a power of N.

0.3 Calibration ledger

Constant	Calibrated from	Value
c	SI definition	$299\,792\,458$ m s^{-1}, exact
r_0	CMB acoustic angle via the resonance budget (§25.11)	1.3735×10^{26} m: $H_0 = 67.35 \pm 0.06$ km s^{-1} Mpc^{-1}, $N = 4.538 \times 10^{122}$
F	G: $F = c^4/G$	1.2103×10^{44} N
ℓ_{P}	$\hbar$: $\ell_{\mathrm{P}}^2 = \hbar G/c^3$	1.6163×10^{-35} m
α	CODATA 2022	$1/137.035\,999\,177$
μ_ℓ, δ_ℓ	m_e, m_μ (§42)	$\mu_\ell^2 = 313.856$ MeV, $\delta_\ell = 0.222\,222\,0$; then $m_\tau = 1776.969$ MeV
μ_ν, δ_ν	NuFIT 6.1 with JUNO (§42.4)	$\Delta m_{21}^2 = 7.53 \times 10^{-5}$, $\Delta m_{31}^2 = 2.529 \times 10^{-3}$ eV^2
M_*	dark-matter mass m_δ (§37.2)	open; $\leq 7.5 \times 10^8 M_\odot$ (Lyman-α); sets the defect length L_* and the potential scale ε_V; $V_0 = 0$ (§38.6)
ϕ_i	dark-matter density (§37.7)	open; 10^{-3}–10^{-2} of $f_\delta = m_{\mathrm{P}}c^2/\sqrt{8}$
n	SPARC, ephemerides (§22.6–22.7)	tail order $1.62 \leq n \leq 1.91$
$\alpha_s(M_Z)$, v_{EW}, λ_H	measured (§43D, §50.9)	0.1180, 246.22 GeV, 0.129

Everything else is derived, including $\Omega_b/\Omega_c = 0.18697$ (§25.7) and $\Omega_\Lambda/\Omega_c = 2.580$ (§25.11). Fitting r_0 to the acoustic angle makes the local H_0 a prediction (§26.7). H_0-dependent values are quoted at 67.35 (horizon $r, M_{\mathrm{H}}, E_{\mathrm{H}} \propto H_0^{-1}$, $N \propto H_0^{-2}$, temperature and $a_0 \propto H_0$).

0.4 The three primitives and the identifications

Postulate I (Seed)**.** Light emitted at an event reaches, after a time t, every point at distance ct, in every direction and for every inertial observer: with height measured as light-travel time, $h \equiv ct = r$, i.e. $ds^2 = 0$ (§1.10).

Postulate II (Generators)**.** Three equivalent generators generate the structure; their cyclic symmetry C_3 prefers none, and its minimal realization is the equilateral triangle (§1.1–1.4).

Postulate III (Force). $FG = c^4$, with F the force of the whole horizon sphere; one light cone carries $F/4$ (§8).

Identifications, which read a geometric quantity as a physical one, carry the risk:

I1	§8.2	the factor 4 of the maximum force is the four-cone closure $V_s = 4V_c$
I2	§4.4	the defect cone $h = \sqrt{3}\,r$ is the radiation era's sound cone
I3	§18.1	the deficit–mass map's reduction length is the horizon scale r
I4	§22.1	Poisson activation on the cycle, hazard linear in the load; Gamma exposure of order n
I5	§25	$\Omega_\Lambda/\Omega_c = 5/2$, the defect power ratio; withdrawn, disfavoured at 4.6σ by the CMB spectra; the resonance fixes the ratio (§25.11, §52.6)
I6	§42.1, §42.8	each generation's amplitude is its generator's invariant raised by the conjugate pair Φ; Koide's cone is then the vacuum $\Xi = \lvert\Phi\rvert^2 = 2$
I7	§42.4	the lepton doublet is the conjugate pair (neutrino conjugate branch)
I8	§36.13	the rotation's phase turns with the horizon's spin at the final rate $H_\infty = c\sqrt{\Lambda/3}$; derived up to an integer, $\Omega_0 = jH_\infty$, $j = 1$ measured (§36.14)

Hypotheses, tested rather than assumed: H1, the present lies at the three-to-one condition (§26.5); H2, the conjugate-branch neutrino lies along the electron generator (§42.5); H3, the present is the resonance of the horizon's C_3 mode with the spin (§36.7; the opening principle), which fixes the dark ratio (§25.11). Everything else is a theorem or a calibration.

0.5 Status tags

DERIVED	derived from the primitives and established physics, every step shown
STANDARD	established physics recovered in GRUT's variables, credited
IDENTIFICATION	a posited correspondence (I1–I8)
CALIBRATED	a value set once from data (§0.3)
HYPOTHESIS	a testable conjecture (H1–H3)
PREDICTION	a number that can fail, tested in Appendix C
DATA-CONSTRAINED	a shape parameter constrained by data (the tail order n)
MEASURED	a fit to data in this paper
OPEN	an open problem, with its target

0.6 What GRUT adds

GRUT's claim is structural: one geometry produces all its relations, not all of them new.

(A) New results: $a_0 = (\sqrt{3}/\pi^2)\,cH_0$ from the first C_3 eigenvalue, independent of the exposure law and tracking $H(z)$ (Theorem 4; §22, §36.15; compare [51, 50]); the family μ_n, its transition and tail decoupled (§22.3–22.7); the maximum-power epoch $\Omega_m = \frac{1}{3} \iff q = -\frac{1}{2} \iff \dot{E}_\mathrm{H} = c^5/4G$, the defect ladder and the master rate (Theorems 8, 9; §2.8); the closure mass (Theorem 2) and $F_\mathrm{cone} = F/4$ from $V_s = 4V_c$ (§3.12, §8.2); the resonance budget, $\Omega_\Lambda = 27/4\pi^2$ now (§25.11); Koide's cone as the seed cone and the vacuum, with the conjugate branch and TM1 (§42); the time-rate function and $L_\mathrm{crit}(r_s) = c^5/4G$ (§43B); $1/\alpha = 4\pi^3 + \pi^2 + \pi - \frac{\alpha}{24}(1 + e^{-2\pi}) = 137.035\,999\,152$, 1.2σ from CODATA 2022 (§43A/C, §43F); the scalar-sector closure (§37–38); the spin of the horizon, its birefringence angle and the rotation's rate $\Omega_0 = H_\infty$ (§1.11, §36.10–36.14); first starlight from the CMB's twist β, $H_\infty t_\star = (\pi/3)\,\beta$ (§48.8); the dark sector as one geometry, $w = -1 + \Theta$ (Θ the excited fraction), with lensing's dark matter the dilation $\mathcal{S} = g/g_\mathrm{bar} = 1/\cos_n\theta$, $\tan\theta = a_0/g$, which leaves the background and H_0 unmoved (§53).

(B) Known relations re-derived, credited where they appear: maximum force [28, 48]; $E_\mathrm{H} = k_B T S$, for de Sitter holographic equipartition [97]; the Hubble horizon's generalized second law [49]; 10^{-122} as a horizon count [95]; $w = -\frac{1}{3}$ per defect dimension [44, 56]; Koide's 45° cone [11, 47]; the neutrino negative root [7]; TM1 [58, 5]; the thermal four-sphere and first law of de Sitter [29]; Noether-charge entropy [134]; the area step $8\pi\ell_\mathrm{P}^2$ [136, 137]; zitterbewegung spin [131]; the Josephson–Anderson relation [146]; horizon-synchronized condensates [148].

(C) Standard physics recovered: Lorentz group [46], Einstein equations [36, 33], Born rule [30], Schrödinger equation [42], Maxwell's equations, Boltzmann distribution [34].

0.7 The derivation map

Columns run from a primitive through derived structure to predictions (Appendix C):

Seed $h = r$, null cone	C_3 **triangle**, $\delta_j = \pi/3$, $\delta_{\rm tot} = \pi$	**Axiom** $FG = c^4$, $F_{\rm cone} = F/4$
Lorentz group, $V_s = 4V_c$, Born $p = \frac{1}{2}$ Koide cone, conjugate branch, TM1	C_3 spectrum, $a_c = k_3\, cH$ μ_n, $a_0 = \frac{4}{3}a_c$, three-arc completion	$G, \hbar, N$; $M_{\rm H}, E_{\rm H}, S, T$; $E_{\rm H} = k_B TS$, flatness master rate, ladder, max-power epoch
m_τ, $\sum m_\nu$, $\sin^2\theta_{12}$, $\delta_{\rm CP}$	a_0, RAR shape, planetary tail	Ω_m, $H_0 = 67.35$, q_0, $z_{\rm acc}$, $z_{3:1}$

The C_3 generators also build the generation-space cone (§42) and the budget (§25.11); H sets a_c (§5).

0.8 Scope

Gravity, defect scalar	the action of §41.4: Einstein–Hilbert gravity, condensate, vacuum; with the gauge, fermion and Higgs sectors, one action (§54.5)
Galactic dynamics	nonrelativistic AQUAL law (§22); covariant completion open (Appendix F); with the condensate as source the Solar System sees only the Galactic tide (§46–47)
Horizon thermodynamics	from the axiom and the Gibbons–Hawking temperature (§17–24, §34)
Cosmology	Friedmann background; CAMB's ΛCDM transfer functions plus the dilation's growth (§53.9); full defect perturbations open; primordial spectrum from the terminal de Sitter phase and the C_3 pair convolution (§45)
Electromagnetism	Maxwell from the phase gradient, Coulomb from three dimensions, charges from anomaly cancellation ($N_c = 3$), neutral condensate, massless photon (§31)
Interactions	four features of one shape, one group $(SU(3)\times SU(2)\times U(1))/\mathbb{Z}_6$, one action, one running (§54)
Quantum mechanics	recovered (§32–33), not generated by the action
Generations	Koide cone and TM1: identifications and a hypothesis (§42); fermions on the tetrad with its spin connection (§41.7)
Black holes	standard solutions and thermodynamics (§40); no new scalar solution

GRUT and TRSD [63] are one theory in two notations (§44); their shared results (gravity law, SPARC fit, scalar-sector closure) are consistency, not independent confirmation.

1 The Geometry

Geometric origin. The cosmic horizon is generated by two conjugate C-shaped arcs spinning about perpendicular axes and crossing as they sweep the sphere. Their threefold wobble, the beat of the fold against the counter-turning conjugate arc (§1.11), creates three conical defects at equal azimuthal separations, each of deficit $\pi/3$, closing into the C_3 cycle (the equal-frequency figure first proposed has no nodes and is withdrawn, §52.5). The spin is physical: its rims move at c, and its fixed rate, derived from the final horizon's thermal geometry, sets the final horizon, the cosmological constant and the present (§36.9–36.10).

1.1–1.4 Point, conjugate pair, third point, triangle

A point separates into the conjugate pair $p_\pm$; a third point off their line spans a plane, the smallest structure with an area. No vertex is preferred (as for the generators, §4.2), so the generating triangle is equilateral, with angles $\pi/3$ and rotation group C_3.

1.5 The recursion

Each vertex generates its own pair and third point: after d levels there are $N_d = \sum_{j=0}^{d} 3^j = (3^{d+1} - 1)/2 = 1$, 4, 13, 40, 121, … units, the same triangle at every scale.

1.6 The surface

A triangle rotated about its altitude sweeps a right circular cone, apex at the upper vertex. The right isosceles triangle gives $h = r$ (apex 90°), the light cone (§1.10); the equilateral one gives $h = \sqrt{3}\, r$ (apex 60°), the defect cone (§4.4).

1.7 The defect

Unrolled, the lateral surface is a sector of radius the slant height l and arc $2\pi r$. The missing angle is the deficit,

$$\delta = 2\pi\Big(1 - \frac{r}{l}\Big), \tag{2}$$

the rotation of a vector parallel-transported once around the apex: the cone's curvature, concentrated there.

1.8–1.9 The fold and the 3-sphere

The solid cone is a 3-ball. The conjugate pair's two balls, glued along their boundary, close into the 3-sphere; a single ball folded onto itself, opposite boundary points identified, is the rotation group (the rotation by $\psi \le \pi$ about $\hat{n}$ is the point $\psi\hat{n}$, and those by π about $\pm\hat{n}$ coincide):

$$B^3_+ \cup_{S^2} B^3_- \;\cong\; S^3 \;\cong\; SU(2), \qquad B^3/(x \sim -x \text{ on } \partial B^3) \;\cong\; \mathbb{RP}^3 \;\cong\; SO(3). \tag{3}$$

The map $S^3 \to \mathbb{RP}^3$ is two-to-one: the pair double-covers the fold. This is the geometry of spin: a 2π rotation closes in $SO(3)$ but not in $SU(2)$, so a spin-$\frac{1}{2}$ state needs 4π. S^3 is finite (volume $2\pi^2R^3$), with no centre and no boundary.

1.10 The seed

With height measured in light-travel time, $h \equiv ct$, light emitted at the apex event reaches every point at distance ct, in every direction and for every inertial observer: the seed is the null condition,

$$r = ct = h \quad\Longleftrightarrow\quad ds^2 = -c^2dt^2 + dx^2 + dy^2 + dz^2 = 0. \tag{4}$$

The seed and the constancy of c are one statement: c converts the height, a time, into a length. In an expanding universe the seed holds exactly in conformal time $\eta = \int dt/a$: with $ds^2 = a^2(-c^2d\eta^2 + d\chi^2)$, light obeys $\chi = c\eta$ (§26.2).

1.11 The threefold beat

DERIVED Per turn of the arc p_+, the conjugate p_- turns once in the opposite sense and the fold twice, since the pair double-covers it (§1.8): phases $e^{i\theta}$, $e^{-i\theta}$, $e^{2i\theta}$. Two superposed motions of counts n_1, n_2 stay in step under $\theta \to \theta + \psi$ only if $e^{i(n_1-n_2)\psi} = 1$, so they keep $\mathbb{Z}_{|n_1-n_2|}$:

$$e^{i\theta} + e^{-i\theta} \text{ (pair)}: \;\mathbb{Z}_2, \text{ nodes } \pi \text{ apart}; \qquad e^{2i\theta} + e^{-i\theta} \text{ (fold against arc)}: \;\mathbb{Z}_3, \text{ nodes } \tfrac{2\pi}{3} \text{ apart}, \tag{5}$$

the latter the three-petalled rose $2\cos(3\theta/2)\,e^{i\theta/2}$. For the complex scalar $\Phi = \phi\, e^{i\theta}$ of the action,

$$|\Phi^2 + \Phi^*|^2 = \phi^4 + \phi^2 + 2\phi^3\cos 3\theta, \tag{6}$$

whose interference term $2\,\mathrm{Re}\,\Phi^3$ is the $\mathbb{Z}_3$ harmonic leading the phase potential of §45.1; the pair alone, $|\Phi+\Phi^*|^2 = 2\phi^2(1+\cos 2\theta)$, gives the $\mathbb{Z}_2$ harmonic, the elementary cut of the deficit π (§50.2) and the π crossing of the coil's lobes (§44.5). The two beats are the $\mathbb{Z}_2$ and $\mathbb{Z}_3$ of the horizon's closure (§50.10), together $\mathbb{Z}_6$.

IDENTIFICATION The threefold wobble is the second beat, realizing Postulate II without replacing it: that its three nodes are the horizon's three defects is the reading of the generators as phases (§15.3).

2 The Method

2.1–2.7 Equation, algebra, bootstrap

An equation relates an object's invariant to r and π; calculus enters when a quantity changes with scale ($dE_{\mathrm{H}}/dr = F_{\mathrm{geom}}$, $dN/dr = 4\pi r/\ell_{\mathrm{P}}^2$), and the algebra is that of the axiom (§8) and the cell count (§7). An observation calibrates one anchor (§0.3), which predicts the next quantity, checked against a different observation; no calibrating value counts as a test, and every value downstream of the calibrations is one.

2.8 Dynamic coupling

DERIVED Every horizon quantity is a geometric constant times a power of r, $X \propto r^k$: $k = 1$ for r, $M_{\rm H}$, $E_{\rm H}$; $k = 2$ for N, $S_{\rm GH}$, $L_{\rm H}$; $k = -1$ for $T_{\rm GH}$, a_c, a_0, H. With $r = c/H$, the deceleration parameter $q = -\ddot{a}a/\dot{a}^2$ and $\dot{H} = -(1+q)H^2$, one master rate drives them all:

$$\dot{r} = c\,(1+q), \qquad \frac{\dot{X}}{X} = k\,(1+q)\,H. \tag{7}$$

The anchors c, F, $\ell_{\rm P}$ do not flow, so $G = c^4/F$ and $\hbar$ are constant, as lunar laser ranging requires ($|\dot{G}/G| \lesssim 10^{-13}\,{\rm yr}^{-1}$ [32]). The horizon energy $E_{\rm H} = Fr/2$ grows at

$$\dot{E}_{\rm H} = \tfrac{1}{2}F\dot{r} = 2(1+q)\,P_{\max}, \qquad P_{\max} = F_{\rm cone}\,c = \frac{c^5}{4G}, \tag{8}$$

today $0.95\,P_{\max}$ ($q_0 = -0.526$, §25.11). As $1+q = \frac{3}{2}(1+w_{\rm eff})$, with $w_{\rm eff}$ the total equation of state, $\dot{E}_{\rm H} = P_{\max}$ exactly at $w_{\rm eff} = -\frac{2}{3}$, $\Omega_m = \frac{1}{3}$: the maximum-power epoch (§26.5). The cell count grows, $\dot{N} = 2(1+q)HN \geq 0$ for $q \geq -1$ (§26.6).

3 The Seed and the Cone

3.1–3.11 Measures of the seed cone

For $h = r$: apex angle 90°, slant $l = \sqrt{2}\,r$, base area $A_b = \pi r^2$, total area $A_c = (1+\sqrt{2})\pi r^2$, volume $V_c = \frac{1}{3}\pi r^3$, apex solid angle $2\pi(1-h/l) = (2-\sqrt{2})\pi = 1.8403$ sr. Against the sphere of the same radius, $V_s/V_c = A_s/A_b = 4$.

3.12 Why four cones fill the sphere

DERIVED At height $z \in [0, r]$ above the apex, the slices of the cylinder and hemisphere of radius r and of the seed cone obey (cylinder − cone = hemisphere)

$$\pi r^2 - \pi z^2 = \pi(r^2 - z^2). \tag{9}$$

A cone of height h has the slice $\pi z^2 r^2/h^2$, so the identity holds at every z only for $h = r$. Integrated slice by slice (Cavalieri), it gives Archimedes' theorem,

$$V_c : V_{\rm hemi} : V_{\rm cyl} = 1 : 2 : 3, \qquad V_s = 2V_{\rm hemi} = 4V_c, \tag{10}$$

and $h = r$ is the unique cone for which it holds slice by slice: the seed closes the sphere by volume.

4 The Defect and the Fold

4.1–4.4 Three equal generators and the value of the deficit

DERIVED The three vertices generate three defects δ_j, equal because no vertex is preferred. They meet at the apex as three equilateral faces of angle $\pi/3$, so the total angle there is π and

$$\delta_{\rm tot} = 3\delta_j = 2\pi - \pi = \pi, \qquad \delta_j = \frac{\pi}{3}: \tag{11}$$

each deficit is the triangle's interior angle, the total its angle sum. Three results confirm the value. (i) By (2), $\delta = \pi$ requires $l = 2r$, i.e. $h = \sqrt{3}\,r$, the cone of the equilateral triangle (§1.6); it unrolls to a half-disk and is intrinsically the three-face apex. (ii) Four such apexes, the vertices of the regular tetrahedron, close a sphere: by Descartes' theorem (polyhedral Gauss–Bonnet),

$$4\,\delta_{\rm tot} = 4\pi = \oint_{S^2} K\,dA. \tag{12}$$

The light cone closes the sphere by volume (§3.12), the C_3 cone by curvature: both closures take four cones.

(iii) DERIVED (I2) With height as light-travel time, the $h = \sqrt{3}\,r$ cone is a front moving at $c/\sqrt{3}$, the sound speed of radiation, whose stress is traceless:

$$T^{\mu}{}_{\mu} = -\rho c^2 + 3p = 0 \;\Rightarrow\; p = \tfrac{1}{3}\rho c^2 \;\Rightarrow\; c_s = \frac{c}{\sqrt{3}}. \tag{13}$$

The epochs the cone describes, the radiation era and the conformally invariant reset boundary, are traceless, so the defect cone is the early universe's sound cone, imprinted as the CMB's acoustic scale (§39.1). The seed cone, of deficit $(2-\sqrt{2})\pi$ and front speed c, is causal, not defect-carrying.

4.5–4.6 The fold; the 3-sphere and its radius

The pair closes into $S^3 \cong SU(2)$, the fold into $\mathbb{RP}^3 \cong SO(3)$ (3). An S^3 spatial section of radius R enters the Friedmann equation as $H_0^2 = \frac{8\pi G}{3}\rho - c^2/R^2$, so

$$\Omega_{\text{tot}} = 1 + \left(\frac{r}{R}\right)^2. \tag{14}$$

PREDICTION The horizon condition makes the section flat, $R \to \infty$ (§12.2, §50.7); a closed section survives only as a correction beyond it, bounded by $\Omega_K = 0.0007 \pm 0.0019$ [39] to $R > 18\,r$ (95%) and tested by surveys reaching $|\Omega_K| \sim 10^{-3}$. An S^3 of radius r ($\Omega_{\text{tot}} = 2$) is the group manifold $SU(2)$ of §1.8, not space.

5 The Cycle C_3

DERIVED The generators define the cycle $1 \to 2 \to 3 \to 1$ on the base circle, with arcs $\ell = 2\pi r/3$. The discrete Laplacian $(\Delta f)_n = \ell^{-2}(2f_n - f_{n+1} - f_{n-1})$ has on $f_n = e^{2\pi i m n/3}$ the eigenvalues

$$\lambda_m = \frac{2}{\ell^2}\left[1 - \cos\frac{2\pi m}{3}\right]: \qquad \lambda_0 = 0, \qquad \lambda_1 = \frac{3}{\ell^2} = \frac{27}{4\pi^2 r^2}. \tag{15}$$

On n nodes $\lambda_1 = \frac{n^2}{2\pi^2 r^2}\left[1 - \cos\frac{2\pi}{n}\right] \to 1/r^2$, the continuum value; C_3, the coarsest closed cycle, has $\lambda_1 r^2 = 27/4\pi^2 = 0.6839$, the present Ω_Λ at the resonance (§25.11). Light on the cycle has $\omega_1 = c\sqrt{\lambda_1} = 3\sqrt{3}\,c/2\pi r$ and period $\tau_1 = 2\pi/\omega_1$.

Definition. The critical acceleration brings a body from rest to c in one period (the centripetal $c\,\omega_1$ is 2π larger):

$$a_c = \frac{c}{\tau_1} = \frac{3\sqrt{3}}{4\pi^2}\frac{c^2}{r} = k_3\, cH_0, \qquad k_3 = 0.131620, \tag{16}$$

(κ is the surface gravity, §8.2); at $H_0 = 67.35$, $a_c = 8.612 \times 10^{-11}\ \text{m s}^{-2} = \frac{3}{4}a_0$ (§22).

6 Probability and the Born Rule

6.1–6.3 Effective solid angle and $P = \frac{1}{2}$

DERIVED A line defect through the observer removes its deficit from the azimuth:

$$\Omega_{\text{eff}} = \int_0^{\pi} \sin\theta\, d\theta \int_0^{2\pi - \delta_{\text{tot}}} d\phi = 4\pi - 2\delta_{\text{tot}} = 2\pi, \qquad P = \frac{\Omega_{\text{eff}}}{4\pi} = \frac{1}{2}. \tag{17}$$

The seed gives it independently: $P = \cos^2 45°$, with $45°$ its half-apex angle, is the probability that a system prepared along one axis passes a test at $45°$ (Malus's law).

6.4–6.5 The Born rule; conjugate points

STANDARD The conjugate pair with equal weights, $|\psi\rangle = (|+\rangle + e^{i\varphi}|-\rangle)/\sqrt{2}$, has amplitudes $\cos 45°$ and probabilities $P_\pm = \frac{1}{2}$, $P_+ + P_- = 1$. The general rule is Gleason's theorem [30]: in a Hilbert space of dimension three or more, every probability assignment additive over orthogonal outcomes is $P = \text{Tr}(\rho\,\Pi)$ (ρ the density operator, Π the outcome's projector).

IDENTIFICATION The three generators supply exactly that minimal dimension, three.

7 Effective Area and Cell Count

DERIVED The effective horizon area is $r^2\Omega_{\text{eff}} = 2\pi r^2$; in cells of the Planck area $\ell_{\text{P}}^2 = \hbar G/c^3$ the horizon carries

$$N = \frac{2\pi r^2}{\ell_{\text{P}}^2} = \frac{2\pi c^5}{\hbar G H_0^2} = 4.538 \times 10^{122} \quad (H_0 = 67.35). \tag{18}$$

The same number measures the vacuum: against the Planck density $c^5/\hbar G^2$,

$$\frac{\rho_c}{\rho_{\rm P}} = \frac{3H_0^2}{8\pi G}\cdot\frac{\hbar G^2}{c^5} = \frac{3}{4N} = 1.65\times10^{-123}, \tag{19}$$

so the famous ratio is a count of horizon cells: the holographic reading of the vacuum-energy problem [95], with its coefficient, 3/4, fixed exactly.

8 The Axiom

8.1 The force axiom

$$F\,G = c^4, \qquad F = \frac{c^4}{G} = 1.2103\times10^{44}\ \text{N}, \tag{20}$$

the force of the whole universe. It is the Planck force: two Planck masses one cell apart attract with $Gm_{\rm P}^2/\ell_{\rm P}^2 = c^4/G$.

8.2 The factor of four: one cone

STANDARD A mass M has horizon radius $r_s = 2GM/c^2$ and surface gravity $\kappa = c^4/4GM$; the force on it at its own horizon, $F_{\rm cone} = M\kappa = Mc^2/2r_s = c^4/4G = F/4 = 3.0256\times10^{43}$ N, is the same for every M: the maximum force of a horizon [28, 48].

DERIVED (I1) The four is the closure of §3.12: a uniformly filled sphere gives each of its four cones a quarter of its mass, $M_{\rm cone} = \rho\,\pi r^3/3 = M_{\rm H}/4$, so $F = 4F_{\rm cone} \Leftrightarrow V_s = 4V_c$. General relativity supplies the value $c^4/4G$; the axiom supplies its geometric reading and the normalization of F.

8.3 The electric axiom

DERIVED Two elementary charges repel with $F_e = \alpha\hbar c/d^2$ and two Planck masses attract with $F_g = Gm_{\rm P}^2/d^2 = \hbar c/d^2$, so $F_e/F_g = \alpha$ at every separation d; at $d = \ell_{\rm P}$, where $F_g = F$,

$$F_e\,G = \alpha\,c^4, \qquad F_e = \alpha F. \tag{21}$$

8.4 Geometric force

DERIVED With $M_{\rm H} = Fr/2c^2$ (§12.2) and $E_{\rm H} = M_{\rm H}c^2 = Fr/2$, $F_{\rm geom} \equiv dE_{\rm H}/dr = F/2 = c^4/2G$ and $E_{\rm H}/2r = F/4 = F_{\rm cone}$: the horizon energy over the horizon diameter is the one-cone force.

9 The Zero Equation

9.1–9.2 Smooth count and zero equation

STANDARD The zeros of $\zeta(s)$ with $0 < \operatorname{Im} s \le T$ number $N(T) = \vartheta(T)/\pi + 1 + S(T)$, with $\vartheta(T) = \arg\Gamma(\frac{1}{4} + \frac{iT}{2}) - \frac{T}{2}\log\pi = \frac{T}{2}\log\frac{T}{2\pi} - \frac{T}{2} - \frac{\pi}{8} + O(1/T)$ and $S(T) = O(\log T)$, so $N(T)/N_{\rm smooth}(T) \to 1$ with

$$N_{\rm smooth}(T) = \frac{T}{2\pi}\log\frac{T}{2\pi e} + \frac{7}{8}, \tag{22}$$

the $\frac{7}{8}$ being the pole at $s = 1$ less the Gamma factor's phase $\frac{1}{8}$ ($N(1000) = 649$, $N_{\rm smooth}(1000) = 648.6$).

9.3–9.4 Cone connection and offset

The cone has the same form: $A_c/V_c = 3(1+\sqrt{2})/r \to 0$ is its smooth part, and the deficit the constant that keeps the surface from closing flat. The leading count $\frac{T}{2\pi}(\log\frac{T}{2\pi} - 1)$ is the phase-space area of the Hamiltonian xp, which generates scale transformations [19]; in GRUT the same generator, $r\partial_r$, has the horizon quantities of §2.8 as eigenfunctions, $r\partial_r X = kX$. A scale operator whose spectrum is the zeros would complete the link (the Hilbert–Pólya problem); this is an analogy, not yet a derivation.

10 Derivation of r

10.1 From H_0

CALIBRATED $r = c/H_0 = 1.3735 \times 10^{26}$ m $= 14.52$ Gly at $H_0 = 67.35$, set through the budget by the CMB acoustic angle (§0.3, §16).

10.2 From G: the horizon condition

DERIVED The horizon is the Schwarzschild radius of the mass it contains,

$$r = \frac{2GM_{\rm H}}{c^2} = \frac{2c^2 M_{\rm H}}{F}, \tag{23}$$

a condition equivalent to flatness (§12.2).

10.3 The radius partition

DERIVED Being linear in mass, the Schwarzschild radius divides among the contents by density fraction: with $M_i = \Omega_i M_{\rm H}$,

$$r = \sum_i r_i, \qquad r_i = \frac{2GM_i}{c^2} = \Omega_i\, r, \tag{24}$$

$r_m = 0.316\, r$ and $r_\Lambda = 0.684\, r$ at the resonance budget. The observable horizon, the distance light has travelled in the space frame, is the conformal radius $c\eta_0 = 3.18\, r = 46.1$ Gly (§26.3).

11 Derivation of N

11.1–11.3 From the cell count and from the action

DERIVED Besides (18): the horizon energy $M_{\rm H}c^2$ acting for the light-crossing time r/c is the action $M_{\rm H}cr$, in units of $\hbar$

$$Q = \frac{M_{\rm H}cr}{\hbar} = \frac{Fr^2}{2\hbar c} = \frac{c^3 r^2}{2\hbar G} = \frac{N}{4\pi} \tag{25}$$

quanta, the 4π being the full sphere's. The two routes agree, $2\pi r^2 c^3/\hbar G = 4\pi M_{\rm H} cr/\hbar$, exactly when $M_{\rm H} = c^2 r/2G$: the horizon condition (23).

12 Derivation of G

12.1 From the axiom

$G = c^4/F = 6.6743 \times 10^{-11}$ m^3 kg^{-1} s^{-2}, constant because F is (§2.8).

12.2 From the horizon mass

DERIVED The horizon mass

$$M_{\rm H} = \frac{4\pi}{3} r^3 \rho_c = \frac{c^2 r}{2G} = \frac{Fr}{2c^2} \tag{26}$$

satisfies $2GM_{\rm H}/c^2 = r$; conversely, $r_s(M) = r$ for $M = \frac{4\pi}{3} r^3 \rho$ gives $\frac{8\pi G}{3}\rho = c^2/r^2 = H^2$, the flat Friedmann equation. The horizon condition and flatness are one statement, $2GM_{\rm H}/c^2 = r \iff \rho = \rho_c \iff \Omega_{\rm tot} = 1$. Mass is linear in r; the power r^2 belongs to the information currency, $N \propto r^2$ (§24).

13 Derivation of $\hbar$

DERIVED From the cell count, the axiom and (26),

$$\hbar = \frac{2\pi r^2 c^3}{GN} = \frac{2\pi F r^2}{Nc} = \frac{4\pi M_{\rm H} cr}{N}, \tag{27}$$

constant as r flows since $N \propto r^2$. In the units (1) it is $2\pi/N$ units of action: the horizon action shared among the cells.

14 Derivation of F

DERIVED From §8: the universe carries $F = c^4/G = 1.2103 \times 10^{44}$ N and each of its four cones $F/4 = 3.0256 \times 10^{43}$ N, $F_{\rm geom} = F/2$, and the maximum power is $P_{\max} = F_{\rm cone} c = c^5/4G = 9.071 \times 10^{51}$ W.

15 The Fine-Structure Constant α

15.1–15.2 From the force ratio; value

CALIBRATED By (21), $\alpha = F_e/F = F_e G/c^4$, calibrated at $1/137.035\,999\,177(21)$ (CODATA 2022). This is the zero-momentum value, from which the running to $1/128.9$ at the Z mass (on shell) follows (§31.5).

15.3 Geometric origin

IDENTIFICATION The generators' phases 1, $e^{2\pi i/3}$, $e^{4\pi i/3}$ form $\mathbb{Z}_3$, in the Standard Model the centre of colour $SU(3)$, whose tie to hypercharge puts quark charges in thirds: the gauge group is $[SU(3)\times SU(2)\times U(1)]/\mathbb{Z}_6$ [17]. Closure gives the singlet rule: with $\pi/3$ per generator and closure at multiples of π, one generator ($\pi/3$) or two ($2\pi/3$) do not close, a generator with its conjugate (0) or three generators (π) do—the meson and baryon combinations; the two that fail are never seen free. This is a correspondence with triality, not a proof of confinement.

DERIVED The number of colours follows without it: Witten's global anomaly needs an even number of $SU(2)$ doublets per generation, N_c+1, so N_c is odd [116], and asymptotic freedom with six flavours needs $11N_c > 12$; the smallest allowed N_c is 3, the order of C_3 (§5). Anomaly cancellation then derives $Q_u = \frac{2}{3}$, $Q_d = -\frac{1}{3}$, $Q_e = -1$, $Q_\nu = 0$ (§31.4).

IDENTIFICATION The value of α is the conjecture of §43A/C, solved self-consistently,

$$\frac{1}{\alpha} = 4\pi^3 + \pi^2 + \pi - \frac{\alpha}{24}\left(1 + e^{-2\pi}\right) = 137.035\,999\,151\,7,$$

1.2σ below CODATA 2022; no other prediction rests on it.

16 Derivation of H_0

PREDICTION $H_0 = c/r$. With the resonance budget (§25.11) the CMB acoustic angle fixes $H_0 = 67.35 \pm 0.06$ km s^{-1} Mpc^{-1}, at which every H_0-dependent number in this paper is quoted (r, $M_{\rm H}$, $E_{\rm H} \propto H_0^{-1}$; N, $L_{\rm H} \propto H_0^{-2}$; $T_{\rm GH}$, $a_0 \propto H_0$).

16.1 H_0 against the 2025–26 measurements

MEASURED Pulls of 67.35; after the slash, with the local-rate scatter below added to the three direct ladder measurements:

measurement	value	pull
Local Distance Network [79]	73.50 ± 0.81	-7.6 / -6.1
SH0ES [40]	73.04 ± 1.04	-5.5 / -4.8
CCHP TRGB [80]	70.39 ± 1.94	-1.6 / -1.5
TDCOSMO lensing [79]	$71.6^{+3.9}_{-3.3}$	-1.3
DESI DR2 BAO + BBN, flat ΛCDM [79]	68.51 ± 0.58	-2.0
CMB, ΛCDM [79]; Planck 2018 [39]	67.24 ± 0.35; 67.36 ± 0.54	$+0.3$; -0.1
DESI Lyα + BBN, CMB-independent [65]	66.5 ± 1.3	$+0.6$

The ladder lies 9% above 67.35; 99.5% of the gap is the CMB calibration GRUT shares with ΛCDM (Appendix D).

DERIVED GRUT has one expansion rate, the background's 67.35. The dilation leaves the background and its sound horizon unmoved and changes the local rate only through the local density contrast (§53.9). It shifts clocks and

rulers only by the gravitational potential, under a part in 10^5, so no dilation of clocks can carry the ladder's 9%. Nor can a local void: the supernova Hubble diagram shows no change of H_0 across $0.023 < z < 0.15$ to 0.60% (§53.9).

HYPOTHESIS On the unbound web the dilated flows widen the scatter of the local rate to 0.88% on 300 Mpc (0.41% in ΛCDM, §53.9), which leaves the network 6.1σ from 67.35 (7.6σ bare), SH0ES 4.8σ and CCHP 1.5σ.

PREDICTION Every distance ladder calibrated alike converges on 67.3 within that scatter; the CMB-independent Lyman-α value already lies 0.6σ from it.

17 Energy

Values in §§17–21 are at $H_0 = 67.35$; r, M_H, E_H scale as H_0^{-1} and N, S_GH, L_H as H_0^{-2}.

17.1 From mass

$E_\mathrm{H} = M_\mathrm{H}c^2 = Fr/2 = c^5/2GH_0 = 8.312 \times 10^{69}$ J.

17.2 From force

$E_\mathrm{H} = \int_0^r F_\mathrm{geom}\, dr' = \frac{1}{2}Fr$, with $F_\mathrm{geom} = F/2$ (§8.4).

17.3 From the axiom

DERIVED The factor $\frac{1}{2}$ is the horizon condition $2GM_\mathrm{H}/c^2 = r$ (§10.2) and the active fraction $p = \frac{1}{2}$ of the force (§6): $E = Frp = Fr/2 = E_\mathrm{H}$, and one cone carries $E_\mathrm{cone} = F_\mathrm{cone}\, r\, p = Fr/8 = E_\mathrm{H}/4 = 2.078 \times 10^{69}$ J, in the volume ratio $V_s = 4V_c$.

18 Mass

$M_\mathrm{H} = E_\mathrm{H}/c^2 = Fr/2c^2 = c^3/2GH_0 = 9.248 \times 10^{52}$ kg, and one cone carries $M_\mathrm{cone} = M_\mathrm{H}/4 = 2.312 \times 10^{52}$ kg; in the units (1), $M_\mathrm{H} = \frac{1}{2}$ and $M_\mathrm{cone} = \frac{1}{8}$.

18.1 Mass as deficit, and the closure mass

STANDARD A straight defect carrying mass M along a length L has deficit angle $\delta = 8\pi GM/c^2L$ [44]: that of a point mass M in 2+1 gravity with the reduced coupling $G_3 = G/L$, $\delta = 8\pi G_3 M/c^2$ [25].

DERIVED (I3) The horizon's cone carries the C_3 deficit π (second axiom) and the mass $M_\mathrm{cone} = c^2r/8G$ (I1), so $\pi = 8\pi GM_\mathrm{cone}/c^2L$ gives $L = r$: the reduction length is the horizon radius, and

$$\delta(M) = \frac{8\pi GM}{c^2 r} = 4\pi\, \frac{M}{M_\mathrm{H}}. \tag{28}$$

Theorem 2 (Closure mass). DERIVED *A spatially closed section of the reduced geometry carrying point masses M_i requires $\sum_i \delta(M_i) = 4\pi$, that is $\sum_i M_i = M_\mathrm{H}$: the horizon mass is the closure mass. The regular tetrahedron realizes it with four masses M_cone, each carrying the deficit $\delta_\mathrm{tot} = \pi$ of one C_3 apex.*

Proof. A closed surface flat except at conical points has total curvature $2\pi\chi = 4\pi$, the sum of its deficits (Gauss–Bonnet; Descartes, (12)); use (28). A vertex of the regular tetrahedron has defect $2\pi - 3 \cdot \frac{\pi}{3} = \pi$ (§4.4). □

The horizon condition (flatness, §12.2), the curvature closure $4\delta_\mathrm{tot} = 4\pi$ (§4.4) and the four-cone partition $M_\mathrm{H} = 4M_\mathrm{cone}$, which I1 reads as $V_s = 4V_c$ (§3.12), are therefore one statement.

19 Momentum

The four-momentum $p^\mu = (E/c, \vec{p})$ has $p^\mu p_\mu = -E^2/c^2 + |\vec{p}|^2 = -m^2c^2$. For $m = 0$, $E = |\vec{p}|\, c$: the cone $h = r$ with height E/c and radius $|\vec{p}|$, so the seed in momentum space is the dispersion relation of light. A massive

particle lives on the hyperboloid inside the cone,

$$E = \sqrt{p^2c^2 + m^2c^4} = mc^2 + \frac{p^2}{2m} - \frac{p^4}{8m^3c^2} + \cdots . \tag{29}$$

The horizon's momentum scale is $M_{\mathrm{H}}c = Fr/2c = 2.773 \times 10^{61}$ kg m s^{-1}.

20 Information

$$I = \frac{N}{2} = \frac{\pi r^2}{\ell_{\mathrm{P}}^2} = \frac{4\pi r^2}{4\ell_{\mathrm{P}}^2} = S_{\mathrm{GH}} = 2.269 \times 10^{122} \text{ nats } (3.275 \times 10^{122} \text{ bits}). \tag{30}$$

STANDARD This is the Bekenstein–Hawking entropy of the full horizon sphere [18, 31, 29]: the full area is twice the effective area, $A = 2A_{\mathrm{eff}} = 2N\ell_{\mathrm{P}}^2$, so $A/4\ell_{\mathrm{P}}^2 = N/2$.

21 Angular Momentum

DERIVED The horizon's mass carried around its radius at light speed,

$$L_{\mathrm{H}} = M_{\mathrm{H}}\, c\, r = \frac{Fr^2}{2c} = \frac{\hbar N}{4\pi} = 3.808 \times 10^{87} \text{ J s}, \tag{31}$$

is the horizon action of §11 ($L_{\mathrm{H}}/\hbar = Q$) and the Noether charge of the horizon's spin (§36.11): $S_{\mathrm{GH}} = 2\pi L_{\mathrm{H}}/\hbar$ and $E_{\mathrm{H}} = (c/r)\, L_{\mathrm{H}} = k_B T_{\mathrm{GH}} S_{\mathrm{GH}}$.

22 Critical Acceleration and a_0

Rotating galaxies obey the radial acceleration relation (RAR): the observed acceleration g is one function of the baryons' Newtonian acceleration g_{bar}, Newtonian at high acceleration and $g \to \sqrt{a_0\, g_{\mathrm{bar}}}$ below $a_0 \approx 1.2 \times 10^{-10}$ m s^{-2} [51, 37], a scale of order $cH_0/6$. GRUT derives the coefficient from the C_3 spectrum (§5) and the function from a threshold response on the cycle; the one modelling input, the exposure law, does not affect the scale (Theorem 4).

22.1 Exposure and survival

DERIVED (I4) A mode on the horizon cycle is activated by a local acceleration a as a Poisson process of rate βx per unit affine parameter s, linear in the load $x \equiv a/a_c$ ($a_c = c/\tau_1 = k_3 cH_0$, §5; derived in §22.10), with β the rate constant of the relaxation operator $d/ds + \beta$ of the eikonal flow. The exposure is one memoryless stage per arc: on a cycle of n arcs it is Erlang, $p_n(s) = (n\beta)^n s^{n-1} e^{-n\beta s}/(n-1)!$, with mean $1/\beta$.

Lemma 3 (Survival)**.** DERIVED *A mode survives unactivated with probability* $\int_0^\infty p_n(s)\, e^{-\beta x s} ds = (1 + x/n)^{-n}$, *independent of* β*; its active fraction is*

$$f_n(x) = 1 - \left(1 + \frac{x}{n}\right)^{-n}. \tag{32}$$

For $n = 1$ the survival $a_c/(a + a_c)$ is the TRSD threshold distribution; a fixed exposure gives e^{-x}. The index n (a Gamma exposure for non-integer n) is the order of the Newtonian tail: a coordinate on shapes, constrained by data (§22.6–22.7), not a count.

22.2 Activation and response

IDENTIFICATION The response along a direction is the load times the active fraction, $j_n(x) = x\, f_n(x)$.

22.3 Isotropic average

DERIVED The radial field averages the response over directions: with $u = \cos\theta$ the projected load is xu, its radial projection carries a second factor u, and isotropy weights u uniformly on $[0, 1]$; the factor 3 makes a linear response return itself:

$$\sigma_n(x) = 3\int_0^1 j_n(xu)\, u\, du, \qquad \mu_n(x) \equiv \frac{\sigma_n(x)}{x} = 3\int_0^1 u^2\, f_n(xu)\, du. \tag{33}$$

With $w = 1 + xu/n$ the integrals close (checked against quadrature to 10^{-12}):

$$\mu_1(x) = 1 - \frac{3}{2x} + \frac{3}{x^2} - \frac{3\ln(1+x)}{x^3}, \tag{34}$$

$$\mu_2(x) = 1 - \frac{24}{x^3}\Big[\frac{x}{2} + \frac{x/2}{1+x/2} - 2\ln\Big(1+\frac{x}{2}\Big)\Big], \tag{35}$$

$$\mu_3(x) = 1 - \frac{81}{x^3}\Big[\ln\Big(1+\frac{x}{3}\Big) + \frac{2}{1+x/3} - \frac{1}{2(1+x/3)^2} - \frac{3}{2}\Big], \tag{36}$$

$$\mu_\infty(x) = 1 - \frac{3}{x^3}\Big[2 - e^{-x}\big(x^2+2x+2\big)\Big], \tag{37}$$

and at small load $\mu_n = \frac{3}{4}x - \frac{3(n+1)}{10n}x^2 + \frac{(n+1)(n+2)}{12n^2}x^3 - \cdots$.

Theorem 4 (Universality of $\frac{3}{4}$)**.** DERIVED *If (a) activation is a Poisson process whose hazard is linear in the load, (b) the exposure has a finite mean, used as the unit, (c) the response is the load times the active fraction, and (d) directions are averaged as in (33), then $\mu(x) = \frac{3}{4}x + O(x^2)$ for every exposure distribution.*

Proof. The survival $1-f$ is the Laplace transform of the exposure density in units of its mean, so $f(y) = y + O(y^2)$ and $\mu(x) = 3\int_0^1 u^2(xu)\,du + O(x^2) = \frac{3}{4}x + O(x^2)$. □

Outside the class (a)–(d) the slope changes, and (a) is itself required by data:

Proposition 5 (Linear hazard)**.** DERIVED *A hazard growing as x^p gives $\mu \simeq \frac{3}{3+p}x^p$ and, in the deep regime, $g \propto g_{\rm bar}^{1/(1+p)}$: rotation curves are asymptotically flat, with $V^4 \propto M$, only for $p = 1$, which the baryonic Tully–Fisher relation [67] therefore fixes. If the exposure mean diverges, so does $f'(0)$, and μ is not linear at small x.*

Proposition 6 (Transition and tail)**.** DERIVED *For real $n \neq 1, 2, 3$, with $\xi = x/n$ and $w = 1 + \xi$,*

$$1 - \mu_n(x) = \frac{3}{\xi^3}\Big[\frac{w^{3-n}-1}{3-n} - \frac{2(w^{2-n}-1)}{2-n} + \frac{w^{1-n}-1}{1-n}\Big], \tag{38}$$

so the tail is $1 - \mu_n \simeq \frac{3n^n}{3-n}\,x^{-n}$ for $n < 3$, while the transition moves little: $\mu_n(1) = 0.4206,\ 0.4515,\ 0.4557,\ 0.4601,\ 0.4623$ for $n = 1,\ 1.62,\ 1.75,\ 1.91,\ 2$. In a one-exponent form such as $z/(1+z^n)^{1/n}$ the one exponent sets both the transition width and the tail, $1 - \mu \simeq z^{-n}/n$.

Proof. Substitute $w = 1+\xi u$ in $3\int_0^1 u^2(1+\xi u)^{-n}du$; the leading term at large ξ is $3\xi^{-n}/(3-n)$. The one-exponent tail is the expansion of $(1+z^{-n})^{-1/n}$. □

22.4 Field equation

STANDARD The Lagrangian $\mathcal{L} = -\frac{a_c^2}{8\pi G}\,\mathcal{F}\big(|\nabla\Phi|^2/a_c^2\big) - \rho\Phi$ with $\mathcal{F}'(y) = \mu(\sqrt{y})$ gives the AQUAL form [52], here with a derived μ,

$$\nabla\cdot\Big[\mu\Big(\frac{|\nabla\Phi|}{a_c}\Big)\nabla\Phi\Big] = 4\pi G\rho; \tag{39}$$

in spherical symmetry Gauss's theorem gives $\mu(g/a_c)\,g = g_{\rm bar}$ exactly.

22.5 The scale

DERIVED In the deep regime $\mu \simeq \frac{3}{4}\,g/a_c$, so $\frac{3}{4}g^2/a_c = g_{\rm bar}$ and $g = \sqrt{\frac{4}{3}a_c\,g_{\rm bar}}$:

$$a_0 = \frac{4}{3}\,a_c = \frac{\sqrt{3}}{\pi^2}\,cH_0, \qquad \frac{\sqrt{3}}{\pi^2} = 0.175493, \tag{40}$$

the coefficient $k_3 = 3\sqrt{3}/4\pi^2$ of the first C_3 eigenvalue ($a_c = c/\tau_1$, §5) times the isotropic $\frac{4}{3}$. Neither factor is adjustable, and neither depends on the exposure law (Theorem 4).

Prediction. $a_0 = 1.148 \times 10^{-10}\ \mathrm{m\,s^{-2}}$ at the resonance budget's $H_0 = 67.35$ (§25.11); $a_0 \propto H_0$, so 1.245×10^{-10} at the SH0ES 73.04. The RAR measures $1.20 \pm 0.02\,(\mathrm{stat}) \pm 0.24\,(\mathrm{sys}) \times 10^{-10}$ [37], so both agree within the systematic band; read in reverse it gives $H_0 = 70.4 \pm 1.2 \pm 14$. Once the systematic falls below 5%, the same measurement will also choose between the calibrations of H_0.

22.6 The radial acceleration relation

MEASURED Against the 2696 points of 147 SPARC galaxies [37, 62], with mass-to-light ratios $\Upsilon_{\rm disk} = 0.5$ and $\Upsilon_{\rm bul} = 0.7$, a_c fixed by H_0 and nothing fitted, the residuals $\log_{10}(g_{\rm obs}/g_{\rm pred})$ have mean -0.007, $+0.016$, $+0.024$ dex and rms 0.133, 0.134, 0.136 for μ_1, μ_2, μ_3 (computed at $H_0 = 67.12$; at 67.35 every mean is lower by less than 0.001 dex, at 73.04 by 0.012–0.013). The observed scatter is 0.13 dex and the best empirical fit reaches 0.133: with no parameter adjusted the family reproduces the galaxies to within 0.024 dex (6%) in the mean (Figure 1). Fitting the scale alone (errors from a bootstrap over galaxies) gives $a_0 = 1.105 \pm 0.071$ (μ_1) and 1.301 ± 0.079 (μ_3), the empirical fit returning $g_\dagger = 1.161 \pm 0.071$ against the published 1.20; through (40) the galaxies measure $H_0 = 64.8 \pm 4.2 \pm 13$ (μ_1) or $76.3 \pm 4.6 \pm 15$ (μ_3), bracketing 67.35. The family deviates from the empirical function ($g_\dagger = 1.20$) by at most 0.022, 0.038 and 0.069 dex for $n = 1$, 3, ∞.

The rotation-curve likelihood. MEASURED The RAR points take mass-to-light ratios, distances and inclinations as known. The full likelihood [66] fits the velocities of 153 galaxies (quality flag below 3, inclination $\geq 30°$; 3168 points) with 490 per-galaxy nuisance parameters: $\log \Upsilon_{\rm disk}$ and $\log \Upsilon_{\rm bul}$ with lognormal priors of 0.1 dex about 0.5 and 0.7, and distance and inclination with Gaussian priors from the SPARC table, minimized galaxy by galaxy while the global scale is scanned. The best χ^2 is 14370 for 3167 degrees of freedom, $\chi^2_{\rm red} = 4.54$, so errors are scaled by $\sqrt{\chi^2_{\rm red}} = 2.13$ (median per-galaxy reduced χ^2: 2.1). The threshold family with real n and TRSD's original one-exponent Torridity form $z/(1+z^n)^{1/n}$ [63] are compared, with the empirical function as a benchmark:

form	n	a_0 [10^{-10}]	$\Delta\chi^2/\chi^2_{\rm red}$	cost of $a_0 = 1.144$
threshold	1	1.063	4.7	3.5
threshold	1.38 (optimum)	1.045	0	5.5
threshold	1.75 (mid-window)	1.037	2.2	6.7
threshold	2	1.034	5.1	7.2
threshold	3	1.029	18.6	8.3
Torridity	0.97 (optimum)	1.057	4.4	4.0
Torridity	$\pi/3$	1.081	9.2	2.2
Torridity	$2\pi/3$	1.201	397	2.7
benchmark	–	1.009	-13.6	–

Scale errors are ± 0.04 (statistical, scaled). The fourth column is measured from the best threshold member; the last is $\Delta\chi^2/\chi^2_{\rm red}$ for fixing the scale at the theory value of $H_0 = 67.12$, measured from the member's own best scale. At 67.35 the theory value is 1.148 and each pull, the square root of that cost, grows by 0.1 (falls by 0.1 for $2\pi/3$).

The shape parameter is not the physics: n is a coordinate on shapes, and the two forms assign different values, 1.38 and 0.97, to nearly the same transition. The galaxies constrain the width of the transition, the planets the tail (§22.7). The threshold family fits better than any Torridity exponent (by 4.4). The empirical function leads it by 13.6, but its form was chosen on these same data [37]: it is a benchmark derived from nothing, not a competing prediction.

The scale depends little on the shape: $a_0 = 1.034$–1.045 for $1.38 \leq n \leq 2$. With the nuisance parameters fixed at their nominal values the likelihood returns $g_\dagger = 1.206$, reproducing the published 1.20; profiling them lowers every scale by about 16%, while moving the $\Upsilon_{\rm disk}$ prior centre to 0.4 or 0.6 shifts it by only ± 1.5%. With half the fixed-versus-profiled shift as the systematic, the window of §22.7 gives $a_0 = 1.04 \pm 0.04\,({\rm stat}) \pm 0.10\,({\rm nuisance})$: the prediction 1.148 is 1.0σ high (2.7σ on the statistical error alone; 1.9σ for 1.245 at $H_0 = 73.04$). The agreement is conditional on the nuisance model.

22.7 Newtonian recovery and the transition–tail decoupling

DERIVED At high acceleration $1 - \mu_n = 3\int_0^1 u^2(1 + xu/n)^{-n}du$, and the tails differ:

$$1 - \mu_1 \simeq \frac{3}{2x}, \qquad 1 - \mu_2 \simeq \frac{12}{x^2}, \qquad 1 - \mu_3 \simeq \frac{81}{x^3}\Big[\ln\frac{x}{3} - \frac{3}{2}\Big], \qquad 1 - \mu_\infty \simeq \frac{6}{x^3}. \tag{41}$$

For $n = 1$ the anomaly $\Delta g \equiv g - g_{\rm bar}$ tends to the constant $\frac{3}{2}a_c = 1.29 \times 10^{-10}\ {\rm m\,s^{-2}}$ at every planet. An inward anomaly $\Delta g \propto d^{\,k}$ (d the heliocentric distance) turns a perihelion back by $\Delta\varpi = -(k+2)\pi\,\Delta g\, a^2\sqrt{1-e^2}/GM_\odot$ per orbit (a, e the semi-major axis and eccentricity; exact for $k = 0$, leading order in e otherwise): -1.7×10^3 mas/cy for Mercury and -2.8×10^3 mas/cy for the Earth, against the ephemeris residuals -2.0 ± 3.0 and 0.19 ± 0.19 mas/cy [55]. The single-stage tail is excluded by factors of 6×10^2 and 1.5×10^4.

A tail x^{-n} has $k = 2n - 2$. Both perihelia within 2σ of their residuals (the Earth's binds) require $n \geq 1.62$ for the threshold family and $n \geq 1.53$ for the Torridity form; the galaxies give 1.38 ± 0.20 and 0.97 ± 0.04 (§22.6).

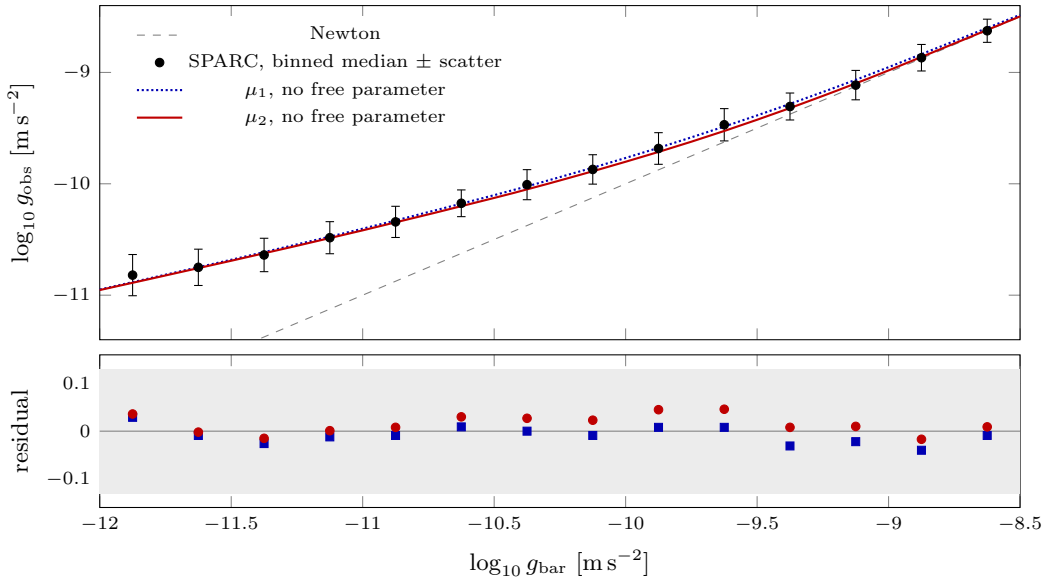


Figure 1: The SPARC radial acceleration relation against GRUT at $H_0 = 67.35$, no parameter fitted. Top: binned medians of the 2696 points (bars: scatter per bin). Bottom: median residuals per bin, μ_1 (squares) and μ_2 (circles); shaded, the observed scatter of 0.13 dex.

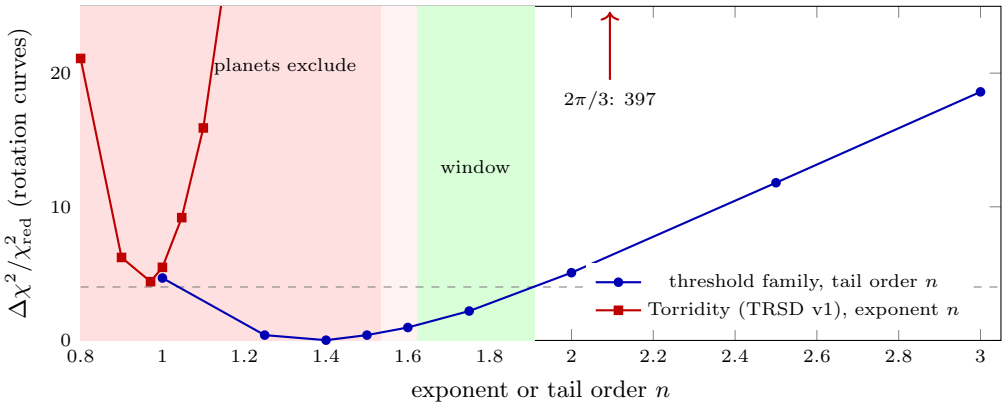


Figure 2: Rotation-curve likelihood against the shape parameter of the two forms, from the best threshold member. Red: excluded by the perihelia of Mercury and the Earth (both forms below 1.53, the threshold family below 1.62). Green: the window where the threshold family meets both. Dashed: 2σ. Wherever the planets allow the Torridity form, the galaxies exclude it.

DERIVED The one-exponent form therefore fails for every exponent: the galaxies and the planets ask for different values of its one parameter, which is why TRSD's first version needed a screening factor. Its original exponent $\pi/3$ lies at $\Delta\chi^2/\chi^2_{\rm red} = 9.2$ and turns the perihelia back by 6×10^2 (Mercury) and 1.1×10^3 mas/cy (Earth); its disk-galaxy exponent $2\pi/3$ passes the planets and lies at 397 (Figure 2). The threshold family separates transition and tail (Proposition 6) and meets both in the window $1.62 \le n \le 1.91$, whose upper edge is $\Delta\chi^2/\chi^2_{\rm red} = 4$. At $n = 1.75$ the perihelion shifts are -0.004 (Mercury) and -0.028 mas/cy (Earth); at the lower edge the Earth's reaches -0.20 mas/cy, the present sensitivity.

Proposition 7 (Running exponent)**.** DERIVED *Let the local exponent $m(z)$ at $z = g/a_0$ be the exponent of the one-parameter form through the same point, $z/(1+z^m)^{1/m} = \mu_n(\frac{4}{3}z)$. For the threshold family with $n < 3$, $m \to 1$ as $z \to 0$ and $m \to n$ as $z \to \infty$.*

Proof. At small z, $\mu_n = z - b_n z^2 + O(z^3)$ with $b_n = 8(n+1)/15n > 0$ (§22.3), while the one-parameter form is $z - z^{1+m}/m + \ldots$; matching gives $m - 1 = \ln(b_n m)/\ln z \to 0$. At large z, $1 - \mu_n \simeq \frac{3n^n}{3-n}(\frac{4}{3}z)^{-n}$ (Proposition 6) against z^{-m}/m, so $m \to n$. □

There is therefore no universal exponent: it runs from 1 far out to the tail order close in, and each system reads it at its own acceleration:

	$\log_{10} g$	m ($n = 1.62$)	m ($n = 1.75$)	m ($n = 1.91$)
far out	-14	1.014	1.017	1.020
galaxy outskirts	-11.5	1.032	1.038	1.045
galaxy transition	-10.5	1.059	1.070	1.082
inner galaxies	-9.5	1.135	1.161	1.190
Saturn	-4.19	1.507	1.612	1.740
the Earth	-2.23	1.534	1.646	1.782
Mercury	-1.40	1.542	1.656	1.794

Galaxies (10^{-12}–10^{-9} m s^{-2}) read the exponent between 1.03 and 1.28, the planets between 1.51 and 1.79. A one-parameter form must use one value at every acceleration, and fails; the running exponent meets both. The planets' requirement on a one-parameter form, $m \geq 1.534$ at the Earth, is what the lower edge of the window reads there: the two planetary conditions are one, to 0.003. TRSD's original exponent $\pi/3 = 1.047$ is the running exponent of galaxy outskirts ($m = 1.04$–1.06 at $g = 10^{-11}$ m s^{-2} across the window), measured where it holds; the regime dependence TRSD anticipated with multiples of $\pi/3$ is realized by the running exponent, but $2\pi/3$ is not what disks read.

A multiplicative screening factor cannot play this role: $1 - \mu_{\rm scr} = (1 - \mu_1)(1 + x)^{-\beta_{\rm s}}$ gives $\mu_{\rm scr} \simeq (\frac{3}{4} + \beta_{\rm s})x$ at small x, changing the deep-regime coefficient and with it a_0. TRSD's $\beta_{\rm s} = \pi/2$ would lower the scale to $a_c/(\frac{3}{4} + \frac{\pi}{2}) = 0.43\, a_c$ and move the fixed-scale fit to SPARC by +0.18 dex (rms 0.235). Every exposure law leaves $\frac{3}{4}$ untouched.

DERIVED The Solar-System test that probes μ near $x \sim 2$ is the quadrupole induced by the Galactic external field [53], which Cassini ranging constrains tightly [54]: across the window it is $(3.0$–$3.2) \times 10^{-26}$ s^{-2} for the baryon-only equation, excluded at 16σ (§49.11). In the complete equation (§46–47) and the rotation reading (§23) the planets see no bound medium, and the quadrupole is the Galactic tide alone, at -0.9σ (§49.10).

22.8 Status of the TRSD regime scales

TRSD's dwarf and cluster scales, written as (pure number)$\times 10^{-10}$ m s^{-2}, depend on the choice of the metre and the second and cannot be framework statements (Theorem 1). GRUT retains the single scale $a_0 = (\sqrt{3}/\pi^2)\, cH_0$; a dwarf or cluster scale would enter as a dimensionless multiple of a_0 derived from the geometry.

22.9 The dynamic exponent

DERIVED Each symmetry-preserving level j of the cascade (§44, TRSD) is a closed C_3 structure with $N_a = 2j + 1$ arcs enclosing $N_s = 2j$ regions; its arcs are exponential exposure stages shared equally by its regions. Dividing the level's exposure, Gamma(N_a), into N_s independent equal parts takes the unique root $(1 - it)^{-N_a/N_s}$ of its characteristic function $(1 - it)^{-N_a}$, so each region is exposed with order

$$n_j = \frac{N_a}{N_s} = \frac{2j + 1}{2j} = \tfrac{3}{2},\ \tfrac{5}{4},\ \tfrac{7}{6},\ \ldots \tag{42}$$

IDENTIFICATION An element is exposed at level j in proportion to the fraction of structure still open there, the closure weight $\kappa_{2j-1} = 1/[2j(2j + 1)] = \frac{1}{2j} - \frac{1}{2j+1}$ of the full ladder; over the symmetric ladder these sum to $1 - \ln 2$, so

$$w_j = \frac{1}{(1 - \ln 2)\, 2j(2j + 1)} = 0.543,\ 0.163,\ 0.078,\ \ldots \tag{43}$$

Every level is exposed over the same horizon period (§22.10), so the levels differ in order, not in scale:

$$\boxed{f(y) = \sum_{j \geq 1} w_j\, f_{n_j}(y), \qquad \mu(x) = 3 \int_0^1 u^2 f(xu)\, du = \sum_{j \geq 1} w_j\, \mu_{n_j}(x),} \tag{44}$$

each μ_{n_j} from (38). Nothing is fitted: the orders are the arc-to-sphere ratios and the weights the closure weights.

The order is dynamic. The boost is carried by the surviving elements (§46.6), whose exposure order at load x is

$$\bar{n}(x) = \frac{\sum_j w_j n_j\, [1 - \mu_{n_j}(x)]}{\sum_j w_j\, [1 - \mu_{n_j}(x)]}, \qquad \bar{n}(0) = \sum_j \frac{1}{4j^2(1 - \ln 2)} = \frac{\pi^2}{24(1 - \ln 2)} = 1.340; \tag{45}$$

it falls to 1.307 at $x = 10$ and 1.239 at $x = 100$: under strong load only the deepest levels, of order near one, still carry survivors. In TRSD's Torridity form the same law has the running exponent of Proposition 7, here $n(z)$:

$$\mu = \frac{z}{\left(1 + z^{\,n(z)}\right)^{1/n(z)}}, \qquad z = \frac{g}{a_0}, \tag{46}$$

$\log_{10} g$ [m s^{-2}]	−13	−12	−11.5	−11	−10.5	−10	−9.5	−9	−8
$n(z)$	1.008	1.011	1.014	1.018	1.026	1.040	1.065	1.095	1.144

It tends to one in the deep regime, where the first correction to μ is analytic in x, rises through the transition, passes $\pi/3$ at $g = 1.27\,a_0$—TRSD's exponent is the dynamic exponent at the transition—and peaks at 1.158 near $10^3 a_0$.

Limits. Every level has unit mean exposure, so $\mu = \frac{3}{4}x + O(x^2)$ (Theorem 4) and $g^2 = a_0 g_{\rm bar}$ in the deep regime. Toward the Newtonian regime $1-\mu$ falls only slightly faster than $1/x$ ($x(1-\mu) = 0.63$, 0.41, 0.29 at $x = 10^2$, 10^3, 10^4), because the levels of order near one dominate the approach. At planetary loads such a tail is excluded (§22.7); capture switches it off there, the Sun holding none of the medium ($\Gamma = 7\times10^{-9}$, §49.10).

PREDICTION The boost is $g/g_{\rm bar} = 3.67$, 1.58, 1.067 and 1.019 at $g_{\rm bar} = 0.1$, 1, 10 and $30\,a_0$, and the excess $g - g_{\rm bar}$ peaks at $0.70\,a_0$ at $g_{\rm bar} = 4.7\,a_0$ (§47.4).

The MOND-like equation. With the capture fraction Γ of §47 the equation is

$$\boxed{g\left[1-\Gamma\big(1-\mu(g/a_c)\big)\right] = g_{\rm bar}, \qquad \mu(x) = \sum_{j\geq1} w_j\,\mu_{n_j}(x), \qquad a_c = \frac{3\sqrt{3}}{4\pi^2}\,cH_0.} \tag{47}$$

Full capture gives $g^2 = a_0 g_{\rm bar}$ in the deep regime and $g \to g_{\rm bar}$ in the Newtonian one; no capture gives Newton at every acceleration; partial capture, $0 < \Gamma < 1$, gives a constant enhancement $g \simeq g_{\rm bar}/(1-\Gamma)$ in the deep regime instead of the square-root law. The exponent is no longer an input: the cascade levels that still carry survivors set it at each acceleration, and the equation changes with acceleration, environment and epoch through $n(z)$ and Γ (§22.6–22.7 test fixed orders). At the horizon's own load, $z_H = GM_{\rm H}/(r^2 a_0) = \pi^2/2\sqrt{3}$, the exponent is 1.0654 and fixes the baryon ratio (§25.7).

22.10 The map $a_c = c/\tau_1$ and the exposure law

DERIVED An accelerated clock approaches its horizon at the rate a/c (§43B.8). With activations at this rate and the exposure one period of the fundamental C_3 mode, $\tau_1 = 2\pi/\omega_1 = 4\pi^2 r/(3\sqrt{3}\,c)$ (§5), the mean number of events in an exposure is

$$\lambda = \frac{a\,\tau_1}{c} = \frac{a}{a_c}, \qquad a_c = \frac{c}{\tau_1} = \frac{3\sqrt{3}}{4\pi^2}\,cH, \tag{48}$$

so the map $a_c = c/\tau_1$ states that the threshold is one event per cycle ($\tau_1 = 110.3$ Gyr and $c/\tau_1 = 8.612\times10^{-11}$ m s^{-2} today) and that the hazard is linear in the load: hypothesis (a) of Theorem 4, derived, as the Tully–Fisher slope requires (Proposition 5). If the cycle completes in n exponential stages, the exposure U in units of τ_1 is Gamma-distributed with unit mean, and the survival $\langle e^{-\lambda U}\rangle = (1+x/n)^{-n}$ is the threshold family; its order is set by the cascade (§22.9), and a fixed order is confined to the window of §22.7.

DERIVED *The exposure law from the Unruh response.* A mode carried with proper acceleration a sees the vacuum as the Rindler thermal state [117]. A detector of gap ω coupled with strength ϵ to a massless field is excited at the rate per unit proper time

$$\dot F(\omega) = \frac{\epsilon^2}{2\pi}\,\frac{\omega}{e^{2\pi c\,\omega/a}-1} \xrightarrow{\omega\to0} \frac{\epsilon^2 a}{4\pi^2 c}, \tag{49}$$

so a gapless mode is excited at a rate linear in the acceleration, that is, in the load. At weak coupling the transitions in the thermal state are independent (Fermi's rule): their count is Poisson and the stages between them memoryless. One activation per unit of horizon approach, $a\,d\tau/c$, is the normalization $\epsilon^2 = 4\pi^2$, and it follows from the mode's extent: the activated mode is the whole C_3 cycle, at unit radius the circle $S^1(2\pi)$ of the thermal factor (§43F), and with unit coupling per unit length, the only normalization that introduces no further scale, its coupling is the cycle's length, $\epsilon = 2\pi$. I4 is derived.

23 The Rotation of the Universe

HYPOTHESIS The universe rotates as a whole: the phase θ of the conjugate pair $\phi_\pm = \phi\,e^{\pm i\theta}$ turns in time at one rate, the final expansion rate $H_\infty = c\sqrt{\Lambda/3}$, one quantum of the horizon's spin (§36.14). Today, at the resonance (H3, §36.7), it equals the frequency of the horizon's lowest C_3 mode, of eigenvalue $\lambda_1 = 27/4\pi^2 r^2$:

$$\Omega_0 = H_\infty, \qquad \Omega_0 = c\sqrt{\lambda_1} = \frac{3\sqrt{3}}{2\pi}H_0 = 0.827\,H_0 \ \text{ today.} \tag{50}$$

DERIVED *An internal rotation.* Space does not rotate. On the background $\theta = \Omega_0 t$, and the flow $u_\mu = \partial_\mu\theta/\sqrt{X}$ (§23.1) is a gradient, so its vorticity vanishes identically: the CMB bound on vorticity and vector shear, $(\sigma_V/H)_0 < 4.7 \times 10^{-11}$ at 95% [115], which a rotation of space at $0.827H_0$ would exceed by ten orders of magnitude, is met exactly. Nor do windings carry angular momentum: the phase is single-valued on the closed horizon (a section of a trivial line bundle), so its integer windings around the three defects sum to zero and, equal by the C_3 symmetry, each vanishes, and with them the phase's momentum density (§52.2), $T_{\hat{t}\hat{\varphi}} = \chi_0^2\, n\, \Omega_0/(c\varrho)$ at distance ϱ from the axis. A global frame dragging (§52.2) is excluded by the vorticity bound; the extra gravity of galaxies is carried by the rotation's medium bound around the baryons (§23.1).

23.1 The cosmic share: the rotation held at its rate

DERIVED The fixed rate is a constraint. With a Lagrange multiplier λ,

$$S_{\rm rot} = \int d^4x\,\sqrt{-g}\,\lambda\,(X - X_0), \qquad X = -g^{\mu\nu}\partial_\mu\theta\,\partial_\nu\theta, \qquad X_0 = \Big(\frac{\Omega_0}{c}\Big)^2 = \frac{\Lambda}{3}, \tag{51}$$

with no new parameter (I8, derived up to an integer in §36.14). Varying λ holds $X = X_0$; varying θ gives the conserved flow $\nabla_\mu(\lambda\,\partial^\mu\theta) = 0$, so $\lambda \propto a^{-3}$; on the constraint $T_{\mu\nu} = 2\lambda\,\partial_\mu\theta\,\partial_\nu\theta$, pressureless dust:

$$\rho_{\rm rot} = 2\lambda X_0 \propto a^{-3}, \qquad p = 0, \qquad c_s^2 = 0 \text{ on every background.} \tag{52}$$

Its amount is a constant of integration: dark matter without a particle (mimetic dark matter [110]), with cold-dark-matter perturbations. The constraint vanishes on its solutions, adding no vacuum energy: dark energy remains space, the capacity energy of §48.10. The kinetic function $K(X-X_0)^2$ [109] is the same physics with finite stiffness, (51) its rigid limit. Without the constraint the scalar's own kinetic function, $P = f^2X + f^4X^2/8$ with the radial mode integrated out (f its normalization), has its only extremum at a spacelike gradient, $X = -4/f^2$, where $\Xi = \phi_+\phi_- = 0$: the constraint is what holds the time rotation.

IDENTIFICATION The dust is the cosmic share Ω_c of the budget; the Boltzmann fit of §26.7 treats it as pressureless dust of zero sound speed.

DERIVED *Dwarf profiles.* Around a cored baryonic system the medium's effective density follows from the law alone, with no Jeans length. For a Plummer sphere (mass M, scale b) in the deep regime, $g = \sqrt{a_0 g_N}$ and $4\pi G(\rho_b + \rho_{\rm ph}) = r^{-2}\,d(r^2 g)/dr$ give

$$\rho_{\rm ph} \simeq \frac{5}{8\pi}\sqrt{\frac{Ma_0}{G}}\,b^{-3/2}\,r^{-1/2} \quad (r \ll b), \qquad \rho_{\rm ph} \simeq \frac{1}{4\pi}\sqrt{\frac{Ma_0}{G}}\,r^{-2} \quad (r \gg b)\,: \tag{53}$$

a cusp of logarithmic slope $-\frac{1}{2}$ inside the baryonic core, isothermal outside, in place of the Jeans-length cores of §46.5.

DERIVED *One quantity at two scales.* The dust is the modulation the baryons tune ($\delta\omega_b$, §49.2): unbound, cold dust; bound in a galaxy, a polarization of permittivity μ (§46.6) whose effective density is the phantom density of §46 (§49.11); bound in a cluster, the infall share of §47. Pressureless, it follows the galaxies through a cluster collision and leaves the gas, each stream a branch of the multivalued phase. Dust and vacuum are spacetime excited and unexcited, two states of the spin's quanta; what lensing infers as dark matter is the dilation (§53.4), and the Bullet Cluster's offset follows from the dust's geodesic flow (§53.6).

23.2 What this means for the missing mass

No dark particle, no new force: the extra gravity is the universe's rotation, turning everywhere with zero windings, modulated by the baryons and bound around them (§23.1, §49.11).

23.3 What this is

General relativity with a scalar whose phase turns at a fixed rate; the carrier of the extra gravity is the universe's own rotation, which the baryons tune rather than generate.

23.4 The dark-sector sections, read through the rotation

The condensate of Sections 37, 46 and 47 is the rotation's modulation held by (51), not a population of particles: their equations stand with the rotation as carrier, and the capture and relaxation of §47 are the bound medium's. The radial mode keeps its mass m_δ (§37) as an excitation of the field.

24 The Three Currencies

24.1 Mass, energy, information

With $F = c^4/G$, $r = c/H_0$ and N the horizon's cell count,

$$M_{\mathrm{H}} = \frac{Fr}{2c^2}\ (\propto r), \qquad E_{\mathrm{H}} = \frac{Fr}{2}\ (\propto r), \qquad I = \frac{N}{2} = \frac{\pi r^2}{\ell_{\mathrm{P}}^2}\ (\propto r^2). \tag{54}$$

24.2 Exchange rates

The horizon has the Gibbons–Hawking temperature [29], $k_B T_{\mathrm{GH}} = \hbar c/2\pi r = \hbar H_0/2\pi$ ($T_{\mathrm{GH}} = 2.653 \times 10^{-30}$ K), and energy and information exchange at exactly that rate:

$$k_B T_{\mathrm{GH}}\, I = \frac{\hbar c}{2\pi r} \cdot \frac{\pi r^2 c^3}{\hbar G} = \frac{c^4 r}{2G} = E_{\mathrm{H}}. \tag{55}$$

So $E_{\mathrm{H}}/M_{\mathrm{H}} = c^2$ and $E_{\mathrm{H}}/I = k_B T_{\mathrm{GH}} = 3.662 \times 10^{-53}$ J per nat; a bit costs $k_B T_{\mathrm{GH}} \ln 2 = 2.539 \times 10^{-53}$ J, Landauer's erasure energy [35] at the horizon temperature, of mass 2.82×10^{-70} kg.

STANDARD For the de Sitter horizon the exchange law is holographic equipartition [97]: the Komar energy $2E_{\mathrm{H}}$ equals $\frac{1}{2} N_{\mathrm{sur}} k_B T_{\mathrm{GH}}$ with $N_{\mathrm{sur}} = 4\pi r^2/\ell_{\mathrm{P}}^2 = 2N$. GRUT reads it as the exchange rate between mass and information.

24.3 The cone rate

The TRSD rate $\hbar H_0/8\pi$ is the one-cone value $k_B T_{\mathrm{GH}}/4$, matching $M_{\mathrm{cone}} = M_{\mathrm{H}}/4$.

24.4 Consistency

$$M_{\mathrm{H}} c^2 = E_{\mathrm{H}} = k_B T_{\mathrm{GH}}\, I. \tag{56}$$

Mass, energy and information are one quantity in three units, exchanged at c^2 and at the horizon temperature. The exchange is the horizon's spin: $E_{\mathrm{H}} = \omega L_{\mathrm{H}}$ with $\omega = c/r$ and $L_{\mathrm{H}} = c^3 r^2/2G$; $k_B T_{\mathrm{GH}} = \hbar\omega/2\pi$ and $I = 2\pi L_{\mathrm{H}}/\hbar$ (§36.11).

25 The Cosmic Budget

25.1–25.3 The C_3 budget

IDENTIFICATION The baryon ratio is TRSD's product $\Omega_b/\Omega_c = \alpha_{\mathrm{TRSD}}\, n$ of the acceleration coefficient $\alpha_{\mathrm{TRSD}} = \sqrt{3}/\pi^2$ ($a_0 = \alpha_{\mathrm{TRSD}} c H_0$, §22.5) and the Torridity exponent n: $\sqrt{3}/3\pi = 0.18378$ at $n = \pi/3$, 0.18697 as derived (§25.7). The dark ratio was taken from the C_3 geometry too (I5), as the defect power ratio $\Omega_\Lambda/\Omega_c = C_3/C_2 = 5/2$ (§39.3). *That power ratio is withdrawn* (§52.6): the CMB spectra disfavour it at 4.6σ (§25.9), the dynamics leaves the dark amount free (§23.1), and the resonance gives 2.580 (§25.11).

DERIVED *The dark sectors by axial charge.* The three-defect source (§39.3) admits at lowest order the quadrupole $\ell = 2$, $m = 0$, invariant under every rotation about the axis, and the octupole $\ell = 3$, $m = \pm 3$, invariant only under C_3. The internal rotation is a continuous symmetry, so the cosmic share is the axial quadrupole; the three-arc cells of space carry the discrete pattern, so the vacuum is the triangular octupole. The assignment fixes harmonics, not densities.

25.4 Matter as the frame

By the partition (24), matter holds the fraction Ω_m of the horizon radius and the vacuum holds the rest.

25.5 Totals

$M_{\mathrm{total}} = M_{\mathrm{H}}/4 = c^3/8GH_0$, $E_{\mathrm{total}} = E_{\mathrm{H}}/4 = c^5/8GH_0$, and $N = 2\pi c^5/\hbar G H_0^2$: the one-cone values.

25.6 Comparison with data

The framework's matter density (§25.11) against the data:

Dataset	Ω_m measured	GRUT 0.3160
Planck 2018 [39]	0.3153 ± 0.0073	$+0.1\sigma$
Pantheon+ supernovae [20]	0.334 ± 0.018	-1.0σ
DES-Y5 supernovae [22]	0.352 ± 0.017	-2.1σ
DESI DR2 BAO [23]	0.2975 ± 0.0086	$+2.2\sigma$

and -0.5σ from DESI Lyman-α. Fitted to the CMB (§26.7), the budget matches the acoustic angle by construction, with ω_b at $+0.9\sigma$ and ω_c at $+0.3\sigma$. The largest variance, DESI DR2's, is ΛCDM's own: Planck's Ω_m lies $+2.1\sigma$ from it and -2.2σ from DES-Y5 (Appendix D.5).

25.7 The baryon ratio

A derivation must tie baryons, which carry $\mathbb{Z}_3$ triality, to the cosmic share. Four mechanisms fail (at $\Omega_m \approx \frac{1}{3}$): (i) the baryons' phantom mass at the horizon, M_b/μ at the load $g/a_c = \Omega_m/2k_3 = 1.27$ ($a_c = \frac{3}{4}a_0$, $k_3 = 3\sqrt{3}/4\pi^2$), gives $\Omega_b/\Omega_c \approx 0.9$–$1.0$, five times too large: cosmic dark matter is not horizon-scale phantom mass, as the CMB's third peak (a component clustering before recombination) also says; (ii) deep-MOND scaling, $M_{\rm D}^2 = a_0 r^2 M_b/G$, gives $\Omega_c^2 = 2\alpha_{\rm TRSD}\Omega_b$ and 0.595; (iii) equal number densities need a 5 GeV particle, not the ultralight field of §37; (iv) equipartition over the three C_3 modes gives 1:2.

DERIVED *(v) The dynamic exponent at the horizon's own load.* The exponent runs with the acceleration (§22.9); the universe reads it at its horizon's load. By the horizon condition $r = 2GM_{\rm H}/c^2$ (§10.2), with $r = ct_H$ on the seed cone,

$$g_H = \frac{GM_{\rm H}}{r^2} = \frac{c^2}{2r} = \frac{r}{2t_H^2}, \qquad z_H = \frac{g_H}{a_0} = \frac{\pi^2}{2\sqrt{3}} = 2.8491, \tag{57}$$

a pure number in every epoch because the scale tracks the horizon, $a_0 \propto H$ (§36.15), as the rising scale of high-redshift rotation curves requires (a scale fixed by the spin would make z_H run).

IDENTIFICATION Reading the exponent there, the dynamic law of §22.9 gives $n(z_H) = 1.06540$ and

$$\boxed{\frac{\Omega_b}{\Omega_c} = \frac{\sqrt{3}}{\pi^2}\, n\Big(\frac{\pi^2}{2\sqrt{3}}\Big) = 0.18697,} \tag{58}$$

against Planck's $\omega_b/\omega_c = 0.1864 \pm 0.0022$ ($+0.3\sigma$; $\sqrt{3}/3\pi$ lay at -1.2σ). Nothing is fitted: the load is the horizon condition, the exponent the cascade's, the ratio TRSD's product. Both components are made at the conformal reset from one boundary energy, by defect nucleation and pair production, so the ratio must also be that energy's branching between the radial (neutral) and winding (charged) channels (§25.8); until that is computed, a data-driven alternative such as $3\sqrt{6}/4\pi^2 = 0.18614$ (Appendix A) has equal standing.

25.8 The baryon ratio from the reset

DERIVED The rotation cannot supply the asymmetry: as a chemical potential for baryon number through the anomaly, $\mu_B = \hbar\dot\theta = \hbar H_\infty = 1.2 \times 10^{-33}$ eV is 9×10^{-45} of the temperature at electroweak freeze-out ($T \simeq 130$ GeV); even $\dot\theta = 0.827H$ would give $n_B/s \sim 5 \times 10^{-18}$ ($\mu_B/T = 1.5 \times 10^{-16}$), seven orders below the observed 8.7×10^{-11}. The amount is the reset's, fixed by the budget: $\omega_b = 0.02248$ and the measured CMB temperature give

$$\frac{n_b}{n_\gamma} = 6.16 \times 10^{-10}, \qquad \frac{n_B}{s} = 8.75 \times 10^{-11}, \tag{59}$$

$+0.7\sigma$ in ω_b from Planck's 6.13×10^{-10} and $+0.4\sigma$ from deuterium burning's $\omega_b = 0.02233 \pm 0.00036$ [151]. The 5/2 budget's $+2.5\sigma$ in ω_b was the dark ratio's.

DERIVED Sakharov's conditions hold without a new coupling: the reset is out of equilibrium (§45); the electroweak anomaly (charges in §31) violates baryon number, sphalerons turning a lepton asymmetry into $B = \frac{28}{79}(B - L)$; CP violation enters through the mixing matrix (§42), its sign set by the two-C/S handedness (§39.4), the circulant phase δ_ℓ (§42.7) carrying none (conjugation, $\delta_\ell \to -\delta_\ell$, only relabels the generations).

IDENTIFICATION The reset's branching realizes the asymmetry: a lepton asymmetry made out of equilibrium and converted by sphalerons, in the budget's amount and with the ratio of §25.7(v).

OPEN Open is the computation of the branching: Bogoliubov coefficients of the boundary action with the three-defect boundary condition and the radial mode's amplitude (§37.7), with the Majorana masses' lepton-number violation (§42.4) and the mixing's CP phase (§42.5), out of equilibrium. No prediction waits on it.

25.9 The linear cosmology computed

DERIVED At linear order the dust (§23.1) is cold matter and the capacity energy a constant vacuum: the linear cosmology is flat ΛCDM with derived parameters. Its CMB spectra (CAMB [61]; primordial amplitude, tilt and optical depth refitted, since the framework does not predict them; a Planck-like likelihood calibrated to Planck's uncertainties [39]) fix, with the derived baryon ratio and the acoustic angle matched,

$$\left.\frac{\Omega_\Lambda}{\Omega_c}\right|_{\rm CMB} = 2.61 \pm 0.02, \tag{60}$$

at $\Delta\chi^2 = 0.2$ from Planck's best fit: the baryon ratio passes. I5's 5/2 lies 4.6σ away (its budget, $\omega_b = 0.02275$, $\omega_c = 0.12168$, $H_0 = 67.12$, cost $\Delta\chi^2 = 17.9$, 3.5σ, through ω_b, $+2.5\sigma$, and ω_c, $+1.4\sigma$), the two-C/S value 2.29 further. Short of the 5σ removal threshold set in advance (TRSD [63]), I5 is withdrawn because the resonance supplies the ratio, 2.580, at $\Delta\chi^2 = 1.6$ (§25.11). The full Planck likelihood decides. PREDICTION Register row 65.

25.10 The budget with the measured ratio

DERIVED The cosmic share's amount being an integration constant (§23.1), the dark ratio is boundary data, as N is, unless the present is selected; the framework's own content in the budget is then flatness (§50.1), the baryon ratio and the acoustic angle. With the CMB's ratio 2.611 these land on Planck's densities, $\omega_b = 0.02239$ and $\omega_c = 0.1198$ ($H_0 = 67.44$), and $\Omega_m = 0.3139$ lies 1.9σ from DESI's (5/2: 3.0σ; ΛCDM's CMB fit: 2.1σ). The resonance selects the present, and the ratio with it (§25.11).

25.11 The dark ratio from the resonance

DERIVED With a constant vacuum in a flat universe $\Omega_\Lambda(t) = H_\infty^2/H(t)^2$ exactly, so the cosmic share's integration constant (§23.1) decides when the resonance falls, not the budget there. With the rotation at the spin, $\Omega_0 = H_\infty$ (§36.14), and the present at the resonance (H3, §36.7), $\Omega_\Lambda(\text{now}) = 27/4\pi^2$. Flatness, the derived baryon ratio and the neutrino masses of §42.4, counted inside Ω_m as everywhere in the budget, fix $\Omega_\Lambda/\Omega_c = 2.580$ (2.568 without the neutrinos), and the acoustic angle fixes the physical densities (§26.7). The framework's reference values:

	resonance (H3)	data
Ω_Λ/Ω_c	2.580	2.611 ± 0.024 (CMB spectra, §25.9)
Ω_Λ	0.6839	0.6847 ± 0.0073 [39]
Ω_m	0.3160	0.2975 ± 0.0086 (DESI) [23]
H_0 ($\mathrm{km\,s^{-1}\,Mpc^{-1}}$)	67.35	67.36 ± 0.54
ω_b	0.02248	0.02237 ± 0.00015
ω_c	0.12022	0.1200 ± 0.0012
S_8	0.833	0.832 ± 0.013
n_b/n_γ (10^{-10})	6.16	6.13
a_0 (10^{-10} $\mathrm{m\,s^{-2}}$)	1.148	$1.04 \pm 0.04 \pm 0.10$ (§22.6)
t_0 (Gyr)	13.79	
$K = m_p/m_*$	19.759	
β (§49.12)	0.314°	$0.277° \pm 0.057°$

It lies 1.3σ below the CMB's ratio ($\Delta\chi^2 = 1.6$), 0.1σ from Planck's Ω_Λ, $+0.7\sigma$ in ω_b, $+0.2\sigma$ in ω_c, $+0.1\sigma$ in S_8 (Ω_m: §25.6): the 5/2 budget's material variances, ω_b and DESI's Ω_m, fall to $+0.7\sigma$ and to ΛCDM's level (Appendix D.7). The rest follows: $q_0 = -0.526$, $z_{\rm acc} = 0.630$, maximum power at $z = 0.027$, 0.38 Gyr ago (§26.5); final cell count $N_\infty = N/\Omega_\Lambda = 6.63 \times 10^{122}$ and horizon $r_\infty = c/H_\infty = 17.56$ Gly; $\rho_\Lambda = \frac{3}{4}\rho_P/N_\infty = 1.130 \times 10^{-123}\rho_P$; 73.4% of the aeon's conformal time elapsed; first starlight (§48.8) at 100.6 Myr.

DERIVED Hence I5 is withdrawn: the dark ratio is not the defect power ratio. The sectors' harmonics (§25.1–25.3) stand without a density; 5/2 is kept in §52.6 as the single-triple limit of the defect pattern. PREDICTION Register row 74.

26 The Arrow of Time: Three of Space, One of Time

26.1 The 3:1 ratio

Space has three dimensions and time has one. In an expanding universe a volume grows as a^3 and a length as a:

$$\frac{d\ln V}{dt} = 3H, \qquad \frac{d\ln a}{dt} = H. \tag{61}$$

Space, measured by volume, expands at rate 3 while the ruler expands at rate 1.

26.2 Two frames

DERIVED The two frames are cosmic time t, kept by matter, and conformal time $\eta = \int dt/a$, in which the seed $h = r$ is exact (§1.10); light covers the comoving distance $c\eta$. For matter, $\rho \propto a^{-3}$ and $H^2 \propto \rho$ give $a \propto t^{2/3}$ and

$$\eta_0 = \int_0^{t_0} \frac{dt}{a} = t_0^{2/3} \int_0^{t_0} t^{-2/3}\, dt = 3\, t_0. \tag{62}$$

In d spatial dimensions $a \propto t^{2/d}$ and $\eta_0 = \frac{d}{d-2}\, t_0$: the 3 is the dimension of space, and the space frame runs three times the matter frame.

26.3 Ages

For pure matter $t_0 = 2/3H_0$ and $\eta_0 = 3t_0$ exactly. With the full contents—radiation shortens the conformal age, the vacuum lengthens both—the budget gives $t_0 = 13.79$ Gyr and $c\eta_0 = 46.1$ Gly $= 3.18\, c/H_0$, the radius of the observable universe, so $\eta_0 = 3.34\, t_0$. Both ages are correct; they belong to different frames.

26.4 Acceleration

STANDARD Gravity is sourced by the active mass density, counting the one time component and the three space components of the stress:

$$\frac{\ddot a}{a} = -\frac{4\pi G}{3}\Big(\rho + \frac{3p}{c^2}\Big) = -\frac{4\pi G}{3}\big(\underbrace{\rho}_{\text{1 of time}} + \underbrace{3\,p/c^2}_{\text{3 of space}}\big). \tag{63}$$

With $w_{\rm eff} \equiv p_{\rm tot}/\rho_{\rm tot}c^2$, a flat universe has

$$q = \frac{1 + 3w_{\rm eff}}{2}, \qquad 1 + q = \frac{3}{2}\,(1 + w_{\rm eff}), \tag{64}$$

so tensions in the three spatial directions that outweigh the one temporal energy accelerate the expansion. For matter plus vacuum $w_{\rm eff} = -\Omega_\Lambda$ and

$$q = \frac{\Omega_m}{2} - \Omega_\Lambda = \frac{1 - 3\,\Omega_\Lambda}{2}\,: \tag{65}$$

the universe accelerates exactly when $3\,\Omega_\Lambda > 1$, three of space outweighing one of time. Today $q_0 = -0.526$ (Planck: -0.527).

The tension comes from the defect: one extended in n spatial dimensions has tension $-\rho c^2$ along each and none across, so averaged over orientations [44, 56]

$$p = -\frac{n}{3}\,\rho c^2, \qquad w = -\frac{n}{3}, \qquad \rho + \frac{3p}{c^2} = (1 - n)\,\rho. \tag{66}$$

Strings ($n = 1$) coast, walls have $w = -\frac{2}{3}$, and a defect filling space has $w = -1$ and active mass -2ρ: the horizon defect spread through space is the vacuum, one of time minus three of space.

26.5 Matter creates time: the maximum-power epoch

Matter's stress has one entry, the time–time energy; the vacuum's has four equal entries, one of time and three of space. The time-like share of all stress in a flat matter-plus-vacuum universe is therefore

$$\tau = \frac{\Omega_m + \Omega_\Lambda}{\Omega_m + 4\,\Omega_\Lambda} = \frac{1}{1 + 3\,\Omega_\Lambda}. \tag{67}$$

HYPOTHESIS (H1) Matter is the time-carrying content of the universe, and the present lies near the moment when matter's share of the budget equals the time-like share of the stress, $\Omega_m(z) = \tau(z)$. With $\Omega_\Lambda = 1 - \Omega_m$,

$$3\,\Omega_m^2 - 4\,\Omega_m + 1 = 0 \quad \Longleftrightarrow \quad (3\,\Omega_m - 1)(\Omega_m - 1) = 0, \tag{68}$$

whose non-trivial root is $\Omega_m = \frac{1}{3}$ (in d spatial dimensions, $1/d$). The condition has a dynamical meaning:

Theorem 8 (Maximum-power epoch). DERIVED *For a flat universe of matter and vacuum the following are equivalent:*

(a) $\Omega_m = \tau$*: matter's share equals the time-like share of the stress;*
(b) $\Omega_m = \frac{1}{3}$*;*
(c) $q = -\frac{1}{2}$*;*
(d) $\dot{r} = c/2$*: the Hubble radius recedes at half the speed of light;*
(e) $w_{\rm eff} = -\frac{2}{3}$*: the universe as a whole has the equation of state of a domain-wall network,* $n = 2$ *in (66);*
(f) $\dot{E}_{\rm H} = P_{\max} = c^5/4G$*: the horizon energy grows at exactly the maximum power of the axiom.*

Proof. (a) $\Longleftrightarrow$ (b) is shown above. With $\Omega_\Lambda = 1 - \Omega_m$, equation (65) reads $q = (3\Omega_m - 2)/2$, so (b) $\Longleftrightarrow$ (c). The master rate gives $\dot{r} = c(1 + q)$ (equation (7)), so (c) $\Longleftrightarrow$ (d). Equation (64) gives (c) $\Longleftrightarrow$ (e). Finally $\dot{E}_{\rm H} = 2(1 + q)P_{\max}$ (equation (8)), so (c) $\Longleftrightarrow$ (f). □

Matter's carrying time is therefore not a counting rule: it marks the horizon's growth passing through the axiom's maximum power, from $3P_{\max}$ in pure matter toward zero in de Sitter. $P_{\max}$ bounds the luminosity of any localized source (§43B.6); the horizon's growth is not a luminosity, so it may exceed $P_{\max}$ early, and the crossing is a distinguished event, not a violation. With the budget it lies at $z = 0.027$, 0.38 Gyr ago ($z = 0.028$ with Planck's Ω_m), and today $\dot{E}_{\rm H}/P_{\max} = 2(1 + q_0) = 0.948$: we live just after the maximum-power epoch. The cosmic coincidence problem becomes a condition, and the condition a statement about the axiom.

26.6 The arrow

DERIVED The cell count scales as $N \propto r^2 \propto H^{-2}$. For a flat universe the Friedmann and continuity equations give $\dot{H} = -4\pi G(\rho + p/c^2)$, so

$$\frac{dN}{dt} = -2N\frac{\dot{H}}{H} = \frac{8\pi G N}{H}\left(\rho + \frac{p}{c^2}\right) \;\geq\; 0 \tag{69}$$

whenever $\rho + p/c^2 \geq 0$. The horizon's cell count, and with it the information $N/2$, never decreases; it stays constant only for pure vacuum. This is the arrow of time in GRUT and the generalized second law of the Hubble horizon [49].

26.7 Consequences of the budget

PREDICTION From $\Omega_m = 0.3160$, $\Omega_\Lambda = 0.6839$ (§25.11) and the flat Friedmann equation:

$$q_0 = -0.526, \qquad z_{\rm acc} = \left(\frac{2\Omega_\Lambda}{\Omega_m}\right)^{1/3} - 1 = 0.630, \tag{70}$$

$$z_{m\Lambda} = \left(\frac{\Omega_\Lambda}{\Omega_m}\right)^{1/3} - 1 = 0.293, \qquad z_{3:1} = \left(\frac{\Omega_\Lambda}{2\Omega_m}\right)^{1/3} - 1 = 0.027, \tag{71}$$

and $H_0 t_0 = \frac{2}{3\sqrt{\Omega_\Lambda}}\,\mathrm{arcsinh}\sqrt{\Omega_\Lambda/\Omega_m} = 0.9501$.

The Hubble constant. The CMB measures the acoustic angle θ_* to 0.03% [39]. With $\Omega_\Lambda = 27/4\pi^2$, the baryon ratio and $\sum m_\nu = 59.34$ meV (§42.4) fixed, CAMB [61] is solved for the H_0 that reproduces Planck's measured $100\theta_* = 1.04110 \pm 0.00031$:

$$H_0 = 67.35 \pm 0.06\,(\theta_*)\ \mathrm{km\,s^{-1}\,Mpc^{-1}}, \tag{72}$$
$$\omega_b = 0.02248\ (+0.7\sigma), \qquad \omega_c = 0.12022\ (+0.2\sigma),$$

with $r_{\rm drag} = 146.9$ Mpc and $t_0 = 13.79$ Gyr; $a_0 = (\sqrt{3}/\pi^2)\,cH_0 = 1.148 \times 10^{-10}$ m s^{-2} lies inside the measured band (§22.5). H_0 coincides with Planck's ΛCDM value and 5.5σ below SH0ES (4.8σ with the dilated scatter of the local rate): GRUT predicts that the tension resolves at the low value, every distance ladder calibrated alike converging on 67.3 (§53.9). The 5/2 budget's $\omega_b = 0.02275$ ($+2.5\sigma$) was its material variance, set by the dark ratio (Appendix D).

Curvature. The budget is flat; a spatial S^3 (§4.6) must have radius $R > 18\,r$.

26.8 The defect ladder of cosmic history

Theorem 9 (Defect ladder). DERIVED *In any flat Friedmann universe the horizon energy grows at*

$$\dot{E}_{\mathrm{H}} = 3\,(1 + w_{\mathrm{eff}})\,P_{\max}. \tag{73}$$

When the effective equation of state equals the value of an n-dimensional defect, $w_{\mathrm{eff}} = -n/3$, the horizon grows at $\dot{E}_{\mathrm{H}} = (3-n)\,P_{\max}$ and recedes at $\dot{r} = \frac{1}{2}(3-n)\,c$.

Proof. Combine $\dot{E}_{\mathrm{H}} = 2(1+q)P_{\max}$ and $\dot{r} = c(1+q)$ with $1+q = \frac{3}{2}(1+w_{\mathrm{eff}})$, equation (64). □

Radiation, $w = +\frac{1}{3}$, is $n = -1$ of the same formula. Cosmic history is the ladder $n = -1, 0, 1, 2, 3$:

stage	n	w_{eff}	q	$\dot{E}_{\mathrm{H}}/P_{\max}$	$\dot{r}/c$	epoch (budget)
radiation	-1	$+\frac{1}{3}$	1	4	2	early universe
matter	0	0	$\frac{1}{2}$	3	$\frac{3}{2}$	$z \gg 1$
strings: acceleration begins	1	$-\frac{1}{3}$	0	2	1	$z = 0.630$, 6.1 Gyr ago
walls: maximum-power epoch	2	$-\frac{2}{3}$	$-\frac{1}{2}$	1	$\frac{1}{2}$	$z = 0.027$, 0.38 Gyr ago
vacuum: de Sitter	3	-1	-1	0	0	future

For matter plus vacuum the crossings sit at $\Omega_\Lambda(z) = n/3$, that is $(1+z_n)^3 = (\Omega_\Lambda/\Omega_m)(3-n)/n$: the dark-energy era is divided into thirds, one per spatial dimension the defect fills, and the present lies between the second and third rungs (Figure 3).

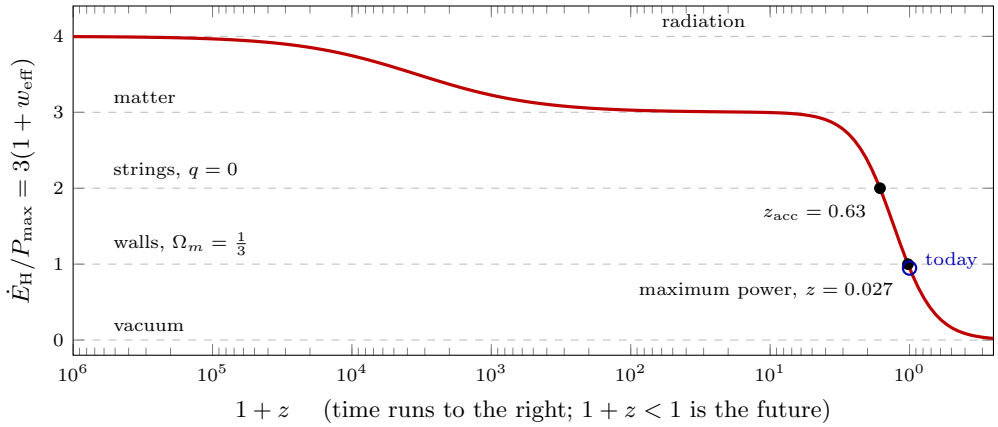


Figure 3: The defect ladder (Theorem 9) for the resonance budget with radiation: the horizon energy grows at $4, 3, 2, 1, 0$ times the maximum power through the radiation, matter, string, wall and vacuum values of the effective equation of state. Filled points: the onset of acceleration and the maximum-power epoch; open point: today, $\dot{E}_{\mathrm{H}} = 0.948\,P_{\max}$.

27 The Light Cone and the Speed of Light

27.1 c given

$c = 299\,792\,458$ m s^{-1}, exact.

27.2 Cone as light cone

The seed is the light cone (4); the causal structure of spacetime is that cone placed at every event.

27.3 Terminal state

Massless quanta move at c at every epoch, $v^2 = c^2$. Massive bodies lose peculiar momentum as the universe expands, $p \propto 1/a$, and settle into the comoving frame. If the dark-sector length L_* grows with the horizon, as in the cosmic reset, the mass $m_\delta = \sqrt{8}\,\hbar/cL_*$ falls to zero and that sector joins the massless one. The terminal state is then the light-cone state, with every remaining degree of freedom on the null cone.

28 The Lorentz Structure

28.1 From the cone

STANDARD Place the seed cone at every event. The Alexandrov–Zeeman theorem [46]: in dimension three or more, every bijection of spacetime that preserves the order defined by the cones ("y lies inside the future cone of x"), in both directions, is a Lorentz transformation composed with a translation and a dilation. No continuity or linearity is assumed. The Lorentz group follows from the cone alone.

The dilation is fixed by the anchors of §0.2: the cone does not choose a length, the anchor r does. The count $3+1$ comes from stability. In $d \geq 4$ spatial dimensions no bound orbit is stable, and in $d \leq 2$ the potential does not decay, so $d = 3$ is the unique dimension with both stable orbits and free separation (Ehrenfest [26]).

28.2 Invariant interval

$$ds^2 = -c^2 dt^2 + dx^2 + dy^2 + dz^2, \tag{74}$$

with the seed as $ds^2 = 0$. For the closed universe of §4.6, $ds^2 = -c^2 dt^2 + a^2(t) R^2 [d\chi^2 + \sin^2\chi (d\theta^2 + \sin^2\theta \, d\phi^2)]$.

29 Newtonian Gravity

29.1 Inverse square

STANDARD The field of a point mass carries the same flux through every sphere around it (Gauss). In d spatial dimensions the sphere's area grows as r^{d-1}, so $g \propto r^{-(d-1)}$, and in three dimensions

$$F = \frac{GMm}{r^2}, \qquad g = \frac{GM}{r^2}. \tag{75}$$

The $1/r$ potential is one of only two central potentials whose bound orbits all close (Bertrand), so Kepler's ellipses are a consequence of $d = 3$.

29.2 G from the axiom

$G = c^4/F$ (§12).

29.3 Equivalence principle

Gravity is carried by a massless spin-2 field on the light cone. Weinberg [45] proved that Lorentz invariance of the amplitude for emitting such a quantum forces every particle to couple to it with the same strength, so inertial and gravitational mass are equal. MICROSCOPE confirms this to $|\eta| \lesssim 10^{-15}$ [43].

30 General Relativity

30.1 Curvature from the defect

STANDARD Curvature is what the deficit measures (§1.7). Lovelock's theorem [36]: in four dimensions the only symmetric, divergence-free tensor built from the metric and its first two derivatives is $aG_{\mu\nu} + bg_{\mu\nu}$. The second-order field equations with local conservation are therefore

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu} = \frac{8\pi}{F} T_{\mu\nu}, \tag{76}$$

the last form using the axiom: geometry couples to matter with strength 8π divided by the universe force, or 2π divided by the one-cone force. Jacobson [33] obtains the same equation from the horizon relation $\delta Q = T \, dS$ with $S = A/4\ell_{\mathrm{P}}^2$, the entropy of §20.

30.2 Λ from the vacuum

$\Lambda = 8\pi G\rho_\Lambda/c^2 = 3\Omega_\Lambda H_0^2/c^2 = 3\Omega_\Lambda/r^2$, which is 1.09×10^{-52} m^{-2} with Planck values.

30.3 Schwarzschild and the horizon

$r_s = 2GM/c^2 = 2c^2M/F$, and for the horizon mass $r_s = r$ (23). The defect form of mass (28) is exact for straight defects in $3+1$ dimensions, $\delta = 8\pi G\mu/c^2$, and for point masses in $2+1$ dimensions, $\delta = 8\pi G_3 M/c^2$ [25].

31 Electromagnetism

31.1 The electric $U(1)$

DERIVED The three generators carry the phases of $\mathbb{Z}_3 \subset U(1)$, and the fermion generations of §42 carry the generators. Letting the phase vary continuously from point to point, $\psi \to e^{iq\Lambda(x)/\hbar}\psi$, requires a connection, $D_\mu = \partial_\mu - \frac{iq}{\hbar}A_\mu$ with $A_\mu \to A_\mu + \partial_\mu\Lambda$, so that $D_\mu\psi \to e^{iq\Lambda/\hbar}D_\mu\psi$. This $U(1)$ acts on the fermions, not on the defect condensate. Were the condensate charged, the Higgs mechanism would give the photon the mass $m_\gamma = \sqrt{2}\,q_\delta e f_\delta \phi_0$ (§41.7), and the bound $m_\gamma < 10^{-18}$ eV [13] would require $|q_\delta| < 4\times10^{-46}$. Charges come in units of $e/3$ (§31.4), and the smallest of them would give $m_\gamma \sim 10^{27}$ eV: DERIVED the condensate is exactly neutral and the photon exactly massless.

31.2 Maxwell from the phase gradient

The gauge-invariant field strength is $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$. The lowest-order Lorentz- and gauge-invariant Lagrangian, $-\frac{1}{4\mu_0}F_{\mu\nu}F^{\mu\nu} - J^\mu A_\mu$, gives Maxwell's equations,

$$\partial_\mu F^{\mu\nu} = \mu_0 J^\nu, \qquad \partial_{[\lambda}F_{\mu\nu]} = 0. \tag{77}$$

STANDARD The homogeneous pair is the Bianchi identity of $F = dA$, and the $U(1)$ symmetry conserves the current, $\partial_\mu J^\mu = 0$, as $\partial_\nu\partial_\mu F^{\mu\nu} = 0$ requires. Gauge invariance, $A_\mu \to A_\mu + \partial_\mu\Lambda$, is the continuous closure of the cycle's phase.

31.3 Coulomb's law and the electric axiom

DERIVED In the three spatial dimensions of §26, Gauss's law $\oint \mathbf{E}\cdot d\mathbf{A} = q/\varepsilon_0$ gives the field of a point charge, $E = q/4\pi\varepsilon_0 d^2$, so two elementary charges repel with

$$F_e = \frac{e^2}{4\pi\varepsilon_0 d^2} = \frac{\alpha\hbar c}{d^2}, \qquad \alpha \equiv \frac{e^2}{4\pi\varepsilon_0\hbar c}. \tag{78}$$

This is the electric axiom of §8.3: the inverse square is the three-dimensionality of space, and the coupling is α. In D spatial dimensions the force would fall as d^{1-D}.

31.4 Charge quantization

DERIVED The C_3 cycle fixes the number of colours, $N_c = 3$ (§15.3, IDENTIFICATION), and §42.2 supplies each generation as weak doublets and singlets with one Higgs doublet. Requiring the $U(1)$ to be free of gauge anomalies, and the fermion masses to be gauge invariant, fixes every charge. With the left-handed fields $Q\,(N_c, 2)$, $u^c, d^c\,(\bar{N}_c, 1)$, $L\,(1, 2)$ and $e^c\,(1, 1)$, the mass terms QHu^c, $QH^\dagger d^c$ and $LH^\dagger e^c$ give $Y_u = -Y_Q - Y_H$, $Y_d = -Y_Q + Y_H$ and $Y_e = -Y_L + Y_H$, so the $[SU(N_c)]^2U(1)$ condition holds identically; the $[SU(2)]^2U(1)$ condition gives $Y_L = -N_c Y_Q$; and the gravitational and cubic conditions reduce to

$$Y_H - N_c Y_Q = 0, \qquad -(N_c Y_Q - Y_H)^3 = 0, \tag{79}$$

both solved by $Y_Q = Y_H/N_c$. With $Q = T_3 + Y$ and the Higgs normalized to $Y_H = \frac{1}{2}$,

$$\boxed{Q_u = \tfrac{1}{2}\Big(1 + \frac{1}{N_c}\Big) = \tfrac{2}{3}, \qquad Q_d = -\tfrac{1}{2}\Big(1 - \frac{1}{N_c}\Big) = -\tfrac{1}{3}, \qquad Q_e = -1, \qquad Q_\nu = 0.} \tag{80}$$

The proton, uud, has charge $2Q_u + Q_d = 1$, and PREDICTION hydrogen is exactly neutral, $q_p + q_e = 0$, against the bound $|q_p + q_e| < 10^{-21}e$ [13]. The thirds of §15.3 are therefore derived, not only matched to triality: the colour singlets carry integer charge because the $\mathbb{Z}_3$ phases close.

31.5 The running of α

DERIVED Charged fermions screen the charge. With the charges (80) and the lepton masses, m_τ from §42.3, the one-loop leptonic shift of α to the Z mass is

$$\Delta\alpha_{\text{lep}} = \frac{\alpha}{3\pi} \sum_{\ell=e,\mu,\tau} \left[\ln \frac{M_Z^2}{m_\ell^2} - \frac{5}{3} \right] = 0.03142, \tag{81}$$

to which higher orders add 0.25%. The quark contribution below a few GeV is non-perturbative and is taken from e^+e^- data, $\Delta\alpha_{\text{had}} \simeq 0.028$ [13]; together they give $1/\alpha(M_Z) \simeq 128.9$ in the on-shell scheme. The running from 1/137.036 follows once the zero-momentum value is given, so a geometric value of α must be that Thomson-limit coupling.

31.6 The value of α

The coupling is $\alpha = F_e/F$ (§8.3, §15), calibrated at $1/137.035\,999\,177$. Its geometric conjecture and exact target, $K = 0.041741(3)$, are §43A/C and §43F.

32 Quantum Mechanics

32.1 Wavefunction from cone modes

The modes of the cycle, $f_n = e^{2\pi i m n/3}$, are the discrete prototype: complex amplitudes whose phases wind around the cone. The wavefunction is their continuum version, a complex amplitude on configuration space.

32.2 $\hbar$

$\hbar = 2\pi F r^2/Nc$ (§13).

32.3 Born rule

$P = |\langle a|\psi\rangle|^2$, by Gleason's theorem in dimension ≥ 3 (§6.4).

32.4 Commutator

STANDARD Space is homogeneous, so translations $T(a)\psi(x) = \psi(x-a)$ form a continuous unitary group. Stone's theorem [42] gives $T(a) = e^{-ia\hat{p}/\hbar}$. Differentiating at $a = 0$ gives $-\psi' = -\frac{i}{\hbar}\hat{p}\psi$, so $\hat{p} = -i\hbar\,\partial_x$ and

$$[\hat{x}, \hat{p}]\psi = x(-i\hbar\psi') + i\hbar(\psi + x\psi') = i\hbar\,\psi. \tag{82}$$

33 The Schrödinger Equation

STANDARD Time is homogeneous, so time translations form a continuous unitary group, $U(t) = e^{-i\hat{H}t/\hbar}$ (Stone), and

$$i\hbar\,\partial_t\psi = \hat{H}\psi. \tag{83}$$

The kinetic term comes from the mass hyperboloid inside the cone (§19). With $\hat{p}_\mu = -i\hbar\partial_\mu$ the relation $p^\mu p_\mu = -m^2c^2$ becomes the Klein–Gordon equation $(\Box - m^2c^2/\hbar^2)\phi = 0$. Writing $\phi = \psi\, e^{-imc^2t/\hbar}$, the rest-energy terms cancel, and for slow motion ($|\partial_t^2\psi| \ll (mc^2/\hbar)|\partial_t\psi|$)

$$i\hbar\,\partial_t\psi = -\frac{\hbar^2}{2m}\nabla^2\psi + V\psi, \tag{84}$$

in agreement with the expansion (29). The potential V is the interaction of the field in question: gravity through $m\Phi$ (§37), electromagnetism through qA_0.

34 Thermodynamics

34.1 Entropy

$S = A/4\ell_{\rm P}^2 = \pi r^2/\ell_{\rm P}^2 = N/2$ (in units of k_B), equal to the information I of §20.

34.2 Temperature

A horizon with surface gravity κ (an acceleration) has temperature $T = \hbar\kappa/2\pi c k_B$. For the cosmic horizon $\kappa = cH = c^2/r$, so

$$T_{\rm GH} = \frac{\hbar H_0}{2\pi k_B} = 2.878 \times 10^{-30}\ \text{K (73.04)}, \quad 2.654 \times 10^{-30}\ \text{K (67.36)}. \tag{85}$$

34.3 Second law

$dN/dt \geq 0$ whenever $\rho + p/c^2 \geq 0$, equation (69): the horizon moves to ever larger counts.

34.4 First law

The horizon energy is exactly temperature times entropy, $E_{\rm H} = k_B T_{\rm GH} S$ (55). Its differential form on the apparent horizon, with the matter energy flowing across it, reproduces the Friedmann equations [21].

34.5 Potentials, heat capacity and the Bekenstein bound

With $S \propto H^{-2}$ and $T \propto H$, the entropy is $S \propto T^{-2}$, so

$$C = T\frac{\partial S}{\partial T} = -2S. \tag{86}$$

The heat capacity is negative, as for any self-gravitating system. Using $E_{\rm H} = k_B T_{\rm GH} S$ (§24) and $PV = -\rho_\Lambda c^2 \cdot \frac{4}{3}\pi r^3 = -E_{\rm H}$ for $w = -1$ in the de Sitter limit $\rho_\Lambda = \rho_c$:

$$F_{\rm Helm} = E_{\rm H} - k_B T_{\rm GH} S = 0, \qquad G_{\rm Gibbs} = E_{\rm H} - k_B T_{\rm GH} S + PV = -E_{\rm H}. \tag{87}$$

The horizon has zero Helmholtz free energy. The Bekenstein bound is saturated exactly:

$$\frac{2\pi k_B r E_{\rm H}}{\hbar c} = \frac{2\pi k_B}{\hbar c} \cdot \frac{c}{H_0} \cdot \frac{c^5}{2GH_0} = \frac{\pi k_B c^5}{\hbar G H_0^2} = S. \tag{88}$$

Read as a spin, the bound is $S \leq 2\pi L_{\max}/\hbar$, with $L_{\max} = ER/c$ the largest angular momentum an energy E inside R can carry, reached only when it moves at light speed on the rim; the horizon saturates it because the rim of its turn moves at c (§36.11). The Euler relation does not close on its own: $k_B T_{\rm GH} S - PV = 2E_{\rm H} \neq E_{\rm H}$. It closes with a term conjugate to the deficit,

$$E_{\rm H} = k_B T_{\rm GH} S - PV + \mu\,\delta_{\rm tot}, \qquad \mu = -\frac{E_{\rm H}}{\pi} = -\frac{c^5}{2\pi G H_0}, \tag{89}$$

so the deficit behaves as a conserved charge with its own chemical potential. All four statements use the sphere quantities; each cone carries a quarter of $E_{\rm H}$ and half the area, and the same relations hold with those factors.

35 Statistical Mechanics

35.1, 35.3 Partition function and Boltzmann distribution

STANDARD Maximizing $S = -\sum p_i \ln p_i$ at fixed normalization and fixed mean energy U, with multipliers λ_0 and β, gives $-\ln p_i - 1 - \lambda_0 - \beta E_i = 0$, so

$$p_i = \frac{e^{-\beta E_i}}{Z}, \qquad Z = \sum_i e^{-\beta E_i} \tag{90}$$

[34].

35.2 Ensembles

The horizon has $\Omega = e^S = e^{N/2}$ microstates. Each bit occupies $4\ln 2\,\ell_{\rm P}^2 = 2.77\,\ell_{\rm P}^2$ of horizon area, and each of the N effective cells carries half a nat. The number of distinct horizon configurations is $e^{N/2} = 2^{N/2\ln 2}$.

35.4 Ergodicity

The cycle C_3 visits its three states in turn, so time averages equal the uniform average over the cycle.

36 Cosmology

36.1 H_0

$H_0 = c/r$.

36.2–36.3 Density

$$\rho = \frac{M_{\rm H}}{V_s} = \frac{Fr/2c^2}{\frac{4}{3}\pi r^3} = \frac{3F}{8\pi c^2 r^2} = \frac{3H_0^2}{8\pi G} = \rho_c. \tag{91}$$

36.4 Flatness

$\Omega_{\rm tot} = 1$ is the horizon condition (23). With the S^3 of §4.6 as the spatial section, $\Omega_{\rm tot} = 1 + (r/R)^2$.

36.5 Scale bridge

$a_0 = (\sqrt{3}/\pi^2)\,cH_0$, with $\alpha_{\rm TRSD} = \sqrt{3}/\pi^2 = 0.175493$.

36.6 The selection of the present epoch

MEASURED With the budget, today is $t_0 = 13.82$ Gyr and the maximum-power crossing ($\Omega_m = \frac{1}{3}$, Theorem 8) was at 13.56 Gyr: $|\ln(t_0/t_\times)| = 0.019$. For an observer placed at random in log time over two decades of cosmic history, the chance of lying this close to the crossing is 0.8%. HYPOTHESIS A principle of extremal horizon power would place the observer at $\Omega_m = \frac{1}{3}$ exactly; the resonance budget puts today 0.38 Gyr past it, at $\Omega_m = 0.316$. The coincidence is quantified; no principle that selects the epoch is derived, and the hypothesis stays.

36.7 Now: where the horizon meets the final rate

HYPOTHESIS The internal rotation is held at a constant rate (§23), and the one constant rate of the late universe is the final expansion rate H_∞, set by the final horizon; the rotation turns at H_∞. DERIVED The horizon's lowest C_3 mode has frequency $(3\sqrt{3}/2\pi)H(t)$, which falls as the universe expands, and it meets the rotation's rate once:

$$\boxed{\frac{3\sqrt{3}}{2\pi}\,H_{\rm now} = H_\infty, \qquad \Omega_\Lambda({\rm now}) = \left(\frac{H_\infty}{H_{\rm now}}\right)^2 = \frac{27}{4\pi^2} = 0.6839.} \tag{92}$$

This is the present: the moment the horizon's own mode comes into resonance with the rate of the universe's final state, the future's rate meeting the present horizon. It is the same statement as the rotation's turning at $0.827H_0$ today (§23). It fixes when now is, $t_{\rm now} = 0.786/H_\infty = 0.950/H_{\rm now}$, and with the derived baryon ratio it gives $\Omega_m = 0.3160$ and $\Omega_\Lambda/\Omega_c = 2.580$, with the neutrinos of §42.4 counted inside Ω_m as everywhere in the budget (2.568 without them); the full budget is §25.11. Planck's ΛCDM fit gives $\Omega_\Lambda = 0.6847 \pm 0.0073$, 0.1σ from (92); the full-spectrum fit with the derived baryon ratio (§25.9) prefers 2.611 ± 0.024, 1.3σ; DESI's $\Omega_m = 0.2975 \pm 0.0086$ lies 2.2σ away, as it does from Planck. The maximum-power rule (§36.6) placed now within 3% of the age; the resonance places it at the measured value. IDENTIFICATION Passing through resonance turns the phase of a driven mode by π: before now the horizon's mode runs ahead of the rotation, after it runs behind. The present is that sign flip, the cosmic instance of the crossings at which a falling rate meets a fixed one—the horizon of a black hole, where the time and radial directions exchange, and the conformal reset, where one aeon's future is glued to the next one's past. OPEN The identification of the rotation's rate with H_∞ was made with the

measured Ω_Λ in view. It is recorded as a hypothesis (H3), tested by the dark ratio as the CMB and BAO sharpen (register row 66); its grounds are set out in §36.8, and with the spin derived they reduce to the identification I8 (§36.13), itself derived up to an integer that the data fix (§36.14). H3 is the opening principle's statement of the present, and with it the dark ratio needs no identification of its own (§25.11).

36.8 Why the rotation turns at H_∞

DERIVED The constraint holds the phase at a constant rate (§23), so the rate is a constant of the whole solution, and its value can be read where the solution is simplest, in the final state. There nothing is left but the final horizon: matter has thinned away, and the only scale is H_∞, set by the boundary datum N_∞ (§44.6). The phase is an angle, so its rate is a clock rate, and the final state offers one clock.

IDENTIFICATION That clock is the final horizon's own frequency, its surface gravity over c, $\kappa_\infty/c = H_\infty$. Two routes in the framework give it. The horizon's temperature is $T_\infty = \hbar H_\infty/2\pi k_B$, and its thermal circle, the circle $S^1(2\pi)$ of the electric coupling's calculation (§43F), has the period $2\pi/H_\infty$: a phase in equilibrium with the horizon advances by 2π per thermal period. And the final state's spatial spheres, at their narrowest of radius c/H_∞, carry the conformal scalar's lowest mode at frequency $c/R = H_\infty$, the mode that supplies the weight of the electric coupling's correction.

DERIVED With the rate fixed at H_∞, the resonance of §36.7 follows without further assumption: the present is the moment the current horizon's C_3 cycle oscillates at the final horizon's own frequency, and $\Omega_\Lambda = 27/4\pi^2$. OPEN The equilibrium of the phase with the final horizon is the one premise not yet derived from the action; the measured Ω_Λ was known when it was formulated. The spinning-arc principle of §36.9 gives the rate without it. Both routes above are read again in §36.10: the thermal circle is the turn of the final horizon's arcs, and its rate is fixed by the smoothness of the horizon; the premise becomes I8, that the internal phase turns with that spin (§36.13), which the quantum of the spin derives up to an integer measured to be one (§36.14).

36.9 The spinning-arc principle

DERIVED (A hypothesis in release 1.11.0; derived for the final horizon in §36.10–36.11.) The horizon is not a static sphere: it is swept out by C-arcs spinning about perpendicular axes, vertical and horizontal, crossing one another (§1). The spin is physical. Nothing acts on the arcs, so their rate is fixed, and on a horizon, a surface receding at the speed of light, the arcs' rims move at c:

$$\boxed{\omega R = c.} \tag{93}$$

DERIVED *The cosmic sphere.* The universe's horizon grows toward the radius at which the fixed spin's rims reach light speed, and no further: $R_\infty = c/\omega$, so $H_\infty = \omega$ and

$$\Lambda = \frac{3\omega^2}{c^2}. \tag{94}$$

The cosmological constant is the signature of the spin. The internal rotation is the spin itself (I8, §36.13; derived in §36.14), so it turns at H_∞, with no equilibrium premise, and the present is the moment the current horizon's C_3 cycle has slowed to the spin, $\Omega_\Lambda = 27/4\pi^2$ (§36.7).

DERIVED *The wobble meets the spin.* The threefold wobble of the arcs, which creates the three generator defects (§1; its count is the beat of §1.11), is the horizon's C_3 mode: its frequency $(3\sqrt{3}/2\pi)\,c/R$ falls as the horizon grows, while the spin is fixed. The present is the moment the wobble falls into step with the spin.

The spin is measured. The birefringence of the CMB is the spin's phase accumulated since last scattering, $\beta = \kappa\alpha\,\omega(t_0 - t_{\rm rec})/2\pi$ (§49.12), so the measured angle measures the spin: $0.277° \pm 0.057°$ gives $\omega = (0.73 \pm 0.15)H_0$, against $H_\infty = 0.827H_0$ from the expansion, 0.6σ. Equivalently the twist of the oldest light measures the cosmological constant, $\beta = (\kappa\alpha/2\pi)\sqrt{\Lambda c^2/3}\,(t_0 - t_{\rm rec})$.

DERIVED *The quantum sphere.* The same rule holds for a particle. A sphere swept at frequency ω has radius c/ω; with energy $\hbar\omega = mc^2$ that radius is $\hbar/mc$, the reduced Compton wavelength, and with the zero-point energy $\hbar\omega/2$ it is half that. The internal clock de Broglie gave every particle, at $mc^2/\hbar$ [129], and the light-speed trembling Schrödinger found in the electron's motion, at $2mc^2/\hbar$ and radius $\hbar/2mc$ [130, 131], are this rule at the smallest scale. From the electron to the edge of the universe each closed sphere is swept by arcs whose rims move at light speed, and the Hubble quantum at the foot of the ladder, $\hbar H_\infty$, is the cosmic sphere's quantum of spin (§50.2). PREDICTION Register rows 67–71.

36.10 The spin derived: the thermal geometry of the final horizon

STANDARD In the final state only the vacuum remains ($w = -1$, $V_0 = 0$, §38), and the universe is de Sitter space

with horizon radius $R_\infty = c/H_\infty$. A static observer writes it, with ϱ the distance from the observer and the horizon at $\varrho = R_\infty$, as

$$ds^2 = -\Big(1 - \frac{\varrho^2}{R_\infty^2}\Big)c^2 dT^2 + \frac{d\varrho^2}{1 - \varrho^2/R_\infty^2} + \varrho^2 d\Omega^2. \tag{95}$$

Its thermal form, $T \to -i\tau$, is the four-sphere of radius R_∞ [29]:

$$X_0 + iX_4 = \sqrt{R_\infty^2 - \varrho^2}\, e^{iH_\infty \tau}, \qquad \vec{X} = \varrho\, \hat{n}, \qquad X_0^2 + X_4^2 + |\vec{X}|^2 = R_\infty^2. \tag{96}$$

Static time is an angle: advancing τ turns the four-sphere in the (X_0, X_4) plane at the rate H_∞, and the horizon, $\varrho = R_\infty$, is the axis of the turn. DERIVED This turn is the spinning-arc picture, exactly.

The arcs. Hold the angle and follow one line of sight, $\hat{n}$, through the observer to both sides: the points $\vec{X} = s\,\hat{n}$, with s running from $-R_\infty$ to R_∞, form a semicircle of radius R_∞ from a horizon point, through the observer, to the opposite horizon point—a C-arc. The turn spins it about the axis through its two ends, and it sweeps the two-sphere $X_0^2 + X_4^2 + s^2 = R_\infty^2$. Lines of sight at right angles, vertical and horizontal, give arcs spinning about perpendicular axes, and at each instant all the arcs cross at the observer. The two halves of each great circle, half a turn apart, are a conjugate pair of C-arcs.

The rim moves at light speed. A point of the arc at distance ϱ from the observer runs on a circle of radius $\sqrt{R_\infty^2 - \varrho^2}$ at the rate H_∞, so its speed is

$$v(\varrho) = c\sqrt{1 - \varrho^2/R_\infty^2}, \tag{97}$$

the rate at which a static clock at ϱ runs: light speed at the observer, rest at the horizon, where time stands still. The rim is the observer, and

$$\boxed{\omega = H_\infty, \qquad \omega R_\infty = c, \qquad \Lambda = \frac{3\omega^2}{c^2}.} \tag{98}$$

Every observer stands on the rim of the arcs of its own horizon.

The rate is fixed by the geometry. At proper distance σ from the horizon the plane of the turn is $\sigma^2(H_\infty d\tau)^2 + d\sigma^2$, a plane in polar coordinates with angle $H_\infty \tau$. It closes smoothly only if that angle runs once around: the arcs meet the horizon without a defect only at the rate H_∞, and a turn at any other rate ω' leaves at their ends a conical defect $2\pi(1 - H_\infty/\omega')$. That defect would lie in the plane of the turn; the C_3 deficit π lies across it, on the horizon's sphere (§7). The thermal period is one turn, $2\pi/\omega$, and the temperature of the final horizon is the frequency of the turn over 2π,

$$k_B T_\infty = \frac{\hbar\omega}{2\pi}. \tag{99}$$

In real time. The same Killing flow is a boost. The horizon's generators are light rays, and the flow advances their rapidity at the rate $\kappa_\infty/c = H_\infty$, the surface gravity over c: in real time the rims of the arcs are the light rays that make the horizon.

Why the spin is fixed. STANDARD The surface gravity of a stationary horizon is the same over the whole horizon, the zeroth law [132], so all the arcs turn together. And $\Lambda = 3\omega^2/c^2$ cannot change while matter is separately conserved: the contracted Bianchi identity, $\nabla^\mu G_{\mu\nu} = 0$, with $\nabla^\mu T_{\mu\nu} = 0$, forces $\partial_\nu \Lambda = 0$. In the unimodular form of gravity Λ is a constant of integration, conjugate to the four-volume and conserved as energy is conserved by time translation [133]. DERIVED Nothing in the action exchanges energy with the vacuum: the scalar's vacuum constant vanishes (§38.6) and the rotation's constraint adds none (§23.1). Nothing acts on the arcs, so their rate does not change. The rate is the framework's boundary datum read as a frequency,

$$\omega\, t_{\rm P} = \frac{\ell_{\rm P}}{R_\infty} = \sqrt{\frac{2\pi}{N_\infty}} = 9.66 \times 10^{-62}, \qquad t_{\rm P} = \frac{\ell_{\rm P}}{c}, \tag{100}$$

so that "reality is the spin" and "N_∞ is the boundary datum" (§44.6) are one statement.

What is derived and what is not. DERIVED For the final horizon the spinning-arc principle is a theorem of general relativity read in the framework's geometry: the arcs, their perpendicular axes and their crossing at the observer, the fixed rate H_∞, the light-speed rim, $\Lambda = 3\omega^2/c^2$, and the temperature as one turn. OPEN The value of ω is not derived; it is N_∞ (§44.6). The derivation holds only if the final state is de Sitter: evolving dark energy established at 5σ removes the final horizon and the fixed spin with it (register row 6). At every earlier epoch the current horizon has its own rate $c/r = H(t)$, larger than the spin and falling toward it; on the current horizon the fixed spin's rim moves at $\omega r = c\sqrt{\Omega_\Lambda(t)}$, $0.82\,c$ today, and reaches c only at R_∞ (§36.9).

36.11 The charge of the spin

STANDARD The entropy of a Killing horizon is 2π times the Noether charge of its Killing field taken at unit surface gravity [134]. DERIVED For the final horizon that charge is the angular momentum of §21, read at its radius:

$$\boxed{L = \frac{Ac^3}{8\pi G} = M_{\rm H}\, c\, R = \frac{\hbar N}{4\pi}, \qquad S = \frac{2\pi L}{\hbar}, \qquad E_{\rm H} = \omega L = k_B T\, S.} \tag{101}$$

§21's $L_{\rm H} = M_{\rm H} cr$, the horizon's mass carried around its radius at light speed, is the charge of the turn; the entropy is the action of one turn in units of $\hbar$; and the exchange law of §24, $E_{\rm H} = k_B T S$, is the energy of a turn whose rim moves at c, $E = \omega L$, the relation of light carried on a circle ($E = pc$, $L = pR$). The three were derived separately (§20, §21, §24); they are one statement about the spin. The same identities hold at every epoch for the current horizon with its own rate c/r.

The spin is not a rotation of space. STANDARD A body with mechanical spin s obeys a lower entropy bound, $S \le 2\pi\sqrt{(ER/c)^2 - s^2}/\hbar$ [135]: a rotation that carried the horizon's energy around at light speed, $s = ER/c$, would leave it no entropy. DERIVED The horizon has the largest entropy its energy allows, $2\pi E_{\rm H} r/\hbar c$ (§34.5), and so no mechanical spin: its spin is the turn of its time. This is the vorticity bound of Section 23 seen from the other side: the universe's rotation is internal because a rotation of space would cost the horizon its entropy. The Bekenstein bound itself reads as a spin, $S \le 2\pi L_{\max}/\hbar$, with $L_{\max} = ER/c$ the largest angular momentum an energy E inside R can carry, reached only on a rim moving at light speed. With $S = 2\pi L/\hbar$ and $E = \omega L$ the horizon saturates it exactly when $\omega R = c$: the entropy alone fixes the rim.

The quantum of the spin. DERIVED The turn is an angle of period 2π, so the charge conjugate to it takes the values $L = n\hbar$, and by (101)

$$\boxed{S = 2\pi n, \qquad A = 8\pi n\, \ell_{\rm P}^2, \qquad N = 4\pi n, \qquad \Delta E = \hbar\omega \text{ per quantum.}} \tag{102}$$

Each quantum of spin adds 2π nats, $8\pi\ell_{\rm P}^2$ of area and $4\pi\ell_{\rm P}^2$ of effective area. IDENTIFICATION That is the surface of one Planck sphere, the quantum sphere of the Planck mass, whose rim moves at light speed and which carries $\hbar$ (§36.12): the horizon's spin is the sum of the spins of the Planck spheres that tile its effective surface. DERIVED The identity $\hbar A_{\rm eff}/4\pi\ell_{\rm P}^2 = L$ holds exactly when $A_{\rm eff} = 2\pi r^2$; with the full area the sum would be twice the horizon's spin, so the additivity of spin is a further route to the deficit π (§4.4, §6, §18.1). STANDARD The spacing $8\pi\ell_{\rm P}^2$ is the one Bekenstein proposed for the horizon area as an adiabatic invariant [136]; Maggiore obtained it from the damping of black-hole quasinormal modes [137], Kothawala, Padmanabhan and Sarkar found the entropy step 2π in every Lanczos–Lovelock theory [138], and the periodicity of Euclidean time has been read as its origin [139]. Here it is the quantum of the spin. For the cosmic horizon the quantum is $\hbar\omega = \hbar H_\infty = 1.18 \times 10^{-33}$ eV, the Hubble quantum at the foot of the ladder (§50.2), and the final horizon holds $L_\infty/\hbar = N_\infty/4\pi = 5.36 \times 10^{121}$ of them.

36.12 The quantum sphere and black holes

DERIVED A rim moving at light speed carries $E = \omega L$, so a sphere whose energy turns at its own frequency carries $L = E/\omega$. For a particle, $E = mc^2$. At de Broglie's clock, $\omega = mc^2/\hbar$, the rim is the Compton radius $\hbar/mc$ and $L = \hbar$. At the zero-point energy, $mc^2 = \frac{1}{2}\hbar\omega$, the turn is Schrödinger's trembling at $2mc^2/\hbar$, the rim is $\hbar/2mc$, and

$$L = \frac{\hbar}{2}: \tag{103}$$

the half of the zero-point energy is the half of the spin, the sign that needs 4π to close (§1.8). STANDARD This is the zitterbewegung reading of the electron's spin [131]. For the Planck mass the rim is $\ell_{\rm P}$ and $L = \hbar$: the Planck sphere of §36.11.

DERIVED A black hole is a closed sphere of the same kind. Its thermal form is a cigar whose turn has the rate $\omega = \kappa/c = c^3/4GM$ and whose widest circle, far from the hole, has radius $c/\omega = 2r_s$, where the rim moves at light speed. Its entropy is again 2π times the charge of the turn [134]; a quantum $\hbar\omega = 2\pi k_B T_{\rm BH}$ changes it by 2π through the first law, so the area quantum is $8\pi\ell_{\rm P}^2$ for every horizon, and a quantized hole absorbs and emits at $\omega_n = n\,\omega$, shifted by $m\Omega_{\rm H}$ for a rotating hole [140, 141]. STANDARD The overtones of its ringing are damped in steps of the same rate: the imaginary parts of the Schwarzschild quasinormal frequencies are spaced by κ/c at high overtone [143, 144]. For a $60\,M_\odot$ remnant $\omega = 846$ rad s^{-1} (135 Hz), in the band where ringdowns are measured; the LIGO–Virgo ringdowns give the area quantum $\alpha = 15.6^{+20.5}_{-13.3}$ in units of $\ell_{\rm P}^2$ [142], against $8\pi = 25.1$, $+0.5\sigma$.

36.13 What the spin predicts

The rotation turns with the spin. IDENTIFICATION (I8) The internal phase of Section 23 is the turn of §36.10 read in real time: its constraint (51) holds $X_0 = \Lambda/3$, the phase advances one radian per e-fold of the final expansion, $\theta = \ln a$, and the rotation turns at $\omega = H_\infty$. The constraint carries no scale of its own (§0.2), and of the framework's lengths only R_∞ gives a constant rate of the size the birefringence measures, so X_0 is a pure number k times $\Lambda/3$, and the spin is $k = 1$. MEASURED Two measurements bear on it. The birefringence gives $k = 0.88 \pm 0.18$; with the present at the resonance of §36.7 (H3), Planck's Ω_Λ gives $k = 0.9994 \pm 0.0053$. With I8 the equilibrium premise of §36.8 is not needed, and the resonance is the wobble meeting the spin. DERIVED The quantum of the spin fixes the rate up to an integer, $\Omega_0 = jH_\infty$, and the two measurements select $j = 1$ (§36.14): I8 is derived.

The phase turned since the beginning. DERIVED With the rate H_∞ and the present at the resonance, the phase turned since the beginning is a pure number. In a flat universe of matter and vacuum $H_\infty t_0 = \frac{2}{3}\,\mathrm{artanh}\sqrt{\Omega_\Lambda}$, so

$$\theta_0 = \omega\, t_0 = \tfrac{2}{3}\,\mathrm{artanh}\,\frac{3\sqrt{3}}{2\pi} = 0.7857 \text{ rad}, \tag{104}$$

and 0.78545 with radiation, with no Hubble constant in it. OPEN This lies 5×10^{-5} rad from $\pi/4$, the half-apex angle of the seed (§3, §6): the spin has turned an eighth of a turn since the beginning. The match was noticed after the computation; for Ω_Λ anywhere in $[0.60, 0.75]$ one this close to an angle $p\pi/q$ with $q \le 12$ has a chance of 0.15%. No derivation connects them, and were it exact it would place the present at $\Omega_\Lambda = 0.68388$, 4×10^{-5} from $27/4\pi^2$, closer than any measurement can separate.

The birefringence in closed form. PREDICTION The CMB's polarization turns by the phase turned since last scattering, weighted by the anomaly, $\beta = (\kappa\alpha/2\pi)\,\omega(t_0 - t_{\rm rec})$ (§49.12), so with $\kappa = 6$

$$\boxed{\beta = \frac{2\alpha}{\pi}\,\mathrm{artanh}\,\frac{3\sqrt{3}}{2\pi} = 0.3137^\circ} \tag{105}$$

(0.3136° with radiation and the light's departure at recombination): the fine-structure constant, the anomaly of the cone triples and the C_3 ratio, with nothing measured on the sky and no Hubble constant. The joint ACT and Planck angle, $0.277^\circ \pm 0.057^\circ$ [85], lies 0.6σ below. §49.12 and §49.13 give 0.314° with the budget's expansion history and the C_3 rate read at H_0; at the resonance budget both read $\Omega_\Lambda = 27/4\pi^2$, and they agree within 0.003°.

The acceleration scale from the cosmological constant. DERIVED At the present the C_3 rate is the spin, so the acceleration scale of §22 is set by Λ alone:

$$a_0 = \frac{\sqrt{3}}{\pi^2}\,cH_{\rm now} = \frac{2}{3\pi}\,c\,\omega = \frac{2}{3\pi}\,c^2\sqrt{\frac{\Lambda}{3}}, \tag{106}$$

1.149×10^{-10} m s^{-2} with Planck's Λ. STANDARD Milgrom noted $a_0 \approx c^2(\Lambda/3)^{1/2}$ up to a factor of order one [145]; the spin fixes the factor, $2/3\pi$. PREDICTION It ties galaxies to the cosmological constant without a Hubble constant (register row 71); it is not independent of rows 8 and 66.

Status. Derived: the spin of the final horizon, its rate, its rim, its constancy, its charge and its quantum (§36.10–36.12), and the threefold count as a beat (§1.11). Identified in release 1.13.0, derived in release 1.14.0 up to an integer that the data fix (§36.14): I8, the internal phase turning with the spin. Hypothesis: H3, the present as the resonance. Predictions: register rows 68–73.

36.14 The quantum of the rotation is the quantum of the spin

DERIVED The rate of the rotation is the one constant the action leaves free: the constraint (51) holds $\dot\theta = \Omega_0 = c\sqrt{X_0}$ for any X_0, and I8 set $X_0 = \Lambda/3$. The quantum of the spin fixes it up to an integer. On the cosmological background the rotation sector vanishes on its solutions (§23.1), so the energy of its dust in a comoving volume is its Hamiltonian,

$$E = p_\theta\,\dot\theta = \Omega_0\,Q, \tag{107}$$

with $Q = p_\theta$ the momentum conjugate to the phase, the charge of the conjugate pair. The phase is an angle, so Q takes the values $n\hbar$ and the dust comes in quanta of rest energy $\hbar\Omega_0$: the Josephson–Anderson relation of a condensate whose phase turns at its chemical potential [146]. STANDARD A comoving quantum carries a Killing energy equal to its rest energy: in the flat slicing of de Sitter space the static observer's Killing field is $\xi = \partial_t - Hx^i\partial_i$, and $-u\cdot\xi = c^2$ per unit mass on every comoving worldline. Every comoving quantum but the observer's own leaves the static patch, and the horizon takes up its energy as heat: by the first law of the cosmological horizon [29], energy δE carried out across it raises its entropy by $\delta S = \delta E/k_B T_\infty$. DERIVED The entropy of the final

horizon changes only in steps of 2π, one quantum of its spin (§36.11), so it takes up energy only in multiples of $2\pi k_B T_\infty = \hbar\omega$. Each quantum of the rotation therefore carries a whole number of quanta of the spin:

$$\boxed{\hbar\Omega_0 = j\,\hbar\omega, \qquad \Omega_0 = j\,H_\infty, \qquad X_0 = j^2\,\frac{\Lambda}{3}, \qquad j = 1, 2, 3, \dots} \tag{108}$$

($j = 0$ is no rotation: the constraint's gradient would not be timelike.) The rate is a harmonic of the spin, fixed as Dirac's argument fixes electric charge by the monopole [147]: a classical constant made consistent with a quantized boundary. STANDARD The same structure holds a scalar condensate in equilibrium with a rotating black hole: its frequency is locked to the horizon's, $\omega = m\Omega_{\rm H}$ with m an integer [148]. Here the integer counts the quanta of the spin carried by one quantum of the rotation.

MEASURED The birefringence measures the integer. With the anomaly weight $\kappa = 6$ of the cone fermions (§49.12) it gives $\Omega_0/H_\infty = 0.88 \pm 0.18$: $j = 1$ lies 0.7σ away, $j = 2$ is excluded at 6.2σ and $j = 0$ at 4.9σ (the angle alone fixes $\kappa j = 5.3 \pm 1.1$, and $\kappa = 6$ is derived). With the present at the resonance (H3), Planck's Ω_Λ gives 0.9994 ± 0.0053, independently of κ. Hence $j = 1$: *each quantum of the universe's rotation is one quantum of the final horizon's spin*, and crossing the horizon it adds 2π to its entropy and $8\pi\ell_{\rm P}^2$ to its area. I8 is derived up to an integer, and the integer is measured, as the number of light neutrinos is (§42.2). DERIVED The rotation's quanta sit at the floor of the final universe, $\hbar\Omega_0/c^2 = \hbar/cR_\infty$, the mass whose Compton length is the final horizon (§48.1). The quantum is that of the charge: the dust's density is the multiplier λ of (51), which carries no gradient energy, so no de Broglie length enters its dynamics and its sound speed is zero on every scale (§23.1), unlike a condensate of free particles of the same mass, whose Jeans length would exceed the horizon. In the words of the opening principle: reality is the spin, the universe's rotation is made of its quanta, and the present (H3) is the moment one quantum of the horizon's wobble equals one of them. PREDICTION Register row 72.

36.15 The acceleration scale tracks the horizon, not the spin

DERIVED The threshold of §22.10 is one activation per period of the current horizon's fundamental C_3 mode, $\tau_1 = 4\pi^2 r/3\sqrt{3}\,c$, so the critical acceleration follows the horizon, $a_c \propto H(t)$, and at redshift z

$$a_0(z) = a_0\,E(z), \qquad E(z) = \frac{H(z)}{H_0}. \tag{109}$$

The spin is fixed. The two meet only at the resonance, so the relations that equate them—$a_0 = (2/3\pi)\,c^2\sqrt{\Lambda/3}$, equation (106), $\Omega_0 = (3\pi/2)\,a_0/c$ (§49.13), and with them $\beta = \frac{9}{2}\alpha\,a_0 t_0/c$—hold at the present epoch, which is where they are tested. Were the spin to set the scale at every epoch, a_0 would not evolve. The baryon ratio of §25.7 needs the same reading: the horizon's load $z_H = g_H/a_0$ is a pure number at every epoch only because a_0 tracks H.

MEASURED The data separate the two readings. A radial acceleration relation of 79 star-forming galaxies at $0.33 < z < 1.44$ from the MUSE Hubble Ultra Deep Field [149] has a scale that rises with redshift: 1.99, 2.25, 2.48 and 2.71×10^{-10} m s^{-2} in four bins near $z = 0.47$, 0.70, 0.95 and 1.27, and $2.38^{+0.12}_{-0.10} \times 10^{-10}$ at $z \simeq 1$; a constant scale is excluded. With $a_0 = 1.148 \times 10^{-10}$ (the resonance budget, §25.11), equation (109) gives 1.49, 1.72, 2.00 and 2.40: the measured values lie 33, 31, 24 and 13 per cent above, and within the survey the scale rises by a factor 1.36 where $H(z)$ rises by 1.61. The survey's own linear fit extrapolates to $a_0(0) = 1.00 \pm 0.04$, near the rotation-curve likelihood's 1.04 (§22.6), and its mass-to-light ratios lie below the local calibration's: the offset and the slope fall on opposite sides of (109). Rotation curves at $z \simeq 2$ all but exclude a scale four times today's [150]; (109) gives three. PREDICTION A scale constant in time, or one departing from $a_0 E(z)$ at 5σ in a single method spanning $z = 0$ to 1.5, falsifies the horizon reading (register row 73).

37 Dark Matter

Reading (Section 23). In the rotation reading the condensate of this section is the modulation of the universe's rotation (§23.1), not a population of particles: the equations stand, and their carrier is the rotation. The radial mode keeps its mass m_δ as an excitation of the field.

37.1 Residual modes

The radial mode of the scalar invariant $\Xi = \phi_+\phi_-$ oscillates about the vacuum $\Xi = 2$.

37.2 Mass

The potential $V = \frac{1}{2}(\phi^2-2)^2 + V_0$ has $V''(\sqrt{2}) = 8$. DERIVED Canonically normalized (§41.4), the radial mode has $m_\delta^2 c^2/\hbar^2 = 8\varepsilon_V/f_\delta^2$, so it takes the form below exactly when the potential scale is the defect tension spread over L_*^2, $\varepsilon_V = f_\delta^2/L_*^2$ [63]:

$$m_\delta = \frac{\sqrt{8}\,\hbar}{cL_*}, \qquad L_* = \frac{8\pi G M_*}{c^2} = \frac{8\pi M_* c^2}{F}, \tag{110}$$

with L_* the defect length of a reference mass M_* (§18.1). Hence the scale duality

$$M_*\, m_\delta = \frac{\sqrt{8}}{8\pi}\, m_{\mathrm{P}}^2. \tag{111}$$

Calibration. $M_* = 1.5\times10^{11} M_\odot$ gives $L_* = 0.180$ pc and $m_\delta c^2 = 1.003\times10^{-22}$ eV. The Lyman-α forest requires $m_\delta c^2 > 2\times10^{-20}$ eV [41], which means $M_* < 7.5\times10^{8} M_\odot$: the reference mass is a dwarf-galaxy mass. DERIVED The bound applies to this mode. It assumes [41] a scalar that is all the dark matter and evolves linearly as a free field of mass m. Here the condensate is the whole of Ω_c (§25); about the ring the potential is $4\eta^2 + 2\sqrt{2}\,\eta^3 + \frac{1}{2}\eta^4$ in field units, which shifts the oscillation frequency by $\Delta\omega/\omega = -\frac{3}{8}\eta^2$, below 4×10^{-5} for the displacements $\eta \sim 10^{-3}$–10^{-2} of §37.7 and falling as a^{-3}, far from the large-misalignment regime that changes the transfer function [70]; the self-coupling is gravitational in strength (§37.3); and a radial oscillation at fixed phase does not populate the massless Goldstone mode. If the condensate were only a fraction of the dark matter the bound would weaken [68, 69].

37.3 Self-coupling

DERIVED The coefficient 12 of the quartic term is in field units. For the canonical field $\chi = f_\delta\phi$ the quartic coupling is $\lambda = 12\varepsilon_V/f_\delta^4$; with $\varepsilon_V = f_\delta^2/L_*^2$ and, in natural units, $f_\delta^2 = \delta M_{\mathrm{P}}^2/8\pi$,

$$\lambda = \frac{12\pi}{\delta}\,\frac{m_\delta^2}{M_{\mathrm{P}}^2}, \qquad g_\chi = \frac{\lambda\hbar^3}{8m_\delta^2 c} = \frac{3\pi}{2\delta}\,\frac{\hbar^3}{m_{\mathrm{P}}^2 c} \overset{\delta=\pi}{=} \frac{3\hbar^3}{2m_{\mathrm{P}}^2 c} = 1.24\times10^{-95}\ \mathrm{J\,m^3}, \tag{112}$$

independent of m_δ. (Versions up to 10 wrote $3\hbar^3/2m_\delta^2 c$, larger by $(m_{\mathrm{P}}/m_\delta)^2 \sim 10^{100}$.) In a halo of density 10^{-21} kg m^{-3} the self-interaction energy is 2×10^{-9} of the gradient energy at 1 kpc: the condensate is effectively free.

37.4 Gross–Pitaevskii

$$i\hbar\,\partial_t\psi = -\frac{\hbar^2}{2m_\delta}\nabla^2\psi + m_\delta\Phi\psi + g_\chi|\psi|^2\psi. \tag{113}$$

37.5 Quantum pressure

$$Q = -\frac{\hbar^2}{2m_\delta^2}\,\frac{\nabla^2\sqrt{\rho_\chi}}{\sqrt{\rho_\chi}}. \tag{114}$$

37.6 Solitonic cores

$\lambda_{\mathrm{dB}} = 2\pi\hbar/m_\delta v = 0.60$ kpc for 10^{-22} eV at 200 km s^{-1}, scaling as $1/m_\delta$; at the Lyman-α bound, 2×10^{-20} eV, it is 3.0 pc, so the predicted cores are parsec-scale.

37.7 The initial displacement

STANDARD The relic density of the condensate is fixed by misalignment [57]: a field of mass m displaced by ϕ_i starts oscillating at $3H = m$ and dilutes as a^{-3} afterwards, giving, in the sudden-onset estimate,

$$\Omega_c h^2 \simeq 0.12\left(\frac{\phi_i}{8.5\times10^{16}\ \mathrm{GeV}}\right)^2\left(\frac{m}{10^{-22}\ \mathrm{eV}}\right)^{1/2}. \tag{115}$$

Requiring the observed $\Omega_c h^2 = 0.120$:

m	ϕ_i required	in units of $f_\delta = M_{\rm P}/\sqrt{8} = 4.32 \times 10^{18}$ GeV
10^{-22} eV	8.5×10^{16} GeV	2.0×10^{-2}
2×10^{-20} eV (Lyman-α)	2.3×10^{16} GeV	5.3×10^{-3}
10^{-18} eV	8.5×10^{15} GeV	2.0×10^{-3}

The exact radiation-era solution, $\phi = \phi_i \Gamma(\frac{5}{4})(mt/2)^{-1/4} J_{1/4}(mt)$, leaves 0.403 times the sudden-onset relic density, so each ϕ_i above rises by a factor 1.576 (1.35×10^{17} GeV at 10^{-22} eV); the conclusion is unchanged.

Conversely, a geometric displacement $\phi_i = \sqrt{2} f_\delta$ would require $m = 3.8 \times 10^{-30}$ eV, ten orders of magnitude below the Lyman-α bound and in the range where the field acts as dark energy rather than dark matter.

Constraint. The condensate can be the dark matter only if its initial displacement is 10^{-3} to 10^{-2} of the defect scale. The geometry supplies no such suppression, so this is a second calibration in the dark sector alongside L_*, and a target for the same reset calculation that would give the baryon ratio (§25.7).

37.8 The dual of the horizon mass

DERIVED Evaluated at the horizon mass $M_{\rm H} = c^3/2GH_0$, the scale duality gives

$$m_U c^2 = \frac{\sqrt{8}}{8\pi} \frac{m_{\rm P}^2 c^2}{M_{\rm H}} = \frac{\sqrt{2}}{2\pi} \hbar H_0 = 0.2251\, \hbar H_0, \tag{116}$$

3.2×10^{-34} eV at $H_0 = 67.35$: the dual of the whole horizon has the Hubble mass scale.

38 Dark Energy

38.1 Vacuum defect

The horizon defect spread through the three dimensions of space (§26.4).

38.2 Equation of state

$w = -n/3$ with $n = 3$, so $w = -1$.

38.3 Density

$$\rho_\Lambda = \Omega_\Lambda\, \rho_c = \Omega_\Lambda \frac{3H_0^2}{8\pi G} = \Omega_\Lambda \frac{3F}{8\pi c^2 r^2}, \qquad \rho_\Lambda c^2 = 5.24 \times 10^{-10}\ \mathrm{J\,m^{-3}}\ \text{(Planck)}. \tag{117}$$

38.4 Relation to a_0

Eliminating H_0 with $a_0 = (\sqrt{3}/\pi^2) c H_0$:

$$\rho_\Lambda c^2 = \Omega_\Lambda \frac{\pi^3}{8} \frac{a_0^2}{G}, \tag{118}$$

an energy density: a_0^2/G has units of $\mathrm{J\,m^{-3}}$.

38.5 Ratio

$\Omega_\Lambda/\Omega_c = 5/2$ (§25).

Prediction. $w = -1$ exactly. The DESI DR2 BAO analysis preferred evolving dark energy at 2.8–4.2σ [23]; adding the Lyman-α full shape, the preference is 2.7σ with the CMB and 3.2σ with supernovae [65]. The next releases test this directly.

38.6 The vacuum constant vanishes

Theorem 10. DERIVED *With $\varepsilon_V = f_\delta^2/L_*^2$ (§37.2), a vacuum constant V_0 of order one exceeds the observed vacuum energy by more than 10^{21}; the vacuum energy is carried by the space-filling defect of §38.1, not by V_0.*

Proof. At $\delta = \pi$, $f_\delta^2 = F/8 = 1.51 \times 10^{43}$ N. For $L_* = 0.180$ pc, $\varepsilon_V = 4.9 \times 10^{11}$ J m^{-3}, so $V_0 = 2$ gives 9.8×10^{11} J m^{-3} against $\rho_\Lambda c^2 = 5.2 \times 10^{-10}$ J m^{-3}, a factor 1.9×10^{21}. The Lyman-α bound shortens L_* below 9.0×10^{-4} pc and raises ε_V above 1.9×10^{16} J m^{-3}, so $V_0 < 3 \times 10^{-26}$. □

GRUT therefore takes $V_0 = 0$, the choice already made at the reset.

39 The CMB

39.1 Acoustic peaks from the sound cone

The defect cone is the sound cone (§4.4). Before recombination the photon–baryon fluid carries sound at $c_s = c/\sqrt{3(1+R)}$, with R the baryon loading; at $R = 0$ this is the $h = \sqrt{3}\,r$ cone. The comoving sound horizon $r_s = \int c_s\,dt/a$ at recombination subtends $\theta_* = r_s/D_M = 0.0104109$ [39], which sets the acoustic scale $\ell_A = \pi/\theta_* \approx 302$ and places the first peak at $\ell \approx 220$.

39.2 Selection rule

Three defects at 120° on the equator of the horizon sum as $\sum_j e^{-im\phi_j} = 3$ when $3 \mid m$ and 0 otherwise, and their reflection symmetry contributes $(-1)^{\ell+m}$:

$$b_{\ell m} = 0 \quad \text{unless} \quad m \equiv 0 \pmod 3 \ \text{ and } \ \ell + m \text{ even}. \tag{119}$$

39.3 The full spectrum and C_3/C_2

By the addition theorem $\sum_m Y_{\ell m}(\hat n_j) Y^*_{\ell m}(\hat n_k) = \frac{2\ell+1}{4\pi} P_\ell(\hat n_j \cdot \hat n_k)$, the three defects give

$$C_\ell = \frac{1}{2\ell+1} \sum_m |b_{\ell m}|^2 = \frac{3A^2\delta_j^2}{4\pi}\Big[1 + 2P_\ell\big(-\tfrac{1}{2}\big)\Big], \tag{120}$$

since $\hat n_j \cdot \hat n_k = 1$ three times and $-\frac{1}{2}$ six times. With $P_2(-\frac{1}{2}) = -\frac{1}{8}$ and $P_3(-\frac{1}{2}) = \frac{7}{16}$,

$$\frac{C_3}{C_2} = \frac{1 + \frac{7}{8}}{1 - \frac{1}{4}} = \frac{5}{2}, \tag{121}$$

and the dipole vanishes, $C_1 = 0$. The formula gives every multipole of the defect pattern:

ℓ	2	3	4	5	6	7	8	9	10
C_ℓ/C_2, defect pattern	1	2.50	0.563	1.09	2.20	0.738	1.14	2.05	0.831
C_ℓ/C_2, primordial plateau	1	0.50	0.300	0.200	0.143	0.107	0.083	0.067	0.055

The defect pattern adds to the primordial plateau ($C_\ell \propto 1/\ell(\ell+1)$) with an amplitude ε to be fitted, and the observed map carries it smoothed by the transfer function. The mod-3 / parity search at $\ell = 4$–10 on the Planck Commander map found no excess over the isotropic expectation, which bounds ε for the point-source pattern. Under isotropy the share of power in the admissible modes ($3 \mid m$, $\ell + m$ even, about one axis) has mean $f_\ell = |\mathcal{M}_\ell|/(2\ell+1) = 0.111, 0.182, 0.231, 0.133, 0.176, 0.211, 0.143$ for $\ell = 4, \ldots, 10$; a defect component of fraction ε raises it to $\varepsilon + (1-\varepsilon)f_\ell$, so the null result bounds ε at those multipoles. The smoothed pattern is the next target (Appendix F).

39.4 Octupole handedness

The two-C/S structure carries a handedness. For a real sky $a_{\ell,-m} = (-1)^m a^*_{\ell m}$, so the moduli $|a_{3,\pm3}|$ are equal and the handedness lives in the phase: the orientation of the triangular octupole, and its parity-odd counterpart in polarization, the EB correlation. Isotropic cosmic birefringence is measured at $\beta = 0.34° \pm 0.09°$ [27]; deriving the sign and size of β from the two-C/S rotation is the test (Appendix F).

39.5 Quadrupole–octupole alignment

The defects fix one axis: about it the quadrupole is $m = 0$ and the octupole $m = \pm3$. Because the usual alignment statistic (the axis maximizing $\sum_m m^2 |a_{\ell m}|^2$) points 90° away from the axis of a pure $m = 0$ quadrupole, the defect axis is best compared with the octupole axis directly.

39.6 The projected defect pattern

DERIVED Place the three defects at distance D and project their potential onto the last-scattering sphere of radius $r_{\rm LS}$. For $r_{\rm LS} < D$,

$$\frac{1}{|r_{\rm LS}\hat{n} - D\hat{n}_j|} = \sum_\ell \frac{r_{\rm LS}^\ell}{D^{\ell+1}} \frac{4\pi}{2\ell+1} \sum_m Y_{\ell m}(\hat{n}) Y^*_{\ell m}(\hat{n}_j),$$

so

$$C_\ell^{\rm proj} \propto \frac{(r_{\rm LS}/D)^{2\ell}}{(2\ell+1)^2} \Big[1 + 2P_\ell\big(-\tfrac{1}{2}\big)\Big], \tag{122}$$

with $D/r_{\rm LS}$ in place of $r_{\rm LS}/D$ when the defects lie inside the sphere. The selection rule is untouched—projection acts on ℓ, not on m—and the ratios fall: $C_3/C_2 = \frac{5}{2}(\frac{5}{7})^2(r_{\rm LS}/D)^2 = 1.276\,(r_{\rm LS}/D)^2$.

$D/r_{\rm LS}$	C_3/C_2	C_4/C_2	C_5/C_2	C_6/C_2	C_7/C_2
1	1.276	0.174	0.226	0.325	0.082
1.25	0.816	0.071	0.059	0.054	0.009
1.5	0.567	0.034	0.020	0.013	0.001

The admissible-power test at $\ell = 4$–10 on the Commander maps tests m only and is unchanged; the projection adds a prediction for how the defect component falls with ℓ. A physical defect profile and the integrated Sachs–Wolfe effect of a decaying potential remain.

40 Black Holes and Information

40.1 Extreme cones

STANDARD For radial light in Schwarzschild spacetime, $dr/dt = \pm c(1 - r_s/r)$, which vanishes at $r = r_s$. The cones close up at the horizon, and inside it every future cone points to $r = 0$.

40.2 Horizon

$r_s = 2GM/c^2 = 2c^2M/F$; for the cosmic horizon, $r_s(M_{\rm H}) = r$.

40.3 Entropy

$S = A/4\ell_{\rm P}^2$ with $A = 4\pi r_s^2$; for the cosmic horizon, $S = N/2$.

40.4–40.5 Information and the Page curve

The cell count is preserved by the global structure. The semiclassical "island" calculations reproduce the unitary Page curve of an evaporating black hole [38, 15, 16], which is the result GRUT's information conservation requires.

41 Gravity and the Scalar Action

41.1–41.2 The cone at every scale

One cone governs every scale. Its opening is the light cone (§1.10, §28), its deficit is curvature (§1.7, §30), its modes are quantum amplitudes (§32), and its horizon cells are information (§20).

41.3 Graviton

A massless spin-2 field coupled to energy–momentum must couple universally (Weinberg [45]), and with consistent self-coupling it reproduces general relativity (Deser [24]).

41.4 The gravitational–scalar action

$$S = \int d^4x\sqrt{-g}\left[\frac{c^4R}{16\pi G} - \frac{\delta c^4}{16\pi G}\, g^{\mu\nu}\partial_\mu\phi_+\partial_\nu\phi_- - \varepsilon_V\, V(\Xi)\right], \tag{123}$$

$$\Xi = \phi_+\phi_-, \qquad V(\Xi) = \tfrac{1}{2}(\Xi-2)^2 + V_0. \tag{124}$$

With the axiom, $c^4/16\pi G = F/16\pi$, so the gravitational and kinetic coefficients are $F/16\pi$ and $\delta F/16\pi$. The potential enters with a minus sign ($\mathcal{L} = T - V$) and carries the energy-density scale ε_V. Canonically normalizing the radial mode ties that scale to the dark-matter mass: $m_\delta^2c^2/\hbar^2 = 4\varepsilon_V/K$, with $K = \delta F/16\pi = f_\delta^2/2$; with $m_\delta = \sqrt{8}\hbar/cL_*$ this is $\varepsilon_V = f_\delta^2/L_*^2$ (§37.2). The vacuum energy $\varepsilon_V V_0$ vanishes (Theorem 10). The action covers gravity and the defect scalar; electromagnetic, fermionic and quantum structure enter as standard physics (§0.8), and §54.5 writes them with it as one action.

41.6 Two readings of the pair

Two structural readings of $\phi_\pm$ are worth recording, both consistent with §1.8.

Spinor. The conjugate pair are the two chiralities: the exchange $\phi_+ \leftrightarrow \phi_-$ is σ_1, a 2π rotation exchanges the lobes and 4π restores them, which is the double cover $SU(2) \to SO(3)$ of §1.8 and the signature of spin-$\frac{1}{2}$. Then $\Xi = \phi_+\phi_-$ is the Dirac mass bilinear, and $V(\Xi)$ is a potential for it, of Nambu–Jona-Lasinio type: the mass is a condensate and $V'' = 8$ is the gap. Making this a derivation requires first-order (Dirac) equations with a tetrad; the cone supplies two null directions, which is half of one.

String. The kinetic normalization $f_\delta^2 = \delta c^4/8\pi G$ has the units of a tension, $\mathrm{J\,m^{-1}}$, and is the only tension in the framework. Read as a worldsheet tension $T = f_\delta^2$ it gives the string length $\ell_s^2 = \hbar c/2\pi T = 4\hbar G/\pi c^3 = (4/\pi)\,\ell_{\mathrm{P}}^2$ at $\delta = \pi$, of order the Planck area, with the pair as the left- and right-moving modes. This is a dimensional identification, not a derivation of a string theory.

41.5 The two equations

Geometry: $h = r$. Closure: $V_s = 4V_c$ (§3.12) and $4\delta_{\mathrm{tot}} = 4\pi$ (§4.4).

41.7 The fermion and gauge sectors

DERIVED Spinors exist on the geometry: $S^3 \cong SU(2)$ double-covers $SO(3)$ (§1.8–1.9), so the fermion sector is

$$S_\psi = \int d^4x\, e\left[\bar\psi_k\, i\gamma^a e_a{}^\mu\Big(\partial_\mu + \tfrac{1}{4}\omega_\mu{}^{bc}\gamma_{bc}\Big)\psi_k - \big(\bar\psi_{Lj}\, M_{jk}\,\psi_{Rk} + \mathrm{h.c.}\big)\right], \tag{125}$$

with tetrad $e_a{}^\mu$ and spin connection $\omega_\mu{}^{bc}$. The C_3 symmetry of generation space makes the square-root mass matrix circulant, $\sqrt{M} = a\mathbb{1} + b\,P + b^*P^2$ with P the cyclic shift and $b = |b|e^{i\delta}$; its eigenvalues are $a + 2|b|\cos(\delta + 2\pi k/3)$, with sum $3a$ and sum of squares $3a^2 + 6|b|^2$. The seed cone $h = r$ is $|b| = a/\sqrt{2}$, and then

$$Q = \frac{\sum_k m_k}{\big(\sum_k \sqrt{m_k}\big)^2} = \frac{3a^2 + 6|b|^2}{9a^2} = \frac{2}{3} \tag{126}$$

identically in δ, for signed roots: Koide's relation is derived from the C_3 symmetry and the seed cone, and a negative root is the conjugate branch of the neutrinos (I7).

DERIVED The defect $U(1)$ is not electromagnetism. Were the condensate electrically charged, the Higgs mechanism would give the photon a mass $m_\gamma = \sqrt{2}\,e\,f_\delta\phi_0 \sim 10^{18}$ GeV, against the bound $m_\gamma < 10^{-18}$ eV [13]. Electromagnetism therefore enters as an independent $U(1)$, $S_A = -\frac{1}{4}\int d^4x\, e\, F_{ab}F^{ab}$ on the same tetrad, with the ψ_k charged under it; its charges are derived in §31.4 and the value of its coupling is §43F.

42 Three Generations on the Seed Cone

The same two primitives that build the cosmology—the seed cone $h = r$ and the three C_3 generators—have a second home in particle physics. The Standard Model has exactly three generations of fermions. Placing them on the seed cone yields three predictions for the lepton sector, each testable by experiments running now.

42.1 Generation space and the seed cone

Represent a fermion family by the vector of square-root masses in generation space,

$$\mathbf{v} = (\sqrt{m_1}, \sqrt{m_2}, \sqrt{m_3}). \tag{127}$$

The C_3 that cycles the three generators cycles the three axes, and its fixed direction is the axis $\hat{u} = (1,1,1)/\sqrt{3}$. The seed cone about $\hat{u}$ is the set of vectors at 45° to it: the cone $h = r$, with axial and radial components equal. The angle θ between $\mathbf{v}$ and $\hat{u}$ obeys

$$\cos^2\theta = \frac{(\mathbf{v}\cdot\hat{u})^2}{|\mathbf{v}|^2} = \frac{\left(\sum_k \sqrt{m_k}\right)^2}{3\sum_k m_k} = \frac{1}{3Q}, \qquad Q \equiv \frac{\sum_k m_k}{\left(\sum_k \sqrt{m_k}\right)^2}. \tag{128}$$

So the family lies on the seed cone, $\theta = 45°$, exactly when

$$Q = \frac{2}{3}. \tag{129}$$

IDENTIFICATION (I6) This is Koide's relation [11]. Foot observed that it states a 45° angle to the axis $(1,1,1)$ [47]; GRUT obtains it as the seed condition in generation space: the 45° cone is the light cone $h = r$ of §1.10. DERIVED The cycle supplies the form in which the cone is a light cone. With L the Laplacian of the C_3 cycle (§9), whose doublet eigenvalue is $3/\ell^2$,

$$Q(\mathbf{x}) = \Big(\sum_n x_n\Big)^2 - \ell^2\,\mathbf{x}^{\mathsf{T}} L\,\mathbf{x} = 3\big(|\mathbf{x}_s|^2 - |\mathbf{x}_d|^2\big) \tag{130}$$

has signature $(1,2)$, singlet against doublet, and Koide's relation is $Q(\sqrt{\mathbf{m}}) = 0$: the seed condition in the cycle's own form. Equivalently, $\sum_{\text{edges}}(\sqrt{m_n} - \sqrt{m_{n+1}})^2 = (\sum_n \sqrt{m_n})^2$, which the charged leptons satisfy as 2824.640 against 2824.658 MeV, a difference of 6.6×10^{-6} within the error of m_τ. IDENTIFICATION What remains identified is that the $\sqrt{m}$ vector obeys the seed condition. (The seed condition follows from the vacuum of the axiom's potential, given each generation's amplitude as its generator's invariant raised by the pair, §42.8.) The three generators at 120° parametrize the cone:

$$\sqrt{m_k} = \mu\Big[1 + \sqrt{2}\cos\Big(\delta + \frac{2\pi k}{3}\Big)\Big], \qquad k = 0,1,2. \tag{131}$$

The axial component of $\mathbf{v}$ is $\mu\sqrt{3}$, and the radial component is $\mu\sqrt{2}\cdot\sqrt{3/2} = \mu\sqrt{3}$. They are equal, which is $h = r$. The scale is μ, and δ is the orientation of the generator triangle on the cone.

For the charged leptons (PDG masses [13]),

$$Q_\ell = 0.6666645 \pm 0.0000051, \qquad \theta_\ell = 44.99991°, \tag{132}$$

with $\mu_\ell^2 = 313.851$ MeV and $\delta_\ell = 0.22224$ (the τ at $k = 0$), within 2×10^{-5} of $\frac{2}{9}$ [7]. The charged-lepton family lies on the seed cone to one part in 10^5.

42.2 The group structure

The primitives of §1 map onto the structure of the Standard Model:

GRUT primitive	Group	Standard Model role
conjugate pair, glued (§1.8)	$S^3 \cong SU(2)$	weak isospin doublets, e.g. (ν, e)
three generators (§4.1)	$\mathbb{Z}_3$	three generations; centre of colour $SU(3)$
continuous phase (§31)	$U(1)$	hypercharge, electromagnetism
tetrahedral closure (§4.4)	A_4	the flavour group of tri-maximal mixing

Colour-singlet families carry no $\mathbb{Z}_3$ triality and lie on the seed cone. Quark families carry triality and lie off it: with PDG running masses the up-type family sits at 51.2° ($Q = 0.849$) and the down-type family at 47.5° ($Q = 0.731$). Exactly three generators means exactly three light neutrinos, in agreement with the Z width, $N_\nu = 2.996 \pm 0.007$ [8].

42.3 Prediction 1: the tau mass

Given m_e and m_μ, the seed cone (129) fixes the third charged lepton:

$$m_\tau = 1776.969 \text{ MeV}. \tag{133}$$

The measured value is 1776.93 ± 0.09 MeV [13], 0.4σ away. The uncertainty of the prediction from m_e and m_μ is below 10^{-3} MeV, so the test is limited only by the tau-mass measurement.

Test. BESIII, Belle II, and a future super tau–charm factory aiming at ±0.01–0.05 MeV. A converged value more than 3σ from 1776.969 MeV falsifies the seed cone in generation space. The relation holds for pole masses; explaining why the pole mass is the right one is part of the open work.

42.4 Prediction 2: the neutrino masses

The lepton doublet (ν, e) is the conjugate pair, so its two members share the seed cone: $Q_\nu = \frac{2}{3}$. With the measured splittings, $\Delta m^2_{21} = (7.53 \pm 0.10) \times 10^{-5}$ eV^2 and $\Delta m^2_{31} = (2.529 \pm 0.021) \times 10^{-3}$ eV^2 (NuFIT 6.1, including JUNO [64, 9]), the cone fixes the absolute masses. With all components positive, Q_ν cannot reach $\frac{2}{3}$: its maximum is 0.585 in normal and 0.500 in inverted ordering. The solution puts the lightest state on the conjugate branch, $\sqrt{m_k} < 0$ in equation (131). That is allowed, since the physical mass is the square, and it is the double cone of the conjugate pair. Brannen first applied Koide's relation to neutrinos this way [7]. Each ordering gives one discrete spectrum:

Branch	m_1	m_2	m_3	$\sum m_\nu$	m_β	$m_{\beta\beta}$
normal	0.365	8.685	50.29	59.34	8.97	1.33–4.07
inverted	49.165	49.925	2.917	102.01	48.85	17.2–48.4

All masses in meV. The $m_{\beta\beta}$ range spans the unknown Majorana phases, using the mixing of §42.5.

The 1σ uncertainty of the splittings moves $\sum m_\nu$ by about ±0.2 meV.

Tests.

- *Ordering.* NuFIT 6.1 prefers the normal ordering at $\Delta\chi^2 = 9.4$ [64]; JUNO resolves it within a few years.
- *Cosmology.* DESI DR2 with the CMB bounds $\sum m_\nu < 64$ meV at 95% in ΛCDM [23], which favours the normal branch. Coming surveys with $\sigma(\sum m_\nu) \approx 15$–20 meV separate 59.3 from 102.0.
- *Neutrinoless double beta decay.* The normal branch lies below the reach of next-generation experiments. A detection between 5 and 17 meV falsifies both branches.

The normal-branch neutrino density, $\Omega_\nu h^2 = \sum m_\nu / 93.14 \text{ eV} = 6.35 \times 10^{-4}$, is the value already used in the H_0 derivation of §26.7, so the lepton and cosmology sectors of GRUT close on each other.

42.5 Prediction 3: the mixing column

In flavour space the radial directions of the three generators are $(2, -1, -1)/\sqrt{6}$, $(-1, 2, -1)/\sqrt{6}$ and $(-1, -1, 2)/\sqrt{6}$, one per charged lepton.

Hypothesis. (H2) The conjugate-branch neutrino ν_1—the one with $\sqrt{m_1} < 0$ in §42.4—points along the electron generator's radial direction:

$$(|U_{e1}|, |U_{\mu 1}|, |U_{\tau 1}|) = \frac{1}{\sqrt{6}}(2, 1, 1). \tag{134}$$

This is the tri-maximal TM1 pattern [5]. It arises naturally from the A_4 symmetry of the tetrahedron [6], which is GRUT's closure group. From $|U_{e1}|^2 = \cos^2\theta_{12}\cos^2\theta_{13} = \frac{2}{3}$,

$$\sin^2\theta_{12} = 1 - \frac{2}{3\cos^2\theta_{13}} = 0.3180, \tag{135}$$

with $\sin^2\theta_{13} = 0.02248$ (NuFIT 6.1): $+1.4\sigma$ from the global 0.3088 ± 0.0067 [64] and $+1.0\sigma$ from JUNO's 0.3092 ± 0.0087 [9]. From $|U_{\mu 1}|^2 = \frac{1}{6}$ the CP phase is tied to θ_{23}:

$$\cos\delta_{\text{CP}} = \frac{\frac{1}{6} - \sin^2\theta_{12}\cos^2\theta_{23} - \cos^2\theta_{12}\sin^2\theta_{23}\sin^2\theta_{13}}{2\sin\theta_{12}\cos\theta_{12}\sin\theta_{23}\cos\theta_{23}\sin\theta_{13}}. \tag{136}$$

$\sin^2\theta_{23}$	0.45	0.47	0.50	0.53	0.56	0.59
$\lvert\delta_{\rm CP}\rvert$	102.6°	97.5°	90.0°	82.5°	74.8°	66.6°

Maximal θ_{23} forces maximal CP violation. The T2K+NOvA value $\sin^2\theta_{23} = 0.56^{+0.03}_{-0.05}$ [14] gives $|\delta_{\rm CP}| \approx 75°$. At the NuFIT 6.1 value $\sin^2\theta_{23} = 0.470$ the relation gives $\delta_{\rm CP} = 262.5°$ (or 97.5°); the global normal-ordering fit, 212^{+26}_{-36} degrees [64], lies 1.9σ from the first.

The other geometric alignment—ν_2 along the cone axis $(1,1,1)/\sqrt{3}$, the TM2 pattern—gives $\sin^2\theta_{12} = 0.341$ and is excluded at 4.8σ by NuFIT 6.1. The generator direction, not the axis, is the one that survives.

Tests. JUNO's sub-percent measurement of $\sin^2\theta_{12}$ against (135). DUNE and Hyper-Kamiokande on the $(\theta_{23}, \delta_{\rm CP})$ correlation (136). A value of $\sin^2\theta_{12}$ more than 3σ from 0.318, or a $(\theta_{23}, \delta_{\rm CP})$ pair off the curve, falsifies the alignment.

42.6 What is new

Koide's relation [11], its reading as a 45° cone [47], the neutrino negative root [7] and tri-maximal TM1 mixing [58, 5] are prior results. GRUT contributes four things:

- the identification of Foot's 45° cone with the spacetime seed cone $h = r$ of §1.10: one primitive acting in two spaces;
- the reading of the negative root as the conjugate lobe of the pair (§1.8), which makes it required rather than chosen: with every root positive, Q_ν cannot exceed 0.585 (normal) or 0.500 (inverted), so the seed cone forces the conjugate branch;
- hypothesis H2, which ties the two neutrino facts together—the conjugate-branch state lies along the electron generator—so that the mass spectrum and the TM1 column stand or fall together;
- the joint prediction set with 2025–26 inputs (JUNO, NuFIT 6.1, PDG 2024), each member with a falsification condition in Appendix C.

The lepton predictions share no fitted parameter with the cosmological ones; they share the primitives. They stand or fall together with §26.7 only in that sense.

42.7 The charged-lepton triangle and the quark cones

MEASURED Fitting $\sqrt{m_k} = \mu_\ell\,[1 + \sqrt{2}\cos(\delta_\ell + 2\pi k/3)]$ to the pole masses gives $\delta_\ell = 0.222226 \pm 0.000006$ (the error from m_τ) and $\mu_\ell^2 = 313.851 \pm 0.013$ MeV: $\delta_\ell = 2/9$ holds at 0.7σ. Koide's ratio is $Q = 0.6666645$ for the pole masses and 0.6679287 for the running masses at M_Z [83]. DERIVED The ratio is invariant under a flavour-universal rescaling $m_k \to Z m_k$, so universal running preserves it; QED running depends on flavour through $\ln(\mu/m_k)$ and breaks it at $O(\alpha)$. The cone is a statement about physical masses, which is why pole masses obey it. For the quarks, with the PDG masses [13], the triplet (c, b, t) gives $Q = 0.6692$, on the cone to 0.4%, within the scheme dependence of quark masses; (d, s, b), (u, c, t), (u, d, s) and (s, c, b) give 0.731, 0.849, 0.567 and 0.458. DERIVED The phase is fixed far more sharply by the electron and muon than by the τ. With the cone's amplitude $\sqrt{2}$ exact, m_e and m_μ give $\delta_\ell = 0.222\,222\,047(1)$, that is $2/9 - 1.75 \times 10^{-7}$: within 8×10^{-7} of 2/9, but 450σ from it, so $Q = \frac{2}{3}$ and $\delta_\ell = \frac{2}{9}$ cannot both be exact. With $\delta_\ell = 2/9$ exact instead, the amplitude is $1.414\,213\,26$ and $Q = \frac{2}{3}(1 - 2.1 \times 10^{-7})$. Both branches give $m_\tau = 1776.97$ MeV, 0.4σ from 1776.93(9) [13]. The cone with $3\delta_\ell = Q$ fixes the masses from one scale: with both relations exact, $m_\mu/m_e = 206.7703$ against the measured 206.7683, 10 parts per million, and $m_\tau/m_e = 3477.47$ against 3477.37 ± 0.18. The two numbers hold together to 2×10^{-7}, and one of them carries a correction of that size; a phase landing that close to a simple fraction by chance has a probability near 10^{-5}. They are one statement. Since $\sum_k \cos^3(\delta + 2\pi k/3) = \frac{3}{4}\cos 3\delta$, the cubic invariant of the masses fixes $\cos 3\delta_\ell = (\sqrt{2}/3)\sum_k x_k^3$ with $x_k = \sqrt{m_k}/\mu_\ell - 1$, and the observed pair is

$$3\,\delta_\ell = Q = \tfrac{2}{3}\,: \tag{137}$$

the cubic invariant's angle equals the seed cone's ratio. IDENTIFICATION With the cone (I6) the relation gives $\delta_\ell = 2/9$; the relation itself is identified. The scale, $\mu_\ell^2 = 313.856$ MeV from m_e and m_μ, is the lepton sector's input: no combination of the framework's scales tested—$m_p/3$, $\alpha\,\alpha_{\rm TRSD}\,v_{\rm EW}$, $m_W/256$—matches it within its error.

42.8 The cone is the vacuum

DERIVED The seed condition in generation space follows from the vacuum of the axiom. The three generators carry the phases $\omega^k = e^{2\pi i k/3}$ on the unit circle (§15.3), and the conjugate pair $\Phi = \phi\, e^{i\theta}$ lies in the same plane, in the

units in which the potential is $V(\Xi) = \frac{1}{2}(\Xi - 2)^2$ with $\Xi = |\Phi|^2$ (§37.2). Let each generation's amplitude be the rise of its generator's invariant when the pair is added to it,

$$A_k = |\omega^k + \Phi|^2 - |\omega^k|^2 = \Xi + 2\sqrt{\Xi}\,\cos\Big(\theta - \frac{2\pi k}{3}\Big), \qquad k = 0, 1, 2. \tag{138}$$

The two invariants of the triple are then fixed by Ξ alone, $\sum_k A_k = 3\Xi$ and $\sum_k A_k^2 = 3\Xi^2 + 6\Xi$, so with $\sqrt{m_k} \propto A_k$

$$\boxed{Q = \frac{\sum_k A_k^2}{\big(\sum_k A_k\big)^2} = \frac{\Xi + 2}{3\Xi}}, \tag{139}$$

and $Q = \frac{2}{3}$ exactly when $\Xi = 2$: *Koide's relation is the vacuum.* The parametrization (131) follows with its amplitude $\sqrt{2} = 2/\sqrt{\Xi_0}$, its scale $\mu \propto \Xi_0$, and its phase $\delta = \theta$, the orientation of the pair against the generators.

The geometry. The circle $|\Phi| = \sqrt{2}$ is the director circle of the generators' unit circle: from each of its points the unit circle subtends a right angle, the apex angle of the seed cone $h = r$ (§3). Foot's 45° and the vacuum $\Xi_0 = 2$ of the potential are one number: the vacuum sees the generators under the seed's angle. A_k is the power of the point Φ with respect to the unit circle centred at $-\omega^k$, negative when Φ lies inside that circle; the negative root that the neutrinos require (§42.4) is the pair inside the circle of one generator, the conjugate branch read in the plane.

MEASURED Read backwards, Koide's ratio measures the pair's invariant, $\Xi = 2/(3Q - 1)$. The charged leptons give

$$\Xi_\ell = 2.000013 \pm 0.000031, \tag{140}$$

the vacuum to 1.5 parts in 10^5 (0.4σ, the error from m_τ). The families that carry triality sit off it: (c, b, t) at $\Xi = 1.985$, (d, s, b) at 1.676 and (u, c, t) at 1.293, within the scheme dependence of quark masses.

STANDARD Mechanisms proposed for Koide's relation place the masses in the vacuum structure of a flavour scalar [152] or protect the relation with a family gauge symmetry [153]; the geometric readings are Foot's cone [47] and Kocik's configurations of circles crossing at $\arccos\frac{2}{3}$ [154]. DERIVED Here the potential is the axiom's own and the minimum is its vacuum. IDENTIFICATION What I6 now identifies is (138), the amplitude of a generation as its generator's invariant raised by the pair; the null condition of §42.1 is no longer assumed. The phase δ is a relative angle: mass ratios constant to 10^{-16} per year require it fixed, so the generators turn with the pair, and the universe's rotation enters the masses only as their common phase, which the chiral rotation of §49.12 removes and which gives κ. OPEN The relation $3\delta_\ell = Q$ (§42.7) and its 2×10^{-7} correction are not explained by (138): in this form the branch with $\delta_\ell = 2/9$ exact places the charged leptons 4.2×10^{-7} above the vacuum, $\Xi_\ell = 2(1 + 4.2 \times 10^{-7})$, and the branch with the cone exact leaves the phase 1.75×10^{-7} below 2/9.

43 Causal Time and the Electric Coupling

This section answers the requests of working notes 42A–42C. Those notes ask four questions:

1. Do the existing primitives produce a bounded causal-time variable Θ?
2. Does the inward/outward momentum picture have an invariant form?
3. Does radiation pressure follow from the electromagnetic stress tensor?
4. Does the same geometry fix α?

The first three are answered below from the primitives and established relativity. The fourth is stated with its exact target.

43B.1 The time-rate function

For a particle of mass m and energy E in a local inertial frame, $d\tau^2 = -ds^2/c^2$ and the mass shell (§19) give

$$\Theta \equiv \frac{d\tau}{dt} = \frac{mc^2}{E} = \sqrt{1 - \frac{v^2}{c^2}} = \operatorname{sech}\zeta, \tag{141}$$

where ζ is the rapidity, the hyperbolic angle between the worldline and the time axis. Hence $0 < \Theta \le 1$ for $m > 0$, with $\Theta = 1$ at rest. On the seed cone itself ($m = 0$, $\zeta \to \infty$), $\Theta = 0$. So Θ measures how deep a worldline lies inside the seed cone. The front of the defect (sound) cone, speed $c/\sqrt{3}$, sits at $\Theta = \sqrt{2/3}$.

43B.2 The unified expression

Write a static spacetime as $ds^2 = -N^2c^2dt^2 + h_{ij}dx^i dx^j$. Here t is the Killing time normalized at infinity, which serves as the reference time $t_{\rm ref}$, and N is the lapse.

Consider a particle moving at speed v relative to the local static observer. That observer's clock advances by $N\,dt$, and the particle covers $d\ell = vN\,dt$, so $ds^2 = -N^2c^2dt^2(1 - v^2/c^2)$ and

$$\boxed{\Theta = \frac{d\tau}{dt} = N\sqrt{1 - \frac{v^2}{c^2}} = N\,\frac{mc^2}{E_{\rm loc}} = N^2\,\frac{mc^2}{E_\infty}} \tag{142}$$

Here $E_{\rm loc}$ is the energy measured by the static observer, and $E_\infty = NE_{\rm loc}$ is the conserved Killing energy. For static observers the product of the kinematic and gravitational factors is therefore exact, not an assumption.

The four requested limits follow:

Limit	Condition	Θ
special relativity	$N = 1$	$mc^2/E = \sqrt{1 - v^2/c^2}$
gravitational time dilation	$v = 0$	$N = \sqrt{1 - 2GM/rc^2}$
null trajectory	$m = 0$	0
horizon	$N \to 0$	0

No new object is introduced. The seed cone supplies $\operatorname{sech}\zeta$, the mass shell supplies mc^2/E, and the horizon condition supplies N through $r_s = 2GM/c^2 = 2Mc^2/F$.

43B.3 Two horizons and the balance radius

Add the vacuum of §26.4 ($w = -1$, $\Lambda = 3\Omega_\Lambda H_0^2/c^2$). Static observers around a mass M then have the Schwarzschild–de Sitter lapse

$$N^2 = 1 - \frac{r_s}{R} - \frac{R^2}{r_\Lambda^2}, \qquad r_\Lambda = \frac{r}{\sqrt{\Omega_\Lambda}}. \tag{143}$$

Θ vanishes at both horizons: the black-hole horizon, which is the inward endpoint, and the cosmic horizon, which is the outward endpoint. It is largest where $dN^2/dR = r_s/R^2 - 2R/r_\Lambda^2 = 0$:

$$R_* = \Big(\frac{r_s r_\Lambda^2}{2}\Big)^{1/3} = \Big(\frac{3GM}{\Lambda c^2}\Big)^{1/3}, \qquad \Theta_{\max} = \sqrt{1 - 3\Big(\frac{r_s}{2r_\Lambda}\Big)^{2/3}}. \tag{144}$$

At R_* the inward pull of the mass and the outward push of the vacuum cancel, and a static observer there is in free fall. This is the invariant home of the inward/outward balance.

R_* is the maximum turnaround radius [3]: no structure of mass M stays bound beyond it. For $10^{12}M_\odot$, $R_* = 1.12$ Mpc with the budget (1.11 Mpc with Planck's Λ), comparable to the Local Group's zero-velocity surface.

43B.4 Momentum direction

The sign of $p^{\hat r}$ depends on the observer. The invariant replacement is the expansion θ_+ of the outgoing radial null congruence orthogonal to the spheres of area $4\pi R^2$. In ingoing Eddington–Finkelstein form, with the outgoing null normal regular across the horizon,

$$\theta_+ = \frac{N^2}{R}. \tag{145}$$

Outward light increases the area where $\theta_+ > 0$. At each horizon $\theta_+ = 0$, and inside a horizon $\theta_+ < 0$ (trapped).

Hypothesis H2 of the notes says the causal boundary occurs where the outward null direction ceases to increase the radial coordinate. That is exactly the marginally-trapped condition $\theta_+ = 0$. In a static spacetime it coincides with the static-observer condition $\Theta = 0$, since both reduce to $N^2 = 0$.

43B.5 Radiation pressure from the electromagnetic stress tensor

Radially streaming radiation of luminosity L crossing a sphere of radius R is a null fluid, $T^{\mu\nu}_{\rm EM} = \frac{L}{4\pi R^2 c}\,k^\mu k^\nu$, with $k^{\hat\mu} = (1, 1, 0, 0)$ in the static observer's orthonormal frame. Its nonzero components are

$$T^{\hat t\hat t}_{\rm EM} = T^{\hat t\hat r}_{\rm EM} = T^{\hat r\hat r}_{\rm EM} = \frac{L}{4\pi R^2 c} \equiv \Phi_{\rm out}. \tag{146}$$

The outward momentum flux of the notes is therefore the radial–radial component of the Maxwell stress tensor for radially streaming light. It is exact in the local frame, not a Newtonian analogy.

43B.6 The inward flux and the critical luminosity

For a uniform sphere in Newtonian hydrostatic equilibrium, the central pressure is $P_c = 3GM^2/8\pi R^4 = \frac{3}{2}\Phi_{\rm in}$. So $\Phi_{\rm in} = GM^2/4\pi R^4$ is the gravitational pressure scale. Balancing $\Phi_{\rm out} = \Phi_{\rm in}$ gives

$$L_{\rm crit} = \frac{GM^2 c}{R^2} = P_{\rm max}\Big(\frac{r_s}{R}\Big)^2, \qquad P_{\rm max} = F_{\rm cone}\, c = \frac{c^5}{4G}. \tag{147}$$

At the horizon, $R = r_s$, the critical luminosity equals the maximum power allowed by the axiom. No luminosity can exceed $P_{\rm max}$, so radiation cannot hold a body up at its own horizon. The outward route closes exactly where the inward route closes. The brightest measured source, GW150914, peaked at 0.4% of $P_{\rm max}$ [60]; on the maximum-luminosity conjecture see [59].

43B.7 Dynamical closure versus causal closure

A static star of uniform density has an exact interior solution (Schwarzschild). Its central lapse and central pressure are

$$N_c = \frac{3}{2}\sqrt{1 - \frac{r_s}{R}} - \frac{1}{2}, \qquad \frac{P_c}{\rho c^2} = \frac{1 - \sqrt{1 - r_s/R}}{3\sqrt{1 - r_s/R} - 1}. \tag{148}$$

Both reach their limit together. As $R \to \frac{9}{8}r_s$, $P_c \to \infty$ and $N_c \to 0$: the central clock reaches $\Theta = 0$ while the surface still has $\Theta = \sqrt{1 - 8/9} = \frac{1}{3}$. Buchdahl's theorem [1] extends this to any static fluid whose density does not increase outward: no static configuration exists below $R = \frac{9}{8}r_s$.

So the two closures sit at different radii:

- *dynamical closure* (static support fails) at $R = \frac{9}{8}r_s$;
- *causal closure* (horizon, $\theta_+ = 0$) at $R = r_s$.

They are related but not identical, as the notes anticipated. Between the two radii a body can only collapse.

43B.8 How the clock stops

Consider a body collapsing through $R = r_s$. A distant static observer receives its signals with a rate that decays at late times as

$$\Theta_{\rm obs} \propto e^{-\kappa t}, \qquad \kappa = \frac{g_{\rm H}}{c} = \frac{c^3}{4GM} = \frac{F_{\rm cone}}{Mc}. \tag{149}$$

The late-time law is therefore $d\Theta/dt = -(F_{\rm cone}/Mc)\,\Theta$: the external clock freezes at a rate set by the one-cone force divided by the mass. For $10M_\odot$ the e-folding time is 0.197 ms.

The kinematic endpoint behaves the same way. A body with constant proper acceleration a has $\Theta = \mathrm{sech}(a\tau/c) \to 2e^{-a\tau/c}$, and an accelerated observer sees sources crossing its horizon redshift as $e^{-a\tau/c}$. Both endpoints are approached exponentially at the rate (acceleration)$/c$: proper acceleration for motion, surface gravity for gravity. That is the equivalence principle applied to the two routes.

This answers the question $d\Theta/d\tau = ?$ for the approach to each endpoint. The full trajectory between the endpoints needs the field equations with matter: the TOV equation for static bodies, or a collapse solution.

43B.9 The time-rate of a system

For a composite system, $\Theta_{\rm sys} = M_{\rm inv}c^2/E_{\rm tot}$. A closed system conserves its four-momentum, so $\Theta_{\rm sys}$ is constant; it changes only when energy leaves the system.

Isotropic radiation keeps its invariant mass: a box of light weighs E/c^2. Only a beam in a single direction has zero invariant mass. So "radiating all rest mass as light" drives $\Theta_{\rm sys} \to 0$ only for fully collimated emission, as in a photon rocket. This refines the momentum partition of the notes.

43B.10 Status

Θ is derived from existing primitives—the seed cone, the mass shell and the horizon—and it reproduces special and general relativity exactly. It meets every falsification criterion of note 42B.15: no new object, no new calibration, and the correct limits.

This is the reinterpretation level of the notes, not yet a new prediction. The quantitative content GRUT adds is:

- $L_{\rm crit}(r_s) = P_{\max}$, the axiom's power, equation (147);
- the relaxation rate $F_{\rm cone}/Mc$, equation (149);
- the balance radius R_* evaluated with GRUT's vacuum, equation (144).

43A/43C The fine-structure conjecture

Observation. Let $S \equiv 4\pi^3 + \pi^2 + \pi = 137.036\,303\,776$. Against $1/\alpha = 137.035\,999\,177(21)$ (CODATA 2022 [2]),

$$\frac{1}{\alpha} = S - \delta, \qquad \delta = 0.000\,304\,599. \tag{150}$$

The residual is subtracted: S lies 2.22 ppm above $1/\alpha$. Writing $\delta = K\alpha$ gives $K = 0.041\,741\,012$. Because $1/\alpha$ is known to 1.5×10^{-10}, K is known to 7×10^{-5} of itself, so a derivation must return

$$K = 0.041741(3). \tag{151}$$

The nearest simple fraction, $K = 1/24$, gives $1/\alpha = 137.035\,999\,720$, which is 26σ away.

What a derivation must supply.

(a) *The coefficients.* The reading "one π per dimension" does not match geometry. Unit balls have volumes 2, π, $4\pi/3$ and unit spheres have areas 2, 2π, 4π, $2\pi^2$; π^3 is not a three-dimensional volume coefficient. A structural reading that does exist is

$$4\pi^3 = |S^1| \cdot |S^3|, \tag{152}$$

the volume of the unit $S^1 \times S^3$. Since the spatial section is flat and S^3 is the group manifold of $SU(2)$ (§50.7), $S^1 \times S^3$ is the group manifold of $U(1) \times SU(2)$, the electroweak group of §42.2, at unit radii; it is also the Euclidean manifold of a thermal field at temperature $1/2\pi$. A spectral calculation on it—a heat-kernel expansion, which yields a volume term followed by curvature corrections—naturally produces a leading term with lower-order corrections. Whether it yields $+\pi^2 + \pi$ and then $K\alpha$ is the concrete calculation to do.

(b) *The scale.* α runs, from 1/137.036 at zero momentum transfer to about 1/128 at the Z mass, because charged particles polarize the vacuum. A geometric formula must say which scale it describes and reproduce the running from there; the running from the zero-momentum value is §31.5, so the formula must describe that value.

(c) *The cost of looking.* Consider polynomials $a\pi^3 + b\pi^2 + c\pi + d$:

- with $a \le 4$ and $|b|, |c|, |d| \le 4$, a random target between 100 and 200 is matched to 2.2 ppm 0.5% of the time;
- with coefficients up to 6, the rate is 3.8%.

Many other families have been tried. Wyler's formula matched α to 0.6 ppm before later measurements excluded it [4]. A 2.2-ppm match is suggestive, not evidential, until K is derived.

The link to Θ. A photon has $\Theta = 0$ exactly. The vacuum's static observers have $\Theta_{\rm vac} = \sqrt{1 - R^2/r_\Lambda^2}$ (§43B.3). A "mismatch" between them becomes a number only through a coupling. The physical candidate is the vacuum polarization of the photon on $S^1 \times S^3$, which is the same calculation as in (a).

The spectral route, tested. The decomposition is exact: $4\pi^3 + \pi^2 + \pi = |S^1||S^3| + \frac{1}{2}|S^3| + \frac{1}{2}|S^1|$, bulk volume plus half of each factor. The heat-kernel expansion does not produce it. On $S^1 \times S^3$ with unit radii, $\mathrm{Tr}\, e^{-t\Delta} = (4\pi t)^{-2} \sum_k a_k t^k$ with $a_0 = \mathrm{Vol}$ and $a_1 = \frac{1}{6}\int R$; since $R = 6$ there, $a_1 = a_0$, and every coefficient is proportional to the same volume. The series generates powers of t, not the two lower-dimensional volumes, and no weighting in it halves each factor. Nor does any simple constant reach the target: 1/24 is 26σ away, $1/8\pi$ and $\pi/75$ much further.

The calculated weight. DERIVED The one-loop effective action on $S^1(2\pi) \times S^3(1)$ (§43F) gives the correction its thermal weight: $K = \frac{1}{24}\left(1 + e^{-2\pi}\right)$, so

$$\frac{1}{\alpha} = 4\pi^3 + \pi^2 + \pi - \frac{\alpha}{24}\left(1 + e^{-2\pi}\right) = 137.035\,999\,151\,7. \tag{153}$$

Every piece belongs to $S^1(2\pi) \times S^3(1)$: the volume and the two half factors; half the Casimir coefficient of the S^1 factor, $\frac{1}{24}$; and the first thermal correction at the horizon temperature $1/2\pi$, carried by the one conformally coupled scalar of the reset, the conformal factor of the gluing between aeons (§43F). Against CODATA 2022, 137.035 999 177(21) [2], it lies 1.2σ low; against the electron magnetic moment, 137.035 999 166(15) [111], 0.95σ low; the atom-recoil values 137.035 999 206(11) (Rb) [112] and 137.035 999 046(27) (Cs) [113], which disagree with each other by 5.5σ, lie at -4.9σ and $+3.9\sigma$. With every massless term at $\beta = 2\pi$ the weight is 1.0197 and $1/\alpha = 137.035\,999\,140\,5$, 1.7σ low.

The circle weight. The weight $3/\pi = E_2(i)$, found by inspection in draft 37, returns 137.035 999 177 3, 0.015σ from CODATA, but the calculation does not produce it: it is the circle factor's own number, $-E_2(i)/24 = -1/8\pi$ being the exact energy of a chiral boson on the self-dual torus. Its match is a coincidence at the few-percent level. The two forms differ by 2.6×10^{-8} in $1/\alpha$, which the next generation of measurements resolves.

The half factors. DERIVED A cone of deficit π is the orbifold $\mathbb{R}^2/\mathbb{Z}_2$: the framework's deficit is a $\mathbb{Z}_2$ identification. The electroweak group carries exactly one such $\mathbb{Z}_2$ between its factors, $U(2) = (U(1) \times SU(2))/\mathbb{Z}_2$ with $\mathbb{Z}_2 = \{(1,1),(-1,-1)\}$. Taken from each factor it leaves $U(1)/\mathbb{Z}_2$, a circle of length $\pi = \frac{1}{2}|S^1|$, and $SU(2)/\mathbb{Z}_2 = SO(3) = \mathbb{RP}^3$, of volume $\pi^2 = \frac{1}{2}|S^3|$. Hence

$$S = \mathrm{vol}\big(U(1) \times SU(2)\big) + \mathrm{vol}\big(SO(3)\big) + \mathrm{vol}\big(U(1)/\mathbb{Z}_2\big) = 4\pi^3 + \pi^2 + \pi : \tag{154}$$

the electroweak group at unit radii and its two factors, each closed by the deficit π. DERIVED The coefficient $\frac{1}{24}$ is the generator defect's tension in units of the axiom force. The defect map gives $f_\delta^2 = \delta c^4/8\pi G$, so for $\delta_{\mathrm{gen}} = \pi/3$,

$$\frac{f_{\mathrm{gen}}^2}{F} = \frac{\delta_{\mathrm{gen}}}{8\pi} = \frac{1}{24} \tag{155}$$

[63]; it coincides with the zero-point energy of one chiral boson on the unit circle, $c/24$ with $c = 1$. The correction is therefore built entirely from the framework, $K = (f_{\mathrm{gen}}^2/F)\,(1 + e^{-2\pi})$: the generator tension, dressed by the boundary's calculated thermal factor. IDENTIFICATION That $1/\alpha$ is the volume sum, and that its correction is α times this dressed tension, are the identifications on which the relation rests. The paper's other predictions do not depend on it.

43D The electroweak scale

The hierarchy between the defect scale and the electroweak scale, with $f_\delta = M_{\mathrm{P}}/\sqrt{8} = 4.317 \times 10^{18}$ GeV and $v_{\mathrm{EW}} = 246.22$ GeV, is

$$\Big(\frac{v_{\mathrm{EW}}}{f_\delta}\Big)^2 = 3.25 \times 10^{-33}, \qquad j = \log_4\Big(\frac{f_\delta^2}{v_{\mathrm{EW}}^2}\Big) = 54.0. \tag{156}$$

Read as a cascade depth of 54 levels at a factor of 4 each, this is the measured ratio rewritten in base 4: any ratio returns some j, and the agreement is exact by construction. It becomes a derivation only when a second condition fixes the depth—a termination rule for the cascade, or another observable that lands at the same j.

A second gap sits in the same sector. The Higgs mass requires a quartic $\lambda_H = m_H^2/2v_{\mathrm{EW}}^2 = 0.129$, while the potential of the axiom gives $\lambda_\chi/4! = 0.5$, a factor of 3.9 away. The structure of the weak sector—$SU(2)$ from the two axes, chirality from the two lobes, a doublet Higgs and therefore $\rho = 1$—follows from the geometry (§42.2); the scale and the quartic do not.

43E The full trajectory of Θ

DERIVED Outside a collapsing dust ball the metric is Schwarzschild (Birkhoff), and the surface falls from rest at R_0 on the cycloid $R = \frac{R_0}{2}(1 + \cos\eta)$. A radial photon emitted at R reaches a distant observer at $t_{\mathrm{obs}} = t + [r_*(\infty) - r_*(R)]/c$, with $r_* = r + r_s \ln|r/r_s - 1|$, so the received clock rate is

$$\Theta_{\mathrm{obs}}(R) = \frac{d\tau}{dt_{\mathrm{obs}}} = \frac{1 - r_s/R}{E + \sqrt{E^2 - 1 + r_s/R}}, \qquad E = \sqrt{1 - \frac{r_s}{R_0}}. \tag{157}$$

It runs from $\Theta = E$ at the start—the static redshift, which tends to 1 as $R_0 \to \infty$—to $\Theta = 0$ at the horizon, approached as $(1 - r_s/R)/2E \propto e^{-\kappa t_{\mathrm{obs}}}$ with $\kappa = c/2r_s = c^3/4GM$, the rate of §43B.8. For $R_0 = 10r_s$ the values at $R/r_s = 10, 5, 2, 1.5, 1.1, 1.01$ are 0.949, 0.632, 0.316, 0.196, 0.049 and 0.0052, and the late-time logarithmic slope is $-0.5000\,c/r_s$.

43F The fine-structure constant

MEASURED The natural spectral invariants of $S^1 \times S^3$ at unit radii—the Casimir energies of spin 0, $\frac{1}{2}$ and 1 on S^3 ($\frac{1}{240}$, $\frac{17}{960}$, $\frac{11}{120}$), $\zeta(-1)$, $\zeta(-3)$, and half the S^1 Casimir coefficient, $\frac{1}{24}$—do not return $K = 0.041741(3)$; the nearest, $\frac{1}{24}$, is 25σ away. Its residual, $(K - \frac{1}{24})/\frac{1}{24} = (1.784 \pm 0.072) \times 10^{-3}$ given the uncertainty of K, has the size of the first thermal correction on $S^1_{2\pi} \times S^3$, $e^{-2\pi} = 1.867 \times 10^{-3}$, from which it differs by 1.2σ: not significantly. IDENTIFICATION It equals $E_2(i)\, e^{-2\pi} = (3/\pi)\, e^{-2\pi} = 1.7833 \times 10^{-3}$, where $E_2(i) = 3/\pi$ is the quasi-modular Eisenstein series at the self-dual point of the $S^1(2\pi)$ factor, so

$$K = \frac{1}{24}\left[1 + \frac{3}{\pi}\, e^{-2\pi}\right] = 0.0417410; \tag{158}$$

the value $E_2(i) = 3/\pi$ is exact, from the transformation $E_2(-1/\tau) = \tau^2 E_2(\tau) + 6\tau/\pi i$ at its fixed point $\tau = i$. The relation of §43A then returns $1/\alpha = 137.035\,999\,177\,3$, 0.015σ from CODATA 2022 (equation (153)). The precision of K tests the thermal weight to $\pm 3.7\%$: $3/\pi$ lies at 0.015σ, and the bare weight 1, one conformal scalar's first term, at 1.2σ. The conjecture is a low-energy statement: at M_Z, where $1/\alpha = 127.95$ [13], it would need $K = 0.071$. DERIVED *The calculation.* The one-loop effective action of the GRUT field content on $S^1(2\pi) \times S^3(1)$—the conformal boundary at the horizon temperature—is $\ln Z = -2\pi E_c + \ln Z_{\rm th}$. On $S^3(1)$ a term of order $e^{-2\pi}$ needs a mode with $\omega = 1$, and only a massless conformally coupled scalar has one, with degeneracy 1: each such real scalar adds exactly one unit of $e^{-2\pi}$. Every other field starts higher: a Weyl fermion at $\omega = \frac{3}{2}$, the photon at $\omega = 2$ (degeneracy 6), the graviton at $\omega = 3$; the reset carrier of §45, minimally coupled with $m^2/H^2 \simeq -0.04$, at $\omega \simeq 1.72$; the radial mode, $m_\delta R = \sqrt{8}$, at $\omega = 3$; and the phase, held at its rate by (51), has zero sound speed and no thermal modes. Without a conformal scalar the weight is zero, $K = \frac{1}{24}$, and the relation misses by 26σ; with two it misses by 28σ. The calculation requires exactly one, and the reset supplies exactly one: the conformal factor of the gluing between aeons, which in conformal cyclic cosmology obeys the conformally invariant equation of a conformally coupled scalar with quartic self-coupling [114]—the boundary field χ with $-\frac{1}{12}R\chi^2$ of the companion paper's terminal state [63]. At $\beta = 2\pi$ its full energy is exact,

$$E_\chi = \frac{1}{240} + \sum_{n\geq 1} \frac{n^3}{e^{2\pi n} - 1} = \frac{E_4(i)}{240} = \frac{\Gamma(\frac{1}{4})^8}{5120\,\pi^6} = 0.006\,065\,679, \tag{159}$$

and its complete thermal part, with the photon's, is $\ln Z_{\rm th} = (1.0084 + 0.0113)\, e^{-2\pi}$. The weight is therefore 1, not $3/\pi$, and the relation of §43A gives $1/\alpha = 137.035\,999\,151\,7$ (equation (153)), 1.2σ from CODATA 2022.

DERIVED The mode that carries the weight, at frequency $c/R_\infty = H_\infty$ on the final horizon's three-sphere, is one quantum of the spin (§36.11), the quantum that each quantum of the rotation carries (§36.14), and $e^{-2\pi} = e^{-\hbar\omega/k_B T_\infty}$ is its Boltzmann weight at the horizon's temperature. The correction is the generator tension dressed by the occupation of one quantum of the spin, $K = (f^2_{\rm gen}/F)\,(1 + e^{-\hbar\omega/k_B T_\infty})$: both factors are quantities of the framework. IDENTIFICATION What remains identified is that the volume sum is the inverse coupling at zero momentum; the calculation that would return it is open, and the weight 1 against $3/\pi$ is decided by 2.6×10^{-8} in $1/\alpha$.

44 TRSD and GRUT: the formal bridge

TRSD and GRUT are one framework stated twice: TRSD as a variational principle, GRUT as a geometry with a force axiom. This section proves the equivalence, gives the dictionary that converts every TRSD quantity into the standardized F, and lists exactly what remains an input after both axioms are in place.

44.1 The two axioms

Axiom G (GRUT).

$$F\,G = c^4, \tag{160}$$

with F the force of the whole horizon sphere, together with the geometry of §1–§4: the seed $h = r$ and the C_3 triangle.

Axiom T (TRSD).

$$\delta \int d^4x\sqrt{-g}\left[\frac{c^4 R}{16\pi G} - \frac{\delta c^4}{16\pi G} g^{\mu\nu}\partial_\mu\phi_+\partial_\nu\phi_- - \varepsilon_V V(\Xi)\right] = 0, \qquad \Xi = \phi_+\phi_-. \tag{161}$$

Nothing else is assumed in either column. Every statement below is a consequence.

44.2 Bridge theorem B1: the dictionary

Given Axiom G, every coefficient of Axiom T is a pure number times F, and every TRSD quantity is a pure number times a monomial in c, r, F and ℓ_{P}.

Proof. $G = c^4/F$ by Axiom G. Substituting into the two coefficients of Axiom T gives $c^4/16\pi G = F/16\pi$ and $\delta c^4/16\pi G = \delta F/16\pi$. TRSD's normalization is $f_\delta^2 = \delta c^4/8\pi G = \delta F/8\pi$, and its Einstein coupling is $8\pi G/c^4 = 8\pi/F$. Every other TRSD quantity is built from these with c, r and ℓ_{P} by the unit basis of §0.2. □

TRSD quantity	TRSD form	F-form	Value ($H_0 = 73.04$)
normalization	$f_\delta^2 = \delta c^4/8\pi G$	$\delta F/8\pi$	$F/8$ at $\delta = \pi$
$f_\delta(\pi)$	$\sqrt{c^4/8G}$	$\sqrt{F/8}$	$3.889 \times 10^{21}\ \sqrt{\mathrm{J/m}}$
$f_\delta(\pi/3)$	$\sqrt{c^4/24G}$	$\sqrt{F/24}$	$2.245 \times 10^{21}\ \sqrt{\mathrm{J/m}}$
Einstein coupling	$8\pi G/c^4$	$8\pi/F$	$2.077 \times 10^{-43}\ \mathrm{N}^{-1}$
maximum force	$c^4/4G$	$F/4 = F_{\mathrm{cone}}$	3.026×10^{43} N
total mass	$c^3/8GH_0$	$Fr/8c^2$	2.132×10^{52} kg
total energy	$c^5/8GH_0$	$Fr/8$	1.916×10^{69} J
cell count	$2\pi r^2c^3/\hbar G$	$2\pi Fr^2/\hbar c$	3.858×10^{122}
information	$N/2$	$\pi Fr^2/\hbar c$	1.929×10^{122} nats
temperature	$\hbar H_0/2\pi k_B$	$\hbar c/2\pi k_B r$	2.878×10^{-30} K
dark energy	$\Omega_\Lambda\, 3H_0^2c^2/8\pi G$	$\Omega_\Lambda\, 3F/8\pi c^2r^2$	$5.24 \times 10^{-10}\ \mathrm{J\,m^{-3}}$
defect length	$8\pi GM_*/c^2$	$8\pi M_*c^2/F$	0.18 pc at $1.5 \times 10^{11} M_\odot$
critical acceleration	$3\sqrt{3}\,cH_0/4\pi^2$	$3\sqrt{3}\,c^2/4\pi^2 r$	$9.340 \times 10^{-11}\ \mathrm{m\,s^{-2}}$

Only the labels change: TRSD's "maximum force" is the one-cone force $F/4$, and its "total" mass and energy are the one-cone quarters of the sphere values (§17–§18). No numerical value in TRSD moves.

44.3 Bridge theorem B2: Axiom G follows from Axiom T

Axiom T plus the closure of §3.12 implies $FG = c^4$.

Proof. Varying Axiom T with respect to $g^{\mu\nu}$ gives $G_{\mu\nu} + \Lambda g_{\mu\nu} = (8\pi G/c^4)T_{\mu\nu}$ (§30.1), whose static spherical vacuum solution is Schwarzschild with $r_s = 2GM/c^2$. Its horizon surface gravity is $g_{\mathrm{H}} = c^2/2r_s = c^4/4GM$, so the force on the mass at its own horizon is $Mg_{\mathrm{H}} = c^4/4G$, independent of M: this is F_{cone} (§8.2). The closure $V_s = 4V_c$ of §3.12, proved from the seed by Cavalieri, gives four such cones per sphere, so the sphere carries $F = 4F_{\mathrm{cone}} = c^4/G$, i.e. $FG = c^4$. □

44.4 Bridge theorem B3: Axiom T's geometric sector follows from Axiom G

Axiom G plus §§1–4 fixes both coefficients of Axiom T; it does not fix the potential.

Proof. Lovelock's theorem in four dimensions (§30.1) admits only $aG_{\mu\nu} + bg_{\mu\nu}$, so the gravitational term is $\kappa^{-1}R$ with a single constant; Axiom G fixes it, $\kappa^{-1} = c^4/16\pi G = F/16\pi$. The kinetic coefficient is the defect tension $f_\delta^2/2 = \delta F/16\pi$, with $\delta = \pi$ derived in §4.4 from the three equilateral faces and checked by the curvature closure $4\delta_{\mathrm{tot}} = 4\pi$. The potential $V(\Xi)$ and its scale ε_V are not fixed by Axiom G: the geometry constrains the vacuum to be stable and non-degenerate but selects no particular function. □

So the bridge is exact in the geometric sector and one function short in the scalar sector. The residual freedom is a single potential; of the two numbers it carries, ε_V is fixed by the dark-matter length (§37.2) and V_0 vanishes (§38.6).

44.5 The coil

The coil of the TRSD prototype is the generator of the same geometry. A lobe is

$$\mathbf{r}_\pm(t) = \big(R_0 \sin t \cos(\pm\omega t),\ R_0 \sin t \sin(\pm\omega t),\ \pm pt\big), \qquad t \in [0, \pi], \tag{162}$$

pinching at $t = 0, \pi$. Three statements follow, and one does not.

(i) The pair. The two lobes are the conjugate pair $p_\pm$ of §1.2, so the fold and the double cover of §1.8 apply unchanged.

(ii) The envelope fixes the pitch. The bicone envelope has $\tan\alpha = R_0/p$. The seed cone $h = r$ requires $\alpha = 45°$, hence $p = R_0$; the defect cone requires $\alpha = 30°$, hence $p = \sqrt{3}\,R_0$ (§4.4). The pitch-to-radius ratio is therefore not free: it is 1 for the causal cone and $\sqrt{3}$ for the defect cone.

(iii) Three nodes. The lobes meet where $\sin(\omega t) = 0$, i.e. at $t_n = n\pi/\omega$. Within $t \in [0, \pi]$ there are exactly three such crossings if and only if $2 \le \omega < 3$.

(iv) What does not follow. The azimuth of a crossing is $\omega t_n = n\pi$, so consecutive crossings are separated by π in azimuth for every ω, not by $2\pi/3$. The equal spacing of the three nodes is therefore not a consequence of the crossing condition; it follows from the symmetry statement of §4.2, that no node is preferred, which is the same condition that makes the generating triangle equilateral. The prototype's claim that the crossings are "equally spaced in azimuth by $2\pi/3$" should be replaced by this.

44.6 Bridge theorem B4: what remains an input

With Axiom G and Axiom T both in place, the irreducible inputs are: two unit conventions, one boundary datum, and the dimensionless constants that neither axiom determines.

Proof. By Theorem 1, three dimensionful anchors are conventions. Axiom G supplies one relation between F and G, so one of the pair is fixed by the other, and c and $\hbar$ (equivalently ℓ_{P}) remain as the two conventions. The scale r is not fixed by either axiom, since both are invariant under $r \to \lambda r$ with the dimensionless content unchanged; one datum—N, equivalently r_0/ℓ_{P}—must be supplied. Every remaining quantity is a pure number times a monomial in c, r, F, ℓ_{P} (B1), so anything not determined is a pure number. □

Input	Why it is not derivable	Fixed by
c, $\hbar$	unit conventions (Theorem 1)	SI definitions
N (equivalently H_0)	scale invariance of both axioms	measurement (§7); by α if H4 holds (§48.12)
α	not in either axiom	measurement; §43A/C target
ε_V	tied to L_* by normalization; $V_0 = 0$	§37.2, §38.6
L_* (or m_δ), ϕ_i	scalar sector scales	§37.2, §37.7
α_s (the strong scale)	not in either axiom	measurement; tied to N_∞ by the nucleon, §50.8–50.10
v_{EW}, λ_H	not in either axiom	measurement, §43D
$\sin^2\theta_W$	not in either axiom	measurement; the pair's closure, §54.7
μ_ℓ, μ_ν, δ_ν	scales and phase of the lepton cones	m_e, m_μ, the splittings; §0.3, §42

The baryon ratio, an input in earlier versions, is derived in §25.7(v), and the dark ratio, identified in earlier versions (I5), follows from the resonance (§25.11). That list, with the ledger of §0.3, is complete: every other number in this paper is derived from the two axioms and the geometry.

44.7 Conditions the bridge imposes on the prototype

Keeping both columns formal requires five statements of the TRSD prototype to be restated. Each is corrected in the sections named.

1. Quantities of the form (pure number)$\times 10^{-10}\,\mathrm{m\,s^{-2}}$—the Torridity scales and their sequences—are unit-dependent and cannot be framework statements (§22.8).
2. The threshold parameter β cancels in the normalized survival function, so no value of it, including 4, is fixed or needed (§22.1).
3. $a_c = c/\tau_1$ is not a centripetal acceleration, whose value would be 2π larger (§5); it is the acceleration at which one activation occurs per cycle of the fundamental mode (§22.10).
4. $\delta = 8\pi GM/c^2$ is dimensionless only for a tension, $\mu = M/r$; the correct form is $\delta(M) = 4\pi M/M_{\mathrm{H}}$ (§18.1).
5. Horizon information is counted in nats; "bits" differs by $\ln 2$ (§20).

With those five restatements, and the dictionary of §44.2, the two columns are the same theory: TRSD supplies the action, GRUT supplies the geometry and the force normalization, and each implies the other by B2 and B3. B2 uses the four-cone reading I1.

Note. The revised prototype, TRSD v2 [63], implements the five restatements and returns five results used here: the SPARC confrontation (§22.6), the normalization $\varepsilon_V = f_\delta^2/L_*^2$ (§37.2), the physical self-coupling (§37.3), the vanishing of V_0 (§38.6), and the reading of its flow equation as relaxation to the vacuum $\Xi = 2$. Because the two papers share these constructions, their agreement on them is consistency, not independent confirmation (§0.8).

45 Primordial Spectrum from the Conformal Reset

The background GRUT budget fixes the horizon scale, the matter budget and the acoustic angle. This section derives the primordial two-point function of the defect scalar from the reset: the primary term from the terminal de Sitter phase of the previous aeon, the interaction term from a chiral pair convolution on the C_3 cycle, and from both the exact scalar tilt and running. Two numbers of the reset remain to be computed—the carrier's mass in units of the terminal Hubble rate, which the tilt measures, and its conversion efficiency to curvature, which the amplitude measures.

The standard dimensionless curvature spectrum is defined by

$$\langle \mathcal{R}_{\mathbf{k}} \mathcal{R}_{\mathbf{k}'} \rangle = (2\pi)^3 \delta^3(\mathbf{k}+\mathbf{k}') \frac{2\pi^2}{k^3} \Delta^2_{\mathcal{R}}(k), \tag{163}$$

so that $\langle \mathcal{R}^2 \rangle = \int d\ln k\, \Delta^2_{\mathcal{R}}(k)$. A sum of terms proportional to k and k^3 cannot itself be a dimensionless primordial spectrum, because its terms carry unequal dimensions. With the standard pivot

$$k_* = 0.05\ \mathrm{Mpc}^{-1}, \qquad x \equiv \frac{k}{k_*}, \tag{164}$$

all powers below are dimensionless. The symbol ν is used for the spectral exponent because n is reserved for the exposure order of Section 22 and the defect dimension of Section 26.

45.0 Completed thermodynamic and geometric reset closure

The reset is a closed thermodynamic–geometric transition when the following distinction is kept exact: the old aeon does not undergo a global contraction of its physical scale factor. Before the junction, $a = b^{-1}$ and $b \to 0^+$, so $a \to \infty$; after the junction the new physical scale factor is $\tilde{a} = |b| \to 0^+$. Thus the reset is not a global compressive shock of the old aeon. It is a conformal identification at which the old physical scale loses its operational meaning and the new aeon begins with a vanishing physical scale. This is the geometry already fixed by (360) and $\tilde{H} = 1/(2\tilde{\tau})$.

R1. Matter-sector conservation and thermodynamic closure

On every regular matter branch let N^μ denote a conserved matter-number current and $T^{\mu\nu}_{\mathrm{m}}$ the total matter stress tensor. In the absence of a localized junction source, the finite-flux conditions are

$$\nabla_\mu N^\mu = 0, \qquad \nabla_\mu T^{\mu\nu}_{\mathrm{m}} = 0, \qquad \boxed{[N^\mu n_\mu]^+_- = 0}, \qquad \boxed{[T^{\mu\nu}_{\mathrm{m}} n_\nu]^+_- = 0}. \tag{165}$$

These equations express material and energy–momentum continuity without requiring that every old composite object survive as the same bound state. Stars, galaxies and black holes are not conserved objects; their conserved charges and stress–energy are redistributed among the degrees of freedom of the thermal state.

Thermalization is represented by the entropy current s^μ. The second law gives

$$\nabla_\mu s^\mu = \sigma_s \geq 0. \tag{166}$$

Consequently the reset may destroy macroscopic structure while preserving conserved currents and increasing coarse-grained entropy. The radiation-dominated opening state is therefore a high-entropy thermal state, but it is not identified with exact global maximum entropy or exact equilibrium: GRUT's reset is explicitly out of equilibrium and its subsequent cooling and baryon asymmetry calculation rely on that fact.

For the relativistic plasma stage,

$$p = \frac{1}{3}\rho c^2, \qquad \rho = a_R T^4, \qquad s = \frac{4\rho}{3T}, \qquad \tilde{H} = \frac{1}{2\tilde{\tau}}, \tag{167}$$

so the new aeon has the required radiation-like equation of state and cools continuously as it expands. The thermodynamic closure is therefore thermalization plus conservation plus the entropy inequality, not a postulated instantaneous equilibrium.

R2. The collisionless phase survives the reset

The phase sector is governed by the already closed junction theorem,

$$\boxed{\mathcal{T}_\chi = 1}, \qquad \boxed{\mathcal{M}_{\mathcal{R}}(k) = 1 \text{ for the adiabatic superhorizon mode}}, \qquad \boxed{\mathcal{M}_{\mathcal{I}}(k) = 1}. \tag{168}$$

Thus the phase current crosses the conformal junction without a junction-localized source, and the specified superhorizon curvature correlations and conformal information modes are transmitted unchanged. The Bullet Cluster provides the intended local dynamical analogy: collisional plasma is shock-heated while the gravitating component is much less affected by drag. It is an analogy for the separation of collisional and collisionless sectors, not a separate derivation of the cosmological junction.

R3. No additional Stückelberg transition is required

The reset does not invoke a Stückelberg mass cancellation or restoration of a gauge $U(1)$. In the existing framework, the dark-sector phase θ is a global phase/clock variable and carries no gauge field. The transition is instead the conformal continuation

$$g_{\mu\nu} = b^{-2}\hat{g}_{\mu\nu} \quad \longrightarrow \quad \tilde{g}_{\mu\nu} = b^2\hat{g}_{\mu\nu}, \qquad b(\eta_c) = 0, \tag{169}$$

with the same regular conformal geometry and $b'(\eta_c) = -\omega$. No new vector mass or gauge-symmetry-breaking sector is introduced by the reset.

R4. Recombination and the next structure hierarchy

After the reset, $\tilde{a}$ grows, the radiation density falls as $\tilde{a}^{-4}$, and the baryonic component cools. The transferred adiabatic phase correlations provide the initial gravitational seed through $\mathcal{M}_{\mathcal{R}} = 1$. Baryons subsequently respond to the resulting gravitational potentials and, after neutral recombination, decouple from the radiation and participate in structure formation.

The statement that the *same phase correlations* seed the next aeon is therefore a closed consequence of the junction transport equations. The stronger statement that every individual baryon or bound object in the old aeon is literally the same particle in the new aeon is not part of the conserved-field description and is not required by the framework. In particular, GRUT's baryogenesis sector permits baryon number violation through electroweak anomaly processes while conserving the appropriate anomaly-free combination; material continuity must therefore be understood as continuity of the conserved field degrees of freedom and fluxes, not identity of every composite particle.

The CMB then has its standard role within the framework's sequence: the radiation field evolves through the hot plasma era and the observed microwave background is the relic radiation released as the baryonic plasma recombines. Structure formation occurs later from the inherited primordial perturbations and the subsequent matter dynamics. No independent creation of an abstract information carrier is required: the conformal information channel and phase correlations are transported by the already derived junction map, while the horizon-cell count itself resets according to $N_\infty \to 0 \to N_\infty$.

R5. Completed reset statement

The physical reset is therefore a single closed sequence:

$$\boxed{\text{terminal dilution} \;\to\; \text{conformal junction} \;\to\; \text{radiation-dominated new aeon} \;\to\; \text{cooling and recombination} \;\to\; \text{structure fo}} \tag{170}$$

with phase-current transmission, adiabatic curvature transmission and conformal information transmission fixed by their stated junction assumptions. Separately, the finite conformal-volume remainder of Auxoviscausal strain matches by (339) when the common conformal spatial metric and identified flow tube are regular; the raw logarithmic strain does not. The reset is thermodynamically irreversible at the coarse-grained level through $\nabla_\mu s^\mu \geq 0$, while the specified conformal transport channels are lossless. This is the scope of the thermodynamic and geometric reset closure used by TRSD/GRUT.

45.1 The reset background and the primary mode

DERIVED The previous aeon ends in de Sitter expansion: the space-filling defect has $w = -1$ and the scalar potential has $V_0 = 0$ (§38), so H approaches a constant, and the conformal reset (§43B) identifies the future boundary of that phase with the next Big Bang. A light scalar of mass m on this background, in its Bunch–Davies state, leaves the horizon with the late-time spectrum STANDARD

$$\Delta^2_{\delta\phi}(k) = \Big(\frac{H}{2\pi}\Big)^2 \Big[\frac{2^{\mu-3/2}\,\Gamma(\mu)}{\Gamma(3/2)}\Big]^2 \Big(\frac{k}{aH}\Big)^{3-2\mu}, \qquad \mu = \sqrt{\frac{9}{4} - \frac{m^2}{H^2}}, \tag{171}$$

an exact power law. Converted to curvature at the reset with efficiency κ, $\mathcal{R} = \kappa\,\delta\phi/\phi_*$, the primary term is

$$\boxed{\Delta^2_{\mathcal{R},\mathrm{prim}} = A_s x^\nu, \qquad \nu = 3 - 2\mu, \qquad \frac{m^2}{H^2} = \frac{3}{2}\nu - \frac{\nu^2}{4},} \tag{172}$$

with $A_s = \kappa^2 (H/2\pi\phi_*)^2 [2^{\mu-3/2}\Gamma(\mu)/\Gamma(3/2)]^2 (k_*/aH)^\nu$. The exponent is the carrier's mass ratio, and while m/H is constant the primary term has no running. A massless carrier gives $\nu = 0$, exact dilation invariance; a red tilt requires a slightly tachyonic one.

DERIVED The carrier is the phase of the defect condensate. At the reset the radial mode is heavy—in the companion paper $m_\delta^2/H^2 = 8$, far above $\frac{9}{4}$—so its super-horizon modes oscillate and decay and it carries no spectrum; the phase, light below the $\mathbb{Z}_3$ barrier, is the one field that can. The three generator defects of the C_3 horizon break its symmetry to $\mathbb{Z}_3$, so the potential contains only the harmonics $\cos 3k\theta$, and its leading term is $V(\theta) = \Lambda_3^4(1-\cos 3\theta)$, with Λ_3 set by the tilt; and the defects are where θ passes between the three minima, through the maxima, where $m^2 = -9\Lambda_3^4/f_\delta^2$. The curvature perturbation is seeded there, with

$$\nu \simeq \frac{2}{3}\frac{m^2}{H^2} = -\frac{6\Lambda_3^4}{f_\delta^2 H^2}. \tag{173}$$

45.2 The reset spectrum

The primary mode and its quadratic conjugate-pair interaction give

$$\boxed{\Delta^2_{\mathcal{R},\mathrm{iso}}(k) = A_s x^\nu + C_s x^{2\nu-1},} \tag{174}$$

DERIVED with the interaction amplitude derived in §45.7,

$$\boxed{C_s = 2A_s^2\epsilon^2 u^2 B(\nu+1,\nu+1), \qquad B(a,b) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)},} \tag{175}$$

and the pair coupling fixed by the cubic vertex of the defect potential (§45.7),

$$\epsilon u = \frac{1}{\kappa}\frac{|V'''|\,\phi_*}{2V''} \le \frac{3\sqrt{2}/4}{\kappa}. \tag{176}$$

Define the relative interaction strength

$$\boxed{r(k) \equiv \frac{C_s}{A_s} x^{\nu-1} = 2A_s\epsilon^2 u^2 B(\nu+1,\nu+1)\, x^{\nu-1},} \tag{177}$$

so that $\Delta^2_{\mathcal{R},\mathrm{iso}} = A_s x^\nu [1 + r(k)]$.

45.3 Exact scalar tilt

With $n_s(k) - 1 = d\ln\Delta^2_{\mathcal{R},\mathrm{iso}}/d\ln k$ and $d\Delta^2_{\mathcal{R},\mathrm{iso}}/d\ln k = \nu A_s x^\nu + (2\nu-1)C_s x^{2\nu-1}$,

$$\boxed{n_s(k) - 1 = \frac{\nu A_s x^\nu + (2\nu-1)C_s x^{2\nu-1}}{A_s x^\nu + C_s x^{2\nu-1}} = \frac{\nu + (2\nu-1)r(k)}{1 + r(k)}.} \tag{178}$$

At the pivot, $r_* = C_s/A_s = 2A_s\epsilon^2 u^2 B(\nu+1,\nu+1)$ and

$$\boxed{n_{s*} - 1 = \frac{\nu + (2\nu-1)r_*}{1+r_*}, \qquad \nu = \frac{(n_{s*}-1) + n_{s*}r_*}{1+2r_*}.} \tag{179}$$

45.4 The tilt measures the reset

Exact dilation invariance of the reset seed gives $\nu = 0$, $r_* = 0$, $n_{s*} = 1$: a massless carrier. DERIVED Because C_s carries A_s^2, the interaction is suppressed by the amplitude itself:

$$r_* = 2A_s(\epsilon u)^2 B \simeq \frac{5\times 10^{-9}}{\kappa^2}, \tag{180}$$

so $\nu = (n_{s*}-1) + (2-n_{s*})r_* + \mathcal{O}(r_*^2)$ equals $n_{s*} - 1$ to that accuracy. The observed red tilt is therefore not a calibrated exponent but a measurement of the carrier's mass, equation (172):

$$\boxed{\frac{m^2}{H^2} = \frac{3}{2}(n_{s*}-1) - \frac{(n_{s*}-1)^2}{4}.} \tag{181}$$

45.5 Running and tensor tilt

With $dr/d\ln k = (\nu-1)r$, differentiating (178) gives the exact running

$$\boxed{\alpha_s(k) \equiv \frac{dn_s}{d\ln k} = \frac{(\nu-1)^2 r(k)}{[1+r(k)]^2}.} \tag{182}$$

It is nonnegative for $C_s > 0$, and a robust negative running over the range of the spectrum would falsify equation (174). PREDICTION By (180), at the pivot $\alpha_{s*} \simeq 1.07\, r_* \simeq 5\times10^{-9}/\kappa^2$: below 5×10^{-5} for any conversion efficiency above one per cent, and within the Planck bound for any $\kappa > 8\times10^{-4}$ (§45.9). The running is predicted to be positive and unobservably small. PREDICTION The graviton is massless in the same exact de Sitter phase, so the tensor spectrum is scale-invariant, $n_t = 0$, whereas single-field slow roll requires $n_t = -r/8$.

45.6 Relation to the C_3 defect component

The isotropic reset spectrum is distinct from the anisotropic C_3 defect pattern of Section 39. Write the full initial curvature perturbation as

$$\boxed{\mathcal{R}(\mathbf{k}) = \mathcal{R}_{\rm iso}(\mathbf{k}) + \lambda_{C_3}\mathcal{R}_{C_3}(\mathbf{k}).} \tag{183}$$

The isotropic component carries the observed scalar amplitude, tilt and running, $\langle|\mathcal{R}_{\rm iso}(k)|^2\rangle = (2\pi^2/k^3)\Delta^2_{\mathcal{R},\rm iso}(k)$. The C_3 component carries the angular selection rule, the mod-3 structure, the preferred axis and the possible parity-odd polarization signature. The two should not be conflated. The point-source C_3 pattern is not detected in the current low-multipole null test, and a transfer-function calculation remains required before a projected CMB claim is made.

45.7 Derivation of the interaction term

DERIVED The interaction term follows from three properties of the reset. (i) The reset modes live on the C_3 cycle and are chiral: the counter-rotating generators select one direction of circulation, so the mode wavenumbers q on the cycle are one-sided, $q > 0$. (ii) The conjugate pair $\phi_+\phi_-$ sources curvature quadratically, $b(k) = \epsilon u \int_0^k a(q)\, a(k-q)\, dq$, and the one-sided convolution of two seed spectra $S(q) = A_s(q/k_*)^\nu$ gives

$$\int_0^k S(q)S(k-q)\, dq = A_s^2 k_*\, x^{2\nu+1} B(\nu+1,\nu+1), \tag{184}$$

the Beta function arising exactly and the factor 2 from the two Wick pairings. (iii) The curvature relaxes to its source at first order, $\dot{\mathcal{R}} + \beta\mathcal{R} = b$, with response $|1/(\beta+ik)|^2$; for $k \gg \beta$ this removes two powers, giving $x^{2\nu-1}$. A three-dimensional convolution would give $x^{2\nu}$ instead, so the exponent $2\nu-1$ is the signature of the one-dimensional chiral cycle, and below the relaxation scale the term steepens to $x^{2\nu+1}$. DERIVED The coupling is the quadratic term of the carrier's mode equation: at second order $\delta\phi_2 = -(V'''/2V'')\,\delta\phi_1^2$, so in curvature $\epsilon u = |V'''|\phi_*/(2V''\kappa)$. About the vacuum of the defect potential, $V'''\phi_*/2V'' = 12\sqrt{2}/16 = 3\sqrt{2}/4$ in field units; at a maximum of the $\mathbb{Z}_3$ phase potential the cubic vertex vanishes and the pair term enters only at higher order. Either way $\epsilon u \le (3\sqrt{2}/4)/\kappa$, which is equation (176).

45.8 The tilt today

data	n_{s*}	ν	m^2/H^2	$\Lambda_3^4/(f_\delta^2 H^2)$
Planck 2018 + BK18 + BAO [39]	0.9649 ± 0.0042	-0.0351	-0.053 ± 0.006	0.0059
P-ACT-LB [73]	0.9743 ± 0.0034	-0.0257	-0.039 ± 0.005	0.0043
SPT-3G D1 + DESI	0.9726 ± 0.0028	-0.0274	-0.041 ± 0.004	0.0046

The carrier's mass is $|m|/H = 0.20$–0.23, tachyonic and light, as the $\mathbb{Z}_3$ maxima require. The newer small-scale data move ν toward zero, the scale-invariant limit.

45.9 The reset coupling

MEASURED The measured running bounds the coupling. Planck finds $\alpha_s = -0.0045 \pm 0.0067$ [82], so $\alpha_s < 0.0089$ at 95%; with $\alpha_{s*} \simeq (\nu-1)^2 r_*$ this gives $r_* < 0.0083$, and with $A_s = 2.1\times10^{-9}$ and $B(\nu+1,\nu+1) \simeq 1$,

$$|\epsilon u| < 1.4\times10^3. \tag{185}$$

Section 45 derives the coupling, $\epsilon u \leq (3\sqrt{2}/4)/\kappa$, so the bound holds for any conversion efficiency $\kappa > 8 \times 10^{-4}$, and the running is predicted below 10^{-4} for $\kappa > 0.01$.

45.10 Status

DERIVED The form of the spectrum is derived: the primary term is the late-time two-point function of the reset carrier in the terminal de Sitter phase, and the interaction term is the chiral pair convolution on the C_3 cycle with its coupling fixed by the defect potential. Both terms are consequences of the GRUT background, not postulates. PREDICTION Three predictions follow without a fitted parameter: the running is positive and below 10^{-4} for $\kappa > 0.01$, the tensor tilt is zero, and the interaction term steepens from $x^{2\nu-1}$ to $x^{2\nu+1}$ below the relaxation scale. Two numbers of the reset remain to be computed: the carrier's mass ratio m^2/H^2, equivalently $\Lambda_3^4/(f_\delta^2 H^2)$, which the tilt measures at -0.04 to -0.05, and the conversion efficiency κ with H/ϕ_*, which the amplitude A_s measures. Computing the $\mathbb{Z}_3$ breaking of the phase potential by the three generator defects, relative to the terminal Hubble rate, would turn the measured tilt into a prediction.

46 The Complete Gravity Equation: Dark-Matter Boosting

Reading (Section 23). In the rotation reading the condensate of this section is the modulation of the universe's rotation (§23.1), not a population of particles: the equations stand, and their carrier is the rotation. The radial mode keeps its mass m_δ as an excitation of the field.

46.1 The baryon-only equation is incomplete

DERIVED The field equation of §22.4 takes the baryons as its only source. With that source no interpolation function holds at both ends: every function that reproduces the rotation curves predicts a Solar-System quadrupole of order 10^{-26} s^{-2} (§49.10, [78]), and Cassini allows 10^{-27}. The law must change with scale, and the change cannot be put into μ without fitting it. GRUT already contains what the equation leaves out: the condensate of §37, which is real matter.

46.2 The complete equation

With the condensate as a source,

$$\nabla^2\Phi = 4\pi G\,(\rho_b + \Gamma\rho_\chi), \tag{186}$$

$$\rho_\chi(\mathbf{k}) = \mathcal{J}(k)\,\rho_{\rm ph}(\mathbf{k}), \qquad \rho_{\rm ph} = \frac{1}{4\pi G}\,\nabla\cdot\Big[\big(\nu(|\nabla\Phi_b|/a_c) - 1\big)\nabla\Phi_b\Big], \tag{187}$$

$$\mathcal{J}(k) = \frac{1}{1+(k/k_J)^4}, \qquad k_J = (16\pi G\bar\rho_\chi)^{1/4}\Big(\frac{m_\delta}{\hbar}\Big)^{1/2}, \tag{188}$$

with Γ the capture fraction of §47 and the bound mass limited by the condensate available to the system. STANDARD Equation (186) is Poisson's equation with all the matter. DERIVED The bound condensate is distributed as the phantom density of the system's own baryons: it is the gravitational polarization of the condensate elements that survive the threshold of §22.1 (Theorem 11). Φ_b is the baryonic potential of the system in its freely falling frame, as the equivalence principle requires of real matter, so a uniform external field does not enter. DERIVED Quantum pressure forbids condensate structure below the Jeans length $\lambda_J = 2\pi/k_J$ [81]; $\mathcal{J}$ is the minimal smooth form of that cutoff, its scale derived and its shape near k_J a model. DERIVED The bound mass cannot exceed the condensate available, the system's share of Ω_c (§25).

46.3 What it gives

Galaxies. On scales above λ_J, with $\Gamma = 1$, $\mathcal{J} \to 1$ and the equation is the quasi-linear form of §22 with the internal field: the rotation-curve results of §22.6 stand, and the radial acceleration relation is the signature of the bound condensate.

The Solar System. The Jeans length is

$\bar\rho_\chi$ [$M_\odot$ pc^{-3}]	0.01	0.1	1
λ_J at $m_\delta = 2\times10^{-20}$ eV	285 pc	160 pc	90 pc
λ_J at $m_\delta = 10^{-19}$ eV	128 pc	72 pc	40 pc

The planets, the Oort cloud and wide binaries lie far inside it, so $\mathcal{J} \to 0$; and the Sun, formed inside the Galaxy, could not have captured the field in any case (§47.2). It carries no bound condensate and its field is Newtonian. Without these conditions it would carry $0.9M_\odot$ within its MOND radius of 7198 AU and $29M_\odot$ within a parsec. The only quadrupole is the Galactic tide, 8.6×10^{-31} s^{-2} (§49.10).

The change with scale. The effective law is Newtonian below λ_J, follows the threshold response across galaxies, and turns Keplerian beyond the reservoir radius, where the bound mass $M_{\rm ph}(< r) \simeq \sqrt{a_0 G M_b}\, r/G$ exhausts the available condensate M_χ:

$$r_t = \frac{M_\chi}{M_b}\sqrt{\frac{GM_b}{a_0}}. \tag{189}$$

For $M_b = 10^{11}M_\odot$, $r_t = 60$ kpc if the galaxy holds its cosmic share, $M_\chi/M_b = \Omega_c/\Omega_b = 5.44$, and 0.3–1.1 Mpc if its halo retains 30–100 times its baryonic mass. The exponent changes as the scale grows because the matter that carries the boost has a Jeans length, a capture condition and a finite supply, not because μ is adjusted.

46.4 What it changes elsewhere

The planets no longer constrain the tail order: the window of §22.7 constrains the baryon-only equation, and in the complete equation the exponent is dynamic, set at each acceleration by the cascade (§22.9). The covariant completion of §46.7 is replaced by a simpler one. Because the boost is carried by matter, the complete theory is general relativity with the condensate as a source: lensing follows the same mass as dynamics, tensor modes move at c, the post-Newtonian parameters are those of general relativity, energy and momentum are conserved, and the linear cosmology is that of §47.8.

46.5 Predictions

PREDICTION Each follows without a fitted parameter:

1. The Solar-System quadrupole is the Galactic tide alone, of order 10^{-30} s^{-2}.
2. Wide binaries and globular clusters, smaller than λ_J and formed inside hot hosts, are Newtonian at every separation and acceleration.
3. Galaxy centres carry cores of size $\sim \lambda_J/2$, 0.02–0.15 kpc for $m_\delta = 10^{-19}$–2×10^{-20} eV.
4. Flat rotation curves and flat galaxy–galaxy lensing profiles end at r_t, equation (189).
5. Lensing mass equals dynamical mass, since the boost is matter.
6. There is no external-field effect of the MOND kind: a satellite's bound halo responds to its own baryons and is cut by tides and by the host's reservoir. Satellites in strong host fields, such as Crater II, test this.

46.6 Derivation of the bound-condensate law

Theorem 11 (Bound-condensate law)**.** DERIVED *Let each condensate element, exposed along a direction $\hat{u}$ with $u = \hat{u}\cdot\hat{g}$, survive the load xu with probability $S(xu)$ (§22.1) and activate otherwise, and let a surviving element carry unit gravitational susceptibility and an activated one none. Then the condensate's polarization is $\mathbf{P} = (1-\mu(x))\,\mathbf{g}/4\pi G$, $x = g/a_c$, and its bound density is exactly the phantom density:*

$$\rho_\chi = -\nabla\cdot\mathbf{P} = \frac{1}{4\pi G}\,\nabla\cdot\big[(1-\mu)\nabla\Phi\big] = \rho_{\rm ph}. \tag{190}$$

Proof. The response along $\hat{g}$ of the elements oriented along $\hat{u}$ is their survival at the load xu, projected back onto $\hat{g}$ by u, with the cosine-weighted measure of §22.2. Its average is

$$3\int_0^1 u^2 S(xu)\,du = 1 - 3\int_0^1 u^2 f(xu)\,du = 1-\mu(x), \tag{191}$$

since $3\int_0^1 u^2 du = 1$ and $f = 1-S$. A medium of susceptibility $\chi = 1-\mu$ has $\mathbf{P} = \chi\,\mathbf{g}/4\pi G$, and with $\mathbf{g} = -\nabla\Phi$ Gauss's law gives the bound density $-\nabla\cdot\mathbf{P} = \nabla\cdot[(1-\mu)\nabla\Phi]/4\pi G$. Then $\nabla^2\Phi = 4\pi G(\rho_b + \rho_\chi)$ is the same equation as $\nabla\cdot(\mu\nabla\Phi) = 4\pi G\rho_b$. □

The identity $1-\mu = 3\langle u^2 S\rangle$ holds to ten digits for every tail order (`condensation.py`). The theorem uses the survival law, which is derived (§22.1, §22.10), the directional average of §22.2, Gauss's law, and one identification

IDENTIFICATION: that μ is the gravitational permittivity of the condensate, so that a fully surviving medium carries unit susceptibility. The dark-matter boost is then the polarization of the surviving condensate: where the load is small most elements survive and the boost is large, and where it is large they activate and the field is Newtonian. The Gross–Pitaevskii dynamics set where that polarization can exist: only above λ_J, only in the freely falling frame of the system, and only where the field has been captured (§47). In the quasi-linear form of (187) the same density reads $\nabla\cdot[(\nu-1)\nabla\Phi_b]/4\pi G$; the two forms agree exactly for spherical systems.

46.7 The covariant completion

DERIVED The defect condensate supplies the two scalar degrees of freedom a completion needs: the massive radial mode is the dark matter (§37), and the phase θ, a massless Goldstone mode, is the natural carrier of the MOND scalar. For a scalar ϑ in potential units with Lagrangian $-\frac{a_c^2}{8\pi G}\,\mathcal{F}(|\nabla\vartheta|^2/a_c^2)$, the quasi-static equation is $\nabla\cdot[\mathcal{F}'(y)\nabla\vartheta]=4\pi G\rho$, so the threshold family fixes the kernel

$$\mathcal{F}(y)=\int_0^y \mu_n\big(\sqrt{y'}\,\big)\,dy'. \tag{192}$$

A scalar alone cannot lens by the potential that governs dynamics. The aether–scalar–tensor construction [71] adds a unit timelike vector, propagates tensor modes at c, and enters the MOND function only through its free function; with that function set to this kernel, the completion consists of the metric, the complex defect scalar—radial mode for dark matter, phase for the MOND scalar—and one unit timelike vector, the single field the GRUT action does not yet contain. Whether the linear-cosmology and stability results of the construction survive with this kernel is to be checked. Its quasi-static Solar-System limit is the law of §22.4, so it would carry the baryon-only quadrupole of §49.10. Section 46 replaces this construction: when the boost is carried by the condensate itself, the completion is general relativity with matter.

46.8 Status

The complete equation is (186)–(188) with the reservoir bound and the capture fraction of §47. The bound-condensate law is derived from the survival law and the identification of μ as the condensate's permittivity; the Jeans cutoff, the equivalence principle for real matter, the reservoir and the capture condition follow from the condensate's dynamics. The Solar System sees only the Galactic tide.

47 Kinematic Evolution and Gravitational Condensation of the Topological Field

Reading (Section 23). In the rotation reading the condensate of this section is the modulation of the universe's rotation (§23.1), not a population of particles: the equations stand, and their carrier is the rotation. The radial mode keeps its mass m_δ as an excitation of the field.

To account for the absence of localized dark-matter phenomena in young, high-redshift systems and its dominant structural presence in mature, low-redshift clusters, the geometric boost cannot be treated as a static scalar potential. It is a dynamic field subject to the kinematic evolution of the expanding universe. The topological defect energy introduced by the primordial spatial tension passes from a free-streaming state to a gravitationally bound state through adiabatic momentum decay.

47.1 Adiabatic decay and the equation of state

STANDARD Expansion redshifts every freely propagating field: the physical momentum of the topological background decays inversely with the scale factor,

$$p(a)=\frac{p_0}{a}. \tag{193}$$

For a population of rest mass $m_{\rm eff}$ this fixes its speed and its equation of state,

$$v(a)=\frac{c}{\sqrt{1+(a/a_{\rm nr})^2}},\qquad w(a)=\frac{P}{\rho}=\frac{1}{3}\,\frac{v^2}{c^2},\qquad a_{\rm nr}=\frac{p_0}{m_{\rm eff}c}. \tag{194}$$

In the early universe ($a\ll a_{\rm nr}$) the momentum dominates, $v\to c$ and $w\to\frac{1}{3}$. The free-streaming length $\lambda_{\rm FS}=\int v\,dt/a$ then vastly exceeds the scale of primordial baryonic fluctuations, and the field's radiation-like pressure resists localized collapse. For $a\gg a_{\rm nr}$, $v=c\,a_{\rm nr}/a\propto a^{-1}$ and $w\to 0$.

DERIVED The condensate of §37, born cold by misalignment, has no momentum of its own; its operative speed is the quantum velocity at its Jeans scale, $\sigma_q = (\hbar/m_\delta)^{1/2}(16\pi G\bar{\rho}_\chi)^{1/4} \propto a^{-3/4}$: 0.09, 0.21 and 0.55 $\mathrm{km\,s^{-1}}$ at $z = 0$, 2 and 10 for $m_\delta = 2\times10^{-20}$ eV. A component born relativistic counts as radiation until $a_{\rm nr} = v_0/c$, so the effective number of neutrino species bounds its density and present speed:

$$\Delta N_{\rm eff} = \frac{\Omega_{\rm top}\, v_0/c}{0.2271\,\Omega_\gamma} < 0.3 \quad\Longrightarrow\quad \Omega_{\rm top}\, v_0 < 1.1\ \mathrm{km\,s^{-1}}\ [39]. \tag{195}$$

47.2 The gravitational capture condition

As the universe expands the field cools, while baryons collapse into wells of depth $|\Phi_{\rm bar}|$. The field becomes bound where its speed falls below the escape speed, $v_{\rm esc} = \sqrt{2|\Phi_{\rm bar}|}$. DERIVED For a Maxwellian population of one-dimensional dispersion σ, with $\langle v^2\rangle = 3\sigma^2$, the captured fraction is the phase-space fraction with negative energy,

$$\Gamma = \mathrm{erf}\, X - \frac{2X}{\sqrt{\pi}}\, e^{-X^2}, \qquad X = \frac{v_{\rm esc}}{\sqrt{2}\,\sigma} = \frac{\sqrt{3|\Phi_{\rm bar}|}}{v}. \tag{196}$$

It approaches full capture exponentially, $\Gamma \simeq 1 - \frac{2X}{\sqrt{\pi}}e^{-X^2}$, when $v^2 \ll 2|\Phi_{\rm bar}|$, and vanishes as $\Gamma \simeq \frac{4}{3\sqrt{\pi}}X^3$ when $v^2 \gg 2|\Phi_{\rm bar}|$. The capture is set by the velocities of the field around a system when it forms: a system that forms in the cold cosmic field captures fully, and a system that forms inside a hot host captures almost nothing.

system	$v_{\rm esc}$ [$\mathrm{km\,s^{-1}}$]	σ of the field around it	Γ
the Sun at its MOND radius (7198 AU)	0.50	165 (Galactic halo)	7×10^{-9}
wide binary, $2M_\odot$ at 5000 AU	0.84	165	4×10^{-8}
globular cluster, $10^6 M_\odot$ in 10 pc	29	165	1.5×10^{-3}
dwarf galaxy forming in the cosmic field	40	$\lesssim 1$	1
galaxy or cluster forming in the cosmic field	500–2000	$\lesssim 1$	1

PREDICTION The Sun carries no bound condensate on kinematic grounds alone, independently of the Jeans cutoff of §46: two separate mechanisms leave only the Galactic tide in the Solar System. Stars, binaries and globular clusters formed inside galaxies are Newtonian; satellites keep the halos they captured before infall, cut by tides.

47.3 The dynamically coupled covariant action

The capture fraction enters the action as a modulation of the defect Lagrangian,

$$S = \int d^4x\,\sqrt{-g}\,\Big[\frac{c^4}{16\pi G}R + \mathcal{L}_b + \Gamma\,\mathcal{L}_{\rm def}\Big], \tag{197}$$

DERIVED and it is covariant in stationary spacetimes when written as the phase-space fraction of the field's quanta with negative Killing energy,

$$\Gamma(x) = \frac{\int_{E<0} f(x,p)\, d^3p}{\int f(x,p)\, d^3p}, \qquad E = -\xi^\mu p_\mu, \tag{198}$$

which reduces to (196) in the Newtonian limit, with f evolving by the collisionless Boltzmann equation on the metric. The source term then follows the history of each system:

- *Young, high-redshift systems.* Where $v(a)^2 \gg 2|\Phi_{\rm bar}|$, $\Gamma \to 0$: the field remains unbound and the dynamics are those of the baryons alone.
- *Mature, low-redshift systems.* Where $v(a)^2 \ll 2|\Phi_{\rm bar}|$, $\Gamma \to 1$: the field freezes out into the well and supplies the bound condensate of §46, flattening rotation curves and confining clusters.

For the cold condensate the cosmic field is slow at every epoch ($\sigma_q < 1\ \mathrm{km\,s^{-1}}$ at $z < 10$), so a system forming in the cosmic field captures fully at any redshift; the redshift dependence of the boost enters through the environment a system forms in and through the time the bound distribution takes to settle. A relativistic-born component adds the adiabatic decay $v \propto a^{-1}$ directly, within the bound of §47.1.

47.4 The modulated field equation and the radial acceleration relation

In the static, non-relativistic limit the action gives equation (186), $\nabla^2\Phi = 4\pi G(\rho_{\rm bar} + \Gamma\rho_\chi)$, with ρ_χ the bound polarization density of Theorem 11. A fraction Γ of the condensate carries susceptibility $\Gamma(1-\mu)$, so under

spherical symmetry, on scales above λ_J,

$$g\left[1-\Gamma\big(1-\mu(g/a_c)\big)\right]=g_{\rm bar}, \tag{199}$$

the Torridity generalization of the MOND relation: $\Gamma=0$ gives $g=g_{\rm bar}$, and $\Gamma=1$ the threshold law $\mu(g/a_c)\,g=g_{\rm bar}$. DERIVED Under complete freeze-out the boost $g-g_{\rm bar}$ reaches a maximum, the maximal topological boost acceleration,

$$g_{\rm top}=\max_{g_{\rm bar}}(g-g_{\rm bar})=0.70\,a_0=8.0\times10^{-11}\ {\rm m\,s^{-2}}, \tag{200}$$

reached at $g_{\rm bar}\simeq4.7\,a_0$ with the dynamic exponent of §22.9: no system, however arranged, receives more than about $0.7\,a_0$ from the bound condensate. The variables are $a_0=(\sqrt{3}/\pi^2)\,cH_0$, the scale from the C_3 triangulation of the horizon; $v(a)$, the speed of the field around a system, $\propto a^{-1}$ for a relativistic-born component and $\sigma_q\propto a^{-3/4}$ for the cold condensate; $|\Phi_{\rm bar}|$, the depth of the baryonic well; and $g_{\rm top}$. The formulation separates young from mature systems: where $v(a)^2\gg2|\Phi_{\rm bar}|$ the capture fraction collapses and the dynamics are baryonic, and where $v(a)^2\ll2|\Phi_{\rm bar}|$ it converges to one and the full boost switches on.

47.5 Galaxy clusters

MEASURED Clusters test where the bound condensate sits. For a representative rich cluster—a β-model atmosphere with $\beta=\frac{2}{3}$, core radius 200 kpc, central density 4.5×10^{-3} cm^{-3} and temperature 6 keV, gas fraction 0.116 at 1.2 Mpc, stars adding 12% of the gas—the complete equation with full capture gives, against the hydrostatic mass,

r [kpc]	250	500	1000	1500
$g_{\rm bar}/a_0$	0.101	0.094	0.065	0.048
$M_{\rm GRUT}/M_{\rm HSE}$	0.28	0.38	0.55	0.68
with member halos	0.32–0.36	0.42–0.48	0.61–0.68	0.75–0.83

DERIVED Two things differ from MOND. Member galaxies keep the halos they captured before infall, because the bound condensate obeys the equivalence principle and there is no external-field effect (§46.5); with halos of 4.5–10 times the stellar mass within 50 kpc they add 4–15% of the hydrostatic mass. And the condensate is not scarce: on cluster scales it is pressureless (§47.8), so a cluster collapses with its cosmic share, $\Omega_c/\Omega_b=5.35$ times its baryons, close to what the hydrostatic masses require (7.7 times the baryons at 1.2 Mpc in this model, against 6.35 for the cosmic share). The bound-condensate law places that share too far out: the polarization profile $\rho\propto r^{-2}$ gives $M(<r)\propto r$, and the reservoir radius of equation (189) lies near 1.9 Mpc. DERIVED The cluster problem is one of arrangement (resolved by the relaxation state of §47.6–47.7: a cluster assembled at $z_f=0.5$ carries 0.74–0.80 of its hydrostatic mass), not of amount: inside 1 Mpc the bound law falls short by a factor of 1.5 (with member halos, at 1 Mpc) to 3.6 (at 250 kpc). It closes if infalling condensate reaches the polarization arrangement more slowly than a cluster assembles, so that clusters, the youngest bound systems, still carry a collapsed and centrally concentrated halo. The relaxation time from infall to the polarization equilibrium is the calculation that decides it; §47.6 formulates it as the relaxation state.

47.6 The relaxation state: support, resistance and the cluster–galaxy difference

The bound-condensate law (Theorem 11) describes the end state of a system's condensate, not every state it passes through. A system's condensate occupies one of three configurations, and its position among them is its relaxation state.

Definition 12 (Support, infall and resistance)**.** DERIVED (i) The *support level* is the polarization equilibrium of Theorem 11: the bound condensate is distributed as the phantom density, the halo is isothermal, $\rho\propto r^{-2}$, and the rotation curve is flat. (ii) The *infall configuration* is the condensate as it arrives: on scales above λ_J it is pressureless (§47.8), so a collapsing system gathers its cosmic share, Ω_c/Ω_b times its baryons, into a collapsed, centrally concentrated halo. (iii) The *resistance level* is the absence of condensate: $g=g_{\rm bar}$ and Keplerian curves, reached only where none is bound—below λ_J (§46.3) or where the capture fraction vanishes (§47.2), as in the Solar System.

Definition 13 (Relaxation state)**.** The relaxation state of a bound system is the polarized fraction $f\in[0,1]$ of its condensate,

$$M_\chi(<r)=f\,M_{\rm pol}(<r)+(1-f)\,M_{\rm inf}(<r). \tag{201}$$

DERIVED The resistance level is not the unrelaxed state of a bound system: a system that has captured its condensate cannot return to Newton, because the condensate is there and gravitates. Reading the unrelaxed state as reduced capture, $\Gamma < 1$ in equation (199), lowers the boost and moves cluster masses the wrong way; the infall configuration raises them. For the cluster of §47.5, as fractions of the hydrostatic mass:

r [kpc]	support ($f = 1$)	reduced capture ($\Gamma = 0.5$)	infall ($f = 0$)	$f = 0.2$
250	0.28	0.13	0.83	0.72
500	0.38	0.17	0.82	0.73
1000	0.55	0.22	0.84	0.78
1500	0.68	0.25	0.82	0.80

The infall configuration—the cosmic share in a halo of concentration $c_{500} = 2.6$ ($c_{200} \simeq 4$)—carries 0.82–0.84 of the hydrostatic mass at every radius; the remainder is the depletion of cluster baryons below the cosmic fraction ($M_{\rm HSE}/M_b = 7.6$ at $r_{500} = 1.29$ Mpc, against 6.35 for the cosmic share), which a collapsed halo shares.

Dynamical age. DERIVED Relaxation proceeds on the system's dynamical time. A disk of radius 20 kpc rotating at 200 km s^{-1} has a crossing time of 0.10 Gyr and has lived about 100 of them since assembly; a cluster crossed at 1000 km s^{-1} over r_{500} has a crossing time of 1.3 Gyr and has lived about four. If polarization takes $N_{\rm rel}$ crossing times, $f = 1 - e^{-N/N_{\rm rel}}$, and galaxies at support ($f \geq 0.95$) and clusters short of it ($f \leq 0.3$) both follow for

$$11 \lesssim N_{\rm rel} \lesssim 34. \tag{202}$$

The cluster–galaxy difference is then a difference in relaxation state, not in the law: galaxies have reached support and clusters have not.

The relaxation time. DERIVED A bound system decouples from the expansion when it forms, and its clock is fixed then: a system collapsed to 200 times the critical density has $GM/R^3 = 100H^2$, so its crossing time is $t_{\rm cross} = 1/10H(t_f)$, a fixed fraction of the Hubble time at formation. DERIVED The polarization of its condensate is renewed once per exposure stage, and a stage is one arc of the horizon's C_3 cycle (§22.9), $\tau_1/3$, read at the horizon of formation: the phase returns to a C_3-equivalent configuration each time it advances by one arc, $2\pi/3$, and the bound medium's polarization, defined relative to the phase, is renewed at that period. Then

$$t_{\rm rel} = \frac{\tau_1(t_f)}{3} = \frac{4\pi^2}{9\sqrt{3}}\,\frac{1}{H(t_f)}, \qquad N_{\rm rel} = \frac{t_{\rm rel}}{t_{\rm cross}} = \frac{40\pi^2}{9\sqrt{3}} = 25.3, \tag{203}$$

inside the window (202) that galaxies and clusters independently require; a full cycle, $N_{\rm rel} = 76$, would leave galaxies short of support. The relaxation state follows from the formation epoch alone,

$$f = 1 - \exp\Big[-\frac{9\sqrt{3}}{4\pi^2}\,H(t_f)\,(t - t_f)\Big], \tag{204}$$

and in the matter era, where $Ht = \frac{2}{3}$, it depends only on $D = t/t_f$, the age of the universe in units of its age when the system formed:

$$\boxed{f = 1 - \exp\Big[-\frac{D-1}{x_H}\Big], \qquad x_H = \frac{2\pi^2}{3\sqrt{3}} = 3.80.} \tag{205}$$

A system relaxes on 3.8 times the age the universe had when it formed, and 3.8 is the horizon load of §25.7: both are half the C_3 period in Hubble units, $\tau_1 H/2$. At the resonance budget ($t_0 = 13.79$ Gyr), with the general form above (the boxed one is its matter-era limit), today:

z_f	t_f [Gyr]	$D = t_0/t_f$	f
0.3	10.25	1.3	0.11
0.5	8.58	1.6	0.17
1	5.84	2.4	0.32
2	3.27	4.2	0.58
3	2.14	6.5	0.77
4	1.53	9.0	0.88
6	0.93	14.9	0.97
10	0.47	29.4	1.00

Half the condensate is polarized for systems formed at $z_f = 1.66$ and 90% for $z_f = 4.26$. The cluster of §47.5, assembled at $z_f = 0.5$, has $f = 0.17$, and its condensate carries 0.74, 0.74, 0.79 and 0.80 of the hydrostatic mass at 250, 500, 1000 and 1500 kpc, against 0.28–0.68 at support; a disk whose halo formed at $z_f = 6$ has $f = 0.97$. DERIVED The renewal once per arc follows from the rotation: the phase returns to a C_3-equivalent configuration

each time it advances by one arc, $2\pi/3$, that is after $\tau_1/3$, and the bound medium's polarization, defined relative to the phase, is renewed at that period (§23.1).

Failed relaxation. DERIVED A major merger returns newly accreted condensate to the infall configuration. With mergers a Poisson process of mean interval $t_{\rm dis}$ and relaxation needing $t_{\rm rel} = N_{\rm rel} t_{\rm dyn}$, the probability that a system is disrupted before it reaches support—a trap—is the probability of at least one merger in $t_{\rm rel}$,

$$p_{\rm trap} = 1 - e^{-t_{\rm rel}/t_{\rm dis}} \tag{206}$$

(if relaxation is itself exponential with mean $t_{\rm rel}$, $p_{\rm trap} = t_{\rm rel}/(t_{\rm rel}+t_{\rm dis})$). Across the window, clusters with $t_{\rm dis} \simeq 4$ Gyr have $p_{\rm trap} = 0.97$–1.00 and never reach support; disk galaxies with $t_{\rm dis} \simeq 10$ Gyr have $p_{\rm trap} = 0.10$–0.28.

Relaxed systems are quiet. DERIVED At fixed baryons the deviation from the support law is $(1-f)\,G(M_{\rm inf}-M_{\rm pol})/r^2$, so the scatter of a population about that law is proportional to the spread of $1-f$: it vanishes for systems at support and is largest for systems in transit.

Terminology. A market has a support level, the floor where buyers step in, and a resistance level, the ceiling where sellers step in, and the price moves between them. Here the observed acceleration has a support level, the polarization equilibrium where the boost is fully developed, and a resistance level, Newton, where no condensate is bound; a bound system starts from its infall halo and moves toward support as it relaxes. A trap is a merger before support is reached, and "stairs up, elevator down" is the quiet of relaxed systems against the violence of disrupted ones. The levels are not constants: they emerge from the condensate's state, as a market's levels emerge from the positions of its participants.

47.7 Predictions of the relaxation state

PREDICTION The ordering does not depend on $N_{\rm rel}$:

1. The mass discrepancy against the support law is ordered by dynamical age since the last major merger: largest in clusters and recent mergers, smallest in isolated disks.
2. Early-assembled systems, such as fossil groups, lie closer to the support law than clusters of the same mass.
3. After a major merger a cluster's condensate is centrally concentrated relative to the support law and relaxes toward it over $N_{\rm rel}$ crossing times.
4. At fixed mass, higher-redshift clusters lie further from support than low-redshift ones.
5. The scatter about the radial acceleration relation is larger for disturbed and interacting galaxies than for isolated disks; 10–30% of disks are trapped.
6. Systems with no bound condensate—the Solar System, wide binaries—sit at resistance whatever their age.
7. The relaxation state depends on $D = t/t_f$ alone (equation (205)): a galaxy formed at $z_f = 6$ is 49% polarized when seen at $z = 2$, 75% at $z = 1$ and 97% today; one formed at $z_f = 3$ is 13% polarized at $z = 2$. Young systems at high redshift carry their infall halos.

47.8 Perturbation-level cosmology

DERIVED On the scales of the CMB and linear large-scale structure ($k \lesssim 0.3$ Mpc^{-1}) the defect sector reduces to ΛCDM with the GRUT budget. The radial mode, with $m_\delta \geq 2\times10^{-20}$ eV, has Jeans scale $k_J(a_{\rm eq}) \simeq 9\,(m_\delta/10^{-22}\,{\rm eV})^{1/2}$ Mpc$^{-1} \geq 127$ Mpc^{-1} and half-mode scale $k_{1/2} \simeq 4.5\,(m_\delta/10^{-22}\,{\rm eV})^{4/9}$ Mpc$^{-1} \geq 47$ Mpc^{-1} [81]: it is pressureless there. The vacuum is the space-filling defect, with $w = -1$ and $V_0 = 0$ (§38). The transfer functions of §26.7 and §39 are therefore GRUT's own on these scales, and the Boltzmann fit of §26.7 is the perturbation-level calculation of the isotropic sector. What remains is the perturbation theory of the modified-gravity sector, which needs the completion of §46.7, and of the anisotropic component of §39.6.

47.9 Cluster mergers: the passage of two excited geometries

DERIVED *Equations.* Varying the action of §23.1 with respect to λ, θ and the metric gives

$$X = X_0, \qquad \nabla_\mu(\lambda\,\partial^\mu\theta) = 0, \qquad T_{\mu\nu} = 2\lambda\,\partial_\mu\theta\,\partial_\nu\theta + g_{\mu\nu}\lambda(X-X_0) \xrightarrow{X=X_0} \rho\,u_\mu u_\nu, \tag{207}$$

with $u_\mu = \partial_\mu\theta/\sqrt{X_0}$ and $\rho = 2\lambda X_0$. The phase equation has no baryonic source: stars and gas act on the phase only through the metric. The flow is pressureless and geodesic, $u^\nu\nabla_\nu u_\mu = -\partial_\mu X/2X_0 = 0$. In the infall state of

§47.6 the excited geometry is collisionless dust in the weak field,

$$\nabla^2\Phi = 4\pi G(\rho_b + \rho_\chi), \qquad \partial_t f_\chi + \mathbf{v}\cdot\nabla f_\chi - \nabla\Phi\cdot\partial_{\mathbf{v}} f_\chi = 0, \qquad \rho_\chi = \int f_\chi \, d^3v. \tag{208}$$

The stress tensor, the geodesic identity and the template below are verified symbolically in Appendix H of the companion.

DERIVED *Cross terms.* For an operator $\mathcal{N}$ acting on a field, the cross term of two sources is $\mathcal{N}_\times[\phi_1,\phi_2] = \mathcal{N}[\phi_1+\phi_2] - \mathcal{N}[\phi_1] - \mathcal{N}[\phi_2] + \mathcal{N}[0]$. It vanishes for a linear $\mathcal{N}$ and is $2\lambda\phi_1\phi_2$ for $\mathcal{N}[\phi] = \lambda\phi^2$. In the infall state $\mathcal{N} = \nabla^2$ is linear, so $\Phi = \Phi_A + \Phi_B$ exactly and the only coupling of two clusters is superposition: $\kappa = \kappa_b + \kappa_A + \kappa_B$, and the contribution of B at the galaxy peak of A is $\kappa_B(\mathbf{x}_A)$, the tail of B's profile. It is ordinary gravity and carries no new term. In the support state the law of §22.9 is nonlinear; in its deep limit

$$\mathcal{N}_\times = \frac{1}{a_c}\nabla\cdot\Big[\,|\nabla\Phi_A + \nabla\Phi_B|(\nabla\Phi_A + \nabla\Phi_B) - |\nabla\Phi_A|\nabla\Phi_A - |\nabla\Phi_B|\nabla\Phi_B\Big] \neq 0, \tag{209}$$

which for collinear fields is $(|\nabla\Phi_A|\nabla\Phi_B + |\nabla\Phi_B|\nabla\Phi_A)/a_c$. This term enters a merger only through the clock below.

IDENTIFICATION *Conditional relaxation closure; clock reset at core passage.* Within the model, a major merger is assumed to return the condensate to the infall configuration (the trap, equation (206)); the clock of equations (203) and (205) is then reset at core passage. Given the first-order closure $\dot f = (1-f)/t_{\rm rel}$ with $f(0) = 0$, the share relaxed onto the baryons present is $f(\Delta t) = 1 - \exp[-(9\sqrt{3}/4\pi^2)H(z)\Delta t]$, with $t_{\rm rel} = 4\pi^2/(9\sqrt{3}H)$. At the merger epochs:

system	z	$t_{\rm rel}$ [Gyr]	$f(0.15)$	$f(0.3)$	$f(1)$	$f(2)$
1E 0657−56	0.296	31.4	0.0048	0.0095	0.031	0.062
MACS J0025.4−1222	0.586	26.4	0.0057	0.011	0.037	0.073
Abell 2744	0.308	31.2	0.0048	0.0096	0.032	0.062
Abell 520	0.199	33.2	0.0045	0.0090	0.030	0.059

At $z = 0.296$, $t_{\rm rel} = 31.4$ Gyr implies $f(0.15\,\mathrm{Gyr}) = 0.0048$ and $f(0.20\,\mathrm{Gyr}) \simeq 0.0064$; $f(1\,\mathrm{Gyr}) = 0.031$ and $f(2\,\mathrm{Gyr}) = 0.062$. A passage lasts $\simeq 0.2$ Gyr (720 kpc at 2.6–4.7×10^3 km s^{-1}), so $t_{\rm rel}/t_{\rm passage} \gtrsim 1.2\times10^2$ and the model's relaxation fraction changes only slightly during the encounter. The value 0.0048 corresponds to 0.15 Gyr, not 0.20 Gyr. Capture is complete for a cluster: $\Gamma = 1$ for $\sigma \lesssim 1$ km s^{-1}, and $\Gamma = 0.973$ at $v_{\rm esc} = 1500$ km s^{-1} even at the web's $\sigma_{1D} = 494$ km s^{-1} of §53.9. The clock depends on the redshift of the merger and not on a single value: one $t_{\rm rel}$ does not serve all clusters.

HYPOTHESIS *The model lensing map and baryon-share assignment.*

$$\kappa(\mathbf{x}) = \kappa_b(\mathbf{x}) + (1-f)\big[\kappa_A(\mathbf{x}-\mathbf{x}_A) + \kappa_B(\mathbf{x}-\mathbf{x}_B)\big] + f\,\kappa_{\rm pol}(\mathbf{x}), \qquad M_{\chi,i} = \frac{\Omega_c}{\Omega_b}\,M_{b,i}^{\rm pre}, \tag{210}$$

where κ_b holds the stars and gas, κ_i is the excited geometry cluster i gathered before the merger, from all its baryons including the gas it has since lost, and $\kappa_{\rm pol}$ is the support-state map of §22.9 applied to the baryons present. Each excited geometry moves on the geodesics of its own galaxies, so the peaks sit on the galaxies, the stripped gas carries none, and the lensing masses of the two peaks stand in the ratio of the pre-merger baryon masses, each $1+\Omega_c/\Omega_b = 6.35$ times its baryons (equation (281)). The excess drifts toward the baryons by $f(\Delta t)$ of their separation. Differentiating the same law gives the centroid drift $\mathbf{v}_{\rm rel} = e^{-\Delta t/t_{\rm rel}}\mathbf{D}/t_{\rm rel}$ for approximately fixed separation; the derivation and mass-conserving flux are given in Section 55. This is a redistribution velocity, not a force-driven acceleration of the geodesic component. The displacement is 0.5% for the Bullet Cluster, at most 3.1% after 1 Gyr and 6% after 2 Gyr.

DERIVED *Computed, on synthetic data.* A particle-mesh evolution of a 10:1 pair, with collisionless galaxies and excited geometry and a two-fluid drag for the gas, conserves momentum to 2×10^{-4} km s^{-1} and energy to 0.2%, reaches 3.3×10^3 km s^{-1} at core passage and, 0.15 Gyr after it, has the galaxies 449 kpc apart, the excess within 20 and 13 kpc of its galaxies (a resolution limit: 7 and 11 kpc on the finer mesh), and the subcluster's gas 340–550 kpc behind them for drag strengths 5–45; the excess observables do not move with the drag. The time step is converged (peak convergence changes by 0.02% over a factor of four). The mesh is not: the peak rises by 9.6% from 64 to 96 cells, and the subcluster expands by 40% in isolation over 1.5 Gyr, so offsets below about 30 kpc are not resolved. The ordinary cross-cluster contribution to the excess convergence is 3.4% at the main galaxies and 50% at the subcluster's (the main halo's wing there).

DERIVED *What a merger tests, on synthetic data.* Maps of 24×24 pixels (125 kpc, smoothing 100 kpc) with correlated shape, large-scale-structure and model covariance, with calibration, zero-point and centroid nuisance

parameters marginalised (centroids by linearisation, checked against Monte Carlo: $\ln B = 48.9$ against 49.2), were drawn from the simulated snapshot with the clock applied. With $T = 2\ln B$ and covariance-preserving nulls of 2×10^4 draws: collisionless against baryon-following $T = 97.7$ (null median -53.6, $p_{\rm MC} < 5 \times 10^{-5}$); own against swapped assignment (A's excess on B's galaxies) $T = 99.1$, $p_{\rm MC} < 5 \times 10^{-5}$; GRUT against particle dark matter $T = -0.08$, $p_{\rm MC} = 0.86$, power 0.053 at 5%. The Bullet Cluster therefore separates both GRUT and particle dark matter from a gravity law computed on the baryons present, and does not separate them from each other. The settled share is bounded $f < 0.14$ (95%, uniform prior) against GRUT's 0.0048. The centroid posteriors are calibrated (400 mocks: z standard deviations 0.99–1.06, 68% coverage 0.66–0.70). Ignoring the pixel correlations overstates the median T by a factor 4.6. A Gaussian profile for the excess fits the simulated map acceptably (ΔAIC = 0.0, whitened residuals consistent with noise) and is not established: the exponential profile ties it (0.01), and in mocks the Gaussian is chosen in 87% of cases when true and in 13–37% when the truth is Lorentzian or exponential.

PREDICTION (i) No offset between the excess and the galaxies beyond $f(\Delta t)D$, with D the gas–galaxy separation ($\sigma/m = 0$). (ii) No galaxy-free peak on the gas holding more than $f(\Delta t) \leq 0.06$ of the excess. On the synthetic maps a peak of $q \simeq 0.10$ of the excess is the 2σ detection limit, so a robust one at that level would falsify this. Abell 520 is therefore a test, not a prediction: the dark core of [184, 185] is absent at comparable significance in [186, 187]. (iii) The merger speeds are those of ΛCDM: the shock at 4740 $\mathrm{km\,s^{-1}}$ and the subcluster at 2600–3980 $\mathrm{km\,s^{-1}}$ [180, 181] carry the same rarity [182], and GRUT gives no relief. Shock heating is standard physics and the same in ΛCDM. Register rows 89–91.

HYPOTHESIS *Physical scope of the merger analogy.* The observed separation of lensing-inferred mass from X-ray gas is consistent with a collisionless gravitating component under the standard lensing/GR interpretation; it does not itself identify that component as GRUT's phase sector. The merger relaxation map below is a conditional model closure, not an identity of gravitational lensing.

STANDARD *Three phases and observational classification.* The infall phase precedes first core passage; the core-passage phase contains shock-heated, decelerated intracluster plasma and more nearly collisionless galaxy/lensing-mass trajectories; the post-passage phase is dynamically disturbed and evolves through shocks, orbital motion and relaxation. The condition $f = 0$ is the initial condition at the post-passage clock reset, $\Delta t = 0$, not a universal statement that every pre-merger system has zero relaxation. Bullet Cluster and MACS J0025.4−1222 show gas–lensing-mass separation (NASA Bullet Cluster overview; NASA MACS J0025 overview). El Gordo is usually reconstructed as an ongoing system shortly after first core passage, with outgoing/returning alternatives (MNRAS dynamical reconstruction); Pandora's Cluster/Abell 2744 is a complex multi-subcluster collision, not a clean relaxed endpoint (NASA overview). These are distinct clusters, not successive snapshots of one merger; projection and phase degeneracies are discussed in the 2026 review of cluster mergers.

The local merger is used here as an analogy for the separation of collisional and collisionless components. It does not, by itself, derive a cosmic conformal junction, a sign flip of the scale factor, or information transfer between aeons. In particular, using the same dimensionless $t_{\rm rel}H$ at cluster and cosmological scales is a framework identification; its microphysical validity is not established by the geodesic equation. Testing is deferred in this revision.

48 Limits and the Finite Clock

Every dimensionful quantity in GRUT is a pure number times a monomial in c, $\hbar$ and r and a power of N (§44.6). The speed of light is therefore not the only limit: the horizon bounds every quantity above and the cell bounds it below, and the bounds move with the horizon. This section sorts the bounds into those that are constant, those that freeze and those that never stop moving, and measures where we are in time.

48.1 Three kinds of bound

DERIVED (i) *Constant limits* contain no horizon and no particular particle or event: c; the total force $F = c^4/G$; the largest force one body can exert on another, $c^4/4G$, reached by two black holes in contact; and the maximum power $c^5/4G$ (§43B.6). They are the same at every epoch. (ii) *Ceilings and floors* contain the horizon. The ceilings are the length r, the time r/c, the mass $M_{\rm H} = c^2 r/2G$, the energy $M_{\rm H}c^2$ and the momentum $M_{\rm H}c$. The floors are ℓ_P and t_P; the mass whose Compton size is the horizon, $\hbar/cr$, below which nothing can be localized inside the universe; the energy $\hbar c/r$ and momentum $\hbar/r$ of one quantum across the horizon; and the force $\hbar c/r^2$, the smallest momentum delivered over the longest time. Each ceiling over its floor is $\sqrt{N/2\pi}$ for length, time and acceleration, $N/4\pi$ for mass, energy, momentum and weight, and $N/2\pi$ for force: the finite cell count is the limit behind every limit. (iii) *Clock readings* are built from a particular event. A quantity that contains the age at recombination or at first starlight falls as $1/t$ and is not a limit: the rest energy of an electron spread over

$t_0/t_{\rm rec} = 37\,100$ Planck lengths, 1.37×10^{17} N today, reads the clock.

48.2 The freeze

DERIVED The cell count is $N = 2\pi r^2/\ell_P^2$ with $r = c/H$, so $N \propto H^{-2}$. With the vacuum of §26.4, $w = -1$, the Hubble rate tends to $H_\infty = H_0\sqrt{\Omega_\Lambda}$, and for a flat universe

$$\boxed{\frac{N(t)}{N_\infty} = \frac{H_\infty^2}{H(t)^2} = \Omega_\Lambda(t)} \tag{211}$$

exactly: the fraction of its final cells the universe holds is its dark-energy fraction. Today it is 0.684, with $N_0 = 4.54 \times 10^{122}$ and $N_\infty = N_0/\Omega_\Lambda = 6.63 \times 10^{122}$; the horizon stops at $r_\infty = r/\sqrt{\Omega_\Lambda} = 17.56$ Gly, against 14.52 Gly now. The ceilings and floors freeze with it, reaching 90%, 99% and 99.9% of their final cell count at 21.3, 35.0 and 48.5 Gyr (resonance budget):

quantity	bound	today	final
length	r (ceiling)	1.37×10^{26} m	1.66×10^{26} m
time	r/c (ceiling)	4.58×10^{17} s	5.54×10^{17} s
mass	$M_{\rm H}$ (ceiling)	9.25×10^{52} kg	1.12×10^{53} kg
mass	$\hbar/cr$ (floor)	2.56×10^{-69} kg	2.12×10^{-69} kg
momentum	$\hbar/r$ (floor)	7.68×10^{-61} kg m s^{-1}	6.35×10^{-61} kg m s^{-1}
force	$\hbar c/r^2$ (floor)	1.68×10^{-78} N	1.15×10^{-78} N
force	c^4/G (constant)	1.21×10^{44} N	1.21×10^{44} N

48.3 The maximum-power point

DERIVED The horizon energy $E_{\rm H} = c^5/2GH$ grows at $\dot{E}_H = (c^5/2G)(1+q)$, which equals the maximum power $c^5/4G$ when the deceleration parameter is $q = -\frac{1}{2}$. With $q = \Omega_r + \frac{1}{2}\Omega_m - \Omega_\Lambda$ and flatness, this is

$$\boxed{\frac{N}{N_\infty} = \Omega_\Lambda = \frac{2}{3} + \frac{\Omega_r}{3}:} \tag{212}$$

the horizon grows at the maximum power when the universe holds two thirds of its final cells. With the resonance budget that point was at $z = 0.027$, 0.38 Gyr ago, and we are past it: $\Omega_\Lambda = 0.684 > \frac{2}{3}$, $q_0 = -0.526$. Before it the horizon grew faster than the maximum power allows any single system to radiate; after it, slower, falling to zero at the freeze.

48.4 The finite aeon

DERIVED Proper time runs without end, but conformal time, $\eta = \int c\,dt/a$, converges in the future because the vacuum keeps the expansion accelerating, and it is the clock the reset carries into the next aeon (§45). The aeon's conformal length is

$$\eta_{\rm tot} = 4.327\,\frac{c}{H_0} = 62.8 \text{ Gly}, \tag{213}$$

of which 46.1 Gly has passed: PREDICTION we are 73.4% of the way through the aeon. The remaining 16.7 Gly is the event horizon, the farthest comoving distance a signal sent today can ever reach.

moment	age	$N/N_\infty = \Omega_\Lambda(t)$	fraction of the aeon's conformal time
recombination, $z = 1090$	371 kyr	1.27×10^{-9}	0.015
first stars, $z = 20$	178 Myr	2.32×10^{-4}	0.166
$z = 2$	3.27 Gyr	0.074	0.458
$z = 0.5$	8.58 Gyr	0.391	0.633
maximum-power point	13.41 Gyr	0.667	0.728
today	13.79 Gyr	0.684	0.734

The conformal fraction depends on Ω_m and Ω_r alone, not on H_0, and falls by 0.35 per unit of Ω_m.

48.5 How this section fails

PREDICTION Each statement has a stated failure:

1. The freeze requires $w = -1$ exactly. A dark energy that thaws toward the future sends $H \to 0$ and $N \to \infty$, so there is no final cell count; a phantom one shrinks the horizon. Evolving dark energy established at 5σ falsifies §48.2–48.4 together with §26.4; the present preference is 2.7σ with the CMB and 3.2σ with supernovae (register row 6).
2. The identity (211) requires flatness: $|\Omega_K| > 0.003$ at 3σ falsifies it.
3. The maximum-power point lies in the past only if $q_0 < -\frac{1}{2}$: a flat, $w = -1$ fit returning $q_0 > -\frac{1}{2}$ at 3σ falsifies §48.3 as a statement about today. GRUT gives -0.526 and Planck's budget -0.527.
4. The finite aeon requires the expansion to keep accelerating: an asymptotic equation of state above $-\frac{1}{3}$ makes the conformal time diverge.
5. The constant limits: a force between two bodies above $c^4/4G$, or a luminosity above $c^5/4G = 9.1 \times 10^{51}$ W, falsifies them; the largest measured, GW150914, peaked at 0.4% of it.
6. These numbers are not independent tests. $\Omega_\Lambda(t)$, the freeze, the maximum-power epoch and the conformal fraction are functions of Ω_m (and H_0 for the dimensional ones), and are tested through the Ω_m and H_0 rows of Appendix D. With the CMB-independent DESI Lyman-α measurement, $\Omega_m = 0.325 \pm 0.018$ [65], the conformal fraction is 0.7310 ± 0.0062, $+0.2\sigma$ from GRUT's 0.7321; Planck's Ω_m, inferred within ΛCDM, gives 0.7344 ± 0.0025 (-0.9σ).
7. A drift of the gravitational constant at $2(1+q)H$ would mean fixed, stretching cells, and reverses §48.6.
8. A measured drift of α that tracks the age reverses §48.7.
9. The first-starlight hypothesis of §48.8 fails if 21-cm coupling is detected above $z = 30.1$, below 45.5 MHz—starlight before $t_\star$—or if the onset is measured well after it: first starlight at $z \approx 20$, 1/78 of the age, seen at 67.6 MHz, excludes it. It also fails if GRUT's condensate cannot form stars by $z_\star$ at any allowed mass (§48.9). With the rotation's birefringence it fails if the onset and the CMB's twist miss (222) by more than 5σ together (§48.8).

48.6 The cells do not stretch

DERIVED Space expands by gaining cells, not by stretching them, and the data decide which. With $N = 2\pi r^2/\ell_P^2$, GRUT's gravitational constant is $G = 2\pi c^3 r^2/\hbar N$ (Section 12). If the cell count were fixed while the horizon grew, each cell would stretch with it, and today

$$\frac{\dot G}{G} = 2\,\frac{\dot r}{r} = 2(1+q_0)H_0 = 6.6 \times 10^{-11}\ \text{yr}^{-1}. \tag{214}$$

Lunar laser ranging finds $\dot G/G = (7.1 \pm 7.6) \times 10^{-14}$ yr^{-1} [86], three orders of magnitude smaller. Holding G fixed and letting $\hbar$ carry the change instead would move the fine-structure constant as r^{-2}, by a factor $E(2)^2 = 9.4$ between redshift 2 and today, against quasar spectra that hold $\Delta\alpha/\alpha$ to about a part per million [88]. PREDICTION The Planck length is therefore fixed and the cell count grows as the square of the horizon, $N(t) = 2\pi r(t)^2/\ell_P^2$, which is what §48.2 uses.

48.7 The fine-structure constant sets when light escapes

DERIVED If α read the clock, $\alpha \propto 1/t$, it would change today at $\dot\alpha/\alpha = -1/t_0 = -7.3 \times 10^{-11}$ yr^{-1}, be 4.2 times larger in the light of quasars at $z = 2$, and 3.7×10^4 times larger at recombination, where no atom could form. Optical clocks hold $\dot\alpha/\alpha = (1.0 \pm 1.1) \times 10^{-18}$ yr^{-1} [87], and quasar spectra hold $\Delta\alpha/\alpha$ to about a part per million [88]: α belongs with the constant limits of §48.1.

DERIVED It is, however, the constant that sets when light escapes. The universe becomes transparent when the photon bath can no longer ionize hydrogen, whose binding energy is $B = \frac{1}{2}\alpha^2 m_e c^2 = 13.6$ eV. The Saha balance,

$$\frac{x_e^2}{1-x_e} = \frac{1}{\eta}\,\frac{\pi^2}{2\zeta(3)}\left(\frac{m_e c^2}{2\pi kT}\right)^{3/2} e^{-B/kT}, \tag{215}$$

fixes the temperature at which half the hydrogen is neutral, and with $1 + z = T/T_0$ and the budget's expansion history

$$\frac{d\ln(1+z_{\rm rec})}{d\ln\alpha} = 2.07, \qquad \frac{d\ln D_{\rm rec}}{d\ln\alpha} = 3.54, \qquad D_{\rm rec} \equiv \frac{t_0}{t_{\rm rec}}. \tag{216}$$

The first exponent lies slightly above 2 because of the logarithm in the Saha balance; the second lies between 3 (matter) and 4 (radiation) because the two share the expansion at $z \simeq 1100$. The ratio of the age of the universe to the moment light escaped, 3.7×10^4 at $z = 1090$, therefore carries $\alpha^{3.5}$: a universe with a different α lets its light out at a different fraction of its age. α is the clock-setter, not the clock.

48.8 First starlight

HYPOTHESIS With α constant and the first starlight a fixed moment $t_\star$, the ratio $t_\star/t$ passes through α once. The hypothesis is that it does so now,

$$\boxed{t_\star = \alpha\, t_0, \qquad t_0 = 137.036\, t_\star,} \tag{217}$$

a selection of the present epoch (§36.6) by the moment stars first shone, as §48.3 places us just past the maximum-power point. PREDICTION At the resonance budget it predicts first starlight at

$$t_\star = 100.6 \text{ Myr}, \qquad z_\star = 29.6, \tag{218}$$

and the onset of the 21-cm signal it drives, when starlight first couples the hydrogen spin temperature to the gas, at

$$\nu_\star = \frac{1420.4 \text{ MHz}}{1 + z_\star} = 46.4 \text{ MHz}. \tag{219}$$

It is consistent with what is seen. The earliest spectroscopically confirmed galaxy, at $z = 14.32$ [91], is seen at 286 Myr, 1/48.3 of the age; the absorption EDGES reported at 78 MHz ($z \simeq 17.2$, 220 Myr) [89], which SARAS 3 does not confirm [90], would require starlight earlier still. HERA, REACH, SKA-Low and radio arrays on the lunar far side are built to find the onset. If $t_\star$ is measured, equation (217) gives the age of the universe independently of the distance ladder.

The two ends. DERIVED The 21-cm photons leave the first-starlight end of the history and arrive at ours. Three unit-free ratios connect the ends:

clock	first starlight	today	link between the ends
cosmic time	100.6 Myr	13.79 Gyr	$t_\star/t_0 = \alpha$ (the hypothesis)
scale factor	wavelength 0.211 m	6.46 m	$a_\star/a_0 = 1/30.6$
conformal time	13.5% of the aeon	73.4%	59.9% of the aeon between them

The shell of first starlight lies at a comoving distance of 3.56×10^{26} m (37.6 Gly), which sets the angular scales a 21-cm map of the onset resolves. The three links are not independent: all follow from equation (217) and the budget, so one measurement of the onset tests them together and counts once. A length at one end set against a clock at the other—the 6.46 m wavelength against the 4.36×10^{26} m conformal radius—carries units and no physics. DERIVED GRUT must itself form stars by $z_\star$. In the rotation reading the cosmic share is dust with zero sound speed (§23.1): it suppresses no halos, minihalos form as in cold dark matter, and the first stars of the observable universe form near $z \simeq 65$ in the rarest peaks [92], well before $z_\star = 29.6$. The bound on m_δ that a condensate would impose does not arise.

Between physical epochs. HYPOTHESIS With the present selected by the resonance (H3, §25.11), first starlight no longer selects it. The hypothesis becomes a relation between two physical epochs, cosmic dawn and the resonance at which the horizon's wobble falls into step with the spin, and needs no observer:

$$t_\star = \alpha\, t_{\rm res}, \qquad H_\infty t_\star = \alpha\, \theta_0 = 5.73 \times 10^{-3}, \tag{220}$$

with $\theta_0 = H_\infty t_{\rm res} = 0.78545$ the phase turned by the spin since the beginning (§36.13). PREDICTION At the resonance budget $t_\star = 100.6$ Myr, $z_\star = 29.6$ and $\nu_\star = 46.4$ MHz; a ±3% timing places the onset at $z = 29.0$–30.2, 45.5–47.3 MHz. Stated at the maximum-power epoch instead (§48.10), $t_\star = \alpha t_{\rm mp} = 97.9$ Myr, $z = 30.2$, 45.6 MHz, 2.7% earlier. MEASURED No starlight before $t_\star$ is known. The earliest confirmed galaxy is at $z = 14.44$, 280 Myr [155]; the claimed $z \simeq 32$ candidate "Capotauro", which would have shone at about 90 Myr, moves on the sky and is a Y dwarf 730 ± 110 pc away [156]. OPEN The hypothesis remains a hypothesis: the onset of cosmic dawn depends on the amplitude of the primordial spectrum and the physics of the first stars, which the framework does not compute (§45.10), so (220) is not derived. It is tested by the 21-cm onset, register row 76.

In cycles DERIVED A frequency is a number of cycles per second, and the onset is read as one: a radiometer counts the cycles of the 21-cm wave as they arrive, while each hydrogen atom emitted $\nu_{21} = 1420.405751768$ MHz, so the observable is a ratio of counts, the cycles received per cycle emitted, $\nu_\star/\nu_{21} = a_\star = 1/(1 + z_\star)$, in which the second cancels. The spin's phase is a count of the same kind. In a flat universe of matter and vacuum $\sinh(\frac{3}{2} H_\infty t) = \sqrt{\Omega_\Lambda/\Omega_m}\, a^{3/2}$, so with (220) and the present at the resonance

$$\frac{\nu_\star}{\nu_{21}} = \Big[\frac{\sqrt{4\pi^2 - 27}}{3\sqrt{3}} \sinh\Big(\alpha \operatorname{artanh} \frac{3\sqrt{3}}{2\pi}\Big)\Big]^{2/3} = \frac{1}{30.81}, \tag{221}$$

46.10 MHz, and 1/30.61, 46.41 MHz, with radiation: the fine-structure constant and the C_3 ratio, as in the birefringence (105), with no clock, and with a Hubble constant only in the radiation's 0.7%.

DERIVED The birefringence counts turns of the same weighted phase. The polarization turns by $\kappa\alpha/2\pi$ for each radian the spin turns while the light is in flight (§49.12), $\beta = (\kappa\alpha/2\pi)\, H_\infty (t_0 - t_{\rm rec})$, and (220) places cosmic dawn where the spin had turned α of its present phase, $H_\infty t_\star = \alpha H_\infty t_0$. The fine-structure constant cancels between them, and so does the present's phase:

$$\boxed{H_\infty t_\star = \frac{2\pi}{\kappa}\,\beta = \frac{\pi}{3}\,\beta, \qquad \frac{\nu_\star}{\nu_{21}} = \left[\sqrt{\frac{\Omega_m}{\Omega_\Lambda}}\,\sinh\frac{\pi\beta}{2}\right]^{2/3},} \tag{222}$$

to 3×10^{-5}, the light's departure at recombination, with β taken by magnitude. The spin's phase at cosmic dawn is the CMB's twist over the anomaly's weight per radian, $\kappa/2\pi = 3/\pi$. Two counts of cycles are tied with no α, no clock and no Hubble constant: the twist of the oldest light, a fraction of a turn, and the onset of the 21-cm signal, a ratio of received to emitted cycles. Ω_m/Ω_Λ enters only as the shape of the expansion, which the resonance fixes; Planck's Ω_m moves the onset by 0.1%.

PREDICTION MEASURED With the measured angle, $0.277^\circ \pm 0.057^\circ$ [85], (222) places the onset at $42.7^{+5.6}_{-6.0}$ MHz, $z_\star = 32.2$, 0.6σ below (221), as far as the angle lies from its closed form. The onset moves as $\beta^{0.66}$: a twist known to $\pm0.02^\circ$ places it to ±2.0 MHz, and one known to $\pm0.01^\circ$ to ±1.0 MHz. Read the other way, first starlight at $z \approx 20$, 67.6 MHz (§48.5), would need $\beta = 0.55^\circ$, 4.8σ above the measured angle. An onset and a twist that miss (222) by more than 5σ together exclude the hypothesis with the rotation's birefringence, and a twist settled at zero, which would place cosmic dawn at the beginning, excludes it outright. STANDARD Correlations between 21-cm fluctuations and the B modes a birefringent scalar induces have been studied [157]; a relation between the CMB's twist and the epoch of cosmic dawn has, to our knowledge, not been proposed. OPEN The relation tests (220) against a measurement already in hand but does not derive it: it states the hypothesis between two counts of the one spin, with α the common weight. Register row 78.

48.9 What the hypothesis tests, changes and does not do

Definition. DERIVED The first starlight $t_\star$ is the onset of cosmic dawn: the moment the global Lyman-α background from stars first couples the hydrogen spin temperature to the gas, seen in the sky-averaged 21-cm signal. It is not the first individual star: in the rarest density peaks of the observable volume a first star forms far earlier, near $z \approx 65$ [92].

The test. PREDICTION Timing the onset to $\pm3\%$ places it at $z = 29.0$–30.2, 45.5–47.3 MHz; to $\pm10\%$, at $z = 27.7$–31.8, 43.3–49.5 MHz. The falsifier is one-sided and sharp: coupling detected below 45.5 MHz means starlight before $t_\star$ and excludes the hypothesis. A later onset disfavors it more softly, because coupling follows the first stars with a lag. Instruments reaching below 50 MHz—LOFAR's low band, NenuFAR, LEDA, MIST and radio arrays on the lunar far side—see the onset itself; SKA-Low, HERA, REACH and EDGES begin near 50 MHz ($z = 27.4$) and test what follows it.

What it changes in GRUT. DERIVED The first stars form in minihalos of 10^5–$10^6\,M_\odot$, and the condensate suppresses halos below its half-mode mass, $M_{1/2} = \frac{4\pi}{3}\bar\rho_m(\pi/k_{1/2})^3$ with $k_{1/2} \simeq 4.5\,(m_\delta/10^{-22}\,{\rm eV})^{4/9}\ {\rm Mpc}^{-1}$ [81]: $4.9\times10^7\,M_\odot$ at the Lyman-α bound $m_\delta = 2\times10^{-20}$ eV. Forming first stars by $z = 29.6$ therefore requires $M_{1/2} \lesssim 10^6\,M_\odot$, that is

$$m_\delta \gtrsim 4\times10^{-19}\ {\rm eV} \qquad (2\times10^{-18}\ {\rm eV\ for\ } 10^5\,M_\odot\ {\rm minihalos}), \tag{223}$$

twenty times the Lyman-α bound. At that mass the Jeans length is 20–64 pc and galaxy cores shrink accordingly; the Solar System stays far inside it (§46.3). The estimate uses the half-mode criterion and is good to an order of magnitude.

What it could settle. PREDICTION The age of the universe, independently of the distance ladder and of the CMB's inference of H_0: $t_0 = 137.036\,t_\star$. A 3% timing of the onset gives H_0 to 3%, $\pm2.0\ {\rm km\,s^{-1}\,Mpc^{-1}}$, and separates the Planck and SH0ES values, which differ by 8.4%, at about 3σ. It also fixes the present epoch, $t_0 = t_\star/\alpha$, as a large-number relation of the kind Dirac noted and Dicke read as a condition on when observers can exist [93]; stated at the maximum-power point instead of the present (§48.10), it needs no observer. Counted in cycles the onset needs no clock either: (222) sets it against the CMB's twist, and a Hubble constant follows from it only through the CMB's physical matter density, which turns the count into years (§48.8).

What it does not do. It proves nothing; it does not derive α, since t_0 is an epoch rather than a constant; and the present agreement is not evidence. With first starlight estimated between 50 and 200 Myr, a ratio drawn log-uniformly from that range falls within 35% of α with probability 0.53, within 10% with 0.15, and within 3% with 0.04. Stated here in advance, an onset measured within 3% of $z_\star$ would be a one-in-twenty-five chance if the hypothesis were false: the level at which the relation stops being plausibly a coincidence.

48.10 What dark energy is

Put together, §26.4, §38, §48.2 and §48.6 say what the dark energy is.

DERIVED (i) The scalar sector carries no vacuum energy, $V_0 = 0$ (§38), so the vacuum of quantum fields does not gravitate as a cosmological constant here, and the dark energy is the space-filling defect of the horizon (§26.4), with $w = -1$. (ii) The cells do not stretch (§48.6), and with $w = -1$ their number stops at N_∞ (§48.2). (iii) Since $H_\infty^2 = 2\pi c^2/N_\infty \ell_P^2$,

$$\boxed{\rho_\Lambda = \frac{3}{4}\frac{\rho_P}{N_\infty}, \qquad \rho_P = \frac{c^7}{\hbar G^2},} \tag{224}$$

exactly: $\rho_\Lambda/\rho_P = 1.130 \times 10^{-123} = 3/4N_\infty$ with $N_\infty = 6.63 \times 10^{122}$. Equivalently, each cell of the final horizon holds one horizon-scale quantum, $\hbar c/4\pi r_\infty = 1.51 \times 10^{-53}$ J, and together they hold the whole energy of the final horizon, 1.01×10^{70} J. (iv) The rise of dark energy is the filling of that capacity, $\Omega_\Lambda(t) = N(t)/N_\infty$. PREDICTION Dark energy is the energy of the universe's finite information capacity: the Planck energy density shared over every cell the horizon will ever hold.

What it answers. The cosmological constant problem is that the vacuum energy lies some 10^{123} below the Planck density [94]. Here that number is $3/4N_\infty$: the vacuum energy is shared over 10^{123} cells rather than held by each. The holographic bound on effective field theory [95] and holographic dark energy [96] reach the same scaling; GRUT gives the coefficient, $\frac{3}{4}$, and the reason no larger vacuum energy competes, $V_0 = 0$. The why-now problem sharpens as well. The present lies 0.38 Gyr after the maximum-power point, where $\Omega_\Lambda = \frac{2}{3}$ (§48.3), and the first-starlight relation can be stated between two physical epochs, with no reference to observers:

$$t_\star = \alpha\, t_{\rm mp} = 97.9 \text{ Myr}, \qquad z_\star = 30.2, \qquad \nu_\star = 45.6 \text{ MHz}, \tag{225}$$

2.7% from αt_0 and inside the ±3% window of §48.9; a 1% timing distinguishes the two readings.

What it does not. The magnitude is not derived from geometry: it is relocated to N_∞, the size of the final horizon in cells, which is the framework's one boundary datum (§44.6). The question why Λ is small becomes why the universe is 10^{61} Planck lengths across. Carried through the reset (§45) as the next aeon's information capacity, N_∞ would be a property of the cycle rather than an accident; that is not yet shown. Equation (224) holds as an identity in any flat universe with a constant vacuum; its content in GRUT is causal—the vacuum energy is set by the capacity, because nothing else is left to set it—and it fails with $w = -1$: an evolving dark energy established at 5σ (register row 6) means dark energy is not the capacity energy.

48.11 The vacuum of every field

DERIVED The capacity bound holds for every field, not only for the defect scalar. The standard estimate gives each Planck volume a Planck energy, $\rho_{\rm vac} \simeq \rho_P$ [94]. Inside the horizon that is an energy $\rho_P \cdot \frac{4}{3}\pi r^3$, while the horizon condition of §8 fixes the whole content at $M_{\rm H}c^2 = c^4 r/2G$. Their ratio is

$$\frac{\rho_P \cdot \frac{4}{3}\pi r^3}{M_{\rm H}c^2} = \frac{8\pi}{3}\frac{r^2}{\ell_P^2} = \frac{4N}{3}, \tag{226}$$

the same 10^{123}. A vacuum of Planck density would put the horizon's content inside a Schwarzschild radius $4N/3$ times the horizon: the region would be a black hole 10^{123} times its own size. Gravity couples to the energy of every field alike, so the horizon condition caps the vacuum energy of all fields together—electrons, photons, quarks and the defect scalar—at the critical density, $\rho_{\rm vac} \le \rho_c(t)$. This is the holographic bound of [95], and here it is the horizon condition itself. At the freeze the vacuum saturates the cap, and the energy per cell is

$$\frac{\hbar c}{4\pi r_\infty} = \tfrac{1}{2}k_B T_\infty, \qquad T_\infty = \frac{\hbar H_\infty}{2\pi k_B} = 2.18 \times 10^{-30} \text{ K}: \tag{227}$$

one quadratic degree of freedom at the horizon temperature per cell, equipartition over the horizon's degrees of freedom [97], at the exchange rate of §24, $E_{\rm H} = T_{\rm GH} S$. PREDICTION The discrepancy of 10^{123} is therefore a counting error in the standard estimate—a Planck energy per volume cell in place of half the horizon temperature per surface cell—for all fields together. DERIVED Mode by mode, the cap is a cutoff. A local field theory in the horizon is consistent only while no state it describes lies inside its own Schwarzschild radius [95]. Its zero-point energy, $\hbar c k_{\rm max}^4/16\pi^2$, reaches the critical density at

$$k_{\rm max} = \frac{(6\pi)^{1/4}}{\sqrt{\ell_P r}}, \qquad \frac{2\pi}{k_{\rm max}} = 0.142 \text{ mm}, \tag{228}$$

so modes shorter than the geometric mean of the Planck length and the horizon—the length of the dark-energy density, $(\hbar c/\rho_\Lambda c^2)^{1/4} = 0.088$ mm—are not independent degrees of freedom and do not gravitate separately.

48.12 The present in Planck cells

HYPOTHESIS (H4) The resonance (H3, §36.7) places the present in the expansion history; the cell count N places it in Planck units, and Bridge theorem B4 (§44.6) leaves N the framework's one boundary datum. H4 fixes it by the electric coupling: the present is the epoch at which the horizon, measured in Planck lengths, satisfies

$$\boxed{\ln\frac{r}{\ell_{\rm P}} = \frac{1}{\alpha} + \ln 8\pi + \frac{3\pi}{2}\,\alpha, \qquad \frac{r}{\ell_{\rm P}} = 8\pi\, e^{1/\alpha+3\pi\alpha/2}.} \tag{229}$$

The 8π is the horizon's solid angle 4π taken over the double cover of the fold, $S^3 \to \mathbb{RP}^3$ (§1.8–1.9), the factor that also sets the area quantum $8\pi\ell_{\rm P}^2$ (§36.11); the correction is the three arcs of C_3 times the conjugate pair's quarter turn, $3 \times \frac{\pi}{2}$, at first order in α. Written $r = 8\pi\ell_{\rm P}\, e^{1/\alpha}$, it has the form of dimensional transmutation, $\Lambda = M\, e^{-2\pi/b\alpha}$, with $b = 2\pi$: the horizon is the length at which the electric coupling, counted from one Planck cell, has run through $1/\alpha$ e-folds.

DERIVED It is a condition on the epoch, not a law for α. As a law it would tie α to the growing horizon, $d(1/\alpha)/dt = -\dot{H}/H = (1+q)H$: a drift $\dot{\alpha}/\alpha = -2.4 \times 10^{-13}$ yr^{-1} today against $(1.0 \pm 1.1) \times 10^{-18}$ yr^{-1} from atomic clocks [87], a shift of 0.4% at $z = 1$, and $1/\alpha = 127$ at recombination; on the constant final horizon it would give 137.226. With α constant the condition holds once, and it selects the present as H3 and $t_\star = \alpha t_0$ (§48.8) do. Teller's $1/\alpha \simeq \ln(\hbar c/Gm_p^2)$ was a relation of this kind read as a law of varying constants [126]; H4 is read as a selection.

PREDICTION With CODATA's α, (229) gives, with no cosmological input,

$$H_0 = \frac{c}{8\pi\ell_{\rm P}}\, e^{-1/\alpha-3\pi\alpha/2} = 67.3765 \pm 0.0007 \ \text{km s}^{-1}\,\text{Mpc}^{-1}, \qquad N = 128\pi^3\, e^{2/\alpha+3\pi\alpha} = 4.534 \times 10^{122}, \tag{230}$$

the error being G's. The CMB fit of the resonance budget gives 67.35 ± 0.06 (§26.7): the two selections of the present, one by the expansion history and one by the Planck cells, agree at 0.4σ, and together they predict the acoustic angle of the budget, $100\theta_* = 1.04124$, against Planck's 1.04110 ± 0.00031. With H3 the cosmological constant follows in Planck units,

$$\Lambda\ell_{\rm P}^2 = \frac{81}{256\pi^4}\, e^{-2/\alpha-3\pi\alpha} = 2.843 \times 10^{-122}, \qquad \frac{\rho_\Lambda}{\rho_P} = \frac{3}{4N_\infty} = \frac{81}{2048\pi^5}\, e^{-2/\alpha-3\pi\alpha} = 1.131 \times 10^{-123}, \tag{231}$$

so the 10^{-122} of the cosmological constant problem is $e^{-2/\alpha}$: the vacuum is as small as the present is old. OPEN The 8π and the $\frac{3\pi}{2}$ were read after the value was known, and other small combinations fall inside the ± 0.0009 that the CMB fit's error allows in $1/\alpha$; a derivation would obtain the coefficient $b = 2\pi$ and the correction from the field equations. PREDICTION H4 fails if the converged H_0 settles away from 67.38 at 5σ, if the budget's acoustic angle settles away from 1.04124 at 5σ, or if α is found to vary (register row 86).

49 The Local Baryonic Coupling

This section follows the rotation of Section 23 into galaxies: the baryons modulate its frequency; the modulation, held at the rotation's rate, is a medium bound around them (Sections 46 and 47); and the bound medium gives the flat rotation curves, the Solar-System limit and the birefringence of the CMB.

49.1 Why the global frequency alone cannot hold a galaxy

If the whole universe rotates at one frequency, the orbital speed in a galaxy depends on where it sits, $v_0(\chi) = \omega_0 R_H \sin\chi$: faster toward one end of the rotation and slower toward the other. That is not flat, and it is not what is observed.

49.2 The baryons modulate the frequency

The baryons do not create the gravity; they modulate the frequency. They make the phase wind faster or slower depending on where the matter is:

$$\omega(\chi) = \omega_0 + \delta\omega_b(\chi), \qquad \theta_{\rm phase} = [\omega_0 + \delta\omega_b(\chi)]\, t + n\varphi. \tag{232}$$

The topology does not change: $\oint d\theta_{\rm phase} = 2\pi n$, and n is still an integer. The global frequency does not change: it is still fixed by the C_3 eigenvalue. What changes is the local rate at which the phase is sampled. Requiring a flat velocity $v_{\rm flat}$ in $v(\chi) = R_H \sin\chi\,[\omega_0 + \delta\omega_b(\chi)]$ gives the response equation

$$\delta\omega_b(\chi) = \frac{v_{\rm flat}}{R_H \sin\chi} - \omega_0, \qquad \omega(\chi) = \frac{v_{\rm flat}}{R_H \sin\chi}, \tag{233}$$

a response, not a new winding number. With a response functional of the baryonic density, $\delta\omega_b = \mathcal{C}[\rho_b]$, the local frequency is $\omega(\chi) = \omega_0 + \mathcal{C}[\rho_b(\chi)]$.

49.3 The flat curve and the transition

Once the modulation reaches the value that gives the flat velocity, the velocity is flat—not fitted, not tuned, but what the response equation demands. The transition between the global rotation and the locally modulated rotation is controlled by a coupling function of the local baryonic acceleration $g_b = GM_b(<r)/r^2$:

$$\omega(r) = \omega_0 + \Gamma_b\left(\frac{g_b}{a_c}\right)\left[\frac{v_{\rm flat}(r)}{R_H\sin(r/R_H)} - \omega_0\right], \qquad v(r) = [1-\Gamma_b]\,\omega_0 R_H\sin(r/R_H) + \Gamma_b\, v_{\rm flat}. \tag{234}$$

In the outer regions of galaxies, where the acceleration is weak, the coupling is strong and the curve is flat, $\Gamma_b \to 1$. In the inner regions and the Solar System, where the acceleration is strong, the coupling is suppressed, $\Gamma_b \to 0$, and the rotation is the tiny global mode: ordinary Newtonian behaviour. For a galaxy $r/R_H \sim 10^{-5}$, so $\omega(r) \simeq v_{\rm flat}/r$; inside the baryonic core r_c the response saturates, $\Gamma_b = \frac{r^2}{r^2+r_c^2}\,\Gamma_{\rm gal}(g_b/a_c)$, so $\omega(0) = \omega_0$ stays finite.

49.4 Tests and independent support

PREDICTION The local inertial frame, compared with the quasar frame by high-precision astrometry, tests the mechanism [100]: with the windings zero (§23.2), the internal rotation predicts no rotation of the inertial frame against the quasars. The coupling function, once derived, must reproduce the rotation curves, satisfy the compact-space constraint, and respect the Solar-System limit, as one function. Independently, a slowly rotating universe has been shown to be viable: a Gödel-inspired rotating model [98] with present angular velocity $\simeq 2\times10^{-3}$ Gyr^{-1}, close to the maximal rotation that keeps the tangential speed below c at the horizon, resolves the Hubble tension [99]. That is a rotation of matter, and the vorticity bound [115] constrains it; the rotation here is internal and carries no vorticity.

49.5 The coupling has two parts

The baryons tune the phase in two ways at once: through the metric they curve, and through the cone they carry.

DERIVED *The metric part, from the ground up.* Put the phase into its own field equation on the metric the baryons curve, $ds^2 = -(1+2\Phi/c^2)c^2dt^2 + (1-2\Phi/c^2)\,d\mathbf{x}^2$. For $\theta = \Omega(\mathbf{x})\,t + n\varphi$ the part of $\nabla_\mu(\chi^2 g^{\mu\nu}\partial_\nu\theta) = 0$ that grows with time is $t\,\chi^2 c\,\nabla^2\Omega$, so $\nabla^2\Omega = 0$, and on the closed S^3 the only regular solution is $\Omega = \Omega_0$: a frequency that varied from place to place would wind the phase tighter every second, and its energy would grow as t^2. What the baryons modulate is the frequency a local clock measures,

$$\Omega_{\rm loc} = \Omega_0\Big(1 - \frac{\Phi}{c^2}\Big), \qquad \Gamma_b^{(g)} = -\frac{\Phi}{c^2}, \tag{235}$$

with coefficient exactly one. This is the redshift of the core: the deeper potential slows the local phase by the same factor that redshifts its light. At the Sun's orbit $|\Phi|/c^2 \simeq (1.2\text{–}1.8)\times10^{-6}$.

DERIVED *The geometric part, by matching.* Requiring the orbit condition $v^2/r = g_b + a_{\rm drag}$ to reproduce the field equation $\nabla\cdot[\mu\nabla\Phi] = 4\pi G\rho_b$ (§22) at every radius fixes

$$\Gamma_b \equiv \frac{a_{\rm drag}}{g_b} = \nu\Big(\frac{g_b}{a_c}\Big) - 1 \;\longrightarrow\; \frac{2}{\sqrt{3}}\Big(\frac{a_c}{g_b}\Big)^{1/2} = \Big(\frac{a_0}{g_b}\Big)^{1/2} \quad (g_b \ll a_c), \tag{236}$$

with ν the inverse of μ: the geometric-mean form, its coefficient $2/\sqrt{3}$ set by the isotropic average of §22. The zero point is then predicted, not anchored:

$$\omega(r) = \frac{v_{\rm flat}}{2r}, \qquad v_{\rm flat}^4 = a_0\,G\,M_b, \qquad a_0 = \tfrac{4}{3}\,a_c. \tag{237}$$

At the Sun's orbit the required value is $\Gamma_b = 0.51$. The metric part is of order $(v/c)^2$; the geometric part carries the flat curve, and its source is the cone each baryonic system carries.

49.6 The baryonic defect cone

HYPOTHESIS Baryons at any scale generate their own defect cone. The cone's deficit is π regardless of scale; its slant depends on the scale. DERIVED A deficit of π fixes the cone's shape: unrolled, the cone is a half-disc of radius L, whose arc πL closes into the base circle $2\pi R$, so $R = L/2$, the half-opening is 30°, and the section through the axis is an equilateral triangle—the C_3 triangle of the second axiom, at every scale. Scale therefore enters only through the slant length L. The defect relation (28) is the 2+1 reduction with length $L = r$ (I3); with each system's own slant as its reduction length, a deficit of π requires

$$\pi = \frac{8\pi GM}{c^2 L_\pi} \quad \Longrightarrow \quad L_\pi = \frac{8GM}{c^2} = 4\, r_s, \tag{238}$$

four Schwarzschild radii for any mass: compactness $GM/c^2 L_\pi = \frac{1}{8}$, a quarter of a black hole's—the strength of almost a black hole, without a black hole. For the horizon's cone mass, $M_{\rm cone} = M_{\rm H}/4 = c^2 r/8G$, it returns $L_\pi = r$, the horizon's own cone; for the Sun it is 11.8 km, and for the Milky Way's baryons, $6\times10^{10} M_\odot$, 4.7×10^3 AU.

A cone of fixed shape supplies one dimensionless coupling, a ratio of lengths, $\Gamma_b = r/\ell$. The matching (236) requires

$$\ell = \sqrt{\frac{GM_b}{a_0}} = \sqrt{L_\pi \cdot \frac{c^2}{8a_0}}, \qquad \frac{c^2}{8a_0} = \frac{\pi^2}{8\sqrt{3}}\, r, \tag{239}$$

the geometric mean of the cone's slant L_π and a fixed horizon length: 7.2×10^3 AU for the Sun and 8.5 kpc for the Milky Way's baryons, the radius at which the system's own pull equals a_0. Inside it the cone's coupling is weak and the motion Newtonian; beyond it the coupling grows as r/ℓ and the curve is flat, with $v_{\rm flat}^4 = a_0 GM_b$.

49.7 The coupling length: gravity in two dimensions

DERIVED The deep-regime field of an isolated mass, $g = \sqrt{GMa_0}/r$, falls as $1/r$: it is exactly the field of the same mass under two-dimensional Poisson gravity,

$$\nabla^2\Phi = 4\pi \frac{G}{L_{\rm red}}\,\Sigma, \qquad g = \frac{2GM}{L_{\rm red}\, r}, \qquad L_{\rm red} = 2\sqrt{\frac{GM}{a_0}}. \tag{240}$$

IDENTIFICATION The cone supplies the reduction. The defect relation (28) is already the 2+1 reduction $G_3 = G/L$ (I3); beyond its crossover, each baryonic system's gravity is that of its cone's two-dimensional section. DERIVED The coupling length then follows in four steps:

1. a deficit of π fixes the cone's shape and its slant, $L_\pi = 8GM/c^2$ (238);
2. inside the crossover the field is three-dimensional, GM/x^2; beyond it, two-dimensional, $2GM/(L_{\rm red}x)$;
3. the two laws meet at $x_c = L_{\rm red}/2$;
4. the crossover lies at the horizon's acceleration (§10): $GM/x_c^2 = a_0$.

Hence

$$\ell = x_c = \sqrt{\frac{GM}{a_0}}, \qquad L_{\rm red}^2 = L_\pi\, R_{a_0}, \qquad R_{a_0} = \frac{c^2}{2a_0} = z_H\, r, \qquad z_H = \frac{\pi^2}{2\sqrt{3}} = 2.849. \tag{241}$$

R_{a_0} is the radius of the horizon whose surface gravity $c^2/2R$ equals a_0, and z_H is the horizon load of §25.7. The reduction length is the geometric mean of the system's own π-cone slant, four Schwarzschild radii, and the horizon whose surface gravity is a_0: the altitude of the right triangle whose hypotenuse is cut into L_π and R_{a_0}. The Tully–Fisher law follows with nothing put in,

$$v_{\rm flat}^2 = \frac{2GM_b}{L_{\rm red}} = \sqrt{GM_b a_0}, \qquad v_{\rm flat}^4 = a_0 GM_b, \tag{242}$$

and the frame field takes the same form: $\omega = v_{\rm flat}/2r$ and $a_{\rm drag} = 2\omega v = 2GM_b/(L_{\rm red} r)$. For the Sun $L_{\rm red} = 2.15\times10^{15}$ m ($\ell = 7{,}199$ AU); for the Milky Way's baryons, $6\times10^{10} M_\odot$, $L_{\rm red} = 5.3\times10^{20}$ m ($\ell = 8.55$ kpc).

49.8 Tests

PREDICTION *The Tully–Fisher zero point.* Gas-rich galaxies, whose baryonic masses are least affected by stellar-population modelling, give $a_0 = (1.24\pm0.14)\times10^{-10}$ m s^{-2} with the slope fixed at four [102]; $a_0 = 1.148\times10^{-10}$ lies 0.7σ below.

PREDICTION *Wide binaries.* The law acts where the rotation's modulation is bound to the system (§49.11). A star's escape speed is far below the motions of the Galaxy's medium, so its capture fraction is negligible (7×10^{-9} for the Sun, §47.2): wide binaries carry no bound medium and are Newtonian at all separations. This agrees with the Newtonian Gaia DR3 analysis, $\alpha_{\rm grav} = -0.021^{+0.065}_{-0.045}$ [105]; a confirmed boost such as the 22^{+10}_{-6}% reported beyond 3000 AU [103, 104] would falsify it. Two alternatives were examined and are excluded: a crossover set by each system's own pull gives boosts of 1.33 and 1.69 at 10 and 20 kAU, above both analyses; a crossover set by the total field gives a 10% boost and, in the Solar System, the quadrupole of §49.11, excluded at 16σ.

49.9 The magnetic field and the chiral current

DERIVED The anomaly coupling $\mathcal{L} \supset -(\kappa\alpha/\pi)\,\theta\,\mathbf{E}\cdot\mathbf{B}$ (§52.1) adds the current of axion electrodynamics [106] to Ampère's law,

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{j} + \frac{1}{c^2}\frac{\partial \mathbf{E}}{\partial t} - \frac{\kappa\alpha}{\pi}\,\frac{\dot\theta}{c}\,\mathbf{B} + \cdots , \tag{243}$$

so the rotating phase drives a current along $\mathbf{B}$. In a conductor it acts as an α-effect, unstable below the wavenumber $k_\theta = (\kappa\alpha/\pi)\,\dot\theta/c$ and fastest at $k_\theta/2$. For the global rotation, $\dot\theta = \Omega_0 = 0.827H$, the fastest-growing wavelength is $4\pi^2 c/(0.827\,\kappa\alpha H)$, 1,090 Hubble radii for $\kappa = 6$ at every epoch; for the Galaxy's rotation at the Sun, $\Omega = v/r = 9.2 \times 10^{-16}$ s^{-1}, it is 2.1 Hubble radii. The anomaly current cannot grow a field inside the horizon. The observed fields are set by ordinary dynamo saturation, at equipartition with the turbulent motions,

$$B_{\rm eq} = \sqrt{\mu_0\rho}\; v_t \sqrt{D/D_{\rm cr} - 1} \;\longrightarrow\; \sqrt{\mu_0\rho}\; v_t, \tag{244}$$

4.6 μG for $\rho = 1.67 \times 10^{-21}$ kg m^{-3} and $v_t = 10$ km s^{-1}, the Milky Way's value (the rotation speed in place of v_t would give 107 μG). In the flat regime the local phase frequency equals the galaxy's angular velocity, $\Omega = v_{\rm flat}/r$ (§49.2), which also drives the differential-rotation dynamo: the field and the frame dragging are driven by the same local Ω.

DERIVED In relativistic chiral matter the rotation carries an axial current, $\mathbf{j}_A = \mu^2\boldsymbol{\omega}/2\pi^2$ for each Dirac species ($\hbar = c = 1$, $T = 0$) [107, 108], so $\mathbf{j}_A = \sum_f N_c\,\mu_f^2\,\boldsymbol{\omega}/2\pi^2$; the anomaly coefficient of the $\theta\,\mathbf{E}\cdot\mathbf{B}$ coupling over three generations is $\sum_f N_c Q_f^2 = 3 + 4 + 1 = 8$. It acts where the fermions are relativistic: the early universe and neutron-star interiors.

49.10 The Solar-System quadrupole

PREDICTION In the Galactic field the Sun's potential can acquire the quadrupole

$$\Phi = -\frac{GM_\odot}{r} - \frac{Q_2}{2}\,x^i x^j\Big(e_i e_j - \frac{1}{3}\delta_{ij}\Big), \tag{245}$$

with $\hat e$ toward the Galactic centre [54]. In the complete equation (§46–47) the Sun carries no bound condensate, for two independent reasons: the Solar System lies far inside the condensate's Jeans length (40–290 pc), and the Sun, formed inside the Galaxy, cannot capture a field streaming past at ~ 165 km s^{-1} with an escape velocity of 0.5 km s^{-1} at its MOND radius (capture fraction 7×10^{-9}, §47.2). The only quadrupole is the Galactic tide,

$$|Q_2| \sim \frac{V_0^2}{R_0^2} = 8.6 \times 10^{-31}\ \text{s}^{-2}, \tag{246}$$

against the Cassini range data reduced with DE440, $(1.6 \pm 1.8) \times 10^{-27}$ s^{-2} [75] (-0.9σ), and the earlier $(3 \pm 3) \times 10^{-27}$ [54].

Why the condensate must be the source. With the baryons as the only source—the quasi-static law of §22—the quadrupole is set by the Sun's phantom density $\rho_{\rm ph} = \nabla\cdot[(\nu-1)\nabla\Phi_N]/4\pi G$ in the Galactic field, and since $P_2(\cos\theta)/r^3$ is harmonic,

$$Q_2 = \frac{3}{4\pi}\int (\nu - 1)\,\mathbf{g}_N \cdot \nabla\Big(\frac{P_2(\cos\theta)}{r^3}\Big)\,d^3y. \tag{247}$$

Evaluated with $\nu - 1 = (1-\mu)/\mu$ free of cancellation and the saddle point resolved (the simple and standard functions give 2.80×10^{-26} and 1.98×10^{-26} s^{-2} at $a_0 = 1.2 \times 10^{-10}$, inside the published range 2.1×10^{-27}–4.1×10^{-26} [76, 77]), the threshold family gives 2.8×10^{-26} at $g_e = 1.9 \times 10^{-10}$ m s^{-2} and 3.1×10^{-26} s^{-2} at $g_e = V_0^2/R_0$, twenty times what Cassini allows. A boost that belongs to the baryons' field, rather than to matter bound to each system, cannot hold in the Solar System; every modified-gravity law that fits rotation curves meets the same bound [78]. That is the incompleteness that §46 repairs.

49.11 Where the law acts

DERIVED *The two-dimensional law is derived.* The deep limit of the law of §22, $\mu \to \frac{3}{4}x$, gives $g = \sqrt{a_0 g_N}$ and therefore the $1/r$ field (240). Reading it as the cone's section (§49.7) gives $L_{\rm red}$ its geometric meaning; the predictions need nothing more.

DERIVED *The law acts where the medium is bound.* The extra gravity of a system is carried by the rotation's modulation bound to it (§23.1): its polarization around the baryons has the permittivity μ of §46.6, a theorem that uses the threshold response and no particle mass. Whether the medium is bound is the capture condition of §47.2, $\Gamma = \mathrm{erf}(X) - (2X/\sqrt{\pi})e^{-X^2}$ with $X = v_{\rm esc}/\sqrt{2}\sigma$, set by the motions of the medium and not by a mass. Galaxies and clusters, formed inside the cosmic medium, bind it ($\Gamma \to 1$); a star does not ($\Gamma = 7 \times 10^{-9}$ for the Sun). Every level of structure drags the frame inside it through its own bound medium: the Galaxy's medium sets the Sun's orbit, and no medium of the Sun's own alters the planets.

PREDICTION *The test that decides it.* Applied instead to the total field of the Galaxy and the Sun, every form of the law of §22 gives the Solar System an external-field quadrupole

$$Q_2 = (3.0\text{–}3.2) \times 10^{-26}\ \mathrm{s}^{-2} \tag{248}$$

(threshold orders 1–3 and the dynamic exponent; the method of Appendix A), twenty times the Cassini value $(1.6 \pm 1.8) \times 10^{-27}$ [75]: excluded at 16σ. With the medium unbound around the Sun, Q_2 is the Galactic tide alone, $9 \times 10^{-31}\ \mathrm{s}^{-2}$, at -0.9σ (variance row 21).

DERIVED *Extended systems.* For a bound system the field equation of §22 is local and conservative, and its isolated-mass solution is (240); extended and non-spherical systems obey it directly.

49.12 Birefringence

DERIVED Held at its rate by (51), the phase advances uniformly, $\theta = \Omega_0 t$, and through the anomaly coupling $-(\kappa\alpha/\pi)\,\theta\,\mathbf{E}\cdot\mathbf{B}$ (§52.1) it rotates the plane of polarization of light by half the coupling times the phase advanced in flight. For the CMB,

$$\beta = \frac{\kappa\alpha}{2\pi}\,\Omega_0\,(t_0 - t_{\rm rec}) = 0.052^\circ\,\kappa, \tag{249}$$

with $\Omega_0(t_0 - t_{\rm rec}) = 0.781$ rad, and its sign is the sense of the rotation, the handedness of the two-C/S structure. DERIVED The coupling is the anomaly of the fermions whose masses the phase enters, those on the cone (§42.3, §42.7). Removing the phase from their mass terms is a chiral rotation of unit weight, whose anomaly gives $\kappa = \sum_f N_c Q_f^2$ over them [106]: $1 + 1 + 1 = 3$ for the charged leptons and $3(\frac{4}{9} + \frac{1}{9} + \frac{4}{9}) = 3$ for (c, b, t), so $\kappa = 6$ and

$$\beta = 0.314^\circ, \tag{250}$$

0.6σ above the joint ACT and Planck value $0.277^\circ \pm 0.057^\circ$ [85]; with the charged leptons alone, $\kappa = 3$ and $\beta = 0.156^\circ$, 2.1σ below. This replaces, in the rotation reading, the frozen phase of §52.1, whose bound $|\beta| < 10^{-3\circ}$ left a confirmed cosmological angle a 4.9σ problem. With the rate set by the spin and the present at the resonance the angle takes a closed form, $\beta = (2\alpha/\pi)\,\mathrm{artanh}(3\sqrt{3}/2\pi) = 0.3137^\circ$ (§36.13). With the first-starlight hypothesis the same weighted phase dates cosmic dawn: α cancels, and $H_\infty t_\star = (2\pi/\kappa)\,\beta$ (§48.8).

49.13 One rotation, from galaxies to the oldest light

DERIVED The acceleration scale and the birefringence angle are set by the same rotation. The critical acceleration is the photon's acceleration over one turn of the C_3 eigenmode, $a_c = c/\tau_1$ with $\tau_1 = 2\pi/\Omega_0$, and $a_0 = \frac{4}{3}a_c$, so

$$\Omega_0 = \frac{2\pi a_c}{c} = \frac{3\pi}{2}\,\frac{a_0}{c}. \tag{251}$$

(The eigenmode is the current horizon's, and its period equals the rotation's turn at the present, the resonance; the relation holds at the present, while a_0 itself tracks the horizon at other epochs, §36.15.) The angle is the phase advanced between last scattering and today, weighted by the anomaly, $\beta = \kappa\alpha\,\Omega_0(t_0 - t_{\rm rec})/2\pi$ (§49.12). Eliminating the rotation, and dropping $t_{\rm rec}/t_0 = 3 \times 10^{-5}$,

$$\boxed{\beta = \frac{3\kappa\alpha}{4}\,\frac{a_0 t_0}{c} = \frac{9\alpha}{2}\,\frac{a_0 t_0}{c}} \qquad (\kappa = 6). \tag{252}$$

Each factor is measured by a different science: β in the polarization of the CMB, α in atomic physics, a_0 in galaxy rotation curves, t_0 in the cosmic age. With the framework's own $a_0 = 1.148 \times 10^{-10}\ \mathrm{m\,s}^{-2}$ and

$t_0 = 13.79$ Gyr the relation gives 0.314°, as §49.12 does by the other route; with the radial acceleration relation's $a_0 = (1.20\pm0.24)\times10^{-10}$ m s^{-2} [37] and $t_0 = 13.80$ Gyr it gives $0.33°\pm0.07°$. The measured angle is $0.277°\pm0.057°$ [85], 0.6σ from both: galaxy rotation curves predict the rotation of the CMB's polarization. Milgrom's coincidence $2\pi a_0 \approx cH_0$ [51] ties a_0 to the expansion; equation (252) ties it, through α, to the polarization of the oldest light, and to our knowledge no relation between the MOND acceleration and cosmic birefringence has been proposed before. PREDICTION The relation tests the whole chain at once: the rate fixed by the C_3 eigenvalue, the rotation's being internal and held at its rate, and the anomaly weight $\kappa = 6$ of the cone fermions. With a_0 known to 5% (now 20%) and β to 0.02°, a ratio β/α more than 5σ from $\frac{9}{2}a_0t_0/c$ excludes it (register row 62).

50 The closure law and the scale ladder

50.1 The law

DERIVED Energy is defect. By the defect map (28) every energy E inside the horizon carries the deficit

$$\delta(E) = \frac{8\pi GE}{c^4 r} = 4\pi\,\frac{r_s(E)}{r}, \qquad r_s = \frac{2GE}{c^4}, \tag{253}$$

4π times the ratio of its Schwarzschild radius to the horizon radius. STANDARD On a closed surface the total curvature is topological, $\sum_i \delta_i + \int K\,dA = 2\pi\chi$ (Gauss–Bonnet). In 2+1 gravity, where all curvature sits in point defects, a closed universe therefore has total deficit exactly 4π and a fixed total mass [25, 118]. DERIVED The horizon is the 2+1 surface on which, by the reduction of (28), every interior energy appears as a defect, and it is closed, $\chi = 2$. Its defects sum to 4π:

$$\boxed{\sum_i r_{s,i} = \frac{\chi}{2}\,r, \qquad \Omega_{\rm tot} = \frac{\chi}{2} = 1.} \tag{254}$$

The Schwarzschild radii of everything inside the horizon add up to the horizon, and the total energy is the Hubble mass, $\sum_i E_i = M_{\rm H}c^2 = c^4 r/2G$. Planck gives $\Omega_{\rm tot} = 1.0007 \pm 0.0019$ [39], 0.4σ. DERIVED Energy conservation inside the horizon is then topological: the total cannot change without changing χ, and it grows only as r grows. Brill noted that the 2+1 closure is analogous to $GM/c^2R \sim 1$ for a closed universe [118]; with the defect map the analogy becomes the exact equality (254), and flatness becomes the Euler characteristic of the horizon. STANDARD The smallest closed universe of 2+1 gravity has three particles on two triangular faces glued at their edges, three being the fewest that define an inertial frame [118]: the C_3 cycle is the minimal closed horizon.

50.2 The ladder

DERIVED By Gauss–Bonnet a defect δ is the solid angle swept by the normals around it, and the information of a defect is its share of the cells, so energy, defect, horizon area and information are one fraction:

$$\frac{E}{M_{\rm H}c^2} = \frac{\delta}{4\pi} = \frac{A}{4\pi r^2} = \frac{I}{N_{\rm full}}, \qquad N_{\rm full} = \frac{4\pi r^2}{\ell_P^2} = 2N. \tag{255}$$

The elementary symmetric cut is the $\mathbb{Z}_2$ of the deficit π. The quotient $S^2/\mathbb{Z}_k$ has orbifold Euler characteristic $2/k$, so by (254) it carries $M_{\rm H}c^2/k$, and the iterated $\mathbb{Z}_2$ cut is the ladder

$$E_n = \frac{M_{\rm H}c^2}{2^n}, \qquad \delta_n = \frac{4\pi}{2^n}, \qquad I_n = \frac{N_{\rm full}}{2^n}. \tag{256}$$

Its rungs are geometric. At $n = 0$ the deficit is 4π, the closed horizon, and the energy the Hubble mass, 5.205×10^{88} eV. At $n = 1$ it is 2π, the critical mass of 2+1 gravity beyond which a universe must close. At $n = 2$ it is π, the total of the C_3 triple, $M_{\rm cone} = M_{\rm H}/4$ (I1). One generator, $\pi/3$, sits at $n = \log_2 12 = 3.585$, so the triadic rungs of C_3 interleave the dyadic ones. The ladder ends at one cell, the deficit ℓ_P^2/r^2 that is the solid angle of one Planck cell:

$$n_{\max} = \log_2 N_{\rm full} = \log_2 2N = 408.47, \qquad E_{\rm cell} = \frac{M_{\rm H}c^2}{2N} = \frac{\hbar H}{8\pi} = \tfrac{1}{4}k_B T_{\rm GH}. \tag{257}$$

The Hubble quantum $\hbar H$ lies at $\log_2(N/4\pi) = 403.82$. Read with the horizon temperature, the top of the ladder is $M_{\rm H}c^2 = \frac{1}{2}Nk_BT_{\rm GH}$, the holographic equipartition of Padmanabhan [119]; here it follows from the closure.

50.3 Why halving: three conservation laws at one step

STANDARD Momentum conservation fixes the daughters' energies only in a two-body split, and when the daughters are identical and massless each carries exactly half the parent's energy: $\pi^0 \to \gamma\gamma$ steps from rung 267.70 to 268.70, and para-positronium and $H \to \gamma\gamma$ do the same. DERIVED In the law this split is the $\mathbb{Z}_2$ cut: the parent's defect divides into two equal cones moving apart, each with half the defect, half the cells, and one new bit, which half it is. Energy is conserved because the defect total is topological, information because the cell count is the same fraction (255), and momentum because the two halves are equal and opposite. The ladder's step is where the three meet.

50.4 The fixed points

DERIVED For any mass the gravitational and quantum lengths are reciprocal, $r_s\lambda_C = 2\ell_P^2$. Read downward the ladder halves r_s from r; read upward from $\hbar H$ it halves λ_C from r; the two readings cross at the Planck energy, rung $201.41 = \frac{1}{2}[\log_2(N/4\pi) - 1]$, for every H, since $m_P^2 = 2M_{\rm H}\,\hbar H/c^2$. The dark-energy scale of the capacity relation (224), $E_\Lambda = (\rho_\Lambda \hbar^3 c^3)^{1/4} = 2.23$ meV, sits at rung $303.52 = n_P + \frac{1}{4}\log_2(4N_\infty/3)$, three quarters of the way down, at the geometric mean of the Planck and Hubble energies [95]. IDENTIFICATION Matter sits near two thirds. The Eddington–Weinberg relation $\hbar^2 H \approx Gcm^3$ [120] is the holographic matching $(r/\lambda_C)^3 = (r/\ell_P)^2$: the horizon volume holds as many Compton cells as its area holds Planck cells [121]. With unit cells the matched mass is 59.8 MeV, at rung 268.88, two thirds of the way down; with the cells of N it is 110.3 MeV; the nucleon lies 3.97 rungs above. The global count leaves its cell coefficient open; the per-particle footprint balance of §50.8 fixes its own coefficient and places the matched mass at 47.5 MeV.

50.5 What the law says about masses

PREDICTION Exact integer steps occur only along chains of symmetric splits; masses formed by other physics carry no preference for rungs. The data agree. For 23 particle and atomic energies the offsets from the nearest rung average 0.247 rung (uniform: 0.250; Rayleigh $p = 0.18$), and for 18 planets, moons and dwarf planets 0.237 ($p = 0.92$). Near-exact single matches occur at the rate chance gives and move with H_0, every rung shifting by 8.8% between 67.12 (the withdrawn 5/2 budget) and 73.04: at 67.12 the closest are the Moon (0.3%), the τ (1.2%) and Ganymede (1.2%); at 73.04 they are Titan (0.0%), Eris (1.3%) and the 21-cm line (2.5%).

50.6 The limits

DERIVED The ceiling is the closed horizon's total, $M_{\rm H}c^2 \propto r$, and the floor is one cell, $\propto 1/r$. Both move with the horizon, one rung each per doubling of r, while the Planck crossing stays fixed. The depth grows as

$$\frac{dn_{\max}}{dt} = \frac{2H(1+q)}{\ln 2} = 1.39\,H_0 \quad \text{today}, \tag{258}$$

and stops as $q \to -1$, where the bulk reaches holographic equipartition [119]. The final depth is $\log_2 2N_\infty = 409.03$ against 408.47 now: the ladder is 99.86% complete, with 0.56 rungs to come, 0.28 at each end. DERIVED Beyond our horizon the law holds for every observer's horizon, each closed and each carrying its own Hubble mass; nothing outside changes the total inside, which changes only as the horizon grows. A conformal gluing preserves topology, so across the reset $\chi = 2$ carries over and each aeon opens with the same total defect, 4π. PREDICTION Register rows 63–64.

50.7 The reference mass and the radius of S^3

DERIVED The horizon condition, $2GM_{\rm H}/c^2 = r$ (§8), is $\Omega_{\rm tot} = 1$ exactly; with (14) it gives $R \to \infty$. The spatial section is flat, and the S^3 of §1.8–1.9 is the group manifold $SU(2)$, not space; the closed option of §4.6 survives only as a correction beyond the horizon condition, bounded by $R > 18r$. For the reference mass, the scale duality and the Lyman-α bound give $M_* \le 7.5 \times 10^8 M_\odot$ (§37), and the relic density, $\Omega \simeq 0.1\,(f_\delta\theta_i/10^{17}\,\text{GeV})^2(m_\delta/10^{-22}\,\text{eV})^{1/2}$ [57] with $f_\delta = 4.3 \times 10^{18}$ GeV, requires the initial displacement $\theta_i \simeq 1.0 \times 10^{-2}\,(m_\delta/2 \times 10^{-20}\,\text{eV})^{-1/4}$. M_* is fixed once the reset (§45) supplies θ_i; a displacement of order one would need $m_\delta \sim 2 \times 10^{-28}$ eV, which the Lyman-α forest excludes.

50.8 Why matter sits at two thirds: the footprint balance

IDENTIFICATION A particle has two holographic sizes: its footprint on the horizon, the patch its defect occupies,

$\delta(m)\,r^2 = 4\pi r_s r$ by (253), and its own quantum boundary, the Compton sphere of area $4\pi\lambda_C^2$. Counted by the rule of the closure law, one bit per Planck area, the two balance when

$$\boxed{r_s\, r = \lambda_C^2, \qquad m_*^3 = \frac{\hbar^2 H}{2Gc}, \qquad \frac{G m_*}{\lambda_C^2} = \frac{cH}{2}\ :} \tag{259}$$

the particle's footprint is its own surface, and gravity at its Compton radius is half the Hubble acceleration. DERIVED Since $\hbar^2 H/Gc = m_P^2\,\hbar H/c^2$, the balance mass lies one third of the way down from the Planck mass to the Hubble quantum, the two-thirds point of the ladder. At the resonance budget, $H_0 = 67.35$, $m_* = 47.5$ MeV. Matter sits near it: the muon at $2.23\,m_*$, the pions at 2.84–$2.94\,m_*$, the nucleon at $19.76\,m_*$, the electron at $0.011\,m_*$. Here it follows from the closure law, with its coefficient fixed; earlier heuristic arguments reached the same scaling without a fixed coefficient—a gravitational cross-section equal to the Compton area [122], a Compton-surface gravity of the order of the Hubble acceleration [123]—as did the holographic count of §50.4 [121].

DERIVED The balance drifts as $H^{1/3}$ while the nucleon mass does not, so the nucleon holds a fixed ratio to m_* at one epoch only, as Dicke observed for the large-number coincidences [93]. Deriving the ratio is deriving the epoch. Here the epoch is selected at the maximum-power point (§36.6), whose expansion rate the capacity relation (224) ties to the final horizon; an exact nucleon ratio at that epoch would fix N_∞, the boundary datum that no internal condition determines (§44.6). The size of the universe would follow from the mass of the nucleon.

The factor is fixed by the framework's own expansion rate. At the resonance budget, $H_0 = 67.35 \pm 0.06$, with $m_* \propto H^{1/3}$,

$$K \equiv \frac{m_p}{m_*} = 19.759 \pm 0.006, \qquad m_* = 47.49 \pm 0.01 \text{ MeV}. \tag{260}$$

The nearest factor built from the framework's constants is $\mathrm{vol}(S^3) = 2\pi^2 = 19.739$, the $SU(2)$ group manifold of §1.8–1.9, 0.10% below, which would require $H_0 = 67.56$: 3.3σ from the framework's own H_0 (0.4σ with Planck's). It was found by search after the ratio was known, and among 1071 factors $2^a 3^b \pi^{c/2}$ one lies this close for 31% of random ratios; the framework's own expansion rate disfavours it. (At the withdrawn 5/2 budget the factor was 19.782.)

STANDARD No geometric number can supply K. The nucleon mass is set by dimensional transmutation, $m_p = C\,\Lambda_{\mathrm{QCD}}$ with C computed in lattice QCD and $\Lambda_{\mathrm{QCD}} \propto \mu\, e^{-2\pi/b_0\alpha_s(\mu)}$, so K is exponential in $1/\alpha_s$: at the Z mass, with $b_0 = 23/3$, $d\ln K/d\alpha_s = 2\pi/b_0\alpha_s^2 = 59$, and fixing K to its 0.035% would take α_s to 6×10^{-6}.

DERIVED The relation is therefore an exact identity,

$$\boxed{m_p^3 = K^3\,\frac{\hbar^2 H}{2Gc},} \tag{261}$$

whose geometric part is the footprint balance and its $\frac{1}{2}$, whose remaining factor is the strong coupling, and whose expansion rate is the budget's at the selected epoch. The Eddington–Weinberg coincidence is therefore not a second unexplained number: with the epoch selected (§36.6) and H set by the final horizon, it is the boundary datum N_∞ written in units of the QCD scale. Stating the nucleon's position and stating the size of the universe are one and the same datum.

50.9 The strong coupling: two rules tested

HYPOTHESIS Two rules carry the geometry of the electric coupling (§43F) over to the strong one, each fixed before any comparison. The volume rule takes $1/\alpha$ as the volumes of the electroweak group and its $\mathbb{Z}_2$ quotients and replaces them by those of $SU(3)$ and its centre $\mathbb{Z}_3$, the C_3 of the horizon. The closure rule notes that the centres of $SU(2)$ and $SU(3)$ are the horizon's two closure symmetries, the $\mathbb{Z}_2$ of the deficit π and the $\mathbb{Z}_3$ of C_3, and sets $\alpha_3 = \alpha_2$ at the defect scale $f_\delta = M_P/\sqrt{8}$, where both close.

The measured couplings at M_Z, run at two loops in the Standard Model with $\alpha_s(M_Z) = 0.1180 \pm 0.0009$, decide both. The volume rule, $1/\alpha_s = \mathrm{vol}(SU(3)) + \mathrm{vol}(SU(3)/\mathbb{Z}_3) = \frac{4}{3}\sqrt{3}\,\pi^5 = 706.7$, holds only near 10^{274} GeV. The closure rule predicts $\alpha_s(M_Z) = 0.191$ at f_δ and 0.219 at M_P, 81σ and 112σ high, because the measured α_2 and α_3 cross at 2.9×10^{16} GeV, a factor 148 below f_δ. Both rules are excluded. The crossing lies near $\alpha f_\delta = 3.15\times10^{16}$ GeV, where the closure rule would give $\alpha_s(M_Z) = 0.1188$, 0.9σ from the measurement; that scale was found after the crossing was known, and among thirty natural scales built from f_δ, M_P, α and π one lands in the 1σ window for 36% of random crossings. It is not evidence.

DERIVED The strong coupling is therefore an input of the present framework, as the electroweak scale is (§43D), and through (261) it carries the nucleon's position. DERIVED It is an input by Bridge theorem B4 (§44.6): it is in neither axiom, and with massless quarks it is not a pure number at all but the strong scale Λ_{QCD} written in units

of another scale, so a derivation needs that scale in Planck units—which (261) supplies only through K and the boundary datum. The strong coupling and N_∞ are therefore one datum only if K is derived; with K measured they are two.

50.10 The nucleon as a closed C_3

DERIVED The nucleon is the horizon's closed C_3 at the strong scale. Its quarks are as many as the order of C_3, as the anomaly argument requires, and they form a colour singlet because their $\mathbb{Z}_3$ phases, $e^{2\pi i/3}$ each, sum to zero, as the windings of the phase on the closed horizon do (Section 23); the smallest closed universe of 2+1 gravity is likewise three particles [118]. STANDARD The three flux tubes meet at a junction, at 120°, where their equal tensions cancel: the configuration lattice QCD finds for the static three-quark potential [124], and the force balance of the horizon's three defects.

DERIVED The closure also fixes the charge. The centres of $SU(3)$ and $SU(2)$, $\mathbb{Z}_3$ and $\mathbb{Z}_2$, are the horizon's two closure symmetries, and together they form the $\mathbb{Z}_6$, the largest quotient the Standard Model's charges allow, $(SU(3)\times SU(2)\times U(1))/\mathbb{Z}_6$ [125]. A closed C_3 cancels the $\mathbb{Z}_3$ part, so every colour singlet carries integer charge: the nucleon's $\frac{2}{3}+\frac{2}{3}-\frac{1}{3}=1$ is its closure.

DERIVED The correspondence fixes ratios and leaves one scale. At the horizon the closure fixes every ratio and leaves one scale, set by G and the boundary datum N (§44.6); at the nucleon it fixes the ratios of the closed C_3 and leaves one scale, the strong tension, which carries α_s through dimensional transmutation. The strong coupling is the nucleon's scale datum, as N is the horizon's.

The closed C_3 offers one ratio without a parameter: if the nucleon closes as the horizon does, its radius in its own Compton units is the full closure over the C_3 total, $4\pi/\pi = 4$. The measured charge radius [2] gives $r_p/\bar{\lambda}_p = 3.9977 \pm 0.0030$, 0.8σ from 4, and muonic hydrogen [128] gives 3.9983 ± 0.0019, 0.9σ. The near-equality $r_p \approx 4\bar{\lambda}_p$ was known before this reading, and among the 288 simple numbers between 2 and 8 one lies within 1σ of a random ratio 27% of the time: the radius is consistent with the closure and is not evidence for it.

51 Status of the Framework

51.1 Status

Nothing in this section is left to calculate. Derived: the two-dimensional law and its reduction length; the Tully–Fisher law and its zero point (-0.7σ); the binding condition; the Solar-System quadrupole (-0.9σ, and 16σ against the total-field alternative); the dust of (51) and its zero sound speed; the field's magnetic and chiral currents; the birefringence formula. Identified: the deficit-π cone of each system (HYPOTHESIS), the dust as the budget's Ω_c, and the two-dimensional field as the cone's section. Decided by observation: wide binaries, Newtonian here; a confirmed low-acceleration boost in them falsifies the reading. Derived: the dark sector as the spin's quanta in two states, its equation of state $w = -1 + \Theta$, and the dilation $\mathcal{S} = 1/\cos_n\theta$ as what lensing infers as dark matter (§53); the dilation on the unbound web, which would set the expansion measured inside the local void, is a hypothesis (§53.9).

51.2 The identifications that remain

Beyond the axioms, every other step of the paper is derived. It rests on the correspondences below; for each, what a derivation would need is stated. Release 1.14.0 worked all seven; each item ends with its status.

- I5: that each dark sector's density is the power per mode of its harmonic—the assignment of the sectors to the axial quadrupole and the triangular octupole is derived (§25); a dynamical link between power and density is needed; the CMB spectra disfavour 5/2 at 4.6σ, short of the 5σ removal threshold, and prefer 2.61 ± 0.02 (§25.9); with the measured ratio the dark ratio is boundary data, the amount of the rotation's dust being an integration constant, and the budget lands on Planck's densities (§25.10); the resonance of the present (§36.7) gives $\Omega_\Lambda = 27/4\pi^2$ and the dark ratio 2.580 (2.568 without the neutrinos); in the rotation reading the dust's amount is an integration constant of the constraint (§23.1), which no field equation fixes. *Status: withdrawn.* The integration constant decides when the resonance falls, not the budget there: with the rotation at the spin (§36.14) and the present at the resonance (H3), $\Omega_\Lambda/\Omega_c = 2.580$ follows with no link between power and density, 1.3σ from the CMB's value, and the budget lies within 1σ of Planck's densities (§25.11). 5/2 is kept in §52.6.
- The form $\Omega_b/\Omega_c = \alpha_{\mathrm{TRSD}}\, n$ and the reset's asymmetry—need the branching of the reset (§25.8). *Status:* the exponent is read at the horizon's load, a pure number because the acceleration scale tracks the horizon

(§36.15), and the ratio passes the CMB spectra ($+0.3\sigma$); the amount of the asymmetry is fixed by the budget, $\eta = 6.16 \times 10^{-10}$ at the resonance budget, $+0.7\sigma$ in ω_b and $+0.4\sigma$ from deuterium burning, and Sakharov's conditions hold without a new coupling (§25.8). Identified: TRSD's product form. Open: the branching that realizes the amount, on which no prediction waits.

- I6: that the $\sqrt{m}$ vector obeys the seed condition in the cycle's Lorentzian form (§42), and $3\delta_\ell = Q$—need a dynamics of generation space that makes the mass vector null; the 2×10^{-7} correction is part of the target. *Status:* the null condition is the vacuum of the axiom's potential. With each generation's amplitude the rise of its generator's invariant when the pair is added, $Q = (\Xi + 2)/3\Xi$, so $Q = \frac{2}{3}$ at $\Xi = 2$, the director circle of the generators, from which they subtend the seed's right angle; the charged leptons measure $\Xi_\ell = 2.000013 \pm 0.000031$ (§42.8). Identified: the dressed amplitude (138). Open: $3\delta_\ell = Q$ and its 2×10^{-7} correction.
- The fine-structure sum, and its correction as α times the generator tension $f_{\rm gen}^2/F = \frac{1}{24}$ dressed thermally (§43A/C)—need a calculation that returns the electroweak volume sum as an inverse coupling, with its first correction. *Status:* the thermal dressing is the Boltzmann weight of one quantum of the spin, the quantum the rotation carries, so both factors of the correction are quantities of the framework (§43F). Identified: the volume sum as the inverse coupling at zero momentum. Open: the calculation that returns it; the weight 1 against $3/\pi$ is decided by 2.6×10^{-8} in $1/\alpha$.
- First starlight (§48.8)—a hypothesis, tested by the register. *Status:* with the present selected by the resonance, first starlight no longer selects it; it relates two physical epochs, $t_\star = \alpha\, t_{\rm res}$, at 100.6 Myr, $z = 29.6$, 46.4 MHz, and needs no observer. It is not derived: the onset of cosmic dawn depends on the primordial amplitude and the first stars, which the framework does not compute. No starlight before it is known; the claimed $z \simeq 32$ galaxy is a Y dwarf (register row 76). Counted in cycles, the onset is a ratio of received to emitted cycles, (221), and α cancels against the birefringence, $H_\infty t_\star = (\pi/3)\beta$: the measured twist places the onset at $42.7^{+5.6}_{-6.0}$ MHz with no clock and no Hubble constant (register row 78).
- The strong coupling α_s: with it the nucleon's position above the footprint balance follows, $K = 19.759 \pm 0.006$ (§50.8), and with the selected epoch the boundary datum N_∞; no geometric factor supplies it ($2\pi^2$ is a post-hoc target, consistent with the CMB's H_0), and the volume and closure rules for α_s are excluded (§50.9). It is the nucleon's scale datum, as N is the horizon's (§50.10). *Status:* an input by Bridge theorem B4, which now lists it (§44.6): it is in neither axiom, and with massless quarks it is the strong scale written in units of another scale. At the resonance budget $K = 19.759 \pm 0.006$, and $2\pi^2$ lies 3.3σ below with the framework's own H_0 (0.4σ with Planck's): the post-hoc target is disfavoured.
- The spinning-arc principle (§36.9), $\omega R = c$ with a fixed spin: the founding picture of the framework, stated as a hypothesis; from it the rotation's rate H_∞ follows, in place of the equilibrium premise of §36.8. For the final horizon it is now derived (§36.10–36.12): the arcs, the rate, the light-speed rim, the constancy, the charge and the quantum of the spin. What remains is I8, that the internal phase turns with the spin, $X_0 = \Lambda/3$—a pure number k, tested by the birefringence ($k = 0.88 \pm 0.18$) and, through the resonance H3, by Ω_Λ ($k = 0.9994 \pm 0.0053$) (§36.13). *Status: derived.* Each quantum of the rotation carries Killing energy $\hbar\Omega_0$ across the final horizon, whose entropy steps by 2π, so $\Omega_0 = jH_\infty$ with j an integer, and the two measurements fix $j = 1$: the rotation's quantum is one quantum of the spin (§36.14). The galactic scale tracks the horizon, not the spin, and the high-redshift rotation curves agree that it evolves (§36.15).

The paper assumes, beyond its primitives: the identifications I1–I4, I6 in the form (138), and I7; the product form of the baryon ratio and the fine-structure identification; and the hypotheses H1–H4, of which H3 is the opening principle's statement of the present and carries the dark ratio, the closed-form birefringence and the phase turned since the beginning, and H4 places the same present in Planck cells (§48.12), fixing the boundary datum N by α. Its inputs are those of B4 (§44.6). Two readings carry no prediction and are not assumptions of the paper: μ as a permittivity (§46.6), since the phantom density follows from the field equation itself, and the cone of every baryonic system (§49.6), since the two-dimensional law is the derived deep limit of §22. The dimensionful anchors and the cell count are not among them either: no dimensionful constant follows from dimensionless geometry, and N is the boundary datum that no internal condition fixes.

52 Superseded and Excluded Results

This section keeps in full the results that later work superseded and the mechanism it excluded, each with what replaced it; nothing in the sections above rests on them. The frozen-phase birefringence bound (formerly §46.13) holds the phase frozen, as the condensate reading does; the rotation reading's angle replaces it (§49.12–49.13). The global frame dragging (formerly §50.3, §50.4 and §50.6), first proposed as the carrier of the extra gravity, is excluded: on the closed horizon the windings of the phase sum to zero (Section 23). Its equations are kept

as proposed, with their units made consistent. The equal-frequency Lissajous count of the geometric origin, withdrawn in release 1.13.0, is kept in §52.5, and the defect power ratio as the dark ratio, withdrawn in release 1.14.0, in §52.6.

52.1 The birefringence angle

Reading note. This computation holds the phase frozen by a tilt-fixed mass, as the condensate reading does. In the rotation reading the phase turns at the rate fixed by the C_3 eigenvalue, and the angle is computed in §49.12–49.13: $\beta = 0.314°$.

DERIVED If the defect phase coupled to electromagnetism through an anomaly,

$$\mathcal{L} \supset \frac{\kappa\alpha}{4\pi}\,\theta\,F_{\mu\nu}\tilde{F}^{\mu\nu},$$

the plane of polarization of light from last scattering would rotate by

$$\beta = \frac{\kappa\alpha}{2\pi}\,\Delta\theta, \tag{262}$$

with $\Delta\theta$ the phase advance between last scattering and today. Three parts of this statement are not derived, and the observation is not settled. (i) The coupling: the condensate is exactly neutral (§31.1), so a $\theta F\tilde{F}$ term arises only if θ is a chiral phase of charged-fermion masses, which the action does not contain. (ii) Granted that, the coefficient follows from the charges of §31.4: rotating away a mass phase θ of a Dirac fermion of charge Q generates $(\alpha Q^2/4\pi)\,\theta F\tilde{F}$, so $\kappa = \sum_f N_c Q_f^2$ over the fermions carrying the phase, 3 for the charged leptons and 8 for every charged fermion. (iii) The phase advance had been taken as one C_3 step. (iv) The measurements—$0.342° \pm 0.094°$ (Planck + WMAP [27]), $0.215° \pm 0.074°$ (ACT DR6 [84]) and jointly $0.277° \pm 0.057°$ (ACT + Planck [85])—are not yet established as cosmological: the ACT team reports its angle as an unexplained instrumental discrepancy, and the joint analysis states that unresolved systematics must be understood first.

The phase advance, computed. DERIVED The phase is not free to advance a step: its mass is fixed by the tilt. With the carrier of §45 read as the phase near a $\mathbb{Z}_3$ maximum (IDENTIFICATION), $V(\theta) = \Lambda_3^4(1-\cos 3\theta)$ gives $m_\theta^2 = 9\Lambda_3^4/f_\delta^2 = |m^2|$, and §45.8 measures $|m^2|/H_\infty^2 = 0.039$–0.053 at the terminal rate $H_\infty = H_0\sqrt{\Omega_\Lambda}$ of the previous aeon, whose potential and budget are those of this one. Hence

$$\frac{m_\theta}{H_0} = \Big(\frac{|m^2|}{H_\infty^2}\,\Omega_\Lambda\Big)^{1/2} = 0.16\text{–}0.19, \qquad \frac{\Lambda_3^4}{\rho_\Lambda} = \frac{\pi}{27}\,\frac{|m^2|}{H_\infty^2} = 0.005. \tag{263}$$

A field this light is held by Hubble friction. Integrating $\ddot{\theta} + 3H\dot{\theta} + \frac{1}{3}m_\theta^2 \sin 3\theta = 0$ from $z = 1090$ to today in the budget's expansion history, from rest at every initial phase between a minimum and a maximum, the largest advance and the angles it allows are

$$|\Delta\theta|_{\max} = (1.2\text{–}1.6)\times 10^{-3}\ \text{rad}, \qquad |\beta| < 3.3\times 10^{-4}\,^\circ\ (\kappa = 3), \qquad |\beta| < 9\times 10^{-4}\,^\circ\ (\kappa = 8). \tag{264}$$

PREDICTION GRUT therefore predicts no measurable isotropic birefringence from the defect phase, $|\beta| < 10^{-3}\,^\circ$. If the measured angle is confirmed as cosmological it lies 4.9σ from this and needs a source outside the defect phase; if it is instrumental, the prediction stands. The phase holds at most $2\Lambda_3^4 \simeq 1\%$ of the dark-energy density and thaws only on a time of order $10^2/H_0$, so $w = -1$ is unaffected.

52.2 From modulation to orbit

A phase that winds at a modulated rate has a momentum density, and that momentum density sources the frame-dragging field. The chain is

$$\begin{aligned}\text{baryons} \;&\to\; \text{modulated phase frequency} \;\to\; \text{momentum density}\\ &\to\; \text{frame dragging} \;\to\; \text{orbital velocity.}\end{aligned} \tag{265}$$

The phase stress tensor is $T_{\mu\nu} = \chi_0^2\big[\partial_\mu\theta_{\rm phase}\partial_\nu\theta_{\rm phase} - \frac{1}{2}g_{\mu\nu}(\partial\theta_{\rm phase})^2\big]$ with $\chi_0^2 = c^4/4G$, and its momentum density is the mixed component, in orthonormal components with $\varrho = R_H\sin\chi$ the distance from the axis,

$$T_{\hat{t}\hat{\varphi}} = \frac{c^4}{4G}\,\frac{n\,[\omega_0 + \delta\omega_b(\chi)]}{c\varrho}. \tag{266}$$

It drives the dimensionless frame field $h \equiv h_{\hat{t}\hat{\varphi}}$ through $\nabla^2 h = -(16\pi G/c^4)\,T_{\hat{t}\hat{\varphi}}$, that is $\nabla^2 h = -4\pi n[\omega_0 + \delta\omega_b(\chi)]/c\varrho$, and the frame field acts on orbits through

$$\Omega_{\rm frame} = -\frac{c\,h}{\varrho}, \qquad a_{\rm drag} = 2\Omega_{\rm frame}v = -\frac{2c\,h}{\varrho}\,v, \qquad \frac{v^2}{r} = g_b + a_{\rm drag}. \tag{267}$$

The gravity was already there, from the global winding. The baryons tune it: they make it locally coherent in a way that produces flat rotation curves.

52.3 The compact space is essential

The geometry is closed, so whatever sources the frame dragging must have a compensating background: $\int_{S^3} \nabla^2 h \, dV = 0$. With $\Lambda_{\rm loc} = 4\pi n[\omega_0 + \delta\omega_b(\chi)]/c\varrho$, the source is

$$\Lambda_{\rm eff} = \Lambda_{\rm loc} - \langle \Lambda_{\rm loc} \rangle_{S^3}, \qquad \int_{S^3} \Lambda_{\rm eff} \, dV = 0. \tag{268}$$

The background is not arbitrary: it is the average of the modulation over the closed space,

$$\rho_{\rm bg} = \frac{\int_0^\pi \sin^2\chi \, \Lambda_{\rm loc}(\chi) \, d\chi}{\int_0^\pi \sin^2\chi \, d\chi} = \frac{8n}{cR_H} \int_0^\pi \sin\chi \, [\omega_0 + \delta\omega_b(\chi)] \, d\chi. \tag{269}$$

This is what makes the equation solvable; without it there is no solution at all. The solution is

$$h(\chi) = \int_{S^3} G_{S^3}(\chi, \chi') \, \Lambda_{\rm eff}(\chi') \, dV', \qquad \nabla^2 G_{S^3} = -\delta_{S^3} + \frac{1}{V}, \qquad h = h_{\rm loc} - \langle h_{\rm loc} \rangle, \tag{270}$$

the additive constant being fixed by the chosen frame.

52.4 The completed system

$$\begin{aligned}
&ds^2 = -c^2 dt^2 + R_H^2 \, [d\chi^2 + \sin^2\chi \, d\Omega_2^2], \\
&\theta_{\rm phase} = [\omega_0 + \delta\omega_b(\rho_b)] \, t + n\varphi, \qquad \delta\omega_b = \Gamma_b(\rho_b) \left[\frac{v_{\rm flat}(\rho_b)}{R_H \sin\chi} - \omega_0 \right], \\
&T_{\hat{t}\hat{\varphi}} = \frac{c^4}{4G} \, \frac{n\,\omega(\chi)}{c\varrho}, \qquad \nabla^2 h = -\frac{16\pi G}{c^4} T_{\hat{t}\hat{\varphi}} + \rho_{\rm bg}, \qquad \int_{S^3} \Big(- \frac{16\pi G}{c^4} T_{\hat{t}\hat{\varphi}} + \rho_{\rm bg} \Big) dV = 0, \\
&\Omega_{\rm frame} = -\frac{c\,h}{\varrho}, \qquad a_{\rm drag} = -\frac{2c\,h}{\varrho} \, v, \qquad v(\chi) = \varrho\,\omega(\chi), \quad \varrho = R_H \sin\chi.
\end{aligned} \tag{271}$$

52.5 The equal-frequency Lissajous count

The geometric origin of release 1.11.0 stated: "The threefold count is the Lissajous figure of the two counter-rotating arcs: two equal-frequency perpendicular motions a quarter turn apart trace on the sphere a closed figure with three nodes at $2\pi/3$, where the counter-rotation leaves no net tangential motion and the curvature concentrates." DERIVED They do not. In the plane the two motions, $\cos\omega t$ and $\sin\omega t$, trace a circle, with no node. On the sphere, two semicircles turning at one rate in opposite senses about perpendicular axes cross, if started together, on a great circle, $(\cos u, \sin u, \cos u)/\sqrt{1+\cos^2 u}$, and, if started a quarter turn apart, on an arc; neither has a node. A point carried by both turns traces a figure-eight with at most one node. At equal rates no figure has three nodes; three require a relative rate of three, which the fold supplies against the conjugate arc (§1.11). The claim is withdrawn, as the companion paper had withdrawn the same chain [63].

52.6 The defect power ratio as the dark ratio

Releases up to 1.13.0 identified the two dark densities with the power per mode of their harmonics in the three-defect pattern (I5, §25.1–25.3, kept there in full): $\Omega_\Lambda/\Omega_c = C_3/C_2 = 5/2$, from $C_\ell \propto 1 + 2P_\ell(-\frac{1}{2})$ (§39.3). The budget built on it gave $\Omega_m = 0.3219$ and $H_0 = 67.12$. MEASURED The CMB spectra disfavour it at 4.6σ (§25.9): it set the two material variances of that budget, the physical baryon density ($+2.5\sigma$) and DESI's matter density ($+2.8\sigma$). DERIVED It is withdrawn because it is not needed: the dynamics leaves the dark amount an integration constant (§23.1), and the resonance fixes the ratio at the present, 2.580 (§25.11). The assignment of the sectors to the axial quadrupole and the triangular octupole stands; the power ratio 5/2 remains the single-triple limit of the defect pattern on the sky, where the selection rule, not the ratio, is the test (§39.2–39.6).

52.7 The cross assignment of mergers and the returning-merger dark core

Two readings of mergers proposed for release 1.19.0 are not adopted. *The cross assignment:* that the lensing mass at a galaxy peak is its stars plus the *other* cluster's condensate. It contradicts §53.6, since the excited

geometry and the galaxies share geodesics, and the synthetic Bullet-like maps exclude it at $T = 99.1$ against the own assignment (§47.9); the true cross term is the superposition tail of §47.9. *The returning-merger dark core as a prediction:* that after a first passage the excess returns to the gas, as in Abell 520. It contradicts the frozen field, $t_{\rm rel} \simeq 31$ Gyr against passages of 0.2–1 Gyr (equation (203)); the settled share that could follow the gas is $f(\Delta t) \leq 0.06$ after 2 Gyr. A galaxy-free peak on the gas is kept as the falsifier of register row 91.

53 The Dark Sector as One Geometry

Dark energy is spacetime, the geometry itself; what is called dark matter is the same spacetime in its excited state, bound to baryonic structure; and what lensing infers as dark matter is the dilation—the excess of the inferred mass over the baryonic mass, smallest at the centre, growing toward the edge, and varying from system to system. There is spacetime and there are baryons: dark matter is not a separate entry. Each statement is derived below from the capacity of the horizon (§48.10), the rotation held at its rate (§23.1) and the gravity law (§22.9).

53.1 Spacetime, unexcited and excited

DERIVED The spin's quantum is $\hbar\omega$ with $\omega = H_\infty$ (§36.11). The final horizon holds $L_\infty/\hbar = N_\infty/4\pi$ of them, and their energy is the final horizon's own, $(N_\infty/4\pi)\,\hbar H_\infty = c^5/2GH_\infty$, the Hubble mass of the final horizon (§36.11, eq. (101)). Spread over the final horizon's volume $\frac{4\pi}{3}R_\infty^3$, this is exactly the vacuum:

$$n_*\,\hbar H_\infty = \rho_\Lambda c^2, \qquad n_* = \frac{N_\infty/4\pi}{\frac{4\pi}{3}R_\infty^3} = \frac{3}{8\pi\,\ell_P^2\,R_\infty}, \tag{272}$$

three quanta per area quantum $8\pi\ell_P^2$ (§36.11) per final-horizon radius. Dark energy is spacetime *unexcited*: the spin's quanta held by the horizon, uniform, constant by the Bianchi identity (§36.10), $w = -1$. The rotation carries quanta of the same kind—each quantum of the rotation is one quantum of the spin (§36.14)—and that is spacetime *excited*: dust of number density $n \propto a^{-3}$, conserved by the rotation's Noether current (§23.1), with $w = 0$ and zero sound speed, which clusters and binds to baryonic structure. Both sit in one stress tensor,

$$\boxed{T^{\rm dark}_{\mu\nu} = \hbar H_\infty\big(n\,u_\mu u_\nu - n_*\,g_{\mu\nu}\big) = 2X_0\big(\lambda\,u_\mu u_\nu - \lambda_*\,g_{\mu\nu}\big), \qquad \lambda_* = \frac{3c^4}{16\pi G},} \tag{273}$$

with $\lambda = n\hbar c^2/2H_\infty$ the multiplier of (51), $u_\mu = \partial_\mu\theta/\sqrt{X_0}$ and $X_0 = \Lambda/3$ (I8, derived in §36.14). One scale, the spin $X_0 = \Lambda/3$; one kind of quantum, $\hbar H_\infty$; two states, unexcited and excited. The vacuum's tension $\lambda_* = (3/16\pi)\,c^4/G$ is a fixed fraction of the force $F = c^4/G$ of §14, 7.22×10^{42} N; the rotation's is 2.80×10^{42} N today, in the ratio 1/2.580 of the resonance budget (§25.11). The inventory is therefore spacetime—0.684 of the critical density unexcited and 0.265 excited, 0.949 together—and the baryons, 0.050, with the neutrinos and the radiation.

53.2 The equation of state of the whole

DERIVED From (273) the dark sector's equation of state is

$$\boxed{w_{\rm dark} = -1 + \Theta, \qquad \Theta = \frac{n}{n+n_*} = \frac{\lambda}{\lambda+\lambda_*} = \frac{\rho_\chi}{\rho_\chi+\rho_\Lambda},} \tag{274}$$

the excited fraction of spacetime's quanta. This is the proposal's $w = -1+\Theta$, with Θ derived rather than posited. Over cosmic time $n/n_* = (\Omega_c/\Omega_\Lambda)\,a^{-3}$, so

$$\Theta(a) = \frac{1}{1+(a/a_{\rm eq})^3}, \qquad a_{\rm eq} = \Big(\frac{\Omega_c}{\Omega_\Lambda}\Big)^{1/3}: \tag{275}$$

a step in $\ln a$ from 1 to 0, centred on dark-sector equality. At the resonance budget, $\Omega_\Lambda/\Omega_c = 2.580$, equality falls at $z = 0.372$, $\Theta = 0.9$ at $z = 1.85$, the dark sector alone stops decelerating ($w_{\rm dark} = -\frac{1}{3}$) at $z = 0.728$, and today $\Theta_0 = 0.279$, $w_{\rm dark,0} = -0.721$.

Locally $\Theta(\mathbf{x}) = n(\mathbf{x})/[n(\mathbf{x})+n_*]$. Inside haloes, where the bound density is 10^2–10^5 times ρ_Λ, $\Theta \simeq 1$ and the dark sector is dust; in voids at $\delta = -0.8$, $\Theta = 0.07$ and $w = -0.93$. The threshold is the vacuum's own: $\Theta = \frac{1}{2}$ where the excited quanta equal the unexcited ones, $\rho_\chi = \rho_\Lambda$. Where the bound share follows the baryons at the cosmic

ratio, that is a baryon density $\rho_b = (\Omega_b/\Omega_c)\,\rho_\Lambda$, $n_b = 0.65$ m^{-3}, 2.58 times the mean: the proposal's threshold in ρ_b, located. Binding is a separate surface: a region is bound inside the zero-acceleration radius, where the mean density is $2\rho_\Lambda$, the balance radius of §43B.3 (register row 18), at matter overdensity $2\Omega_\Lambda/\Omega_m = 4.33$; there $\Theta \geq 0.63$.

53.3 Excited once, gathered since

DERIVED The excitation is not made where baryons are dense. The two states are separately conserved: n_* by the Bianchi identity and the capacity relation (§36.10, §48.10), n by the rotation's Noether current (§23.1); an exchange needs a term the action does not contain. Nor could the vacuum in a halo's own volume supply it: the mean density inside r_{200} is $292\,\rho_\Lambda$, so the vacuum there holds 0.34% of the halo's mass, and the dark density near the Sun, 0.4 GeV cm^{-3}, is $1.2 \times 10^5\,\rho_\Lambda$. The excitation was set once, at the beginning, as the integration constant of §23.1, and gravity has gathered it onto baryonic structure since.

DERIVED Because each state is separately conserved, the expansion is that of the resonance budget, $H^2 = H_0^2(\Omega_\Lambda + \Omega_m a^{-3} + \Omega_r a^{-4})$: the decomposition moves no distance, growth rate or parameter of Appendix D.2; what the dilation does to growth in the unbound web is §53.9. STANDARD Gravity measures only the total dark stress-energy [159], and a single Lagrange-multiplier field that unifies the two components was proposed in [160]; here the multiplier carries no potential, and the vacuum is the horizon's, tied to the dust by the one constant $X_0 = \Lambda/3$. MEASURED Fits that allow an exchange find at most a mild one: with DESI DR2 the simplest coupling is preferred at slightly more than 1σ, with no strong evidence for it [161]; a non-parametric reconstruction finds a coupling that reverses sign with time, at about 2σ at either end, and fits the data as well as evolving dark energy does [162]. The prediction of no exchange therefore stands or falls with the constant $w = -1$ of the vacuum.

53.4 The dilation

DERIVED Lensing and dynamics infer mass with the flat relation between field and source, Poisson's equation: a field g at radius r is read as the mass gr^2/G. The proposal's dilation is the ratio of the inferred mass to the baryonic mass, and the excess is what is called dark matter:

$$\mathcal{S}(r) \equiv \frac{M_{\rm inf}(<r)}{M_b(<r)} = \frac{g}{g_{\rm bar}}, \qquad M_{\rm inf} - M_b = (\mathcal{S}-1)\,M_b. \tag{276}$$

The geometry does not respond flatly. In a relaxed system its response is the gravity law, $\mu\, g = g_{\rm bar}$ exactly in spherical symmetry (§22.4), and in the form whose exponent runs with the acceleration (eq. (46)) the response is a cosine:

$$\boxed{\mathcal{S} = \frac{1}{\mu} = \frac{1}{\cos_n\theta}, \qquad \tan\theta = \frac{a_0}{g}, \qquad \cos_n\theta \equiv \left(1+\tan^n\theta\right)^{-1/n}, \qquad g^2 = g_{\rm bar}\left(g^n + a_0^n\right)^{1/n},} \tag{277}$$

exact at every acceleration, with $n = n(z)$ the dynamic exponent of §22.9. Take the local field g and the horizon's acceleration a_0 (§49.7) as the two components of one field, tilted by θ from the local direction. Projecting the measured field through the tilt gives the baryons' field, $g_{\rm bar} = g\cos_n\theta$; the flat inference takes no tilt and reads the whole measured field as the baryons', so the mass it infers is dilated by the secant. The last identity says the measured field is the geometric mean of the baryons' field and the whole field. In the Euclidean plane, $n = 2$, the cosine is the ordinary one, $\mathcal{S} = \sqrt{1 + a_0^2/g^2} = 1/\cos\theta$ exactly: the proposal's form, whose values 2, 3.86 and 11.5 at $\theta = 60°$, 75° and 85° are the dilation where the field has fallen to 0.58, 0.27 and $0.087\,a_0$. (The proposal's first table attached these angles to the heights $R/2$, $0.87R$ and $0.98R$ of a sliced sphere, where $1/\cos$ is 1.15, 2.03 and 5.03.) The cascade sets the exponent between 1.01 and 1.10 across galaxies (§22.9), close to the ℓ^1 plane, $\mathcal{S} \simeq 1 + a_0/g$; at those three tilts the dilation is 2.67, 4.67 and 12.4. A fixed exponent near the Euclidean one is too sharp at the transition: the rotation curves put $2\pi/3$ at $\Delta\chi^2/S^2 = 397$ (§22.6).

Smallest at the centre, growing toward the edge. Where $g \gg a_0$, $\theta \to 0$ and $\mathcal{S} \to 1$: Newton, with nothing inferred beyond the baryons. Where $g \ll a_0$, around an isolated mass whose field is $\sqrt{GM_b a_0}/r$,

$$\mathcal{S} \simeq \frac{a_0}{g} = \frac{r}{\ell}, \qquad \ell = \sqrt{\frac{GM_b}{a_0}}, \tag{278}$$

the radius in units of the coupling length of §49.7. Around $10^{11}M_\odot$ of baryons, $\ell = 11.0$ kpc and the dilation is 2 at 16 kpc, 4 at 39 kpc and 11 at 116 kpc—the range the proposal gave—and 91 at 1 Mpc. The boost of §22.9, 3.67, 1.58, 1.067 and 1.019 at $g_{\rm bar} = 0.1$, 1, 10 and $30\,a_0$, is the dilation.

The tilt is the horizon's. $a_0 = (\sqrt{3}/\pi^2)\,cH$ (§22.5), and at the present $(2/3\pi)\,c^2\sqrt{\Lambda/3}$ (§36.13). Without a horizon $a_0 = 0$, $\theta = 0$ and $\mathcal{S} = 1$ at every radius: baryons alone, nothing to call dark matter. The dilation is the effect of dark energy, the geometry, on the lensing of the baryons—the proposal's statement, derived.

It varies. Through each system's field, θ rising outward; through the exponent, $n(z)$; through capture, $\mathcal{S} = 1/[1-\Gamma(1-\mu)]$ (eq. (47)), which is 1 for the Solar System and wide binaries, which have captured none (§47.2); and through the relaxation state (§47.6): a relaxed galaxy carries the dilation of the threshold law, an unrelaxed cluster its gathered share (§53.6). MEASURED Weak lensing extends the radial acceleration relation by two decades of baryonic acceleration, in agreement with the MOND-type law, yet early- and late-type galaxies of the same stellar mass differ by at least 6σ [167], which no function of the acceleration alone can give; clusters depart from the galactic relation [163], one CLASH analysis finding the deep slope at a scale seventeen times higher [169]. PREDICTION The dilation differs between systems at the same acceleration as their capture and relaxation differ (register row 82).

What carries it. DERIVED In general relativity light bends only by energy and momentum, and spacetime's excitation carries them, as a gravitational wave does, without any matter. The dilation looks like mass and is no matter: in a relaxed system the excited geometry is distributed exactly as the phantom density of the gravity law (Theorem 11), $\rho_\chi = \rho_{\rm ph} = (4\pi G)^{-1}\nabla\cdot[(1-\mu)\nabla\Phi]$—the mass the flat inference sees and no matter supplies. Because the completion lenses by the dynamical potential (§46.4), the dilation of lensing and the dilation of the rotation curves are one.

How far it grows. OPEN If the dilation draws only on the excitation bound to a system, it stops where that is spent, at the reservoir radius (189): 59 kpc around $10^{11}M_\odot$ holding its cosmic share. MEASURED Weak lensing finds the circular velocities of isolated galaxies flat with no clear decline to 1 Mpc [168], a dilation near 90 there. Either an isolated galaxy binds 30–100 times its baryons in excited spacetime, or the dilation is not drawn from the bound store; the data so far favour a dilation that keeps growing (register row 28).

53.5 The dilation on the sky

DERIVED Lensing sees the dilation through a second one of the same form: the slant of each shell to the line of sight. For a spherical distribution

$$\boxed{\Sigma(b) = 2\int_b^\infty \rho(r)\, S_r(b)\, dr, \qquad S_r(b) = \frac{r}{\sqrt{r^2-b^2}} = \frac{1}{\cos\psi},} \tag{279}$$

with ψ the angle between the line of sight and the normal of the shell of radius r where it crosses: each crossing piles the shell's surface into a smaller patch of sky, 1 at the centre of the shell's disc and rising to the limb. For a thin shell of surface density σ this is the proposal's $\kappa = (\Sigma_{\rm surf}/\Sigma_{\rm crit})\, S$ for each crossing, and $\kappa(b) = 2\sigma S_R(b)/\Sigma_{\rm crit}$ for the two; a shell of thickness Δ caps the rise near $S \simeq \sqrt{2R/\Delta}$, the tangent chord $\sqrt{2R\Delta}$ over the thickness. Over the sky each crossing's mean is 2, so a closed shell's mean surface density is 4σ: Cauchy's ratio of a convex surface to its shadow, $4\pi R^2/\pi R^2$, which the proposal's first form took as the mean dilation; the deficit-π cone, half-apex 30°, has the uniform slant $1/\sin 30° = 2$ and, with its base, the ratio 3. Sliced in parallel planes a sphere's slant averages $\pi/2$ over the heights, and its zones are $4/\pi$ of their cylinders. The four is the framework's own: a ball is the union of cones of height r from its centre to its surface, $V_s = \frac{1}{3}rA_s$, and the cone of §3.12 stands on the shadow, $V_c = \frac{1}{3}r\,\pi r^2$, so the four-cone closure $V_s = 4V_c$, $M_{\rm H} = 4M_{\rm cone}$, is Cauchy's ratio. Exchanging the order of integration in (279),

$$\boxed{\int_0^\infty \Sigma(b)\, 2\pi b\, db = \int_0^\infty 4\pi r^2\rho(r)\, dr = M,} \tag{280}$$

since $\int_0^r 2\pi b\cdot 2S_r(b)\, db = 4\pi r^2$: the sky's slant redistributes toward the limbs and keeps the total, so the excess lensing measures is the field's dilation, carried onto the sky intact. In an aperture the cylinder takes in the outer shells, by $\pi/2$ for an isothermal profile (eq. (53)) and 1.3–1.9 for an NFW halo between $r_s/2$ and $4r_s$. Nor is the excess one number: it is 6.35 for the universe as a whole, $1+\Omega_c/\Omega_b$, reached at R_{200} in clusters, and growing without a fixed mean in galaxies.

At edges the sky's slant sharpens what the field has made. The excited geometry is geodesic and forms the caustics of cold matter; at the splashback radius, where the infalling streams turn round, its density drops sharply [165], and in projection the slant turns that edge into the steepening measured around clusters [166]. PREDICTION The splashback of cold matter, with no self-interaction to blur it (register row 81).

53.6 The Bullet Cluster

MEASURED The lensing peaks of 1E 0657−56 lie within 1σ of the galaxies' light and 8σ from the gas [158]. The gas is the dominant baryonic component in every aperture, 29–48 times the stellar mass at the gas peaks; at the subcluster the visible mass at the gas peak is 1.8 times that at the brightest galaxy, yet $\kappa = 0.20 \pm 0.05$ at the galaxy and 0.02 ± 0.06 at the gas. DERIVED A dilation computed from the baryons present would follow the gas.

The dilation follows the excited geometry that carries it, and that is collisionless: with $X = X_0$ constant its flow is geodesic, $u^\nu \nabla_\nu u_\mu = \nabla_\mu(\partial\theta \cdot \partial\theta)/2X_0 = 0$; it is pressureless (§23.1), and the constraint contains no interaction, so it scatters only gravitationally, $\sigma/m = 0$. A cluster, about four crossing times old, holds its infall configuration (§47.6), so a merger is the passage of two excited geometries through each other with their galaxies, while the gas is shocked and held back. The galaxies sit where the dilation is large and the gas where it is small, as the proposal had it. A system that holds its cosmic share is dilated by

$$\frac{M_{\rm tot}}{M_b} = 1 + \frac{\Omega_c}{\Omega_b} = 6.35, \qquad f_b = 0.158, \tag{281}$$

with the derived baryon ratio (§25.7). MEASURED In the X-COP clusters the baryon fraction reaches the cosmic value at R_{200} [163], and in the Bullet Cluster the dark-plus-galaxy convergence is about seven times the gas's [158]. PREDICTION Zero self-interaction: eleven double-radio-relic mergers give $\sigma/m < 0.63\ \mathrm{cm^2\,g^{-1}}$ (95%) [164]; zero offsets between the excess and the galaxies in mergers. DERIVED The collision is worked in §47.9: each excited geometry stays on its own galaxies and its mass is the cosmic share of its own pre-merger baryons, the gas it has since lost included; the Bullet Cluster separates GRUT and particle dark matter from a law computed on the baryons present, and does not separate them from each other. Register rows 79–80 and 89–91.

53.7 The dilation field

DERIVED The proposal's second form writes the dilation as a field, $S = e^{-\psi}$, with $S \simeq 1 - 2\Phi/c^2$ in the weak field. That is the refractive index of the geometry. With $\Phi = \Psi$ (§46.4) light obeys Fermat's principle in the optical metric $d\ell_{\rm opt} = S\,d\ell$: S dilates the measured path, delays the light by $\int (S-1)\,d\ell/c$ (the Shapiro delay) and bends it by $\nabla_\perp \int \ln S\,d\ell$. It is the conformal factor of the optical metric, not of spacetime, as the proposal says; what it records is spacetime's dilation along the light path: in the weak field time is dilated by $1 - \Phi/c^2$ and space by $1 - \Phi/c^2$, and light, crossing both, is slowed by their product, $S \simeq 1 - 2\Phi/c^2$—half from time and half from space, the two halves that double the bending of light over Newton's. Hence the names: S is the dilation field, and $\mathcal{S}$, its gradient against the baryons', the dilation, S for the scale a dilation changes. Its value differs from one by $2|\Phi|/c^2$: a few parts per million in galaxies and 10^{-5}–10^{-4} in clusters. Even the logarithmic potential of the deep regime adds only $2v_{\rm flat}^2/c^2 \approx 10^{-6}$ per e-fold of radius. Its gradient carries the field, $\nabla \ln S = 2\mathbf{g}/c^2$, and the mass lensing infers inside r is $c^2 r^2 |\partial_r \ln S|/2G$. So the dilation of §53.4 is a ratio of gradients,

$$\mathcal{S} = \frac{\partial_r \ln S}{\partial_r \ln S_b}, \tag{282}$$

where S_b is the index the baryons alone would make. The excess that lensing measures is the part of the path's dilation that the baryons do not make.

The field equation. DERIVED In these variables the gravity law of §22.4 reads

$$\boxed{\nabla \cdot \Big[\mu\Big(\frac{c^2|\nabla \ln S|}{2a_c}\Big)\,\nabla \ln S\Big] = -\frac{8\pi G}{c^2}\,\rho_b, \qquad R_{a_0}\,|\nabla \ln S| = \frac{g}{a_0}, \qquad R_{a_0} = \frac{c^2}{2a_0},} \tag{283}$$

with $\mu \to 1 - \Gamma(1-\mu)$ under capture (eq. (47)). $R_{a_0} = 3.91 \times 10^{26}$ m is the radius of the horizon whose surface gravity is a_0 (§49.7). Where the path's dilation changes by more than one e-fold per R_{a_0}, the geometry is stiff and the equation is Poisson's: the proposal's $(\nabla^2 - \mu_{\rm Y}^2)\ln S = -(8\pi G/c^2)\rho$ at $S \simeq 1$, with its screening scale written $\mu_{\rm Y}$. Where it changes more slowly, the geometry softens and dilates further. The proposal's self-coupling, the S on the right of its equation, therefore acts through the gradient. The value of $\ln S$ is a few parts per million throughout a galaxy and cannot switch anything, while its gradient in units of $1/R_{a_0}$ runs over decades.

Outside a source the proposal's solution, $S = \exp[(2GM_S/c^2)\,e^{-\mu_{\rm Y} r}/r]$, is exact for a point source; an extended source multiplies M_S by its Yukawa form factor, which is 1 for $\mu_{\rm Y} R \ll 1$. Inside, Gauss fixes the gradient, $\partial_r \ln S = -2GM_S(<r)/c^2 r^2$, while the value includes the shells outside r. So $\ln S = 2GM_S(<r)/c^2 r$ holds at the surface and beyond, not within. DERIVED The screening scale has one value available in the rotation reading, whose quanta carry no other scale (§36.14): the horizon's, $\mu_{\rm Y} = H_\infty/c = 1/R_\infty$. The dilation is cut off at $R_\infty = 5.4$ Gpc, and inside 1 Mpc the screening changes $\kappa = \Sigma_{\rm eff}/\Sigma_{\rm crit}$ by less than 10^{-7}. MEASURED Lensing profiles flat to 1 Mpc [168] rule out any screening length below the Mpc scale. PREDICTION The lensing excess has no exponential cut inside the horizon (register row 83).

The switch. DERIVED The proposal's coupling, $\alpha(\rho) = \alpha_0(\rho/\rho_c)^n/[1+(\rho/\rho_c)^n]$, turns the dilation on in dense systems and off in diffuse ones. The framework's switch is the capture fraction of §47.2. A system of radius R in a field of dispersion σ binds the fraction $\Gamma = \mathrm{erf}\,X - (2X/\sqrt{\pi})\,e^{-X^2}$ of the excited geometry streaming past it, with $X^2 = \rho/\rho_c$. As a function of its mean density this is a switch of the same kind,

$$\Gamma \simeq \frac{4}{3\sqrt{\pi}}\Big(\frac{\rho}{\rho_c}\Big)^{3/2} \ \ (\rho \ll \rho_c), \qquad \Gamma \to 1 \ \ (\rho \gg \rho_c), \qquad \rho_c = \frac{3\sigma^2}{4\pi G R^2}, \tag{284}$$

with $n = 3/2$ and $\alpha_0 = 1$ derived. ρ_c is not a constant of nature: it is the density at which the system's escape speed matches the speed of the geometry streaming past it. A galaxy forming in the cold cosmic field has $\rho_c \simeq 0$ and captures fully; the Sun, at its MOND radius in the Galaxy's field, has $\rho/\rho_c = 4.5 \times 10^{-6}$ and $\Gamma = 7 \times 10^{-9}$ (§47.2). The switch gates the excess, not the baryons. The lensing density is $\rho_b + \Gamma\rho_{\rm ph}$ (eq. (186)), so a system that has captured nothing lenses as its baryons, $S = S_b$, never as nothing.

The second form's numbers. DERIVED In its galaxy estimate the exponential is $1 + 2G\alpha_0 M/c^2 R = 1 + O(10^{-6})$. So $\langle \alpha S \rangle_{\rm gal} = \alpha_0$, and the whole excess would be α_0. A constant α_0 at high density is shared by the laboratory, the Sun and the stars, and is absorbed into the measured G. A coupling that rises with density puts the excess where densities are highest, in galaxy centres and the Solar System, and removes it where it is largest, in the outskirts and in low-surface-brightness galaxies [37, 62]. $\alpha \to 0$ would switch off the gas's own gravity, which general relativity does not allow. And a shock compresses gas rather than thinning it. What the second form's §7 asks of α—galaxies lensing, gas not—the collisionless excited geometry gives (§53.6).

53.8 Scales and the background

DERIVED The proposal's last part sets three conditions. Each scale is the coarse-grained output of the scales below it and the fixed input of those above. An intermediate scale B is fixed jointly by two outer scales $A \cong C$, as neither their product nor their union. And the background is the solution of its own consistency condition. The framework meets each.

The background's consistency condition is the closure law (254). Every energy inside the horizon is a defect on it, and summing them returns the horizon, $\sum_i r_{s,i} = r$: coarse-graining the contents gives the background, $\langle g^{(k-dk)} \rangle = g^{(k+dk)}|_{\rm bg}$. The background in turn sets, through a_0, the dilation of everything inside it (§53.4). It is above and below at once, with no external stage.

The rule that does not change with scale is the spinning-arc rule, $\omega R = c$. The final horizon, every black hole and every particle's sphere obey it (§36.9–36.12), so it is the fixed point of the coarse-graining. The ladder of §50 runs from the Hubble mass down to the Hubble quantum, and its two readings cross at the Planck energy, its fixed point (§50.4).

The intermediate scale needs both outer ones, as exact geometric means. In each, the two outer quantities are of one kind and the middle is their midpoint on the logarithmic scale k:

$$\boxed{\ell_P^2 = \tfrac{1}{2}\, r_s \lambda_C, \qquad \ell^2 = \frac{GM}{a_0} = r_s\, R_{a_0}, \qquad g^2 = g_{\rm bar}\, a_0 \ \ \text{(deep regime).}} \tag{285}$$

At the bottom, the Planck length lies between the gravitational and quantum lengths of every mass (§50.4). At the galactic scale, the coupling length lies between the system's own horizon, its Schwarzschild radius, and the horizon whose surface gravity is a_0: 7,200 AU for the Sun, between 3 km and 13 Gpc, and 8.5 kpc for the Milky Way's baryons. In the field, the measured acceleration lies between the baryons' field and the horizon's; at every acceleration this is $g^2 = g_{\rm bar}(g^n + a_0^n)^{1/n}$ (277). The dark-energy scale sits the same way between the Planck and Hubble energies (§50.4). OPEN A sheaf or renormalization-group formulation of the coarse-graining, which the proposal notes a rigorous treatment needs, is not supplied.

53.9 The dilation on the unbound web

DERIVED *The background cannot move.* A homogeneous universe has no gradient of the dilation field, $\nabla \ln S = 0$, so $\mathcal{S} = 1$: the Friedmann equation, the sound horizon and the acoustic angle are those of §25.11, and the CMB's $H_0 = 67.35 \pm 0.06$ stands. A tension between that number and one measured inside structure is another matter, because the dilation acts inside structure. The distance ladder measures the expansion of our own neighbourhood, $0.023 < z < 0.15$. Continuity of a spherical region of mean contrast δ, growing at the rate $f = d\ln\delta/d\ln a$, gives its edge the peculiar velocity $-\frac{1}{3} f H \delta R$, so the expansion rate measured inside it is

$$\frac{H_{\rm loc}}{H} = 1 - \frac{f\,\delta}{3}. \tag{286}$$

The local distance network's 73.50 ± 0.81 [79] needs $f|\delta| = 0.275 \pm 0.036$; SH0ES's 73.04 ± 1.04 [40] needs 0.254 ± 0.047. Undilated growth gives $f = 0.528$ today, which needs a region 52% underdense; whether the dilation can supply the rest is settled by the data at the end of this section.

DERIVED *The web is in the deep regime.* The rms peculiar field of a sphere of radius R is $\frac{1}{2}\Omega_m H_0^2 \sigma(R) R$: between 0.0010 and $0.0024\, a_0$ for every radius from 12 to 500 Mpc, and $0.0079\, a_0$ at a point, summed over the linear modes. Its baryons' share, $0.0012\, a_0$, the law dilates 28.9-fold: 4.5 times the cosmic dilation $1 + \Omega_c/\Omega_b$ that the

share already supplies, the most the web's growth can take. The field cannot say when that boost begins or where it stops. It is $0.0019\,a_0$ at $8\,h^{-1}$ Mpc and $0.0014\,a_0$ at 300 Mpc, so a dilation set by it alone boosts the one nearly as much as the other: a boost B of the growth equation on every scale raises S_8 to 0.89 for $B = 3.6$ from $z = 0.372$, 6.4σ above DES Y6 with both errors counted [174], to 1.00 for $B = 8.1$ from the same epoch and to 1.42 for $B = 5.6$ from $z = 1$. That form is excluded.

DERIVED *The dilation keeps a clock's causes.* A clock runs at $\Theta = N\sqrt{1 - v^2/c^2}$ with $N^2 = 1 - r_s/R - R^2/r_\Lambda^2$ (§43B.2–43B.3): it is slowed by the well it sits in, by its motion, and by the horizon. In a bound system the dilation is the horizon's tilt of the light path, set by what the system captures (§53.4, §53.7); on the unbound web it is the horizon's setting of the clocks, and each cause fixes one part of it.

When. The horizon sets the gradient of a region's clocks only beyond the region's balance surface, where its mean density is below $2\rho_\Lambda$ (§43B.3, §53.2); inside it, the region's own mass does. A system inside its balance surface carries the dilation of §53.4, set by what it captures; the unbound web takes its dilation from the horizon, and only where the horizon sets its clocks. For the universe as a whole that is $q = 0$, $z = 0.630$, the onset of acceleration; for a region 20% underdense today, $z = 0.713$; for one 80% overdense today, $z = 0.378$. At recombination the matter density was 3×10^8 times $2\rho_\Lambda$, and the baryons moved with the photon fluid at $0.45c$: off on every scale, and off through the growth of the first structures, which ran on the clocks of matter.

Where. A region holds the medium where its well slows time more than the medium's motion does: bound when $v^2 < 2|\Phi|$, the capture fraction Γ of eq. (284). The well is the potential, which carries the scale the field lacks: the rms $|\Phi|/c^2$ is 0.9×10^{-6} at 12 Mpc, 8.6×10^{-6} at 100 Mpc and 1.7×10^{-5} at 300 Mpc; per mode it is flat for wavelengths beyond the equality horizon, $1/k_{\rm eq} = 96$ Mpc, and falls inside it, where the radiation era erased the wells. Only peculiar motion counts, since comoving clocks keep cosmic time, and all of it counts: the flows the dilation itself drives, and the motion inside haloes, which hold 77% of the mass with a mass-weighted dispersion of 264 $\mathrm{km\,s^{-1}}$ on the framework's own spectrum. The law acts on the field the matter moves in, the sum of the modes, so with $\mu = 1/28.9$ the dilation exceeds the share where $\Gamma(1-\mu) > \Omega_c/\Omega_m$, $\Gamma > 0.873$. Dispersion and cut close on themselves: the dilated flows give 417 and the haloes 264 $\mathrm{km\,s^{-1}}$, $\sigma_{1D} = 494$ $\mathrm{km\,s^{-1}}$, and the dilation acts above 68 Mpc ($k < 0.025$ $\mathrm{Mpc^{-1}}$), with boosts 1.6, 3.4 and 4.2 at $k = 0.02$, 0.01 and 0.005 $\mathrm{Mpc^{-1}}$ and growth rates today 0.63, 0.95 and 1.06 against 0.53. No input remains. The previous form of this section cut at 29 to 80 Mpc according to which motions it counted; the medium's motion is all of them.

DERIVED *Light crosses the web undilated.* The horizon's part of the lapse, R^2/r_Λ^2, is de Sitter's, and de Sitter spacetime is conformally flat: the horizon dilates clocks and rulers by one factor. A dilation it sets on the web is therefore a conformal factor of spacetime, where the captured dilation of a bound system is the index of the optical metric (§53.7). The index is the ratio of space to time, so the common factor cancels in it:

$$\tilde{g}_{\mu\nu} = \Omega^2 g_{\mu\nu}: \qquad \frac{\tilde{g}_{ij}}{|\tilde{g}_{00}|} = \frac{g_{ij}}{|g_{00}|}, \qquad \frac{\tilde{\nu}_{\rm o}}{\tilde{\nu}_{\rm e}} = \frac{\Omega_{\rm e}}{\Omega_{\rm o}}\,\frac{\nu_{\rm o}}{\nu_{\rm e}}, \qquad \ddot{\mathbf{x}} = -\nabla\Phi - c^2\nabla\ln\Omega. \tag{287}$$

Light's paths are unchanged, a photon picks up the factor only at its two ends, and a massive body, whose clock it slows, feels the dilated pull, $c^2\nabla\ln\Omega = (B-1)\nabla\Phi$ on the held modes. Light sees only the matter the dilation has moved. Had it crossed the web dilated, the late integrated Sachs–Wolfe signal would rise 4,200-fold at $\ell = 10$ and 3,000-fold at $\ell = 20$, its correlation with galaxies reversing sign: the CMB excludes that outright. As derived, the large-scale potentials, fed by growth that now keeps pace with the expansion, decay less, and the late signal falls to 0.28, 0.41, 0.54 and 0.71 of the background's at $\ell = 2$, 4, 10 and 20, taking 222, 97 and 25 $\mu\mathrm{K}^2$ from the power at $\ell = 2$, 4 and 10, in the sense of the low large-angle power the CMB shows [39]. Its correlation with galaxies keeps 0.4–1.0 of the background's at $\ell = 4$–40 for tracers near $z = 1$, which carry most of the 4σ detection [179]; for tracers at $z = 0.3$–0.5 it falls to 0–0.6 at $\ell = 10$–20 and reverses at $\ell = 4$, and from $\ell = 40$ it is unchanged.

MEASURED *What the data say.* The added growth lives above 68 Mpc and below $z = 0.63$. σ_8 rises by 0.16% today and by 0.02–0.08% at $z = 0.2$–0.4, the epochs of clusters and lensing; cosmic shear changes by 0.3% at $\ell = 30$ and not at all from $\ell = 100$. DES Y6, 0.789 ± 0.012 [174], therefore stays 2.8σ below, the difference between that survey and the CMB that the dilation did not make, and KiDS-Legacy, $0.815^{+0.016}_{-0.021}$ [175], 0.9σ. CMB lensing rises by 3.5% at $L = 10$ and 0.8% at $L = 20$ and is unchanged from $L = 40$: the amplitude GRUT predicts at its own neutrino mass, $S_8^{\rm CMBL} = \sigma_8(\Omega_m/0.3)^{1/4} = 0.822$, is the $0.825^{+0.015}_{-0.013}$ that ACT, SPT and Planck measure together [178]. No lensing excess is left for a negative neutrino mass to absorb [172]: $\sum m_\nu = 59.34$ meV lies inside ACT DR6's 82 meV [73] and DESI DR2's 64 meV with the CMB [173] (95%), and the floor's remaining pull rests on DESI's BAO matter density, 0.2975 ± 0.0086 against 0.316 (2.2σ), which the dilation does not touch. The flows are where the dilation shows. Cosmicflows-4 bulk flows of 387 ± 28 and 419 ± 36 $\mathrm{km\,s^{-1}}$ within 150 and $200\,h^{-1}$ Mpc, which ΛCDM expects at 139 and 120 and allows with probabilities below 0.03% and 0.003% [176], fall at the 78th and 90th percentiles of the dilated web's distribution (rms 321 and 288 $\mathrm{km\,s^{-1}}$); the 628 ± 82 $\mathrm{km\,s^{-1}}$ dipole of the 120–$160\,h^{-1}$ Mpc shell, 3.4σ in ΛCDM [177], at its 98th.

PREDICTION *The Hubble rate.* The framework has one expansion rate, the background's 67.35, and the dilation leaves

it, and the sound horizon behind it, alone. Nor can the dilation make the ladder read more. A clock or a ruler is shifted by $|\Phi|/c^2 \lesssim 10^{-5}$, so a 9% excess is not a dilation of clocks; and a local void cannot carry it either. The K-band luminosity density is $(3.09 \pm 0.19) \times 10^8\, h_{70} L_\odot\,\mathrm{Mpc}^{-3}$ in the samples below $z = 0.07$ and 5.40 ± 0.27 beyond $z = 0.09$ [170], a ratio 1.75 ± 0.14; read as a matter deficit and inverted with the framework's own outflow, which lowers the apparent density by $H_{\rm loc}/H$, it is a void 37% deep today (25% when its clocks were set, at $z = 0.79$) whose interior would expand at 74.2 ± 0.6 (74.6 with the full-sky 2M++ sample alone, 72.9 for the authors' summary ratio of 1.5, 72.2 undilated). The supernovae decide it. The deficit ends by $z \approx 0.09$, the ladder's supernovae run to $z = 0.15$, and beyond the edge the rate would fall back by 8.3%; the supernova Hubble diagram shows no change of H_0 across $0.023 < z < 0.15$ to 0.60% [171], which bounds the deficit below 27% at 5σ with undilated growth and near half that with the dilated. What the dilation does change is the scatter of the local rate: its flows give a 300 Mpc sphere an rms offset of 0.88%, against 0.41% in ΛCDM. The prediction is therefore that distance ladders calibrated alike converge on 67.3 within that scatter. The 2026 network lies 6.1σ away (7.6σ bare), SH0ES 4.8σ, the CCHP red-giant and carbon-star ladder [80] 1.5σ, and the lens time delays, $71.6^{+3.9}_{-3.3}$ (TDCOSMO, as quoted in [79]), 1.3σ. The ladder calibrates the supernovae against Cepheids and red giants and the CMB against the sound horizon; JWST distances to the supernova hosts settled at the network's value at 5σ would falsify the framework's background (register row 84).

HYPOTHESIS *Status.* GRUT as built gates the excess with the capture of a store, which is zero in the unbound web, and the web grows undilated. The proposal's reading—the dilation as the geometry rather than a store, the branch the lensing of isolated galaxies already favours (§53.4, register row 28)—lets the web take it from the horizon on a clock's terms: from the onset of acceleration, above 68 Mpc at the fixed point of its own flows, on clocks and not on light (287). So read, it keeps S_8, cosmic shear and CMB lensing at the background's values, lowers the late Sachs–Wolfe signal, and carries the Cosmicflows-4 bulk flows that ΛCDM does not; it leaves the ladder's H_0 at the background's 67.35, on which the framework predicts the ladders converge (register rows 84–85). Unaffected either way: the TM1 solar angle and $a_0(z)$.

53.10 What this changes

DERIVED Derived:

- dark energy as spacetime unexcited and the cosmic share as spacetime excited, the spin's quanta in two states (273), with no separate dark matter;
- the equation of state $w = -1+\Theta$, the vacuum's own density as the threshold, and equality at $z = 0.372$ (274);
- the dilation $\mathcal{S} = 1/\cos_n\theta$, $\tan\theta = a_0/g$ (277), 1 at the centre and r/ℓ toward the edge (278), set by the horizon and varying with capture and relaxation, its excess carried by the excited geometry as the phantom density;
- the dilation field as the refractive index of the optical metric, whose gradient, measured against $1/R_{a_0}$, carries the law (283);
- the proposal's switch as the capture fraction (284);
- the sky's slant (279), which keeps the total (280);
- the Bullet Cluster's offset from the collisionless excitation;
- the background as its own consistency condition, the closure law, with the intermediate scales as geometric means of the outer ones (285);
- the background unmoved by the dilation, and in the proposal's reading the web dilated on a clock's terms—from the onset of acceleration, above 68 Mpc at the fixed point of its own flows, on clocks and not on light (287); the exclusion by S_8 of a dilation set on the web by the field alone, and by the integrated Sachs–Wolfe effect of one that light would see.

OPEN How far the dilation grows is open (register row 28). Unchanged: the background expansion, which has no gradient and so no local dilation, and with it the CMB's H_0, the TM1 solar angle and the acceleration scale at $z \simeq 1$. Changed in the proposal's reading (§53.9): the growth of the web above 68 Mpc after $z = 0.63$, which carries the Cosmicflows-4 bulk flows and lowers the late Sachs–Wolfe signal, with S_8, cosmic shear and CMB lensing at the background's values, so that the neutrino-mass floor's pull rests on the BAO matter density alone. Not changed: the distance-ladder H_0, since the supernova Hubble diagram rules out a local void as the ladder's rate (register rows 84–85). Corrected:

- the first form of this section, which held the tensions of the October 2026 confrontation unchanged: only the background is (§53.9);

- its second form, which read the ladder's H_0 as the rate inside a local void, 71.8 ± 0.8, against the supernova Hubble diagram, and cut the web's dilation at 29–80 Mpc according to the motions it counted;
- the heights of the proposal's first table, and its single mean dilation $\langle \mathcal{S} \rangle = 4$, which is Cauchy's ratio, not the measured excess;
- in the second form, the excess assigned to the value of the dilation field rather than its gradient;
- a coupling that rises with density rather than with capture, and multiplies the baryons rather than the excess;
- the enclosed-mass form of $\ln S$ inside a source.

Dark energy	spacetime, the geometry itself: unexcited, $T_{\mu\nu} = -\rho_\Lambda g_{\mu\nu}$, $w = -1$; excited and bound to baryonic structure, $T_{\mu\nu} = \rho_\chi u_\mu u_\nu$, $w = 0$
Dark matter	the dilation: what lensing infers beyond the baryons, carried by the excited geometry
Inventory	spacetime, $\Omega_\Lambda + \Omega_c = 0.949$, and baryons, $\Omega_b = 0.050$
Dilation	$\mathcal{S} = 1/\cos_n \theta$, $\tan\theta = a_0/g$: 1 at the centre, r/ℓ toward the edge, varying
Dilation field	$S = e^{-\psi} \simeq 1 - 2\Phi/c^2$, the index of the optical metric; $\mathcal{S} = \partial_r \ln S / \partial_r \ln S_b$
Dynamics	$\nabla \cdot [\mu(c^2 \lvert\nabla \ln S\rvert/2a_c)\nabla \ln S] = -(8\pi G/c^2)\rho_b$; screening at R_∞
Switch	the capture fraction Γ: $(\rho/\rho_c)^{3/2}$ below $\rho_c = 3\sigma^2/4\pi G R^2$, 1 above; it gates the excess
Lensing excess	$(\mathcal{S} - 1)M_b$, the phantom density; 6.35 for the universe as a whole
Scales	$\ell_P^2 = \frac{1}{2} r_s \lambda_C$, $\ell^2 = r_s R_{a_0}$, $g^2 = g_{\rm bar} a_0$: the middle scale is the geometric mean of the outer two
Background	the closure law, $\sum r_s = r$: the contents coarse-grained are the horizon that sets their dilation
Tensions	the background unmoved; the web dilated on clocks, not light, from $z = 0.63$ and above 68 Mpc: the measured bulk flows, with S_8, shear and CMB lensing at the background's values and a lower late Sachs–Wolfe signal; H_0 unmoved, a local void excluded by the supernovae

54 The Four Interactions from One Shape

The C/S shape of §1 has four features, and each carries one interaction. This section collects them in one place: the gauge group they form, the chain of scales that sets their strengths, the action they write together, and the content and running that action fixes.

54.1 Four features, four interactions

DERIVED Each feature of the shape carries one symmetry, and each symmetry one interaction:

feature of the shape	symmetry	interaction	strength
the cone $h = r$ and its deficit (§1.10, §18.1)	local Lorentz (§28), spin 2 (§41.3)	gravity	$G = c^4/F$ (§8); in Planck units of the present, H4
the generators' phase, $\mathbb{Z}_3 \subset U(1)$ made continuous (§31.1)	$U(1)$; $Q = T_3 + Y$ unbroken	electromagnetism	α at zero momentum, $S^1(2\pi) \times S^3(1)$ (§43F)
the conjugate pair, glued (§1.8, §42.2)	$S^3 \cong SU(2)$; chirality from the two lobes	weak	$g_2 = e/\sin\theta_W$; $v_{\rm EW}$ (§43D)
the three generators, C_3 (§4.1, §50.10)	$SU(3)$, centre $\mathbb{Z}_3$	strong	$K = m_p/m_*$ (§50.8)

The structure is derived in each row (§28, §31, §42.2, §50.10); the strengths are taken up in §54.3. The electric phase is the phase of the three generators, which the fermion generations carry (§42), so it acts on the fermions. The spin's own phase, θ in $\phi_\pm = \phi\, e^{\pm i\theta}$, is a different phase and a global one: held at its rate it is the rotation that carries the dark sector (§23.1, §53), and it carries no gauge field, since a charged condensate would give the photon a mass near 10^{18} GeV (§31.1, §41.7). The cone with its spin is spacetime, gravity and the dark sector together; the generators' phase, the pair and the triangle are the three gauge interactions.

54.2 One group

DERIVED The shape's two closure symmetries, the conjugate pair's $\mathbb{Z}_2$ (the antipodal identification of the fold, the -1 of $SU(2)$) and the beat's $\mathbb{Z}_3$ (§1.11, the centre of colour), generate the $\mathbb{Z}_6$ that the Standard Model's charges allow. The gauge group is the quotient

$$\boxed{G_{\mathrm{SM}} = \frac{SU(3)\times SU(2)\times U(1)}{\mathbb{Z}_6}, \qquad \frac{t}{3} + T + Y \in \mathbb{Z},} \tag{288}$$

the condition holding for every field of triality t, isospin T and hypercharge Y [125]: the quark doublet ($t=1$, $T=\frac{1}{2}$, $Y=\frac{1}{6}$), the singlets u ($\frac{2}{3}$) and d ($-\frac{1}{3}$), the leptons ($t=0$, $Y=-\frac{1}{2}$, -1) and the Higgs ($Y=\frac{1}{2}$) all satisfy it. With $Q = T_3 + Y$ it is charge quantization: every colour singlet has integer charge, and quarks carry $Q \equiv -t/3 \pmod 1$, $\frac{2}{3}$ and $-\frac{1}{3}$ (§31.4). The interactions are not three groups that happen to coexist but one quotient fixed by the shape, acting on the fields the cone carries.

54.3 One chain of scales

DERIVED Each strength is a ratio of scales on one ladder, from the Planck cell to the horizon:

$$\ell_{\mathrm{P}} \xrightarrow{\times\, 8\pi e^{1/\alpha+3\pi\alpha/2}\ (\mathrm{H4})} r \xrightarrow{(m_*/m_{\mathrm{P}})^3=\ell_{\mathrm{P}}/2r\ (\S50.8)} m_* \xrightarrow{\times K} m_p. \tag{289}$$

Gravity's coupling at the horizon and at the nucleon follow:

$$\boxed{\frac{G\hbar H_0^2}{c^5} = \frac{e^{-2/\alpha-3\pi\alpha}}{64\pi^2}, \qquad \frac{Gm_p^2}{\hbar c} = \frac{K^2}{(16\pi)^{2/3}}\, e^{-2/3\alpha-\pi\alpha} = 5.906\times 10^{-39}.} \tag{290}$$

The hierarchy of the forces is exponential in $1/\alpha$. Gravity is weak at the nucleon because the present is $e^{1/\alpha}$ Planck lengths across, and two protons repel electrically 1.24×10^{36} times more strongly than they attract gravitationally, $\alpha\hbar c/Gm_p^2 = \alpha\,(16\pi)^{2/3}K^{-2}e^{2/3\alpha+\pi\alpha}$, with K the one strong datum.

DERIVED Both rungs are transmutations from the Planck cell. The horizon is the length at which the electric coupling, counted from one cell, has run through $1/\alpha$ e-folds, $r = 8\pi\ell_{\mathrm{P}}\, e^{2\pi/b\alpha+3\pi\alpha/2}$ with $b = 2\pi$ (§48.12). The nucleon is the mass at which the strong coupling, run down from the cell with the coefficient $b_0 = 7$ that the shape's content fixes (§54.6), becomes strong,

$$m_p \propto m_{\mathrm{P}}\, e^{-2\pi/b_0\alpha_s(m_{\mathrm{P}})}, \tag{291}$$

up to the quark thresholds and the hadronic factor of lattice QCD. The measured $\alpha_s(M_Z) = 0.1180$ is $1/\alpha_s(m_{\mathrm{P}}) = 53.0\pm0.1$ at two loops, and the nucleon lies $e^{-44.0}$ below the Planck mass. Nature's two large numbers are therefore one law at the cell, an exponential of an inverse coupling: the size of the universe of $1/\alpha$, the lightness of matter of $1/\alpha_s$. The footprint balance (§50.8) sets the two against each other at the present, and K records where they meet; with H4,

$$K = \frac{m_p}{m_{\mathrm{P}}}\,(16\pi)^{1/3}\, e^{1/3\alpha+\pi\alpha/2} = 19.7566 \pm 0.0002, \tag{292}$$

the error being G's. Of the four strengths, two are fixed by the geometry, α (§43F, 1.2σ from CODATA) and G at the present (H4). The other two are each one number at the closure of their feature, taken today from measurement: the strong scale (K, equivalently $\alpha_s(M_Z) = 0.1180$, with $d\ln K/d\alpha_s = 59$, §50.8–50.9) and the electroweak scale with its angle (v_{EW}, at rung 54.0 of the base-4 ladder, §43D, and $\sin^2\theta_W$). Their closures are §54.7.

54.4 No unification scale

DERIVED The group (288) is the product of the shape's features, not a subgroup of a simple group, so there is no scale at which the three gauge couplings must meet and no gauge boson that turns quarks into leptons. STANDARD The couplings run as the Standard Model's, in which the three do not meet at one scale; the shape fixes them where each feature closes: α at zero momentum, the strong scale at the nucleon, gravity at the horizon. PREDICTION The proton is stable against gauge-mediated decay: $p \to e^+\pi^0$ at the rates grand unification requires is absent, and Super-Kamiokande sets $\tau/B > 2.4\times10^{34}$ yr [127]. Gauge-mediated proton decay, or a free fractionally charged colour singlet, falsifies (288) and this section (register rows 87–88).

54.5 One action

DERIVED The features write one action. On the tetrad of the cone, with the covariant derivative of the three gauge features,

$$D_\mu = \partial_\mu + \tfrac{1}{4}\,\omega_\mu{}^{bc}\gamma_{bc} - ig_3\,G^A_\mu t^A - ig_2\,W^i_\mu T^i - ig_Y\,Y B_\mu, \tag{293}$$

$$\begin{aligned} S = \int d^4x\,e\Big[& \frac{F}{16\pi}\,R - \frac{\delta F}{16\pi}\,g^{\mu\nu}\partial_\mu\phi_+\partial_\nu\phi_- - \varepsilon_V V(\Xi) + \lambda\,(X - X_0) \\ & - \tfrac{1}{4}\,G^A_{ab}G^{Aab} - \tfrac{1}{4}\,W^i_{ab}W^{iab} - \tfrac{1}{4}\,B_{ab}B^{ab} + \bar\psi\,i\gamma^a e_a{}^\mu D_\mu\psi \\ & - (D_\mu H)^\dagger D^\mu H - \lambda_H\big(H^\dagger H - \tfrac{1}{2}v_{\rm EW}^2\big)^2 - \big(\bar\psi_L\,y\,H\,\psi_R + \text{h.c.}\big)\Big]. \end{aligned} \tag{294}$$

Every term is the lowest-order invariant of one feature, and nothing else enters. The first line is the cone and its spin: gravity, with $c^4/16\pi G = F/16\pi$; the conjugate pair's condensate, with $V_0 = 0$ (§41.4); and the spin's phase held at its rate, $X = -g^{\mu\nu}\partial_\mu\theta\,\partial_\nu\theta$ with $X_0 = \Lambda/3$ (§23.1). Dark energy is the horizon's capacity, not a term (§48.10). The second line is the three gauge features with their group (288), and the fermions on the cone's double cover, summed over three generations for the three generators (§41.7, §42). The third is the pair as one Higgs doublet (§43D), and the masses, $M = y\,v_{\rm EW}/\sqrt{2}$ (with $\tilde H = i\sigma_2 H^*$ for the up-type quarks), whose square root the C_3 makes circulant and the seed cone puts on Koide's relation (§41.7, §42.1). The action is general relativity, the Standard Model with its exact global form, and the framework's dark sector, written as one. Its constants are the strengths of §54.3 and the masses and mixings of the fermion families, on which the cones of §42 impose the lepton relations.

54.6 One content, one running

DERIVED The shape fixes the content of (294). Per generator: a quark doublet and two quark singlets in the colour triplet of the triangle, a lepton doublet and a lepton singlet, the doublets from the pair and their handedness from its lobes, every hypercharge fixed by anomaly freedom (§31.4); three generators, three generations (§42.2); one Higgs doublet, the pair (§43D). The content fixes every coefficient of the running. At one loop, $d\alpha_i^{-1}/d\ln\mu = -b_i/2\pi$ with

$$b_3 = -11 + \tfrac{4}{3}n_g = -7, \qquad b_2 = -\tfrac{22}{3} + \tfrac{4}{3}n_g + \tfrac{1}{6} = -\tfrac{19}{6}, \qquad b_Y = \tfrac{20}{9}n_g + \tfrac{1}{6} = \tfrac{41}{6}, \tag{295}$$

from $N_c = 3$ colours, $n_g = 3$ generations and one Higgs doublet, each number from the shape; the two-loop coefficients follow from the same content. Each coupling therefore needs one value at one scale, and the shape names the scale: where its feature closes (§54.4).

DERIVED The signs are the shape's. The triangle's coupling grows toward low energy, $b_3 < 0$, and becomes of order one where the closed C_3 confines its quarks into colour singlets, the nucleon (§50.10). The strong interaction closes in strong coupling, so its natural datum is the scale of the closure, K, equivalent to one value of α_s anywhere on the running. Three generations keep the triangle asymptotically free with room to spare, $\frac{2}{3}n_f = 4 < 11$; it would remain so up to eight. The phase and the pair are one electroweak coupling, $e = g_2\sin\theta_W = g_Y\cos\theta_W$, so $\alpha_w \equiv g_2^2/4\pi = \alpha/\sin^2\theta_W$ at every scale, $1/29.58$ at M_Z [13]: the weak and electric strengths are one strength and one angle.

54.7 What is unified

DERIVED Unified: the four interactions as four features of one shape; one gauge group with the Standard Model's global form and charge quantization; one action, each term the lowest invariant of one feature; one content, which fixes every coefficient of the running; and one chain of scales from the Planck cell, on which the horizon and the nucleon are the transmutations of the electric and the strong coupling and gravity's weakness is $e^{-2/\alpha}$ at the horizon and $e^{-2/3\alpha}$ at the nucleon.

DERIVED Each strength is fixed where its feature closes: the cone's at the horizon, by N, which H4 ties to α (§48.12); the phase's at zero momentum, on the boundary $S^1(2\pi)\times S^3(1)$ (§43F); the triangle's at the nucleon, K (§50.8); the pair's at the electroweak scale, $v_{\rm EW}$ with θ_W and λ_H (§43D). OPEN The first two closures are computed. The other two are the same calculation on the other two features: the one-loop effective action that gives α on the boundary, taken on the closed C_3 for K, whose target (292) is known to one part in 10^5, and on the glued pair for the electroweak scale, its angle and the Higgs quartic. Each strength that calculation returns removes an input of B4 (§44.6).

55 Collision Lag, Geometric Recoil, and CHIRP

This section derives the collision–relaxation chain from the existing GRUT action, merger map, metric equations, and horizon definitions. The key distinction is between (i) the displacement of the excited component relative to collisional baryons and (ii) the expansion of a proper spatial volume. They are connected by the phase current and the metric, but they are not the same velocity. No external streaming-velocity amplitude, new force, or additional matter component is introduced.

1. Collision separates baryonic motion from phase motion

The existing constrained phase sector is

$$S_{\rm rot} = \int d^4x\sqrt{-g}\,\lambda(X - X_0), \qquad X = -g^{\mu\nu}\nabla_\mu\theta\nabla_\nu\theta = X_0. \tag{296}$$

Variation with respect to θ gives $\nabla_\mu(\lambda\nabla^\mu\theta) = 0$; variation with respect to the metric gives

$$T^{(\chi)}_{\mu\nu} = 2\lambda\nabla_\mu\theta\nabla_\nu\theta + g_{\mu\nu}\lambda(X - X_0) \xrightarrow{X=X_0} 2\lambda\nabla_\mu\theta\nabla_\nu\theta. \tag{297}$$

With $u_\mu = \nabla_\mu\theta/\sqrt{X_0}$, the fixed-norm condition gives $u^\mu u_\mu = -1$ and $u^\nu\nabla_\nu u_\mu = 0$. The phase component therefore follows geodesics and has no gas-shock drag term. In the non-relativistic limit its mass current obeys

$$\partial_t\rho_\chi + \nabla\cdot(\rho_\chi\mathbf{v}_\chi) = 0, \qquad (\partial_t + \mathbf{v}_\chi\cdot\nabla)\mathbf{v}_\chi = -\nabla\Phi. \tag{298}$$

The collisional baryonic gas instead obeys

$$(\partial_t + \mathbf{v}_b\cdot\nabla)\mathbf{v}_b = -\nabla\Phi - \frac{\nabla p_b}{\rho_b} + \mathbf{a}_{\rm coll}. \tag{299}$$

Writing $\mathbf{w} = \mathbf{v}_\chi - \mathbf{v}_b$ and subtracting while retaining the distinct convective derivatives gives

$$\boxed{(\partial_t + \mathbf{v}_\chi\cdot\nabla)\mathbf{w} = \frac{\nabla p_b}{\rho_b} - \mathbf{a}_{\rm coll} - (\mathbf{w}\cdot\nabla)\mathbf{v}_b.} \tag{300}$$

The common gravitational acceleration cancels at the same event. Thus the gas can lose bulk momentum in a shock while the phase component continues on its own geodesic; the previous phase density and its stress-energy are not algebraically erased by a change in the baryonic trajectory. The gas terms are those already present in its fluid solution, not fitted velocity inputs.

2. The existing relaxation law derives the component drift

At core passage, the framework resets the relaxation clock. With $t = \Delta t \geq 0$,

$$f(t) = 1 - e^{-t/t_{\rm rel}}, \qquad q(t) = 1 - f = e^{-t/t_{\rm rel}}, \qquad \dot f = \frac{q}{t_{\rm rel}}, \quad \dot q = -\frac{q}{t_{\rm rel}}. \tag{301}$$

The existing merger map is

$$\rho_\chi(\mathbf{x}, t) = q\rho_{\rm inf}(\mathbf{x}, t) + f\rho_{\rm pol}(\mathbf{x}, t). \tag{302}$$

For equal conserved masses in the normalized profiles, let their centroids be $\mathbf{x}_{\rm inf}$ and $\mathbf{x}_{\rm pol}$, and define $\mathbf{D} = \mathbf{x}_{\rm pol} - \mathbf{x}_{\rm inf}$. The mixture centroid is $\mathbf{x}_\chi = q\mathbf{x}_{\rm inf} + f\mathbf{x}_{\rm pol}$, hence $\Delta\mathbf{x}_\chi \equiv \mathbf{x}_\chi - \mathbf{x}_{\rm inf} = f\mathbf{D}$. Differentiating gives

$$\boxed{\mathbf{v}_{\rm rel} = \frac{d\Delta\mathbf{x}_\chi}{dt} = \frac{q}{t_{\rm rel}}\mathbf{D} + f\dot{\mathbf{D}}.} \tag{303}$$

For approximately fixed $\mathbf{D}$,

$$\boxed{\mathbf{v}_{\rm rel} = \frac{e^{-t/t_{\rm rel}}}{t_{\rm rel}}\mathbf{D}, \qquad \Delta\mathbf{x}_\chi = (1 - e^{-t/t_{\rm rel}})\mathbf{D}.} \tag{304}$$

This velocity is derived from the existing displacement and relaxation time; no streaming-speed amplitude is inserted. The framework already fixes $t_{\rm rel} = 4\pi^2/[9\sqrt{3}H(z)]$. For the Bullet Cluster, $t_{\rm rel} \simeq 31.4$ Gyr and $f(0.15\,{\rm Gyr}) \simeq 0.0048$, so the relaxation displacement is 0.48% of D at that epoch. This is a centroid drift of the

excess distribution, not yet the velocity of space itself. Substitution of the framework's own horizon-set clock gives the closed form, for fixed $\mathbf{D}$,

$$\boxed{\mathbf{v}_{\rm rel}(t) = \frac{9\sqrt{3}}{4\pi^2} H(z) \exp\left[-\frac{9\sqrt{3}}{4\pi^2} H(z) t\right] \mathbf{D}.} \tag{305}$$

The velocity scale is therefore derived from the framework's $H(z)$, its fixed relaxation law, and the existing separation $\mathbf{D}$; it is not an externally supplied streaming amplitude.

3. Mass conservation fixes the local relaxation flux

Differentiate the density mixture and use the continuity equations for each profile, $\partial_t \rho_a + \nabla \cdot (\rho_a \mathbf{v}_a) = 0$ for $a \in \{\inf, \mathrm{pol}\}$. Directly,

$$\partial_t \rho_\chi + \nabla \cdot (q\rho_{\rm inf}\mathbf{v}_{\rm inf} + f\rho_{\rm pol}\mathbf{v}_{\rm pol}) = \dot{q}\rho_{\rm inf} + \dot{f}\rho_{\rm pol} \tag{306}$$

$$= \dot{f}(\rho_{\rm pol} - \rho_{\rm inf}). \tag{307}$$

Therefore the transfer flux required by mass conservation obeys

$$\boxed{\partial_t \rho_\chi + \nabla \cdot \mathbf{J}_\chi = 0, \quad \mathbf{J}_\chi = q\rho_{\rm inf}\mathbf{v}_{\rm inf} + f\rho_{\rm pol}\mathbf{v}_{\rm pol} + \mathbf{J}_{\rm rel}, \quad \nabla \cdot \mathbf{J}_{\rm rel} = -\dot{f}(\rho_{\rm pol} - \rho_{\rm inf}).} \tag{308}$$

Under spherical symmetry, regularity at $r = 0$ fixes the radial component:

$$\boxed{J_{{\rm rel},r}(r,t) = -\frac{\dot{f}}{r^2}\int_0^r [\rho_{\rm pol}(s,t) - \rho_{\rm inf}(s,t)] s^2\, ds.} \tag{309}$$

Define the relaxation part of the coarse-grained transport velocity by $v_{{\rm rel},r} = J_{{\rm rel},r}/\rho_\chi$. Its local divergence is then not guessed but follows from the product rule:

$$\boxed{\vartheta_{\rm rel} \equiv \nabla \cdot \mathbf{v}_{\rm rel} = -\frac{\dot{f}(\rho_{\rm pol} - \rho_{\rm inf})}{\rho_\chi} - \mathbf{v}_{\rm rel} \cdot \nabla \ln \rho_\chi.} \tag{310}$$

The sign is determined by the existing profiles and their enclosed-mass difference. Mass conservation allows outward transport in some regions and inward transport in others; it does not imply a positive uniform expansion by itself. Compatibility with the phase action requires the total current $\mathbf{J}_\chi = \rho_\chi \mathbf{v}_\chi$ to satisfy the geodesic Euler equation (298). That equation is the consistency check, not an extra adjustable velocity law.

4. Geometric recoil is the trace of spatial deformation

Let h_{ij} be the proper spatial metric on slices orthogonal to the phase flow. The expansion tensor is $B_{ij} = \frac{1}{2}\mathcal{L}_u h_{ij}$; its symmetric decomposition is $B_{ij} = \sigma_{ij} + \frac{1}{3}\vartheta h_{ij}$, where $\vartheta = \nabla_\mu u^\mu$. A connecting separation ξ^i between neighboring flow lines obeys

$$\frac{D\xi^i}{D\tau} = B^i{}_j \xi^j, \qquad \frac{d\ln V_{\rm proper}}{d\tau} = \mathrm{tr}\, B = \vartheta. \tag{311}$$

The volume identity follows from differentiating the determinant of three independent connecting vectors: $d\ln\det(\xi^i{}_a)/d\tau = \mathrm{tr}\, B$. Therefore

$$\boxed{V_{\rm proper}(\tau) = V_{\rm proper}(\tau_0) \exp\left[\int_{\tau_0}^{\tau} \vartheta(\tau')\, d\tau'\right].} \tag{312}$$

This is the exact geometrical definition of stretching or recoil of a proper spatial volume. It is not the centroid drift in (303). In a locally isotropic patch, $h_{ij} = a^2\delta_{ij}$, $\vartheta = 3H$ and the corresponding recession/recoil speed at proper radius R is

$$\boxed{v_{\rm space}(R) = HR = \frac{R}{3}\vartheta.} \tag{313}$$

For the relaxation contribution, replace ϑ by the component $\vartheta_{\rm rel}$ only when the transfer current is the corresponding part of the phase congruence. The derived $v_{\rm space}$ is therefore a metric-deformation velocity, not the incorrect force impulse $-t_{\rm rel}\mathbf{g}$.

5. The existing action supplies the metric response

Varying the Einstein–Hilbert plus existing field action gives

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T^{\rm eff}_{\mu\nu}, \tag{314}$$

where $T^{\rm eff}_{\mu\nu}$ is the stress-energy of the baryons and the already-present scalar/phase sector. Thus the post-collision metric is sourced by the surviving excited geometry as well as the baryonic gas. In the weak-field, linear regime, harmonic gauge gives

$$\Box \bar{h}_{\mu\nu} = -\frac{16\pi G}{c^4}\,\delta T_{\mu\nu}, \qquad \bar{h}_{\mu\nu}(t,\mathbf{x}) = \frac{4G}{c^4}\int \frac{\delta T_{\mu\nu}(t-|\mathbf{x}-\mathbf{x}'|/c,\mathbf{x}')}{|\mathbf{x}-\mathbf{x}'|}\,d^3x'. \tag{315}$$

The retarded solution depends on the source's past light cone, so the metric response is not an algebraic instruction to erase the pre-collision geometry when the gas shocks. The continuing phase stress-energy and the retarded metric response are the two existing carriers of the lag. The proper-volume recoil is computed from that metric through (311); it is not set equal to a guessed multiple of g.

The same weak-field metric also gives the recoil velocity directly from the changing spatial dilation. With $S \simeq 1 - 2\Phi/c^2$ and $\Phi = \Psi$, the spatial metric coefficient is $h_{ij}/a^2 \simeq S\delta_{ij}$ to first order, while the physical length factor is $a_s = \sqrt{S} = e^{-\psi/2}$. This uses S as the weak-field spatial metric coefficient, not as a conformal rescaling of all spacetime. At a proper radius R corresponding to a fixed coordinate separation, to the weak-field order retained here,

$$\frac{\dot{a}_s}{a_s} = \frac{1}{2}\partial_t \ln S = -\frac{1}{2}\dot{\psi} \simeq -\frac{\dot{\Phi}}{c^2}, \qquad \boxed{v_{\rm rec}(R,t) = R\frac{\dot{a}_s}{a_s} = \frac{R}{2}\partial_t \ln S = -\frac{R}{2}\dot{\psi} \simeq -\frac{R}{c^2}\dot{\Phi}.} \tag{316}$$

This is the spatial-recoil velocity: the time derivative of proper spatial length as the sourced metric changes. It is different from the centroid drift $v_{\rm rel}$ and does not assume the incorrect impulse $-t_{\rm rel}\mathbf{g}$.

For the leading nonrelativistic phase-density contribution to the retarded potential, write $r = |\mathbf{x} - \mathbf{x}'|$ and

$$\Phi_\chi(t,\mathbf{x}) = -G\int \frac{\rho_\chi(t - r/c, \mathbf{x}')}{r}\,d^3x'. \tag{317}$$

Differentiating at fixed field point, then applying the already-derived continuity equation $\partial_t \rho_\chi + \nabla'\cdot \mathbf{J}_\chi = 0$ and integrating by parts (with the existing localized-source boundary condition), gives

$$v_{{\rm rec},\chi}(R,t) = -\frac{R}{c^2}\partial_t \Phi_\chi = \frac{GR}{c^2}\int \frac{\partial_t \rho_\chi(t - r/c, \mathbf{x}')}{r}\,d^3x' \tag{318}$$

$$= \boxed{\frac{GR}{c^2}\int \mathbf{J}_\chi(t - r/c, \mathbf{x}')\cdot\frac{\mathbf{x}-\mathbf{x}'}{|\mathbf{x}-\mathbf{x}'|^3}\,d^3x'.} \tag{319}$$

This expression derives the leading weak-field recoil from the actual phase current generated by the framework's geodesic motion and relaxation map. The complete relativistic answer retains the full retarded $\delta T_{\mu\nu}$ in equation (315); the displayed scalar-source integral is its nonrelativistic phase-density contribution. No streaming-speed amplitude is added. The sign and magnitude follow from the source current and the point at which the metric is evaluated.

For geodesic flow, the Raychaudhuri equation gives the evolution of the same expansion scalar:

$$\boxed{\frac{d\vartheta}{d\tau} = -\frac{\vartheta^2}{3} - \sigma_{\mu\nu}\sigma^{\mu\nu} + \omega_{\mu\nu}\omega^{\mu\nu} - R_{\mu\nu}u^\mu u^\nu.} \tag{320}$$

Tracing (314) in the stated stress-energy units gives $R_{\mu\nu}u^\mu u^\nu = \frac{8\pi G}{c^4}(T_{\mu\nu}u^\mu u^\nu + T/2) - \Lambda$. For the existing pressureless phase sector, this is fixed by its density; it is not an independent recoil force. The sign of the volume response is determined by the metric solution, its initial congruence, and the stress-energy. A positive integrated ϑ leaves a larger proper volume; a negative one contracts it. This is the mathematical distinction between a residual spatial stretch and a label for a moving mass peak.

6. Dilation recoil as a domain expansion rate

In the same weak-field slicing, the isotropic spatial metric including the background scale factor has $dV_{\rm proper} = a^3 S^{3/2} d^3x$. Differentiating its determinant gives the expansion rate of a domain D:

$$\boxed{H_D \equiv \frac{1}{3}\frac{d\ln V_{\rm proper}}{dt} = H + \frac{1}{2}\langle \partial_t \ln S\rangle_D = H - \frac{1}{2}\langle\dot{\psi}\rangle_D \simeq H - \frac{\langle\dot{\Phi}\rangle_D}{c^2}.} \tag{321}$$

The brackets denote the average with the domain proper-volume measure $S^{3/2}d^3x$ in this weak-field slicing (equivalent to coordinate-volume averaging at first order in the weak-field perturbation). This is a kinematic identity linking the existing dilation field to proper-volume growth; the field equation and the retarded source-current expression above determine its value. The relaxation centroid alone is not inserted into this equation as a space velocity.

Using the two expansion rates already tabulated in GRUT, $H_{\rm local} = 73.04$ and $H_{\rm bg} = 67.35\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$, their difference is $\Delta H = 5.69\ \mathrm{km\,s^{-1}\,Mpc^{-1}} = 1.84 \times 10^{-19}\ \mathrm{s^{-1}}$. A homogeneous account of this difference would require

$$\boxed{\delta\langle\vartheta\rangle_D = 3\Delta H = 17.07\ \mathrm{km\,s^{-1}\,Mpc^{-1}} \simeq 5.53 \times 10^{-19}\ \mathrm{s^{-1}}, \qquad \langle\partial_t \ln S\rangle_D = 2\Delta H \simeq 3.69 \times 10^{-19}\ \mathrm{s^{-1}}.} \tag{322}$$

These are derived target values from the framework's stated rates, not fitted recoil inputs. The theory's prediction is the source-current integral's value; it explains the difference only if that calculation returns the required domain average.

7. Horizon information density and the expansion law

The existing Hubble-horizon definitions give

$$I_H = \frac{\pi c^5}{\hbar G H^2}, \qquad V_H = \frac{4\pi}{3}\left(\frac{c}{H}\right)^3, \qquad k_B T_H = \frac{\hbar H}{2\pi}. \tag{323}$$

Dividing information by volume and eliminating H yields

$$\boxed{\iota_H \equiv \frac{I_H}{V_H} = \frac{3c^2 H}{4\hbar G}, \qquad H = \frac{4\hbar G}{3c^2}\iota_H, \qquad V_H = \frac{9\pi c^9}{16\hbar^3 G^3}\,\iota_H^{-3}.} \tag{324}$$

Consequently, the horizon's logarithmic rates obey

$$\boxed{\frac{\dot\iota_H}{\iota_H} = \frac{\dot H}{H}, \qquad \frac{\dot V_H}{V_H} = -3\frac{\dot\iota_H}{\iota_H}, \qquad \frac{\dot T_H}{T_H} = \frac{\dot\iota_H}{\iota_H}.} \tag{325}$$

This is the exact horizon form of the proposed cooling/recoil relation: as H and T_H fall, the horizon information density falls and its Hubble volume grows. It is not an independently fitted streaming amplitude. For the flat-FLRW background already used in GRUT,

$$\dot H = -4\pi G\left(\rho + \frac{p}{c^2}\right), \tag{326}$$

so

$$\boxed{\frac{\dot\iota_H}{\iota_H} = -\frac{4\pi G}{H}\left(\rho + \frac{p}{c^2}\right), \qquad \frac{\dot V_H}{V_H} = \frac{12\pi G}{H}\left(\rho + \frac{p}{c^2}\right).} \tag{327}$$

For $H > 0$ and $\rho + p/c^2 \geq 0$, the Hubble-horizon volume grows while its information density and Gibbons–Hawking temperature fall. Separately, the proper volume of a comoving FLRW region obeys $\dot V_{\rm com}/V_{\rm com} = 3H$, and the same horizon identity gives

$$\boxed{\frac{\dot V_{\rm com}}{V_{\rm com}} = 3H = \frac{4\hbar G}{c^2}\,\iota_H.} \tag{328}$$

Thus the framework derives the exact horizon identities: H is proportional to the Hubble-horizon information density and the Hubble-horizon volume is proportional to its inverse cube. When the solved background has decreasing H, the horizon volume grows while its information density and temperature fall. These identities are not a standalone field equation for H. Auxoviscausality, derived below, identifies the phase-flow integral of the proper-volume expansion as retained logarithmic strain; Raychaudhuri and Einstein determine its local load-dependent rate. The collision/relaxation sector supplies the local metric response through (314)–(320); its contribution to volume is the volume average of the resulting ϑ, not the centroid speed substituted for it.

8. The derived recoil chain

The complete equation chain is

$$\begin{gathered}\text{gas collision } \Rightarrow \text{ baryonic momentum loss but geodesic phase flow,}\\ f = 1 - e^{-t/t_{\rm rel}} \;\Rightarrow\; \mathbf{v}_{\rm rel} = \dot{f}\mathbf{D} + f\dot{\mathbf{D}},\\ \rho_\chi = q\rho_{\rm inf} + f\rho_{\rm pol} \;\Rightarrow\; \nabla\cdot\mathbf{J}_{\rm rel} = -\dot{f}(\rho_{\rm pol} - \rho_{\rm inf}),\\ T^{\rm eff}_{\mu\nu} \;\Rightarrow\; g_{\mu\nu} \;\Rightarrow\; \vartheta = \frac{d\ln V_{\rm proper}}{d\tau},\\ I_H, V_H, T_H \;\Rightarrow\; \iota_H \propto H, \quad V_H \propto \iota_H^{-3}.\end{gathered}$$

No step adds a recoil force or external velocity. The sign and magnitude of a particular system's net proper-volume change are obtained by solving the already-defined profile/current and metric equations with that system's initial and boundary data; the framework does not permit assigning a universal positive sign to every local transfer region independently of those equations.

9. Auxoviscausality: retained geometric memory

Definition and scope. Auxoviscausality is the phase-flow property by which a local gravitationally sourced change in proper spatial volume is recorded as accumulated logarithmic deformation rather than being returned to a reference value by a restoring term. The operational variable is the logarithmic volume strain of a small comoving flow tube, not a new matter density and not a count of newly created particles or topological cells. This definition uses the existing fixed-norm phase field and adds no field, coupling, force, or independent velocity.

On a regular branch, let u^μ be the unit, geodesic, hypersurface-orthogonal phase-flow congruence already defined in equation (296), and write $D \equiv u^\mu\nabla_\mu$ for differentiation along one of its integral curves. Let $V_{\rm prop}$ be the infinitesimal proper three-volume spanned by three neighboring flow lines, with reference value V_*. The volume-determinant identity already derived in equation (311) is

$$D\ln V_{\rm prop} = \vartheta, \qquad \vartheta \equiv \nabla_\mu u^\mu. \tag{329}$$

We therefore define the dimensionless auxoviscausal strain by

$$\boxed{\mathcal{A}_{\rm aux} \equiv \frac{1}{3}\ln\!\left(\frac{V_{\rm prop}}{V_*}\right), \qquad D\mathcal{A}_{\rm aux} = \frac{1}{3}\vartheta.} \tag{330}$$

The factor $1/3$ makes $\mathcal{A}_{\rm aux}$ the logarithmic linear scale associated with a three-volume. Integrating along the same phase-flow worldline gives the causal-memory form

$$\boxed{\mathcal{A}_{\rm aux}(\tau) = \mathcal{A}_{\rm aux}(\tau_0) + \frac{1}{3}\int_{\tau_0}^{\tau}\vartheta(\tau')\,d\tau', \qquad V_{\rm prop}(\tau) = V_* e^{3\mathcal{A}_{\rm aux}(\tau)}.} \tag{331}$$

Only the past segment $\tau' \le \tau$ on that phase-flow curve contributes: causal order is carried by the existing phase flow, not by an added nonlocal force or an acausal boundary condition.

Derivation of the load equation. The phase constraint already makes the congruence geodesic and hypersurface-orthogonal, so its acceleration and vorticity vanish. The existing Raychaudhuri equation, (320), then becomes

$$D\vartheta = -\frac{1}{3}\vartheta^2 - \sigma_{\mu\nu}\sigma^{\mu\nu} - R_{\mu\nu}u^\mu u^\nu, \tag{332}$$

where $\sigma_{\mu\nu}$ is the shear of the phase congruence. Define the signed Ricci-focusing load by $\mathcal{L}_g \equiv R_{\mu\nu}u^\mu u^\nu$, so positive $\mathcal{L}_g$ is a focusing contribution in Raychaudhuri. Substituting $\vartheta = 3D\mathcal{A}_{\rm aux}$ into Raychaudhuri yields the covariant second-order memory equation

$$\boxed{D^2\mathcal{A}_{\rm aux} + (D\mathcal{A}_{\rm aux})^2 + \frac{1}{3}\sigma_{\mu\nu}\sigma^{\mu\nu} = -\frac{1}{3}\mathcal{L}_g.} \tag{333}$$

This is the rigorous local link to gravitational load supplied by the existing geometry: the Ricci projection enters through Einstein's equation and anisotropic deformation enters through the already-defined shear. The Ricci projection alone does not capture arbitrary anisotropic or tidal deformation; the phase congruence records that part through its shear. Positive Ricci load focuses the congruence; the cosmological constant contributes with the opposite sign. The covariant equations use the framework's normalized phase-flow convention (natural units

in Raychaudhuri); explicit factors of c are restored in the FLRW source relation. No independent proportionality coefficient between load and strain is introduced.

For a homogeneous, isotropic FLRW branch, $\sigma_{\mu\nu} = 0$, $V_{\rm prop} \propto a^3$, and $\vartheta = 3H$. Hence

$$\boxed{\mathcal{A}_{\rm aux} = \ln\frac{a}{a_0}, \qquad \dot{\mathcal{A}}_{\rm aux} = H, \qquad \mathcal{L}_g = -3(\dot{H} + H^2) = -3\frac{\ddot{a}}{a}.} \tag{334}$$

With mass density ρ_m, pressure p, and cosmological constant Λ (inverse-length-squared units), the standard Einstein–Friedmann acceleration equation gives

$$\boxed{\mathcal{L}_g = 4\pi G\left(\rho_m + \frac{3p}{c^2}\right) - \Lambda c^2.} \tag{335}$$

Thus the gravitational load drives the expansion history through the existing Einstein/Raychaudhuri system, and Auxoviscausality records that solved history as $\mathcal{A}_{\rm aux}$. It does not replace the field equations with an algebraic proportionality between strain and the instantaneous Ricci projection.

The horizon identities already derived in equation (324) further imply, on the FLRW branch,

$$\boxed{\dot{\mathcal{A}}_{\rm aux} = H = \frac{4\hbar G}{3c^2}\,\iota_H, \qquad \mathcal{A}_{\rm aux}(t) - \mathcal{A}_{\rm aux}(t_0) = \frac{4\hbar G}{3c^2}\int_{t_0}^{t} \iota_H(t')\,dt'.} \tag{336}$$

This is the downstream link between the framework's horizon-information density and retained expansion strain; $\dot{V}_{\rm prop}/V_{\rm prop} = 3H$ remains exactly the previously derived volume law.

Retention condition and limits. Equation (330) contains no restoring term of the form $-\mathcal{A}_{\rm aux}/\tau_r$. If the expansion rate settles to zero after a period of positive expansion, then $D\mathcal{A}_{\rm aux} = 0$ and the previously accumulated value remains; it is not driven back to zero. On an expanding FLRW branch with $H \geq 0$, $\mathcal{A}_{\rm aux}$ is nondecreasing. In a locally contracting region with $\vartheta < 0$, however, the signed geometric strain decreases with the actual proper volume; the present framework does not artificially forbid real contraction. A strictly one-way microscopic damage/plasticity variable, a yield threshold, and positive material entropy production do not follow from this geometric identity and would require an additional irreversible constitutive sector. They are not needed for the Auxoviscausal geometric-memory closure.

9.1. Auxoviscausal memory at the conformal junction

The bulk law (330) is defined using physical proper volume on a regular aeon branch. It must not be matched by setting the two limiting values of $\mathcal{A}_{\rm aux}$ equal: those values diverge with opposite signs at the stated conformal junction. The existing metric identification (360) instead supplies a finite, geometric matching prescription for the conformal-volume remainder, provided the regular conformal spatial metric is nondegenerate on the junction and the same comoving flow tube is identifiable on its two sides.

Let $\hat{h}_{ij}$ be the induced spatial metric of the common conformal representative, and let $\hat{V}$ be the volume of that same coordinate-labelled tube computed with $\hat{h}_{ij}$. The physical induced metrics and volumes are

$$h^{(-)}_{ij} = b^{-2}\hat{h}_{ij}, \qquad h^{(+)}_{ij} = b^{2}\hat{h}_{ij}, \qquad V^{(-)}_{\rm prop} = b^{-3}\hat{V}, \qquad V^{(+)}_{\rm prop} = |b|^{3}\hat{V}. \tag{337}$$

Using the same reference volume V_* in the branchwise definition therefore gives

$$\mathcal{A}^{(-)}_{\rm aux} = -\ln b + \widehat{\mathcal{A}}_{\rm aux}, \qquad \mathcal{A}^{(+)}_{\rm aux} = \ln|b| + \widehat{\mathcal{A}}_{\rm aux}, \qquad \widehat{\mathcal{A}}_{\rm aux} \equiv \frac{1}{3}\ln\left(\frac{\hat{V}}{V_*}\right). \tag{338}$$

The finite matching variable is thus fixed by the existing conformal-volume determinant; it is not an additional field or a new transmission coefficient. Explicitly, under the regularity and tube-identification conditions stated above,

$$\boxed{\lim_{\eta\to\eta_c^-}\left(\mathcal{A}^{(-)}_{\rm aux} + \ln b\right) = \widehat{\mathcal{A}}_{{\rm aux},c} = \lim_{\eta\to\eta_c^+}\left(\mathcal{A}^{(+)}_{\rm aux} - \ln|b|\right)}, \qquad \boxed{[\widehat{\mathcal{A}}_{\rm aux}]^{+}_{-} = 0}. \tag{339}$$

This is the Auxoviscausal junction rule: the divergent isotropic scale contribution is reinitialized by the reciprocal scale-factor continuation ($-\ln b$ on the old branch and $+\ln|b|$ on the new), while the finite conformal-volume remainder is matched. It does *not* make the unregularized $\mathcal{A}_{\rm aux}$ continuous, nor does it imply that every matter degree of freedom or every microscopic record crosses the boundary.

The scope of the result matters. In exactly homogeneous FLRW for a fixed comoving tube, $\hat{V}$ is constant, so $\widehat{\mathcal{A}}_{\rm aux}$ is only the reference-normalized conformal-volume datum; the accumulated $\ln a$ of the old aeon is not preserved as a separate finite scalar. In an inhomogeneous branch, the finite remainder can encode the volume determinant of the regular conformal geometry, but it still is not an independent ledger of the entire past expansion integral. Retaining that entire integral as an extra cross-aeon state would require a new history variable and matching dynamics absent from the present action. The derived result here is deliberately narrower: the geometric volume remainder matches when the existing conformal geometry and the identified flow tube match.

Accordingly, the closed result is the phase-flow law (330), its causal history integral (331), and its load-source form (333). The word *viscoelastic* is an analogy for retained geometric response here; it is not a claim that the existing conservative phase action has derived microscopic viscosity or a material hysteresis law.

9.2. Local Auxoviscausal strain field and the Bullet Cluster

Purpose. The scalar $\mathcal{A}_{\rm aux}$ is local in space-time: it is evaluated on each worldline of the existing phase-flow congruence. It is not a spatially uniform cosmological number, and it is not an independent field with a freely chosen potential. This subsection specifies the local calculation required for a merging cluster without adding a new force or coupling.

Let $h_{ij}(x,\tau)$ be the induced proper spatial metric on hypersurfaces orthogonal to the phase flow, and let $dV_{\rm prop} = \sqrt{h}\, d^3q$ for a fixed infinitesimal coordinate cell d^3q. Relative to the same cell and a reference metric determinant h_*, the local strain is

$$\boxed{\mathcal{A}_{\rm aux}(x,\tau) = \frac{1}{6}\ln\!\left(\frac{h(x,\tau)}{h_*(x)}\right), \qquad D\mathcal{A}_{\rm aux} = \frac{1}{3}\vartheta.} \tag{340}$$

The determinant formula is the local form of the proper-volume definition, not an extra field equation. Along each phase-flow worldline the evolution is fixed by

$$\boxed{D^2\mathcal{A}_{\rm aux} = -\left(D\mathcal{A}_{\rm aux}\right)^2 - \frac{1}{3}\sigma_{\mu\nu}\sigma^{\mu\nu} - \frac{1}{3}\mathcal{L}_g, \qquad \mathcal{L}_g \equiv R_{\mu\nu}u^\mu u^\nu.} \tag{341}$$

This equation fixes the sign convention. $D\mathcal{A}_{\rm aux} > 0$ means the flow-tube volume is currently expanding and the accumulated strain is increasing; $D\mathcal{A}_{\rm aux} < 0$ means it is contracting and the signed strain is decreasing. Positive $\mathcal{L}_g$ contributes negatively to $D\vartheta$ and focuses the congruence, but it does *not* by itself imply that $\mathcal{A}_{\rm aux}$ is already negative: the sign of the strain depends on its reference value and prior expansion history. Shear contributes a non-positive term to $D\vartheta$. A cosmological-constant contribution enters through the signed Ricci projection and has the opposite sign to positive matter focusing under the framework's stated Einstein convention.

In the weak-field, flow-aligned, approximately isotropic slicing already used for the dilation, $h_{ij} \simeq a^2 S\delta_{ij}$, so the same determinant gives

$$\boxed{\mathcal{A}_{\rm aux}(x,t) - \mathcal{A}_{{\rm aux},*} = \ln\!\left(\frac{a}{a_*}\right) + \frac{1}{2}\ln\!\left(\frac{S(x,t)}{S_*}\right), \qquad S \simeq 1 - \frac{2\Phi}{c^2}.} \tag{342}$$

To first order, the spatially varying part is $\delta\mathcal{A}_{\rm aux} \simeq -\delta\Phi/c^2$, and its local rate is $D\mathcal{A}_{\rm aux} \simeq H - \dot{\Phi}/c^2$ only when the chosen coordinates follow the same congruence to the retained weak-field order. This reproduces the sign in the existing recoil relation $v_{\rm rec}/R \simeq -\dot{\Phi}/c^2$; it does not justify treating a static potential offset as a continuing expansion rate. For a general anisotropic metric, equation (340), not the isotropic S approximation, is the definition.

Bullet Cluster prescription. The local strain contrast between a phase-flow tube through a lensing/galaxy peak and a comparison tube through the X-ray gas is

$$\boxed{\Delta\mathcal{A}_{\rm aux}(\tau) = \Delta\mathcal{A}_{\rm aux}(\tau_0) + \frac{1}{3}\int_{\tau_0}^{\tau}\left[\vartheta_{\rm peak}(\tau') - \vartheta_{\rm gas}(\tau')\right]d\tau'.} \tag{343}$$

The two expansion scalars must be computed from the same solved metric and the corresponding phase-flow trajectories. The metric source must include the observed baryonic gas, galaxies, and the framework's existing phase/scalar stress-energy without double-counting any component. The observed lensing convergence and galaxy distribution constrain the gravitating map; X-ray surface brightness and temperature constrain the shocked gas; merger geometry and line-of-sight projection constrain initial and boundary data. The gas is collisional and shock-heated, while the phase congruence is treated according to its existing geodesic equation. The strain is therefore a derived local diagnostic of the solution, not an additional mass component and not a substitute for lensing.

This closes the *definition and sign-consistent evolution equation* for the local Auxoviscausal strain field. A numerical Bullet Cluster strain map is not claimed here: it requires digitized observational maps, a specified merger initial-value reconstruction, and a solved metric/phase-flow solution. In particular, the centroid relaxation fraction $f(\Delta t)$ cannot be inserted as ϑ; centroid drift and proper-volume expansion are different observables. The present equations state exactly what must be computed and what data determine it.

9.3. Cosmological matter, radiation, and thermodynamic closure

The cosmological model must evolve physical matter and radiation through changing thermodynamic regimes, not only a background expansion and a pressureless phase sector. This closure uses the standard baryonic, gauge, fermion, Higgs, electromagnetic, and radiation content already admitted by the framework's total action; it introduces no new fundamental field. Its purpose is to make the background and perturbation equations explicit and to prevent the radiation-era relation $p = \epsilon/3$ or the pressureless phase equation of state from being incorrectly applied to every component and epoch.

Write the total source schematically as $T^{\mu\nu}_{\rm tot} = T^{\mu\nu}_{b} + T^{\mu\nu}_{r} + T^{\mu\nu}_{\chi} + T^{\mu\nu}_{\rm other}$, with each existing scalar/defect contribution included exactly once in $T^{\mu\nu}_{\rm other}$. The metric field equation and the Bianchi identity require

$$\boxed{\nabla_\mu T^{\mu\nu}_{\rm tot} = 0, \qquad \nabla_\mu T^{\mu\nu}_{i} = Q^{\nu}_{i}, \qquad \sum_i Q^{\nu}_{i} = 0.} \tag{344}$$

The Q^{ν}_{i} terms describe transfer among already-present sectors and must be obtained from their interaction terms; they are not adjustable exchange functions. Where the action contains no interaction, the corresponding transfer is zero. The existing separately conserved phase current remains separately conserved unless the action is explicitly amended and re-derived.

For a homogeneous FLRW background, using energy density ϵ_i and pressure p_i, the continuity equations are

$$\boxed{\dot{\epsilon}_i + 3H(\epsilon_i + p_i) = Q_i, \qquad \sum_i Q_i = 0,} \tag{345}$$

where Q_i is the energy-transfer rate per proper volume in the comoving frame. The total equation of state is $p_i = p_i(\epsilon_i, T, n_j, \{x_j\})$ as appropriate to the regime, rather than a universal constant w_i. Radiation in the relativistic limit obeys $p_r = \epsilon_r/3$; a nonrelativistic collisionless phase component obeys its already-derived dust limit; baryons, photons, neutrinos, ionization states, and later neutral gas must use their applicable thermodynamic and kinetic closures.

Composition is tracked by number currents and reaction terms,

$$\boxed{\nabla_\mu (n_j u^\mu_j + J^\mu_j) = R_j,} \tag{346}$$

where J^μ_j is a diffusion/transport current and R_j is the net production rate from the specified reactions. The reaction network must conserve every exactly conserved charge; any allowed violation (for example, an anomaly-mediated process) must follow the existing particle-physics sector rather than an arbitrary source. The equation of state, heat capacities, reaction rates, and transport coefficients are microphysical inputs to be sourced from established physics or derived from the framework's particle sector, not new free cosmological forces.

Radiation and relativistic species are evolved with a covariant kinetic equation,

$$p^\mu \frac{\partial f}{\partial x^\mu} - \Gamma^\alpha_{\mu\nu} p^\mu p^\nu \frac{\partial f}{\partial p^\alpha} = C[f], \tag{347}$$

where $f(x,p)$ is the distribution function and $C[f]$ is the collision operator for the relevant scattering, emission, absorption, and decay processes. In a fluid/radiation moment treatment, the same physics appears as emissivity, opacity, heat flux, and anisotropic stress; moment truncations must be stated and checked in their regime of validity. For thermalized components the entropy current obeys

$$\boxed{\nabla_\mu s^\mu = \sigma_s \geq 0,} \tag{348}$$

with entropy production attributed to actual collisions, diffusion, shocks, chemical reactions, or other specified irreversible processes—not inferred merely from the geometric strain $\mathcal{A}_{\rm aux}$.

The perturbation problem must use the same closure. In particular, pressure perturbations must separate adiabatic and non-adiabatic parts, $\delta p = c_s^2 \delta\epsilon + \delta p_{\rm nad}$, and the calculation must retain anisotropic stress and the relevant radiation/particle hierarchy. The framework's scalar/defect perturbations and the metric response must

then be evolved with these matter perturbations to generate its own transfer functions. Using a precomputed ΛCDM transfer function can remain a comparison baseline, but it is not a derivation of the framework's perturbation predictions.

This is a closed statement of the conservation and kinetic structure, not a claim that every Q_i, reaction rate, opacity, or perturbation kernel has already been derived. The model becomes numerically closed only after each component's equation of state, collision/reaction data, and initial conditions are specified and the coupled background and perturbation system is integrated. Low-temperature molecular opacity and cloud-condensate physics are specialized applications of this same matter/radiation machinery, not ingredients to insert unchanged into the early Universe. No numerical cosmological parameter is changed by this patch.

Update supplement: derivations and the status of the open equations

This supplement integrates the twenty-principle update ledger while keeping a strict distinction between an algebraic consequence, a constitutive closure, an interpretation and a genuinely open matching problem. No field, coupling or boundary-transfer coefficient is added. The merger timeline below is an astrophysical analogy for local collision and relaxation; it is not a derivation of a cosmological junction.

A. The C_3 relaxation scale and the exponential response

From the framework's horizon mode and its one-third-cycle identification,

$$\omega_{C_3} = \frac{3\sqrt{3}}{2\pi} H, \qquad \omega_{C_3} t_{\rm rel} = \frac{2\pi}{3} \quad \Longrightarrow \quad t_{\rm rel} = \frac{4\pi^2}{9\sqrt{3}\, H}. \tag{349}$$

The numerical coefficient is therefore derived once the stated C_3 identification is accepted. The exponential response additionally uses a first-order relaxation closure, $\dot{f} = (1-f)/t_{\rm rel}$, with $f(0) = 0$. Separation of variables gives

$$\int_0^{f(t)} \frac{df'}{1-f'} = \frac{t}{t_{\rm rel}} \quad \Longrightarrow \quad f(t) = 1 - e^{-t/t_{\rm rel}}, \qquad q(t) = 1 - f = e^{-t/t_{\rm rel}}. \tag{350}$$

Thus the exponential is the exact solution of the declared relaxation law, but it is not implied by geodesic flow or stress-energy conservation alone. For a bound system formed at t_f in matter domination, $t_f = 2/(3H_f)$ and the framework's formation-epoch value of $t_{\rm rel}$ give

$$\frac{t_{\rm rel}}{t_f} = \frac{2\pi^2}{3\sqrt{3}} \equiv x_H \simeq 3.80, \qquad f = 1 - \exp\left[-\frac{D_{\rm age} - 1}{x_H}\right], \quad D_{\rm age} \equiv \frac{t}{t_f} \geq 1. \tag{351}$$

For a merger, the clock is reset by definition at core passage and t in (350) means elapsed proper time after that event. The cross-scale use of the same $t_{\rm rel} H$ is a framework identification, not an independently established cluster microphysics result.

B. Early-aeon horizon equations and horizon power

For a spatially flat constant-w epoch, $a \propto \tau^{2/[3(1+w)]}$, hence $H = 2/[3(1+w)\tau]$. With $\mathsf{J} = 1/(2H^2 t_P^2)$ and $N = 4\pi\mathsf{J}$,

$$\mathsf{J} = \frac{9}{8}(1+w)^2 \left(\frac{\tau}{t_P}\right)^2, \qquad N = \frac{9\pi}{2}(1+w)^2 \left(\frac{\tau}{t_P}\right)^2. \tag{352}$$

Matter ($w = 0$) gives $\mathsf{J} = 9(\tau/t_P)^2/8$; radiation ($w = 1/3$) gives $\mathsf{J} = 2(\tau/t_P)^2$ and $N = 8\pi(\tau/t_P)^2$. Since $N_\infty = 2\pi/(\omega^2 t_P^2)$,

$$\frac{\widetilde{N}}{N_\infty} = (2\omega\widetilde{\tau})^2 \qquad \text{(radiation-dominated start)}. \tag{353}$$

The Hubble-horizon energy is $E_H = c^5/(2GH)$. Using $\dot{H} = -\frac{3}{2}(1+w_{\rm eff})H^2$ gives

$$\dot{E}_H = -\frac{c^5}{2G}\frac{\dot{H}}{H^2} = \frac{3}{4}(1+w_{\rm eff})\frac{c^5}{G} = 3(1+w_{\rm eff})P_{\max}, \qquad P_{\max} = \frac{c^5}{4G}. \tag{354}$$

Consequently the radiation-era value is $4P_{\max}$ and the matter-era value is $3P_{\max}$. These are consequences of the Friedmann power-law epoch and the horizon-energy definition; the interpretation as a literal cosmic collision is separate.

C. Local condensation: weak-field scope

For $\Phi = -GM/r$ and $r_s = 2GM/c^2$, the stated weak-field metric identification gives

$$S = e^{-\psi} \simeq 1 - \frac{2\Phi}{c^2} = 1 + \frac{r_s}{r} = 1 + \Omega\frac{R_H}{r}, \qquad \Omega \equiv \frac{r_s}{R_H}. \tag{355}$$

If $a_s = \sqrt{S}$, then to first order in $|\Phi|/c^2$, $\dot{a}_s/a_s \simeq -\dot{\Phi}/c^2$. For inward motion with speed $|\dot{r}|$, $\dot{\Phi} = -g|\dot{r}|$ and therefore

$$v_{\rm rec}(R) \equiv R\frac{\dot{a}_s}{a_s} \simeq \frac{g|\dot{r}|}{c^2}R. \tag{356}$$

Equations (355)–(356) are weak-field, quasi-static relations. They are not exact strong-field formulae, and $v_{\rm rec}$ is a metric-length rate in this approximation, not a new force or a signal velocity.

D. Hawking temperature and the conditional lifetime coefficient

With $R_s = 2GM/c^2$, $\mathsf{J} = R_s^2/(2\ell_P^2) = 2GM^2/(\hbar c)$ and $S = 2\pi\mathsf{J}$,

$$\frac{d(Mc^2)}{dS} = \frac{\hbar c^3}{8\pi GM} \quad\Longrightarrow\quad k_B T_H = \frac{\hbar c^3}{8\pi GM}. \tag{357}$$

For the ideal-blackbody approximation with two photon polarizations, $A = 4\pi R_s^2$ and $\sigma = \pi^2 k_B^4/(60\hbar^3 c^2)$ give

$$P_{\gamma,\,\rm bb} = A\sigma T_H^4 = \frac{\hbar c^6}{15360\pi G^2 M^2}, \qquad \dot{M} = -\frac{P}{c^2}. \tag{358}$$

Integrating from M_0 to zero yields

$$t_{\rm ev}^{\rm bb} = \frac{5120\pi G^2 M_0^3}{\hbar c^4}, \qquad \mathsf{J}(t) = \mathsf{J}_0\left(1 - \frac{t}{t_{\rm ev}^{\rm bb}}\right)^{2/3}, \qquad \dot{\mathsf{J}} = -\frac{1}{1920\pi}\frac{c}{R_s}. \tag{359}$$

The coefficient $1/(1920\pi)$ is therefore the perfect-blackbody, photon-only coefficient. It is not the exact universal Hawking coefficient: the physical flux depends on greybody transmission factors and on all particle species accessible at the black-hole temperature. The semi-classical derivation also fails when $\mathsf{J} = O(1)$, so the endpoint remains open.

E. Merger phases as a local analogue, not a cosmic time-lapse

The observations are interpreted with three physically distinct diagnostics: (i) before first core passage, subclusters approach and their gas is still associated with their own potentials; (ii) during/after core passage, the intracluster plasma shocks and is decelerated while galaxies and the lensing-inferred mass peaks can remain spatially offset; and (iii) later, the disturbed system evolves through orbital motion, shocks, and relaxation. The Bullet Cluster and MACS J0025.4−1222 are examples of lensing–gas separation; El Gordo is a high-redshift ongoing merger usually reconstructed as shortly after first core passage, with outgoing/returning solutions; Pandora's Cluster (Abell 2744) is a multi-subcluster collision, not a clean relaxed endpoint. Sources: NASA on the Bullet Cluster, NASA on MACS J0025, El Gordo dynamical reconstruction, and NASA on Abell 2744.

These objects are different systems observed at different cosmic times and viewing geometries; they are not successive frames of one merger. Projection and time-since-pericentre degeneracies complicate phase assignments (see the 2026 review, Merging Galaxy Clusters and the Search for New Physics of Dark Matter). The data support a collisionless gravitating component relative to the shocked gas under standard lensing/GR interpretations; they do not establish a conformal boundary, a sign flip of the scale factor, or information transfer between aeons.

F. The update-file open items: what is and is not derivable

Open item from the update ledger	Status after derivation	Reason / missing calculation
Whether ℓ_P is the smallest spatial unit	OPEN	The current action uses the Planck cell as a boundary datum; it contains no ultraviolet/quantum-gravity argument proving a minimum length.

Open item from the update ledger	Status after derivation	Reason / missing calculation
Independent temperature of the excited share	OPEN	The pressureless conserved phase sector has no supplied local thermal distribution or equilibrium condition from which a separate temperature can be defined.
Emergence of $[\hat{g}]$ from a more primitive relation	OPEN	The generation relation determines the conformal scale once $[\hat{g}]$ and θ are supplied; it does not derive the conformal class itself.
Auxoviscausal bulk memory	DERIVED (bulk)	The proper-volume determinant and phase-flow expansion give $D\mathcal{A}_{\rm aux} = \vartheta/3$; Raychaudhuri fixes its Ricci-load/shear evolution on each regular branch.
Cross-aeon conformal-volume matching	DERIVED (conditional)	If the common conformal spatial metric is nondegenerate and the same comoving flow tube is identified across the junction, the finite remainder $\widehat{\mathcal{A}}_{\rm aux} = \frac{1}{3}\ln(\hat{V}/V_*)$ matches. The raw strain and the full FLRW expansion integral do not.
Microscopic yield-threshold plasticity	OPEN (material)	The conservative action has no internal damage variable or entropy-producing constitutive law. A one-way material yield mechanism is not implied by the geometric memory equation.
Field that causes and glues the conformal flip	PARTIAL	If $b(\eta_c) = 0$ and $H \to \omega$, then $b' = -H \to -\omega$ and the local linear slope follows. The existence of the zero, signed continuation and junction matching are not fixed by the action.
Transfer of S_∞ across the junction	OK (conformal channel)	The conformal carrier has continuous field and canonical momentum, giving $\alpha_k = 1$, $\beta_k = 0$ and $\mathcal{M}_{\mathcal{I}} = 1$; the cell count itself still resets $N_\infty \to 0 \to N_\infty$.
Adiabatic superhorizon perturbation transport	OK	Smooth conformal spatial geometry gives $[\mathcal{R}]^+_- = 0$; with $\delta p_{\rm nad} = 0$ on each regular branch, $\mathcal{M}_{\mathcal{R}} = 1$ for the specified mode.
Hawking coefficient and $\mathsf{J} = O(1)$ endpoint	PARTIAL	The photon-only blackbody limit yields (359); exact grey-body/species factors and the quantum endpoint need additional QFT/quantum-gravity input.
Black holes changing ε_Λ	NOT GENERATED BY CURRENT ACTION	For fixed Λ, $\varepsilon_\Lambda = \Lambda c^4/(8\pi G)$ and $\dot{\varepsilon}_\Lambda = 0$. A dependence on black-hole history needs an extra coupling or field, which is absent by construction.
Whether na^3V_c crosses the flip	OK	The global phase action has no junction source; continuity of the canonical phase flux gives $[\sqrt{-g}J^\mu n_\mu]^+_- = 0$, hence $n_- a_-^3 V_c = n_+ a_+^3 V_c$ and $\mathcal{T}_\chi = 1$.

The phase-current, curvature and conformal-information rows above are closed under their stated assumptions. The Auxoviscausal bulk law is separately closed in Section 55; its finite conformal-volume remainder has the conditional matching rule (339). The microscopic origin of the zero of b, non-adiabatic/subhorizon transfer, the independent thermal state of the excited share, and a genuinely dissipative plasticity sector remain separate questions.

Junction closure: phase current, curvature perturbation, and conformal information

The conformal junction is now closed at the level of the three transport channels used by the framework. The closure does not add a new field, force, or phenomenological transmission coefficient. It follows from the existing constrained phase action, the smooth conformal metric, and the absence of a junction-localized source or surface action. The physical metrics remain distinct on the two sides:

$$g_{\mu\nu} = b^{-2}\hat{g}_{\mu\nu}, \qquad \widetilde{g}_{\mu\nu} = b^2\hat{g}_{\mu\nu}, \qquad b(\eta_c) = 0, \qquad b'(\eta_c) = -\omega \neq 0. \tag{360}$$

All matching statements below are made in the regular conformal description and then translated back to the physical branches. This avoids incorrectly identifying the divergent pre-flip volume factor with the vanishing post-flip volume factor.

J1. Phase norm and geodesic flow

The phase constraint is

$$X \equiv -g^{\mu\nu}\partial_\mu\theta\partial_\nu\theta = X_0 = \frac{\omega^2}{c^2}, \qquad u_\mu = \frac{\partial_\mu\theta}{\sqrt{X_0}}. \tag{361}$$

Since the Hessian of a scalar is symmetric,

$$u^\nu\nabla_\nu u_\mu = \frac{1}{X_0}\partial^\nu\theta\,\nabla_\nu\partial_\mu\theta = \frac{1}{2X_0}\nabla_\mu(\partial^\nu\theta\partial_\nu\theta) \tag{362}$$

$$= -\frac{1}{2X_0}\nabla_\mu X_0 = 0. \tag{363}$$

Thus the phase congruence is exactly geodesic and hypersurface-orthogonal on every regular branch.

J2. Exact phase-current matching and $\mathcal{T}_\chi = 1$

The phase action already present in the framework is

$$S_\theta = \int d^4x\sqrt{-g}\,\lambda(X - X_0). \tag{364}$$

Variation with respect to θ gives

$$\nabla_\mu J^\mu = 0, \qquad J^\mu \equiv \lambda g^{\mu\nu}\partial_\nu\theta. \tag{365}$$

In a conformal-time foliation, the conserved canonical flux density is

$$\mathcal{F}_\theta(\eta,\mathbf{x}) \equiv \sqrt{-g}\,J^\eta, \qquad \partial_\eta\mathcal{F}_\theta + \partial_i(\sqrt{-g}\,J^i) = 0. \tag{366}$$

Integrate the field equation over an infinitesimal Gaussian pillbox around $\Sigma_c : \eta = \eta_c$. Because the action contains no junction-localized source or surface term, variation of the single global action requires continuity of the canonical momentum conjugate to θ:

$$\boxed{\left[\sqrt{-g}\,J^\mu n_\mu\right]_-^+ = 0} \qquad \Longleftrightarrow \qquad \boxed{\mathcal{F}_{\theta,+} = \mathcal{F}_{\theta,-}}. \tag{367}$$

This is the required junction condition; it is not an additional phenomenological transmission postulate.

For a homogeneous FLRW branch, $J^\mu = nu^\mu$ and $u^\eta \propto a^{-1}$. Hence

$$\mathcal{F}_\theta \propto na^3V_c, \tag{368}$$

where V_c is a comoving volume. Therefore

$$n_-a_-^3V_c = n_+a_+^3V_c \equiv N_\chi. \tag{369}$$

On the pre-flip branch $a_- = b^{-1}$ and $n_- \propto b^3$; on the post-flip branch $a_+ = b$ and $n_+ \propto b^{-3}$. The same conserved canonical flux is obtained from either side. Consequently the phase-quanta transmission coefficient is

$$\boxed{\mathcal{T}_\chi \equiv \frac{\mathcal{F}_{\theta,+}}{\mathcal{F}_{\theta,-}} = 1.} \tag{370}$$

Thus the former partial result $na^3V_c = \text{const}$ on each branch is promoted to an exact cross-junction result, provided the stated action is the complete junction action and no localized source is introduced.

J3. Curvature perturbation transport and $\mathcal{M}_\mathcal{R} = 1$

The previous Bardeen-potential argument is not used. The appropriate matching variable is the comoving curvature perturbation because it is directly the scalar perturbation of the spatial conformal geometry. Write the scalar spatial metric as

$$\gamma_{ij} = a^2e^{2\mathcal{R}}\delta_{ij} + \cdots. \tag{371}$$

A homogeneous conformal rescaling changes a but not $\mathcal{R}$. Therefore the same regular conformal metric on the two sides has

$$\widehat{\gamma}_{ij} = \widehat{a}^{\,2}e^{2\mathcal{R}}\delta_{ij} + \cdots, \qquad [\widehat{\gamma}_{ij}]_-^+ = 0 \quad \Longrightarrow \quad \boxed{[\mathcal{R}]_-^+ = 0}. \tag{372}$$

There is no junction-localized stress tensor in the stated action, so the conformal spatial geometry is the quantity whose finite limit is matched. On each regular branch the pressureless adiabatic phase has

$$\dot{\mathcal{R}} = -\frac{H}{\rho + p}\,\delta p_{\rm nad} + O\!\left(\frac{k^2}{a^2H^2}\right), \qquad \delta p_{\rm nad} = 0, \tag{373}$$

so the superhorizon adiabatic mode is constant away from the junction. Combining branch conservation with (372) gives the exact junction transfer law for that mode:

$$\boxed{\mathcal{R}_+(k) = \mathcal{R}_-(k)}, \qquad \boxed{\mathcal{M}_{\mathcal{R}}(k) = 1} \quad \text{(adiabatic, superhorizon mode)}. \tag{374}$$

This does not assert that every subhorizon, entropy, anisotropic-stress, or additional-field mode has unit transfer. Those modes are outside the closed single-component matching sector.

J4. Conformal information transport and $\mathcal{M}_{\mathcal{I}} = 1$

The information channel is the conformal, massless sector already identified by the framework. For a conformally invariant scalar, the rescaled field $\widehat{\chi}$ has a regular action on the conformal geometry. In the locally flat conformal patch used for the junction calculation,

$$S_\chi = -\frac{1}{2}\int d^4x\sqrt{-\widehat{g}}\,\widehat{g}^{\mu\nu}\partial_\mu\widehat{\chi}\partial_\nu\widehat{\chi}, \qquad \widehat{\Box}\widehat{\chi} = 0. \tag{375}$$

For each Fourier mode, the junction has no delta-function potential because $\widehat{g}_{\mu\nu}$ is smooth and the action has no surface term. The variational junction conditions are therefore

$$[\widehat{\chi}_k]^+_- = 0, \qquad [\partial_\eta\widehat{\chi}_k]^+_- = 0. \tag{376}$$

Using the same global conformal mode basis on both sides,

$$\widehat{\chi}_k = A_k e^{-ik\eta} + B_k e^{ik\eta}, \tag{377}$$

these two equations form an invertible two-by-two system whose unique solution is

$$A_{k,+} = A_{k,-}, \qquad B_{k,+} = B_{k,-}. \tag{378}$$

Hence the local junction Bogoliubov map is

$$\alpha_k = 1, \qquad \beta_k = 0, \qquad \boxed{\mathcal{M}_{\mathcal{I}}(k) = 1}. \tag{379}$$

The map is unitary, so the von Neumann entropy of the complete conformal quantum state is unchanged by the junction. This statement concerns the conformal information channel; it does not identify the reset cell count itself with a conserved entropy, since the framework explicitly has $N_\infty \to 0 \to N_\infty$.

J5. Closed junction theorem

Under the framework's existing assumptions—(i) $b(\eta_c) = 0$ with a smooth conformal metric, (ii) the fixed-norm phase action, (iii) no junction-localized source or surface action, and (iv) a conformally invariant massless information carrier—the junction transport laws are therefore

$$\boxed{\mathcal{T}_\chi = 1, \qquad \mathcal{M}_{\mathcal{R}}(k) = 1 \text{ for the adiabatic superhorizon mode}, \qquad \mathcal{M}_{\mathcal{I}}(k) = 1.} \tag{380}$$

The conformal junction is consequently transparent to the conserved phase current, to the specified adiabatic curvature mode, and to the conformal information channel. The former OPEN/partial junction entries are closed at this level. The separate microscopic origin of the zero of b, any material dissipative plasticity law, and the transfer of non-adiabatic or subhorizon perturbations remain outside this theorem. Auxoviscausal memory remains governed by the bulk volume-flow law; the finite conformal-volume remainder has the conditional determinant-matching rule (339), not an independently postulated transmission coefficient.

J6. Gauge quotient and hierarchy: corrected scope

The global group used by GRUT/TRSD remains

$$G_{\rm SM} = \frac{SU(3)_C \times SU(2)_L \times U(1)_Y}{\mathbb{Z}_6}, \qquad \frac{t}{3} + T + Y \in \mathbb{Z}, \tag{381}$$

where T is the $SU(2)$ representation isospin, not T_3. The quotient is a global-form statement. Within the framework's geometric construction, the $\mathbb{Z}_2$ and $\mathbb{Z}_3$ features are identified with the quotient, but this identification is a model derivation rather than a general theorem of the Standard Model. The absence of a simple unified gauge group means there are no SU(5)-type X/Y gauge bosons and hence no SU(5)-type *gauge-mediated* proton-decay channel. The quotient does not, however, forbid the gauge-invariant dimension-six operator $QQQL/\Lambda^2$; such higher-dimensional baryon violation must be separately suppressed.

For the hierarchy, H4 and the footprint relation give

$$m_*^3 = \frac{\hbar^2 H_0}{2Gc}, \qquad R_H = 8\pi\ell_P \exp\!\left(\frac{1}{\alpha} + \frac{3\pi\alpha}{2}\right), \tag{382}$$

so

$$\boxed{\frac{Gm_p^2}{\hbar c} = \frac{K^2}{(16\pi)^{2/3}} \exp\!\left(-\frac{2}{3\alpha} - \pi\alpha\right)}. \tag{383}$$

Using the framework value $K = 19.7566 \pm 0.0002$ and $\alpha^{-1} = 137.036$ gives 5.9061×10^{-39}, consistent with the framework's quoted 5.906×10^{-39}. The earlier illustrative value $K = 19.759$ would instead give approximately 5.9076×10^{-39}; it is not the value used in the released framework. The hierarchy relation is therefore closed as a conditional consequence of H4 and the footprint balance, not as a consequence of cluster-merger phenomenology.

A Numerical values

Computed with c, $\hbar$ and k_B exact, $G = 6.674\,30\times10^{-11}$, and 1 Mpc $= 3.085\,677\,58\times10^{22}$ m. The companion script `grut_v17_verify.py` (companion file, Verification) reproduces every entry. Entries scale as $r, M_{\rm H}, E_{\rm H} \propto H_0^{-1}$, $N, I, L_{\rm H} \propto H_0^{-2}$ and $T_{\rm GH}, a_0 \propto H_0$.

Quantity	Formula	$H_0 = 67.35$ (resonance budget)
r	c/H_0	1.3735×10^{26} m (14.52 Gly)
N	$2\pi c^5/\hbar G H_0^2$	4.538×10^{122}
$M_{\rm H}$	$Fr/2c^2$	9.248×10^{52} kg
$M_{\rm cone}$	$Fr/8c^2$	2.312×10^{52} kg
$E_{\rm H}$	$Fr/2$	8.312×10^{69} J
$I = S_{\rm GH}$	$N/2$ (nats)	2.269×10^{122}
$T_{\rm GH}$	$\hbar H_0/2\pi k_B$	2.653×10^{-30} K
$k_B T_{\rm GH} I/E_{\rm H}$		1 (exact)
energy per bit	$k_B T_{\rm GH} \ln 2$	2.539×10^{-53} J
$L_{\rm H}$	$Fr^2/2c$	3.808×10^{87} J s
$\rho_c/\rho_{\rm P}$	$3/4N$	1.653×10^{-123}
a_c	$(3\sqrt{3}/4\pi^2)cH_0$	8.612×10^{-11} m s^{-2}
a_0	$(\sqrt{3}/\pi^2)cH_0$	1.1483×10^{-10} m s^{-2}
F (universe)	c^4/G	1.2103×10^{44} N
$F_{\rm cone}$	$F/4 = c^4/4G$	3.0256×10^{43} N
$P_{\rm max}$	$F_{\rm cone} c$	9.071×10^{51} W
$\ell_{\rm P}$	$\sqrt{\hbar G/c^3}$	1.6163×10^{-35} m
$\dot{E}_{\rm H}/P_{\rm max}$ today	$2(1+q_0) = 3(1+w_{\rm eff})$	0.948

Geometry. Seed cone deficit $(2-\sqrt{2})\pi = 1.840302$ rad; defect cone deficit π; $\lambda_1(C_3)\, r^2 = 27/4\pi^2 = 0.683918$ (C_4: 0.810569, C_6: 0.911891, C_∞: 1); $\delta(M_{\rm H}) = 4\pi$ and $\delta(M_{\rm cone}) = \pi$ exactly.

Interpolation family. $\mu_n(1) = 0.42056$, 0.46233, 0.47900, 0.51819 for $n = 1, 2, 3, \infty$; $1 - \mu_3(10^3) = 3.50 \times 10^{-7}$, $1 - \mu_1(10^3) = 1.50 \times 10^{-3}$.

Cosmology (resonance budget, §25.11). $\Omega_\Lambda = 27/4\pi^2 = 0.683918$, $\Omega_m = 0.31600$, $\Omega_\Lambda/\Omega_c = 2.5805$, $\omega_b = 0.02248$, $\omega_c = 0.12022$, $S_8 = 0.833$, $\sigma_8 = 0.811$, $\eta = 6.16 \times 10^{-10}$, $r_{\rm drag} = 146.9$ Mpc; CMB fit to Planck's $100\theta_* = 1.04110$: $H_0 = 67.35 \pm 0.06$; $t_0 = 13.79$ Gyr, $H_0 t_0 = 0.950$; $q_0 = -0.526$; $z_{\rm acc} = 0.630$; $z_{m\Lambda} = 0.293$;

maximum-power epoch $z = 0.027$ (0.38 Gyr ago); conformal age $c\eta_0 = 3.177\,c/H_0 = 46.1$ Gly, 73.4% of the aeon's conformal time; $\Lambda = 1.088 \times 10^{-52}$ m^{-2}; $\rho_\Lambda c^2 = 5.237 \times 10^{-10}$ J m^{-3}; $N_\infty = 6.635 \times 10^{122}$, $r_\infty = 17.56$ Gly; $\rho_\Lambda = \frac{3}{4}\rho_P/N_\infty = 1.130 \times 10^{-123}\rho_P$. The withdrawn 5/2 budget had $\Omega_m = 0.32194$, $H_0 = 67.12$ (§52.6).

Leptons. $Q_\ell = 0.6666645(51)$, $\theta_\ell = 44.99991°$, $\mu_\ell^2 = 313.851$ MeV, $\delta_\ell = 0.22224$; $m_\tau^{\rm pred} = 1776.9690$ MeV; $\sin^2\theta_{12}^{\rm TM1} = 0.3180$ ($\sin^2\theta_{13} = 0.02248$, NuFIT 6.1).

Dark sector. $M_* m_\delta = 5.331 \times 10^{-17}$ kg^2; $L_* = 0.180$ pc at $1.5 \times 10^{11} M_\odot$; exact/sudden misalignment factor 1.576.

Spin (§36.10–36.13; resonance budget). $\omega = H_\infty = 55.70$ km s^{-1} Mpc^{-1} $= 1.8050 \times 10^{-18}$ s^{-1}; $R_\infty = 1.661 \times 10^{26}$ m (17.56 Gly); $\omega t_{\rm P} = \sqrt{2\pi/N_\infty} = 9.73 \times 10^{-62}$; $\Lambda = 3\omega^2/c^2 = 1.088 \times 10^{-52}$ m^{-2}; $T_\infty = \hbar\omega/2\pi k_B = 2.19 \times 10^{-30}$ K; $\hbar\omega = 1.19 \times 10^{-33}$ eV; $L_{\rm H} = 3.808 \times 10^{87}$ J s $= 3.611 \times 10^{121}\hbar$ today, $L_\infty/\hbar = N_\infty/4\pi = 5.28 \times 10^{121}$; area quantum $8\pi\ell_{\rm P}^2 = 6.565 \times 10^{-69}$ m^2; rim of the spin on today's horizon $c\sqrt{\Omega_\Lambda} = 0.827\,c$. At the resonance: $\theta_0 = \frac{2}{3}\,\mathrm{artanh}(3\sqrt{3}/2\pi) = 0.785699$ (0.785447 with radiation), $\beta = (2\alpha/\pi)\,\mathrm{artanh}(3\sqrt{3}/2\pi) = 0.31370°$ (0.31359° with radiation and $t_{\rm rec}$); $a_0 = (2/3\pi)\,c^2\sqrt{\Lambda/3} = 1.148 \times 10^{-10}$ m s^{-2}. Black hole of $60\,M_\odot$: $\omega = c^3/4GM = 846$ rad s^{-1}; electron spheres 3.862×10^{-13} m ($\hbar$) and 1.931×10^{-13} m ($\hbar/2$).

Resonance budget (§25.11; $\Omega_\Lambda = 27/4\pi^2$, derived baryon ratio, neutrinos inside Ω_m). $\Omega_\Lambda = 0.683918$, $\Omega_m = 0.31600$, $\Omega_\Lambda/\Omega_c = 2.5805$; CMB fit: $H_0 = 67.35 \pm 0.06$, $\omega_b = 0.02248$, $\omega_c = 0.12022$, $S_8 = 0.833$, $\sigma_8 = 0.811$, $r_{\rm drag} = 146.9$ Mpc, $t_0 = 13.79$ Gyr; $q_0 = -0.526$, $z_{\rm acc} = 0.630$, maximum power at $z = 0.027$ (0.38 Gyr ago); $a_0 = 1.148 \times 10^{-10}$ m s^{-2}; $H_\infty = 55.70$ km s^{-1} Mpc^{-1}; $N_0 = 4.538 \times 10^{122}$, $N_\infty = 6.635 \times 10^{122}$, $r_\infty = 17.56$ Gly; $\rho_\Lambda/\rho_P = 1.130 \times 10^{-123}$; 73.4% of the conformal time elapsed, event horizon 16.7 Gly; $\eta = 6.16 \times 10^{-10}$; $m_* = 47.49$ MeV, $K = 19.759$; first starlight 100.6 Myr, $z = 29.6$, 46.4 MHz. Rotation (§36.14): $\Omega_0 = jH_\infty$ with $j = 1$, quantum $\hbar H_\infty = 1.19 \times 10^{-33}$ eV. Lepton cone (§42.8): $\Xi_\ell = 2.000013(31)$. First starlight in cycles (§48.8): $\nu_\star/\nu_{21} = 1/30.81$ (46.10 MHz) with matter and vacuum, 1/30.61 (46.41 MHz) with radiation; from the twist, $H_\infty t_\star = (\pi/3)\beta$, the measured $0.277° \pm 0.057°$ gives $z_\star = 32.23$ and $42.74^{+5.64}_{-6.04}$ MHz, with $d\ln\nu_\star/d\ln\beta = 0.66$. Dark sector as one geometry (§53): $n_* = 3/8\pi\ell_P^2 R_\infty = 2.75 \times 10^{42}$ m^{-3} quanta of $\hbar H_\infty = 1.19 \times 10^{-33}$ eV; $\lambda_* = 3c^4/16\pi G = 7.22 \times 10^{42}$ N, $\lambda_0 = 2.80 \times 10^{42}$ N; $\Theta_0 = 0.279$, $w_{\rm dark,0} = -0.721$, equality at $z = 0.372$, $w_{\rm dark} = -\frac{1}{3}$ at $z = 0.728$; threshold $n_b = 0.65$ m^{-3} (2.58 times the mean); $M_{\rm tot}/M_b = 6.35$, $f_b = 0.158$; the vacuum inside r_{200} holds 0.34% of the halo. The dilation: $\mathcal{S} = 3.67$, 1.58, 1.067, 1.019 at $g_{\rm bar} = 0.1$, 1, 10, $30\,a_0$; the Euclidean secant 2, 3.86, 11.47 at $\theta = 60°$, 75°, 85° ($g = 0.577$, 0.268, $0.0875\,a_0$), the dynamic law 2.67, 4.67, 12.37 there; around $10^{11} M_\odot$, $\ell = 11.0$ kpc and $\mathcal{S} = 2$, 4, 11 at 16, 39, 116 kpc, 91 at 1 Mpc; reservoir radius with the cosmic share 59 kpc. The dilation field: $S - 1 = 2|\Phi|/c^2 \approx 10^{-6}$ in galaxies, 10^{-5}–10^{-4} in clusters; $R_{a_0} = c^2/2a_0 = 3.91 \times 10^{26}$ m; $\mu_{\rm Y} = 1/R_\infty = 6.0 \times 10^{-27}$ m^{-1}, a change below 10^{-7} in κ inside 1 Mpc; the Sun at its MOND radius, $\rho/\rho_c = 4.5 \times 10^{-6}$; $\ell = \sqrt{r_s R_{a_0}} = 7{,}190$ AU for the Sun and 8.54 kpc for $6 \times 10^{10} M_\odot$. The web (§53.9): rms peculiar field 0.0010–$0.0024\,a_0$ on 12–500 Mpc, $0.0019\,a_0$ at $8\,h^{-1}$ Mpc, $0.0079\,a_0$ at a point (baryons' $0.0012\,a_0$, dilated 28.9-fold, at most 4.5 against the cosmic dilation); field-only $S_8 = 0.89$, 1.00, 1.42 for $B = 3.6$, 8.1 from $z = 0.372$ and 5.6 from $z = 1$. The clock: horizon-set from $z = 0.630$ (mean), 0.713 (−20%), 0.378 (+80%); $\rho_m/2\rho_\Lambda = 3.0 \times 10^8$ and $c_s = 0.45c$ at recombination; rms $|\Phi|/c^2 = 0.9 \times 10^{-6}$, 8.6×10^{-6}, 1.7×10^{-5} at 12, 100, 300 Mpc; $1/k_{\rm eq} = 96$ Mpc; halo virial dispersion 264 km s^{-1} (77% of the mass), dilated flows 417, $\sigma_{1D} = 494$ km s^{-1}; $\Gamma > 0.873$; cut 68 Mpc ($k = 0.025$ Mpc^{-1}); $B = 1.6$, 3.4, 4.2 and $f_0 = 0.63$, 0.95, 1.06 at $k = 0.02$, 0.01, 0.005 Mpc^{-1}. Data: σ_8 +0.16% ($z = 0$), shear +0.3% at $\ell = 30$, CMB lensing +3.5% at $L = 10$ and unchanged from $L = 40$; $S_8^{\rm CMBL} = 0.822$; late ISW ×0.28, 0.41, 0.54, 0.71 at $\ell = 2$, 4, 10, 20 (light coupled, ×4,200 at $\ell = 10$); bulk flows rms 321, 288 km s^{-1} at 150, $200\,h^{-1}$ Mpc (139, 120 in ΛCDM); local-rate scatter 0.88% on 300 Mpc (0.41%). The KBC ratio 1.75 ± 0.14: a void 37% deep, $H_{\rm loc} = 74.2 \pm 0.6$ inside (72.2 undilated), a step of 8.3% to $0.09 < z < 0.15$, excluded by the supernovae.

B From the draft to the derived form

For updating the draft, each equation whose form changed in the derivation:

Draft	Derived form	Section
$M = r^2$	$M_{\rm H} = Fr/2c^2$ (mass $\propto r$; r^2 is the information scaling)	12, 18
$E = r^2c^2$	$E_{\rm H} = Fr/2$; one cone $Fr/8$	17
$G = c^2/2r$	$G = c^4/F$, constant	12
$FG\alpha = c^4$	$FG = c^4$ (gravity; one cone $F/4$) and $F_e = \alpha F$ (electric)	8
$F = 2rc^2/\alpha$	$F = c^4/G$, $F_{\rm cone} = F/4$, $F_{\rm geom} = F/2$	14
$\hbar = 4\pi r^3 c/N$	$\hbar = 4\pi M_{\rm H} cr/N = 2\pi Fr^2/Nc$	13
$r = c^2/2G$	$r = 2GM_{\rm H}/c^2$ (horizon condition)	10

Draft	Derived form	Section
$R_{\rm obs} = r(1-\Omega_m)$	$r_\Lambda = \Omega_\Lambda r$, $r_m = \Omega_m r$ (radius partition); observable radius $c\eta_0$	10, 25
$\delta_{\rm tot} = \pi$ (half the sphere)	$\delta_{\rm tot} = \pi$ from three equilateral faces; $4\delta_{\rm tot} = 4\pi$ closure	4
S^3 of radius r	spatial S^3 radius R, $\Omega_{\rm tot} = 1 + (r/R)^2$	4.6
$\lvert\psi\rvert^2 = \frac{1}{2}$	$p = \frac{1}{2} = \cos^2 45^\circ$; Born rule from Gleason	6
$S = N/4$	$S = N/2$	33
$\Omega = 2^N$	$\Omega = e^{N/2}$	34
$\rho_{\rm DE} = \rho_c$	$\rho_\Lambda = \Omega_\Lambda \rho_c$	37
space $= 1 + 3t$, time $= 1 + t$	$d\ln V/dt = 3H$; $\eta_0 = 3t_0$	25
32% of matter creates time	$\Omega_m =$ time-like share of stress $\Rightarrow \Omega_m = 1/3$	25.5
acceleration from 3:1	$q = (1 - 3\Omega_\Lambda)/2$; $w = -n/3$	25.4
$\lvert a_{3,3}\rvert \neq \lvert a_{3,-3}\rvert$	handedness in the phase of $a_{3,3}$ and in EB	38.4
$+V(\Xi)$ in the action	$-\varepsilon_V V(\Xi)$	40
$L = r^3 c$	$L_{\rm H} = Fr^2/2c = \hbar N/4\pi$	21
$p = r^2 c$	$E = pc$ on the cone; horizon scale $M_{\rm H}c$	19
$v^2 \to c^2$	massless at c; massive comoving; $m_\delta \to 0$ joins the null cone	26
threefold count as a Lissajous figure	beat of the fold $(+2)$ against the conjugate arc (-1): $\mathbb{Z}_3$, $\cos 3\theta$	1.11
spinning arcs, $\omega R = c$ (picture)	thermal turn of the final horizon: $\omega = H_\infty$, rim at the observer; $S = 2\pi L_{\rm H}/\hbar$	36.10–36.11
$X_0 = \Lambda/3$ (I8, identified)	$\Omega_0 = jH_\infty$ from the quantum of the spin; $j = 1$ measured	36.14
$a_0 = (2/3\pi)\, c^2\sqrt{\Lambda/3}$	at the present only; $a_0(z) = a_0 E(z)$	36.15
$\Omega_\Lambda/\Omega_c = C_3/C_2 = 5/2$ (I5)	the resonance: $\Omega_\Lambda = 27/4\pi^2$, ratio 2.580; I5 withdrawn	25.11, 52.6
Koide's cone $Q = \frac{2}{3}$ (I6)	$Q = (\Xi + 2)/3\Xi$: the vacuum $\Xi = 2$, the director circle of the generators	42.8
$t_\star = \alpha t_0$ (first starlight as a time)	a count of cycles, $\nu_\star/\nu_{21}$, set by the twist: $H_\infty t_\star = (\pi/3)\beta$	48.8
$w(z) = -1 + \Theta(\rho_b)$, a step at a binding density	$\Theta = n/(n + n_*)$, the excited fraction of the spin's quanta; threshold $\rho_\chi = \rho_\Lambda$	53.2
Dark matter is the dilation, $\mathcal{S} = 1/\cos\theta$, rising to the edge	$\mathcal{S} = M_{\rm inf}/M_b = 1/\cos_n\theta$, $\tan\theta = a_0/g$; r/ℓ in the deep regime	53.4
$\langle\mathcal{S}\rangle = 4$ (sphere), 3 (cone)	Cauchy's surface-to-shadow ratio, the four-cone closure; the measured excess varies, 6.35 cosmic	53.5
$S = e^{-\psi}$, $S \simeq 1 - 2\Phi/c^2$	the index of the optical metric; $\mathcal{S} = \partial_r \ln S/\partial_r \ln S_b$	53.7
$(\nabla^2 - \mu^2)\ln S = -(8\pi G/c^2)\alpha\rho_b S$	$\nabla\cdot[\mu(c^2\lvert\nabla\ln S\rvert/2a_c)\nabla\ln S] = -(8\pi G/c^2)\rho_b$; screening $\mu_Y = 1/R_\infty$	53.7
$\alpha(\rho)$, a density switch	the capture fraction: $n = 3/2$, $\alpha_0 = 1$, $\rho_c = 3\sigma^2/4\pi G R^2$; it gates the excess	53.7
Scales: $A \cong C$, B requires both	$\ell_P^2 = \frac{1}{2} r_s \lambda_C$, $\ell^2 = r_s R_{a_0}$, $g^2 = g_{\rm bar} a_0$	53.8
Background as its own consistency condition	the closure law, $\sum r_s = r$	53.8
The dilation explains the tensions; it acts as time dilation does	the background cannot move, $\mathcal{S} = 1$ without a gradient; the web is dilated as a clock is slowed—horizon-set from $z = 0.63$, held above 68 Mpc at the fixed point of its own flows, on clocks and not on light—so the bulk flows are those measured while S_8, shear and CMB lensing keep the background's values and the late Sachs–Wolfe signal falls; the ladder's H_0 is not a local void	53.9

C Prediction register

Every prediction of this draft, with the measurement that tests it and the result that would falsify it.

#	Prediction	GRUT value	Current data	Falsified if
1	Hubble constant (§26.7, §53.9)	67.35 ± 0.06 $\mathrm{km\,s^{-1}\,Mpc^{-1}}$, one rate for every ladder calibrated alike	67.36 ± 0.54 (Planck ΛCDM, 0.0σ); 73.50 ± 0.81 (H0DN 2026, -7.6σ; -6.1σ with the local scatter); 73.04 ± 1.04 (SH0ES, -5.5σ); 70.39 ± 1.94 (CCHP, -1.6σ); 66.5 ± 1.3 (DESI Lyα + BBN, $+0.7\sigma$)	the converged ladder H_0 more than 3σ from 67.35 beyond its 0.88% local scatter
2	Matter fraction (§25.11)	$\Omega_m = 0.3160$	0.3153 ± 0.0073 (Planck, $+0.1\sigma$); 0.2975 ± 0.0086 (DESI DR2 BAO, $+2.2\sigma$); 0.325 ± 0.018 (DESI Lyα AP, -0.5σ)	combined $\Omega_m > 3\sigma$ from 0.3160
3	Maximum-power epoch (§26.5)	$\dot{E}_{\mathrm{H}} = P_{\max}$ at $z = 0.0174$	follows from row 2	row 2 fails
4	Expansion history (§26.7–26.8)	$q_0 = -0.517$, $z_{\mathrm{acc}} = 0.615$	$q_0 \approx -0.53$ (ΛCDM)	reconstructed $z_{\mathrm{acc}} > 3\sigma$ from 0.615
5	Curvature (§4.6)	flat; any spatial S^3 has $R > 18\,r$	$\Omega_K = 0.0007 \pm 0.0019$	$\lvert\Omega_K\rvert > 0.003$ at 3σ
6	Equation of state (§26.4)	$w = -1$ (space-filling defect; $V_0 = 0$)	DESI 2026: 2.7σ (with CMB), 3.2σ (with SN) for evolving dark energy	$w \neq -1$ at 5σ
7	G constant (§2.8)	$\dot{G} = 0$	$\lvert\dot{G}/G\rvert \lesssim 10^{-13}$ $\mathrm{yr^{-1}}$ (LLR)	a measured $\dot{G} \neq 0$
8	Galactic scale (§22.5)	$a_0 = 1.148 \times 10^{-10}$ $\mathrm{m\,s^{-2}}$	likelihood (window): $1.04 \pm 0.04 \pm 0.10$; published RAR $1.20 \pm 0.02 \pm 0.24$ ($\times 10^{-10}$)	a_0 pinned at 5% precision away from 1.148
9	Gravity-law shape (§22.6–22.7)	threshold family, tail order $1.62 \leq n \leq 1.91$	galaxies $n = 1.38 \pm 0.20$; planets $n \geq 1.62$; one-exponent form: no exponent passes both	galaxies and planets disjoint in the threshold family as well
10	Planetary tail (§22.7)	anomaly $< 10^{-18}$ $\mathrm{m\,s^{-2}}$ in the inner solar system	ephemeris residuals consistent with zero	a MOND-type radial anomaly above 10^{-15} $\mathrm{m\,s^{-2}}$
11	Tau mass (§42.3)	1776.969 MeV	1776.93 ± 0.09 MeV ($+0.4\sigma$)	a ± 0.01–0.05 MeV measurement $> 3\sigma$ away
12	Neutrino masses (§42.4)	$\sum m_\nu = 59.34$ (normal) or 102.0 (inverted) meV	$\sum m_\nu < 64$ meV (95%, ΛCDM)	the measured ordering and $\sum m_\nu$ disagree with that branch
13	Double beta decay (§42.4)	$m_{\beta\beta} = 1.3$–4.1 (normal) or 17–48 (inverted) meV	$m_{\beta\beta} < 28$–122 meV (KamLAND-Zen)	a detection between 5 and 17 meV
14	Solar mixing (§42.5)	$\sin^2\theta_{12} = 0.3180$	NuFIT 6.1 0.3088 ± 0.0067 ($+1.4\sigma$); JUNO 0.3092 ± 0.0087 ($+1.0\sigma$)	the final JUNO value is $> 3\sigma$ from 0.318
15	CP phase (§42.5)	$\cos\delta_{\mathrm{CP}}(\theta_{23})$, eq. (136)	$\sin^2\theta_{23} = 0.56^{+0.03}_{-0.05}$; δ_{CP} weakly constrained	DUNE/Hyper-K $(\theta_{23}, \delta_{\mathrm{CP}})$ off the curve
16	Generations (§42.2)	exactly three light neutrinos on the cone	$N_\nu = 2.996 \pm 0.007$	a fourth light neutrino state
17	Maximum luminosity (§43B.6)	$L \leq P_{\max} = c^5/4G = 9.07 \times 10^{51}$ W	peak of GW150914: 3.6×10^{49} W (0.4%) [60]	any source above $c^5/4G$
18	Balance radius (§43B.3)	$R_* = (3GM/\Lambda c^2)^{1/3}$; 1.12 Mpc at $10^{12} M_\odot$	Local Group zero-velocity radius ≈ 1 Mpc	a bound structure larger than $R_*(M)$

#	Prediction	GRUT value	Current data	Falsified if
19	Defect pattern on the sky (§39.3)	power in $3 \mid m$, $\ell + m$ even, about one axis	Commander, $\ell = 4$–10: no excess (bounds ε)	the smoothed pattern also absent at the projected sensitivity
20	Fine structure (§43A/C, §43F)	calculated weight: $1/\alpha = 137.035\,999\,151\,7$	CODATA 2022 -1.2σ; electron $g-2$ -0.95σ	$1/\alpha$ settled more than 5σ from $137.035\,999\,152$
21	Radial-mode self-coupling (§37.3, §23.4)	$g_\chi = 3\hbar^3/2m_{\mathrm{P}}^2 c$ (negligible); the radial excitation, not dark matter	–	a self-interaction of the field above gravitational strength
22	Solar-System quadrupole (§49.10, §46–47)	Galactic tide only, $\sim 9 \times 10^{-31}$ s^{-2}	$(1.6 \pm 1.8) \times 10^{-27}$ (Cassini/DE440), -0.9σ	a quadrupole detected above the tide
23	Charged-lepton phase (§42.7)	$\delta_\ell = 2/9$	0.222226(6), $+0.7\sigma$	$\delta_\ell \neq 2/9$ at 3σ as m_τ improves
24	Quark cone (§42.7)	(c, b, t) on the cone	$Q = 0.6692$	a consistent scheme moving Q from $2/3$ beyond its uncertainty
25	Capture of the field (§47)	no bound condensate in systems formed inside hot hosts	–	a bound dark component around a star, binary or globular cluster
30	Maximal boost (§47.4)	$g_{\mathrm{top}} = 0.70\,a_0$	–	a boost $g - g_{\mathrm{bar}}$ well above $0.7\,a_0$ in any system
31	Running (§45.5)	$\alpha_s > 0$, below 10^{-4}	-0.0045 ± 0.0067 (Planck)	a running measured at $\lvert\alpha_s\rvert > 10^{-3}$
32	Tensor tilt (§45.5)	$n_t = 0$	–	$n_t = -r/8$ measured once tensors are detected
33	Dynamic exponent (§22.9)	$n(z)$ from 1 to at most 1.16; $\pi/3$ at $1.26\,a_0$	–	a local exponent above 1.2 at any acceleration
34	Charges (§31.4)	$\frac{2}{3}, -\frac{1}{3}, -1, 0$; $q_p + q_e = 0$	$\lvert q_p + q_e\rvert < 10^{-21} e$	a free fractional charge outside thirds, or $q_p + q_e \neq 0$
35	Neutral condensate (§31.1)	$q_\delta = 0$, $m_\gamma = 0$	$m_\gamma < 10^{-18}$ eV	a photon mass
36	Birefringence (§52.1, §49.12)	$\beta = \kappa\alpha\Omega_0(t_0 - t_{\mathrm{rec}})/2\pi$ with $\kappa = 6$ from the anomaly of the cone triples: 0.314°	$0.277^\circ \pm 0.057^\circ$ ($+0.6\sigma$), systematics open	β outside 0.312° at 5σ
37	Clusters (§47.5)	bound law short by 1.5–3.6 inside 1 Mpc; cosmic share sufficient	hydrostatic masses	a relaxation time to the polarization equilibrium shorter than cluster assembly
38	Baryon ratio (§25.7)	$\Omega_b/\Omega_c = 0.1870$	0.1864 ± 0.0022	ω_b/ω_c more than 3σ from 0.1870
39	Relaxation state (§47.6)	$t_{\mathrm{rel}} = \tau_1(t_f)/3$, $N_{\mathrm{rel}} = 25.3$; $f = 1 - e^{-(D-1)/x_H}$	hydrostatic masses	a relaxed, early-assembled system carrying the collapsed cosmic share
40	Relaxation ordering (§47.7)	discrepancy ordered by dynamical age	–	fossil groups no closer to the support law than clusters
41	RAR scatter (§47.7)	larger for disturbed systems	–	equal scatter for disturbed and isolated disks
42	The freeze (§48.2)	$N/N_\infty = \Omega_\Lambda(t)$; $N_\infty = 6.63 \times 10^{122}$, $r_\infty = 17.56$ Gly	row 6	evolving dark energy at 5σ
43	Maximum-power point (§48.3)	at $N/N_\infty = \frac{2}{3}$, 0.38 Gyr ago; $q_0 = -0.526$	q_0 from the Ω_m rows	$q_0 > -\frac{1}{2}$ at 3σ
44	Finite aeon (§48.4)	73.4% of the conformal time elapsed; event horizon 16.7 Gly	through Ω_m	future deceleration

#	Prediction	GRUT value	Current data	Falsified if
45	Constant limits (§48.1)	no force above $c^4/4G$, no power above $c^5/4G$	GW150914: 0.4%	either exceeded
46	Cells do not stretch (§48.6)	$\dot{G}/G = 0$; $N \propto r^2$	$(7.1 \pm 7.6) \times 10^{-14}$ yr^{-1} (LLR)	$\dot{G}/G = 2(1+q)H$
47	α sets light escape (§48.7)	α constant; $D_{\rm rec} \propto \alpha^{3.54}$	$\dot{\alpha}/\alpha = (1.0 \pm 1.1) \times 10^{-18}$ yr^{-1}	a drift of α tracking the age
48	First starlight (§48.8–48.9)	$t_\star = \alpha t_0 = 100.6$ Myr, $z_\star = 29.6$, onset 45.5–47.3 MHz ($\pm 3\%$), shell at 37.6 Gly	JWST $z = 14.3$; EDGES disputed	coupling below 45.5 MHz, or onset near $z \approx 20$
49	First stars without a particle (§48.9)	dust of zero sound speed suppresses no halos: first stars near $z \simeq 65$, before $z_\star$	–	no star formation before $z = 29.6$
50	Dark energy (§48.10)	$\rho_\Lambda = \frac{3}{4}\rho_P/N_\infty = 1.130 \times 10^{-123}\rho_P$, $w = -1$	row 6	evolving dark energy at 5σ
51	Cosmic dawn at the maximum-power epoch (§48.10)	$t_\star = \alpha t_{\rm mp} = 97.9$ Myr, $z = 30.2$, 45.6 MHz	–	onset outside 45.5–47.3 MHz; a 1% timing separates it from αt_0
52	Vacuum of every field (§48.11)	$\rho_{\rm vac} \le \rho_c$ for all fields together; $\frac{1}{2}k_B T_\infty$ per cell at the freeze	ρ_Λ	a vacuum energy above the critical density
53	Physical baryon density (§26.7)	$\omega_b = 0.02248$	0.02237 ± 0.00015 (Planck, $+0.7\sigma$)	a full-spectrum fit of the budget excluding 0.02248 at 5σ
54	Rotation of the universe (§23)	internal rotation, $\theta = \Omega_0 t$: zero cosmic vorticity; windings sum to zero	Planck: $(\sigma_V/H)_0 < 4.7 \times 10^{-11}$ [115]	a cosmic vorticity attributable to the rotation
55	Baryonic defect cone and Tully–Fisher zero point (§49.6–49.8)	deficit π, slant $L_\pi = 8GM/c^2$, $L^2_{\rm red} = L_\pi R_{a_0}$; $v^4_{\rm flat} = a_0 G M_b$	$a_0 = 1.24 \pm 0.14$ (gas-rich BTFR, -0.7σ)	a zero point inconsistent with $a_0 = 1.148 \times 10^{-10}$ m s^{-2} at 5σ
56	Wide binaries (§49.8)	Newtonian at all separations (no bound medium)	Gaia DR3: Newtonian [105]; $+22^{+10}_{-6}\%$ [103]	a low-acceleration boost confirmed at 5σ
58	Cosmic share as the rotation held at its rate (§23.1)	constraint $\lambda(X - X_0)$: dust, $\rho \propto a^{-3}$, $c_s^2 = 0$; no particle, no vacuum energy	CMB; clustering; Bullet Cluster lensing	a dark-matter particle detected, or a dark component diluting unlike dust since recombination
59	Solar-System quadrupole, bound-medium reading (§49.11)	tide only, 9×10^{-31} s^{-2}; total-field alternative $(3.0\text{–}3.2) \times 10^{-26}$	Cassini $(1.6 \pm 1.8) \times 10^{-27}$ [75]	$\lvert Q_2 \rvert > 5 \times 10^{-27}$ at 5σ
60	Lepton phase and cone (§42.7)	$3\delta_\ell = Q = \frac{2}{3}$ jointly to 2×10^{-7}; $m_\tau = 1776.97$ MeV	m_e, m_μ; $m_\tau = 1776.93(9)$	m_τ more than 5σ from 1776.97 MeV
61	Baryon-to-photon ratio (§25.8)	$\eta = 6.16 \times 10^{-10}$ from the budget	Planck 6.13×10^{-10} ($+0.7\sigma$ in ω_b)	η settled more than 5σ from 6.16×10^{-10}
62	Birefringence and acceleration (§49.13)	$\beta = \frac{9}{2}\alpha\, a_0 t_0/c$: 0.314° (framework a_0), $0.33^\circ \pm 0.07^\circ$ (RAR a_0)	$0.277^\circ \pm 0.057^\circ$	β/α more than 5σ from $\frac{9}{2}a_0 t_0/c$, a_0 and β measured independently
63	Closure (§50.1)	$\Omega_{\rm tot} = \chi/2 = 1$ exactly; $\sum_i r_{s,i} = r$	Planck 1.0007 ± 0.0019 (0.4σ)	$\lvert\Omega_k\rvert$ nonzero at 5σ
64	Scale ladder (§50.5)	exact rungs only along symmetric splits; other masses uniform in offset	Rayleigh $p = 0.18$ (particles), 0.92 (bodies)	independently formed masses clustered on rungs at 5σ
65	Linear cosmology (§25.9)	$\Omega_\Lambda/\Omega_c = 5/2$ with the derived baryon ratio	CMB spectra: 5/2 at 4.6σ, baryon ratio $\Delta\chi^2 = 0.2$	5/2 beyond 5σ in the full likelihood

#	Prediction	GRUT value	Current data	Falsified if
66	The present as a resonance (§36.7)	$\Omega_\Lambda(\text{now}) = 27/4\pi^2 = 0.6839$	Planck 0.6847 ± 0.0073 (0.1σ); full spectrum with derived baryon ratio 1.3σ (neutrinos counted; 1.8σ in release 1.13.0)	Ω_Λ more than 5σ from 0.6839
67	Spin of the arcs (§36.9)	$\omega = H_\infty = 0.827 H_0$; $\beta = (\kappa\alpha/2\pi)\,\omega(t_0 - t_{\rm rec})$	birefringence: $\omega = (0.73 \pm 0.15) H_0$, 0.6σ	ω from β and $\sqrt{\Lambda c^2/3}$ differing at 5σ
68	Spin of the final horizon (§36.10–36.11)	$\omega = H_\infty = c\sqrt{\Lambda/3}$, rim at c, $k_B T_\infty = \hbar\omega/2\pi$, $S = 2\pi L/\hbar$	row 6	$w \neq -1$ at 5σ: no final horizon, no fixed spin
69	Birefringence in closed form (§36.13)	$\beta = (2\alpha/\pi) \times \operatorname{artanh}(3\sqrt{3}/2\pi) = 0.3137^\circ$	$0.277^\circ \pm 0.057^\circ$ ($+0.6\sigma$), systematics open	β settled more than 5σ from 0.314°
70	Area quantum (§36.11–36.12)	$\Delta A = 8\pi\ell_{\rm P}^2$, $\Delta S = 2\pi$ for every horizon; black-hole lines at $n\,c^3/4GM$	LVK ringdowns: $\alpha = 15.6^{+20.5}_{-13.3}$ ($+0.5\sigma$)	α measured away from 8π at 5σ, or a continuous area spectrum established
71	Acceleration scale from Λ (§36.13)	$a_0 = (2/3\pi)\,c^2\sqrt{\Lambda/3} = 1.149 \times 10^{-10}$ m s^{-2} (Planck Λ)	RAR 1.20 ± 0.24; likelihood 1.04 ± 0.11 ($\times 10^{-10}$)	a_0 and Λ, measured independently, off the relation at 5σ
72	Quantum of the rotation (§36.14)	$\Omega_0 = jH_\infty$ with $j = 1$: one quantum of the rotation is one quantum of the spin	birefringence 0.88 ± 0.18 ($j = 2$ at 6.2σ); Ω_Λ with H3: 0.9994 ± 0.0053	Ω_0/H_∞ settled 5σ from 1
73	Acceleration scale with redshift (§36.15)	$a_0(z) = a_0 E(z)$: 1.49, 1.72, 2.00, 2.40 ($\times 10^{-10}$) at $z = 0.47$–1.27	MUSE-DARK [149]: 1.99–2.71, rising; constant excluded	a constant a_0, or $a_0(z)/a_0$ off $E(z)$ at 5σ in one method over $z = 0$–1.5
74	Budget at the resonance (§25.11)	$\Omega_\Lambda/\Omega_c = 2.580$, $\Omega_m = 0.3160$, $H_0 = 67.35$, $\omega_b = 0.02248$, $\omega_c = 0.12022$	CMB ratio 2.611 ± 0.024 (-1.3σ); ω_b $+0.7\sigma$; DESI Ω_m $+2.2\sigma$	the dark ratio, with the derived baryon ratio, settled 5σ from 2.580
75	Lepton cone as the vacuum (§42.8)	$Q = (\Xi + 2)/3\Xi$ at $\Xi = 2$	$\Xi_\ell = 2.000013 \pm 0.000031$ (0.4σ)	as rows 11 and 60
76	Cosmic dawn at the resonance (§48.8)	$t_\star = \alpha\, t_{\rm res} = 100.6$ Myr, $z_\star = 29.6$, 46.4 MHz	JWST $z = 14.44$ [155]; the $z \simeq 32$ candidate a Y dwarf [156]	coupling below 45.5 MHz, or onset near $z \approx 20$
77	Baryon-to-photon ratio at the resonance (§25.8)	$\eta = 6.16 \times 10^{-10}$	Planck 6.13×10^{-10} ($+0.7\sigma$ in ω_b); deuterium $\omega_b = 0.02233 \pm 0.00036$ ($+0.5\sigma$)	η settled 5σ from 6.16×10^{-10}
78	Cosmic dawn from the twist (§48.8)	$H_\infty t_\star = (\pi/3)\beta$, so the twist fixes the onset ratio $\nu_\star/\nu_{21}$, eq. (222); closed form 46.4 MHz	$\beta = 0.277^\circ \pm 0.057^\circ$ [85]: onset at $42.7^{+5.6}_{-6.0}$ MHz; no onset yet measured	onset and twist off the relation by 5σ together; a twist settled at zero
79	Cluster baryon fraction (§53.6)	$f_b = \Omega_b/(\Omega_b + \Omega_c) = 0.158$; $M_{\rm tot}/M_b = 6.35$	X-COP: cosmic value reached at R_{200} [163]; Bullet Cluster dark-to-gas convergence ≈ 7 [158]	f_b at R_{200} settled 5σ from 0.158 after the depletion correction
80	Dark self-interaction and offsets (§53.6)	$\sigma/m = 0$; no dark–galaxy offset in mergers	$\sigma/m < 0.63$ cm^2 g^{-1} (95%) [164]; Bullet lensing peaks within 1σ of the galaxies	a self-interaction or a dark–galaxy offset established at 5σ
81	Splashback of the excited geometry (§53.5)	the caustic edge of cold dark matter	detected in stacked clusters [166]	no splashback steepening where cold dark matter places it

#	Prediction	GRUT value	Current data	Falsified if
82	The dilation beyond the acceleration (§53.4)	$\mathcal{S}$ set by capture and relaxation as well as by g	weak-lensing RAR: early and late types differ by $\geq 6\sigma$ at fixed stellar mass [167]; clusters off the galactic relation [163, 169]	one $\mathcal{S}(g)$ for every type and scale at 5σ
83	No screening of the dilation (§53.7)	$\mu_{\rm Y} = 1/R_\infty$: no cut inside 5.4 Gpc	weak lensing flat to ≈ 1 Mpc [168]	an exponential cut of the lensing excess at a fixed scale inside the horizon
84	The dilation on the unbound web (§53.9)	horizon-set from $z = 0.63$, held above 68 Mpc at the fixed point $\sigma_{1D} = 494\ \mathrm{km\,s^{-1}}$: bulk flows rms 321 and 288 $\mathrm{km\,s^{-1}}$ within 150 and $200\,h^{-1}$ Mpc; local-rate scatter 0.88% on 300 Mpc; no shift of the ladder's H_0	Cosmicflows-4 387 ± 28 and $419 \pm 36\ \mathrm{km\,s^{-1}}$, at the 78th and 90th percentiles (ΛCDM below 0.03% and 0.003%) [176]; shell dipole $628 \pm 82\ \mathrm{km\,s^{-1}}$, 98th [177]; network 6.1σ from 67.35 [79]; no local void in the supernovae [171]	bulk flows within 150–200 h^{-1} Mpc settled at the ΛCDM expectation at 5σ, or growth enhanced before $z = 0.7$ or below 68 Mpc
85	Light on the web (§53.9)	a conformal factor: light's paths and the late Sachs–Wolfe signal see only the matter, the signal $\times 0.28$–0.71 at $\ell = 2$–20; S_8 +0.16%, shear unchanged from $\ell = 100$ and CMB lensing from $L = 40$; lensing below the dynamical pull above 68 Mpc	DES Y6 0.789 ± 0.012 [174], 2.8σ as for the background; $S_8^{\rm CMBL} = 0.825^{+0.015}_{-0.013}$ [178], -0.2σ; ISW detected at 4σ [179]; the light-coupled form excluded ($\times$4,200 at $\ell = 10$)	an ISW–galaxy correlation at the background's amplitude for $z < 0.5$ tracers at $\ell \leq 20$ at 5σ, or lensing that follows the dilated pull above 68 Mpc
86	The present in Planck cells (§48.12, H4)	$\ln(r/\ell_{\rm P}) = 1/\alpha + \ln 8\pi + \frac{3\pi}{2}\alpha$: $H_0 = 67.3765 \pm 0.0007$, $N = 4.534 \times 10^{122}$, $\Lambda\ell_{\rm P}^2 = 2.843 \times 10^{-122}$; with H3, $100\theta_* = 1.04124$	CMB fit 67.35 ± 0.06 ($+0.4\sigma$); $100\theta_* = 1.04110 \pm 0.00031$ ($+0.4\sigma$); $\dot\alpha/\alpha = (1.0 \pm 1.1) \times 10^{-18}$ $\mathrm{yr^{-1}}$ [87]	the converged H_0 or the budget's θ_* away from the prediction at 5σ, or a measured drift of α
87	One gauge group (§54.2)	$(SU(3) \times SU(2) \times U(1))/\mathbb{Z}_6$ from the shape's $\mathbb{Z}_2 \times \mathbb{Z}_3$; integer charge for every colour singlet	the charges of all known particles	a free fractionally charged colour singlet
88	No unification scale (§54.4)	no gauge boson turns quarks into leptons: no gauge-mediated proton decay	Super-Kamiokande $\tau/B(p \to e^+\pi^0) > 2.4 \times 10^{34}$ yr [127]	gauge-mediated proton decay detected
57	Magnetic fields (§49.9)	no anomaly-driven growth inside the horizon; $B_{\rm eq} = \sqrt{\mu_0 \rho}\, v_t$	Milky Way $\sim 5\ \mu$G	a primordial field grown by the phase rotation on sub-horizon scales
26	Wide binaries, globular clusters (§46.5)	Newtonian at all separations	–	a MOND-like boost in systems smaller than λ_J
27	Dwarf profiles (§23.1)	inner slope $-\frac{1}{2}$ inside the baryonic core, isothermal beyond, eq. (53)	rotation-curve decompositions of dwarfs	a cored (0) or steep (≤ -1) inner slope inside the baryonic core at 5σ

#	Prediction	GRUT value	Current data	Falsified if
28	End of flat curves (§46.3, §53.4)	$r_t = (M_\chi/M_b)\sqrt{GM_b/a_0}$ if the dilation draws on the bound store	weak lensing: flat with no clear decline to ≈ 1 Mpc around isolated galaxies [168]	flat profiles continuing beyond the reservoir
29	External-field effect (§46.5)	none of the MOND kind	–	a MOND external-field effect confirmed in matched isolated and embedded samples
89	Mergers: the excess stays on the collisionless components (§47.9)	offset from the galaxies $\leq f(\Delta t)D$: 0.5% of the gas–galaxy separation D for the Bullet Cluster, $\leq 3.1\%$ after 1 Gyr, $\leq 6\%$ after 2 Gyr; $\sigma/m = 0$	$\sigma/m < 0.63$ cm^2 g^{-1} [164]; MACS J0025 < 4 [183]	an offset above $0.1D$ in a clean merger
90	Each peak carries its own cluster's cosmic share (§47.9)	$(\Omega_c/\Omega_b)M_b^{\rm pre}$, the gas it lost included; peak mass ratio equals the pre-merger baryon mass ratio; the swapped assignment is excluded ($T = 99$ on synthetic maps)	$M_{\rm tot}/M_b = 6.35$, §53.6	a peak mass ratio off the baryon ratio by more than the depletion scatter
91	No galaxy-free peak on the gas (§47.9)	above $f(\Delta t) \leq 0.06$ of the excess; Abell 520 is the test	contested: found by [184, 185], not at that significance by [186, 187]	a $\geq 5\sigma$ galaxy-free gas-coincident peak holding $\gtrsim 10\%$ of the excess

D Variance analysis: prediction against observation

This appendix treats each prediction as a forecast and each measurement as the actual, and decomposes the differences into their drivers.

D.1 Method

The variance is $V = \text{GRUT} - \text{observed}$, and the pull is V/σ with σ the observational error (GRUT's own errors are negligible except for H_0, ± 0.07). Two further labels mark rows that are not independent tests: *calibration* (a calibrated value against the data it was chosen to approach) and *consistency* (a comparison with a fit that shares an input). Materiality follows a three-tier rule: *on track* for $|V/\sigma| < 1$, *monitor* for $1 \leq |V/\sigma| < 2$, and *material* for $|V/\sigma| \geq 2$, which requires a driver-level explanation and a stated test. Rows that share data (the Planck rows) are not independent and are not combined into a single χ^2.

D.2 Variance table

All cosmological rows are at the resonance budget of §25.11 ($H_0 = 67.35$, $\Omega_m = 0.3160$, $\Omega_\Lambda/\Omega_c = 2.580$).

#	Quantity	GRUT	Observed	pull	tier	driver
1	m_τ [MeV]	1776.969	1776.93 ± 0.09	$+0.43$	on track	I6
2	$\sin^2\theta_{12}$ (NuFIT 6.1)	0.3180	0.3088 ± 0.0067	$+1.37$	monitor	H2
3	$\sin^2\theta_{12}$ (JUNO)	0.3180	0.3092 ± 0.0087	$+1.01$	monitor	H2
4	N_ν	3	2.996 ± 0.007	$+0.57$	on track	C_3
5	a_0 (published RAR) [10^{-10} m s^{-2}]	1.148	1.20 ± 0.24	-0.22	on track	$k_3, \frac{4}{3}, H_0$
6	a_0 (likelihood) [10^{-10} m s^{-2}]	1.148	1.037 ± 0.107	$+1.04$	monitor	nuisance
7	Ω_K	0	0.0007 ± 0.0019	-0.37	on track	flatness
8	Ω_Λ/Ω_c	2.580	2.611 ± 0.024 (CMB spectra)	-1.28	monitor	H3
9	Ω_b/Ω_c (§25.7)	0.1870	0.1864 ± 0.0022	$+0.26$	on track	exponent
10	Ω_m (Planck)	0.3160	0.3153 ± 0.0073	$+0.10$	on track	H3
11	Ω_m (Pantheon+)	0.3160	0.334 ± 0.018	-1.00	monitor	H3
12	Ω_m (DES-Y5)	0.3160	0.352 ± 0.017	-2.12	material	as ΛCDM
13	Ω_m (DESI DR2 BAO)	0.3160	0.2975 ± 0.0086	$+2.15$	material	as ΛCDM
14	Ω_m (DESI Lyα AP)	0.3160	0.325 ± 0.018	-0.50	on track	H3
15	H_0 (Planck ΛCDM)	67.35	67.36 ± 0.54	-0.02	consistency	shares θ_*
16	H_0 (SH0ES)	67.35	73.04 ± 1.04	-5.47	material	ladder
17	H_0 (DESI Lyα + BBN)	67.35	66.5 ± 1.3	$+0.65$	on track	budget
18	H_0 (H0DN 2026)	67.35	73.50 ± 0.81	-7.59	material	ladder
19	H_0 (CCHP red giants)	67.35	70.39 ± 1.94	-1.57	monitor	ladder
20	H_0 (TDCOSMO)	67.35	71.6 ± 3.3	-1.29	monitor	lens model
21	Q_2, complete equation [10^{-27} s^{-2}]	0.0009	1.6 ± 1.8	-0.89	on track	tide
22	ω_b (Planck)	0.02248	0.02237 ± 0.00015	$+0.72$	on track	baryon ratio
23	a_0 (gas-rich BTFR) [10^{-10} m s^{-2}]	1.148	1.24 ± 0.14	-0.65	on track	–
24	β (ACT + Planck) [deg]	0.3137	0.277 ± 0.057	$+0.64$	on track	spin, κ
25	ω_c (Planck)	0.12022	0.1200 ± 0.0012	$+0.18$	on track	H3
26	S_8 (Planck)	0.833	0.832 ± 0.013	$+0.05$	on track	–

Fourteen rows are on track, seven are monitored, four are material, and one (row 15, which shares the acoustic angle) is a consistency check rather than a test. With the 0.88% scatter of the local rate (§53.9) added in quadrature, rows 16, 18 and 19 become -4.8σ, -6.1σ and -1.5σ. Row 24 is the closed form of §36.13; the measured angle's systematics are open.

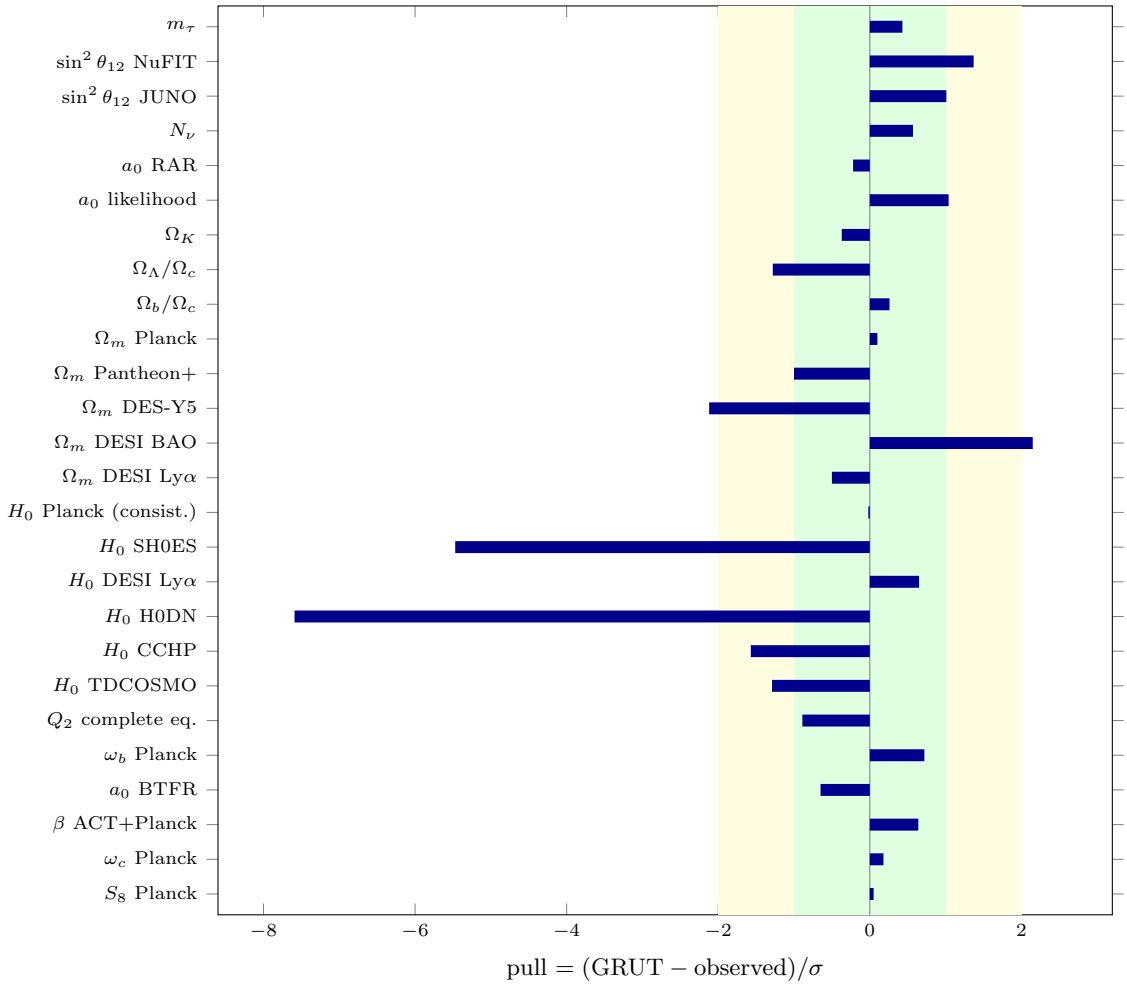


Figure 4: Pulls of the twenty-six comparisons at the resonance budget. Green band: on track; yellow band: monitor.

D.3 Bridge 1: the matter fraction

In the three-component budget $\Omega_m = (1 + r_b)/(1 + r_b + r_\Lambda)$, with $r_b = \Omega_b/\Omega_c$ and $r_\Lambda = \Omega_\Lambda/\Omega_c$. Starting from Planck's ratios ($r_b = 0.18642$, $r_\Lambda = 2.5889$) and replacing one ratio at a time:

Step	$\Delta\Omega_m$	share	cumulative Ω_m
Planck's ratios			0.31426
baryon ratio 0.18697 (§25.7)	+0.00010	12%	0.31436
vacuum ratio 2.5804 from the resonance (§25.11)	+0.00071	88%	0.31507

Restoring radiation and the neutrino masses of §42.4 ($\Omega_\nu = 0.0014$, counted inside Ω_m as in Planck's quoted value) gives the budget's 0.3160, 0.1σ from Planck's 0.3153. The two material Ω_m rows are therefore not the budget's: Planck's own ΛCDM fit lies $+2.1\sigma$ from DESI DR2 and -2.2σ from DES-Y5, and the budget sits where it sits.

D.4 Bridge 2: the Hubble constant

Step	ΔH_0	share	cumulative H_0
SH0ES distance ladder			73.04
CMB acoustic calibration within ΛCDM (Planck)	−5.68	99.8%	67.36
resonance budget at fixed θ_* (§26.7)	−0.01	0.2%	67.35

The offset from the ladder is the offset between the two calibrations, and the budget adds nothing to it. The framework's statement (§53.9) is that there is one expansion rate and that the dilation cannot make a ladder read more: ladders calibrated alike converge on 67.3.

D.5 Variance narratives

Ω_m against DESI DR2 BAO ($+2.15\sigma$) and DES-Y5 (-2.12σ), material. *Driver:* the CMB–late-time split already inside ΛCDM ($+2.1\sigma$ and -2.2σ for Planck's fit); the two samples bracket the budget from opposite sides, and Pantheon+ (-1.0σ) and the DESI Lyman-α full shape (-0.5σ) lie between [65]. *Outlook:* the next BAO releases reach $\sigma(\Omega_m) \approx 0.005$. *Action:* register row 2.

H_0 against the distance ladder (-5.5σ SH0ES, -7.6σ network), material. *Driver:* the calibration of the ladder against the CMB's (Bridge 2). The dilation shifts a clock or ruler by $|\Phi|/c^2 \lesssim 10^{-5}$ and the supernova Hubble diagram excludes a local void (§53.9), so the framework predicts that the ladder converges on 67.3. *Outlook:* JWST distances to the supernova hosts; the red-giant and carbon-star ladder already lies at -1.5σ. *Action:* register rows 1 and 84.

Ω_Λ/Ω_c against the CMB spectra (-1.28σ), monitor. *Driver:* the resonance H3 (§25.11), at $\Delta\chi^2 = 1.6$ from the CMB's ratio; the baryon ratio, derived in §25.7, sits at $+0.26\sigma$. *Action:* the full Planck likelihood decides (register row 65).

$\sin^2\theta_{12}$, $+1.37\sigma$ (NuFIT 6.1) and $+1.01\sigma$ (JUNO), monitor. *Driver:* H2 (the TM1 alignment), with $\sin^2\theta_{13}$ from NuFIT 6.1. *Outlook:* JUNO's six-year precision, about half a percent, makes this a test at more than 5σ if the central value holds: the most decisive near-term row.

a_0 against the rotation-curve likelihood, $+1.04\sigma$, monitor. *Driver:* the per-galaxy nuisance parameters, which move the fitted scale by 16% (§22.6); across the tail-order window it moves by less than 1%. *Action:* if the profiled scale holds at 1.04 with a systematic below 0.03, $a_0 = (\sqrt{3}/\pi^2)cH_0$ at $H_0 = 67.35$ fails at 3σ.

D.6 Summary

No lepton-sector prediction is material. In cosmology the resonance budget lies within 1σ of Planck's densities, Ω_m and S_8; its two material Ω_m rows are ΛCDM's own CMB–late-time split, and its two material H_0 rows are the ladder, whose convergence on 67.3 is the framework's prediction.

D.7 The withdrawn $5/2$ budget

Against the same data the budget of the identification I5 ($\Omega_\Lambda/\Omega_c = 5/2$: $\Omega_m = 0.3219$, $H_0 = 67.12$, $\omega_b = 0.02275$) had two material variances that it drove itself, ω_b at $+2.5\sigma$ and DESI's Ω_m at $+2.8\sigma$, with 99% of its Ω_m offset

from the 5/2. The resonance budget brings them to $+0.7\sigma$ and to ΛCDM's level (rows 13 and 22). I5 is withdrawn (§25.9) and kept in §52.6 as the single-triple limit of the defect pattern.

E Dependencies of the predictions

Each prediction, the inputs it uses, the data already spent on those inputs, and the data that test it independently.

Prediction	Depends on	Data already used	Independent test
$a_0 = 0.1755\,cH_0$	C_3 spectrum, $\frac{3}{4}$ (Theorem 4), I4, H_0	H_0: Planck θ_* via the budget	SPARC likelihood (§22.6)
gravity-law shape	I4, n	SPARC and ephemerides constrain n	new rotation curves, lensing, the external-field quadrupole
$\Omega_m = 0.3160$	H3 (the resonance), Ω_b/Ω_c from the dynamic exponent, $\sum m_\nu$	–	supernovae, DESI BAO and Lyα
$H_0 = 67.35$	Ω_m, θ_*, $\sum m_\nu$	Planck θ_*, ω_b/ω_c	distance ladders, DESI Lyα + BBN
Ω_b/Ω_c	dynamic exponent at the horizon load	–	Planck ω_b/ω_c
H_0 against Planck ΛCDM	shares θ_*	Planck	none (consistency)
maximum-power epoch	Ω_m	as for Ω_m	falls with Ω_m
$w = -1$, $V_0 = 0$	$\varepsilon_V = f_\delta^2/L_*^2$, Ly$\alpha$ bound	Lyα, ρ_Λ	DESI $w(z)$
$m_\delta > 2 \times 10^{-20}$ eV	I6, applicability of the bound (§37.2)	Lyα	cores, other small-scale probes
m_τ	I6 (Koide cone)	m_e, m_μ	Belle II, BES III
$\sum m_\nu$, m_β, $m_{\beta\beta}$	I7 (conjugate branch)	Δm^2_{21}, Δm^2_{31}	cosmology, KATRIN, double beta decay
$\sin^2\theta_{12}$, $\delta_{\rm CP}(\theta_{23})$	H2 (TM1)	θ_{13}	JUNO, DUNE, Hyper-K
selection rule on the sky	three-point source model	Commander $\ell = 4$–10 (null)	smoothed, projected pattern
$L_{\rm crit}(r_s) = c^5/4G$	pressure-balance scale	none	any source above $c^5/4G$
spin of the final horizon	$w = -1$, the thermal section of de Sitter	none	DESI $w(z)$
birefringence in closed form	I8, H3, $\kappa = 6$	α	CMB EB: ACT, SPT, Simons Observatory, LiteBIRD
area quantum $8\pi\ell_{\rm P}^2$	the turn's period, Wald's charge, the first law	none	ringdown spectroscopy (LVK, Einstein Telescope, Cosmic Explorer)
$a_0 = (2/3\pi)\,c^2\sqrt{\Lambda/3}$	§22 scale, H3	Λ (CMB, BAO, SN)	SPARC likelihood, RAR (not independent of rows 8, 66)
$\Omega_0 = jH_\infty$, $j = 1$	quantum of the spin, first law of the cosmological horizon, $\kappa = 6$	β; Ω_Λ with H3	CMB EB (Simons Observatory, LiteBIRD)
$a_0(z) = a_0 E(z)$	§22.10 threshold, $H(z)$	SPARC	rotation curves at $z = 0.3$–2.5
resonance budget	H3, derived baryon ratio, θ_*, $\sum m_\nu$	Planck θ_*	CMB spectra, DESI, supernovae
Koide's cone as the vacuum	I6 as (138), $\Xi_0 = 2$	m_e, m_μ	m_τ (Belle II, BES III)
cosmic dawn at the resonance	H3, first-starlight hypothesis	α	21-cm onset below 50 MHz
cosmic dawn from the twist	first-starlight hypothesis, I8 ($j = 1$), $\kappa = 6$	β; Ω_m/Ω_Λ for the shape	21-cm onset with CMB EB
dark sector as one geometry	§36.11, §36.14, (51)	Ω_Λ/Ω_c (resonance)	the total dark stress-energy only
the dilation $\mathcal{S} = 1/\cos_n\theta$	(47), Theorem 11	a_0	rotation curves, weak-lensing RAR

Prediction	Depends on	Data already used	Independent test
no screening of the dilation	(283), the quanta of §36.14	H_∞	lensing profiles to Mpc scales
the dilation on the unbound web	(277), the lapse of §43B.2–43B.3, (284), (287)	a_0	bulk flows, ISW, S_8, CMB lensing, distance ladder
cluster baryon fraction	derived baryon ratio; the cosmic share held	–	X-ray and lensing masses at R_{200}
$\sigma/m = 0$, no offsets	geodesic dust of (51)	–	merging clusters
mergers: the settled share $f(\Delta t)$ and each peak's own share	(203), (205), (206), (210)	–	lensing of merging clusters: galaxy-free peaks on the gas

The gravity-law results and the scalar-sector closure are shared with TRSD v3; they are counted once across the two papers.

F Next Derivations: the List and Where Each Was Carried Out

The chain above is closed from the anchors down. These are the links still to be built, in the order that would add the most. Each has since been carried out in the section of its topic (map below); its status follows each item.

1. The baryon ratio from the reset: the branching of boundary energy between the radial mode and the winding channel (§25.7). Four simpler mechanisms have been ruled out; the calculation needs a charged coupling and its CP-violating part. *Status (§25.8): the amount follows from the budget and Sakharov's conditions are met without a new coupling; the reset's realization is identified (§51.2). Release 1.14.0: at the resonance budget the amount agrees with the CMB and with deuterium burning,* $\eta = 6.16 \times 10^{-10}$ *(§25.8).*

2. The Galactic external-field quadrupole of the threshold family, across the window, in the inner solar system, against the Cassini bound (§22.7). *Solved (§49.10, §46–47): only the Galactic tide remains, within Cassini's bound.*

3. The dynamical selection of the present epoch: whether a principle—for instance extremal horizon power—places the observer at the maximum-power crossing of Theorem 8 (§26.5). *Status (§36.6): the coincidence has* $p = 0.8\%$*; no principle derived. Release 1.14.0: the present is the resonance (H3, the opening principle), and the resonance fixes the dark ratio (§25.11).*

4. The H_0 prediction (72) against the next distance-ladder, tip of the red giant branch and BAO results (Appendix D). *Status (§16.1):* -7.9σ *from the 2026 ladder consensus,* $+0.5\sigma$ *from DESI Ly*α*.*

5. The map from the C_3 period to the acceleration scale, $a_c = c/\tau_1$, from the field equations of §41, and the exposure law of §22.1 from the same equations. *Derived (§22.10).*

6. The smoothed, projected defect pattern on the sky, for comparison with the Commander maps (§39.3). *Projected (§39.6): the selection rule survives, the ratios fall.*

7. The covariant completion of the gravity law. The field equation of §22.4 is the nonrelativistic AQUAL form; a completion must reduce to it with μ_n in the window, lens by the same potential that governs dynamics, propagate tensor modes at c (GW170817 [72]), give PPN $\gamma = \beta = 1$ in the solar system, conserve energy–momentum, reproduce the CMB and matter spectra at linear order, and be free of ghost and gradient instabilities. A covariant MOND theory meeting these for its own interpolation function exists [71]; the task is one whose weak-field limit is the threshold family and whose scalar is the defect condensate. *Constructed (§46.7): one unit timelike vector is the addition. Superseded by §46: the complete equation is general relativity with the condensate as a source.*

8. The fermion and gauge sectors from the action: a tetrad and spin connection for the generations of §42, and the gauge dynamics of §31. *Status (§41.7): Koide's relation derived from a* C_3 *circulant on the seed cone; electromagnetism shown independent of the defect* $U(1)$*.*

9. The perturbation-level cosmology of the defect sector, replacing the standard transfer functions used in §26.7 and §39. *Status (§47.8): reduced to ΛCDM with the GRUT budget on linear scales.*

10. The reset spectrum of §45: the coupling ϵu and the relaxation rate β on the reset background. *Status (§45.9):* $|\epsilon u| < 1.4 \times 10^3$. *Derived (§45): the spectrum follows from the terminal de Sitter phase and the C_3 pair convolution; the tilt measures the carrier's mass and the interaction is suppressed by A_s.*

11. The fine-structure constant: $K = 0.041741(3)$ from the spectrum on $S^1 \times S^3$, with the running of α (§43A/42C). *Status (§43A/C, §43F): the one-loop calculation gives the thermal weight* 1, *so* $K = \frac{1}{24}(1+e^{-2\pi})$ *and* $1/\alpha = 137.035\,999\,152$, 1.2σ *from CODATA; the half factors are the electroweak $\mathbb{Z}_2$, the deficit π (§43A/C). Release 1.14.0: the thermal dressing is the Boltzmann weight of one quantum of the spin (§43F).*

12. The full trajectory of Θ between its endpoints, from the field equations with matter (§43B.8). *Derived in closed form (§43E).*

13. The sign and size of the birefringence angle from the two-C/S handedness (§39.4). *Status (§49.12–49.13): in the rotation reading the phase turns at* $0.827H_0$ *and* $\beta = 0.312°$, $+0.6\sigma$ *from the measured angle; the frozen-phase bound of §52.1 belongs to the condensate reading.*

14. The reference mass M_* of the scale duality (§37), and the curvature radius R of the spatial S^3 (§4.6). *Status (§50.7): $R \to \infty$ from the horizon condition; M_* fixed by the reset displacement.*

15. The orientation and scale of the charged-lepton triangle, $\delta_\ell \simeq \frac{2}{9}$ and $\mu_\ell^2 = 313.851$ MeV, why pole masses obey the cone, and where the quark cones sit (§42). *Status (§42.7): $\delta_\ell = 2/9$ at 0.7σ; the quark cone is (c, b, t). Release 1.14.0: the cone is the vacuum of the axiom's potential, $Q = (\Xi + 2)/3\Xi$ (§42.8).*

Where each derivation now stands

The list above names fifteen links still to be built. Each has been carried out, in order, and stated as what is derived or computed, the result, and what remains; a result that fails a test is recorded as failed and kept. Each now stands in the section of its topic, as the table gives. The computations are reproduced by `efe_quadrupole.py` and `next_derivations.py`.

Draft	Now	Derivation
46.1	§25.8	The baryon ratio from the reset
46.2	§49.10	The Solar-System quadrupole
46.3	§36.6	The selection of the present epoch
46.4	§16.1	H_0 against the 2025–26 measurements
46.5	§22.10	The map $a_c = c/\tau_1$ and the exposure law
46.6	§39.6	The projected defect pattern
46.7	§46.7	The covariant completion
46.8	§41.7	The fermion and gauge sectors
46.9	§47.8	Perturbation-level cosmology
46.10	§45.9	The reset coupling
46.11	§43F	The fine-structure constant
46.12	§43E	The full trajectory of Θ
46.13	§52.1	The birefringence angle
46.14	§50.7	The reference mass and the radius of S^3
46.15	§42.7	The charged-lepton triangle and the quark cones

The order of release 1.2.1

Release 1.2.1 orders the paper so that every result is read in its final form and nothing read early is corrected later; no result, derivation or number was removed. The rotation that carries the extra gravity is stated first (Section 23), before the sections that use its medium; each derivation of the former Section 46 stands in the section of its topic; the derivation list is kept here; and the superseded results and the excluded mechanism are kept together in Section 52, each with what replaced it. Every section of the previous release has the new number below; within a section the subsection numbers are kept, except where listed.

Was	Now	Was	Now	Was	Now	Was	Now
1	1	2	2	3	3	4	4
5	5	6	6	7	7	8	8
9	9	10	10	11	11	12	12
13	13	14	14	15	15	16	16
17	17	18	18	19	19	20	20
21	21	22	22	23	24	24	25
25	26	26	27	27	28	28	29
29	30	30	31	31	32	32	33
33	34	34	35	35	36	36	37
37	38	38	39	39	40	40	41
41	42	42	43	43	44	44	App. F
45	45	46	distributed	47	46	48	47
49	48	50	23, 49, 51, 52	51	50		

Was	Now	Was	Now	Was	Now	Was	Now
50.1	49.1	50.2	49.2	50.3	52.2	50.4	52.3
50.5	49.3	50.6	52.4	50.7	23.2	50.8	23.3
50.9	49.4	50.10	49.5	50.11	49.6	50.12	49.7
50.13	49.8	50.14	49.9	50.15	49.11	50.16	23.1
50.17	23.4	50.18	49.12	50.19	49.13	50.20	51.1
50.21	51.2	45.9	45.10	47.7	46.8		

G Version history and verification

Release 1.23.2

Patch release: specifies the local Auxoviscausal strain determinant and its sign-consistent evolution; gives the local Bullet Cluster strain-contrast calculation and data requirements; and adds the covariant matter/radiation/thermodynamic conservation and perturbation closure. No new field, force, or fitted parameter is introduced. A numerical Bullet Cluster map and a full cosmological perturbation integration remain explicit computational tasks.

Release 1.22.1

Patch release: derives the conditional Auxoviscausal junction rule from the spatial-volume determinant of the existing conformal metric. The divergent physical strain is not matched directly; the finite remainder $\widehat{\mathcal{A}}_{\rm aux} = \frac{1}{3}\ln(\hat{V}/V_*)$ matches when the conformal spatial metric and the identified flow tube are regular across the boundary. In homogeneous FLRW this remainder carries no separate record of the previous aeon's total $\ln a$. No field or independent transmission coefficient is added.

Release 1.21.0

Minor release: derives Auxoviscausality as the covariant retained logarithmic proper-volume strain of the existing phase-flow congruence. The volume determinant gives $D\mathcal{A}_{\rm aux} = \vartheta/3$; integration along a phase-flow worldline gives the causal memory integral; Raychaudhuri and the Einstein source give the local Ricci-load response, including the shear correction. The FLRW limit reduces to $\mathcal{A}_{\rm aux} = \ln(a/a_0)$ and links its rate to the horizon information density. No new field or coupling is added. The geometric memory law is closed; microscopic yield-threshold plasticity and dissipative entropy production are explicitly not claimed.

Release 1.20.0

Minor release: Section 55 derives collision asymmetry from the existing phase and baryon equations; obtains the relaxation drift and mass-conserving transfer flux by differentiating the existing merger map; derives proper-volume recoil from the congruence expansion scalar; relates that scalar to the existing Einstein/Raychaudhuri equations; derives $v_{\rm rec} = -R\dot{\Phi}/c^2$ from the weak-field spatial metric and retarded phase current; links its domain average to expansion; and derives $H \propto \iota_H$ and $V_H \propto \iota_H^{-3}$ from the horizon definitions. No independent streaming-velocity amplitude or force is introduced. The sign and magnitude of local volume response follow the profile and metric solution. The version history from the first draft to this version (formerly this appendix), and the

verification scripts and their reproduction of every number (formerly Appendix G), are kept unabridged in the companion file `GRUT_v1.19.0_companion.tex`.

References

[1] H. A. Buchdahl, Phys. Rev. 116 (1959) 1027.

[2] CODATA 2022 recommended values of the fundamental physical constants, NIST (2024).

[3] V. Pavlidou, T. N. Tomaras, JCAP 09 (2014) 020.

[4] A. Wyler, C. R. Acad. Sci. Paris A 269 (1969) 743.

[5] C. H. Albright, W. Rodejohann, Eur. Phys. J. C 62 (2009) 599.

[6] G. Altarelli, F. Feruglio, Nucl. Phys. B 720 (2005) 64.

[7] C. A. Brannen, "The lepton masses", preprint (2006).

[8] P. Janot, S. Jadach, Phys. Lett. B 803 (2020) 135319.

[9] JUNO Collaboration, Nature 654 (2026) 343; arXiv:2511.14593.

[10] KamLAND-Zen Collaboration, arXiv:2406.11438 (2024).

[11] Y. Koide, Lett. Nuovo Cim. 34 (1982) 201; Phys. Rev. D 28 (1983) 252.

[12] I. Esteban, M. C. Gonzalez-Garcia, M. Maltoni, I. Martinez-Soler, J. P. Pinheiro, T. Schwetz, JHEP 12 (2024) 216.

[13] Particle Data Group, S. Navas et al., Phys. Rev. D 110 (2024) 030001.

[14] T2K and NOvA Collaborations, Nature (2025), doi:10.1038/s41586-025-09599-3.

[15] A. Almheiri, N. Engelhardt, D. Marolf, H. Maxfield, JHEP 12 (2019) 063.

[16] A. Almheiri, R. Mahajan, J. Maldacena, Y. Zhao, JHEP 03 (2020) 149.

[17] J. Baez, J. Huerta, Bull. Amer. Math. Soc. 47 (2010) 483.

[18] J. D. Bekenstein, Phys. Rev. D 7 (1973) 2333.

[19] M. V. Berry, J. P. Keating, SIAM Rev. 41 (1999) 236.

[20] D. Brout et al. (Pantheon+), Astrophys. J. 938 (2022) 110.

[21] R.-G. Cai, S. P. Kim, JHEP 02 (2005) 050.

[22] DES Collaboration, Astrophys. J. Lett. 973 (2024) L14.

[23] DESI Collaboration, arXiv:2503.14738 (2025).

[24] S. Deser, Gen. Rel. Grav. 1 (1970) 9.

[25] S. Deser, R. Jackiw, G. 't Hooft, Ann. Phys. 152 (1984) 220.

[26] P. Ehrenfest, Proc. Amsterdam Acad. 20 (1917) 200.

[27] J. R. Eskilt, E. Komatsu, Phys. Rev. D 106 (2022) 063503.

[28] G. W. Gibbons, Found. Phys. 32 (2002) 1891.

[29] G. W. Gibbons, S. W. Hawking, Phys. Rev. D 15 (1977) 2738.

[30] A. M. Gleason, J. Math. Mech. 6 (1957) 885.

[31] S. W. Hawking, Commun. Math. Phys. 43 (1975) 199.

[32] F. Hofmann, J. Müller, Class. Quantum Grav. 35 (2018) 035015.

[33] T. Jacobson, Phys. Rev. Lett. 75 (1995) 1260.

[34] E. T. Jaynes, Phys. Rev. 106 (1957) 620.

[35] R. Landauer, IBM J. Res. Dev. 5 (1961) 183.

[36] D. Lovelock, J. Math. Phys. 12 (1971) 498.

[37] S. S. McGaugh, F. Lelli, J. M. Schombert, Phys. Rev. Lett. 117 (2016) 201101.

[38] G. Penington, JHEP 09 (2020) 002.

[39] Planck Collaboration, Astron. Astrophys. 641 (2020) A6.

[40] A. G. Riess et al., Astrophys. J. Lett. 934 (2022) L7.

[41] K. K. Rogers, H. V. Peiris, Phys. Rev. Lett. 126 (2021) 071302.

[42] M. H. Stone, Ann. Math. 33 (1932) 643.

[43] P. Touboul et al. (MICROSCOPE), Phys. Rev. Lett. 129 (2022) 121102.

[44] A. Vilenkin, Phys. Rev. D 23 (1981) 852.

[45] S. Weinberg, Phys. Rev. 135 (1964) B1049.

[46] E. C. Zeeman, J. Math. Phys. 5 (1964) 490.

[47] R. Foot, "A note on Koide's lepton mass relation", arXiv:hep-ph/9402242 (1994).

[48] C. Schiller, Int. J. Theor. Phys. 44 (2005) 1629.

[49] P. C. W. Davies, Class. Quantum Grav. 4 (1987) L225.

[50] E. P. Verlinde, SciPost Phys. 2 (2017) 016.

[51] M. Milgrom, Astrophys. J. 270 (1983) 365.

[52] J. D. Bekenstein, M. Milgrom, Astrophys. J. 286 (1984) 7.

[53] M. Milgrom, Mon. Not. R. Astron. Soc. 399 (2009) 474.

[54] A. Hees, W. M. Folkner, R. A. Jacobson, R. S. Park, Phys. Rev. D 89 (2014) 102002.

[55] E. V. Pitjeva, N. P. Pitjev, Mon. Not. R. Astron. Soc. 432 (2013) 3431.

[56] D. N. Spergel, U.-L. Pen, Astrophys. J. 491 (1997) L67.

[57] L. Hui, J. P. Ostriker, S. Tremaine, E. Witten, Phys. Rev. D 95 (2017) 043541.

[58] Z.-z. Xing, S. Zhou, Phys. Lett. B 653 (2007) 278.

[59] V. Cardoso, T. Ikeda, C. J. Moore, C.-M. Yoo, Phys. Rev. D 97 (2018) 084013.

[60] B. P. Abbott et al. (LIGO Scientific and Virgo Collaborations), Phys. Rev. Lett. 116 (2016) 061102.

[61] A. Lewis, A. Challinor, A. Lasenby, Astrophys. J. 538 (2000) 473.

[62] F. Lelli, S. S. McGaugh, J. M. Schombert, Astron. J. 152 (2016) 157.

[63] J. Mancilla, *Topological Relational System Dynamics (TRSD)*, release 1.13.0 (2026).

[64] NuFIT 6.1 (data to November 2025, including JUNO), www.nu-fit.org; values as quoted in arXiv:2608.19691 and arXiv:2606.18987.

[65] DESI Collaboration (L. Casas et al.), *DESI DR2 Results IV: Alcock–Paczyński measurements from the Lyman alpha forest*, arXiv:2607.27410 (2026).

[66] P. Li, F. Lelli, S. S. McGaugh, J. M. Schombert, Astron. Astrophys. 615 (2018) A3.

[67] F. Lelli, S. S. McGaugh, J. M. Schombert, Astrophys. J. Lett. 816 (2016) L14.

[68] T. Kobayashi, R. Murgia, A. De Simone, V. Iršič, M. Viel, Phys. Rev. D 96 (2017) 123514.

[69] *Lyman-α forest constraints on pure and mixed fuzzy dark matter*, arXiv:2606.06969 (2026).

[70] K.-H. Leong, H.-Y. Schive, U.-H. Zhang, T. Chiueh, Mon. Not. R. Astron. Soc. 484 (2019) 4273.

[71] C. Skordis, T. Złośnik, Phys. Rev. Lett. 127 (2021) 161302.

[72] B. P. Abbott et al. (LIGO, Virgo, Fermi-GBM, INTEGRAL), Astrophys. J. Lett. 848 (2017) L13.

[73] E. Calabrese et al. (ACT), *The Atacama Cosmology Telescope: DR6 constraints on extended cosmological models*, arXiv:2503.14454 (2025).

[74] Tilt values as compiled in arXiv:2606.24131 and arXiv:2607.04504 (2026).

[75] *Improved constraints on modified Newtonian gravity from Cassini radio tracking data*, Phys. Rev. D (2026), doi:10.1103/r7n8-kw38.

[76] L. Blanchet, J. Novak, Mon. Not. R. Astron. Soc. 412 (2011) 2530.

[77] A. Hees et al., *Tests of gravitation at Solar System scales beyond the PPN formalism*, arXiv:1403.1365.

[78] H. Desmond et al., *On the tension between the radial acceleration relation and Solar System quadrupole in modified gravity MOND*, arXiv:2401.04796 (2024).

[79] H0DN Collaboration (S. Casertano et al.), Astron. Astrophys. 708 (2026) A166.

[80] W. L. Freedman et al., Astrophys. J. (2025), doi:10.3847/1538-4357/adce78.

[81] W. Hu, R. Barkana, A. Gruzinov, Phys. Rev. Lett. 85 (2000) 1158.

[82] Planck Collaboration, Astron. Astrophys. 641 (2020) A10.

[83] Z.-z. Xing, H. Zhang, S. Zhou, Phys. Rev. D 86 (2012) 013013.

[84] P. Diego-Palazuelos and E. Komatsu, Cosmic birefringence from the Atacama Cosmology Telescope Data Release 6, arXiv:2509.13654 (2025).

[85] J. R. Eskilt, Cosmic birefringence from a joint analysis of ACT and Planck, arXiv:2608.06480 (2026).

[86] F. Hofmann and J. Müller, Relativistic tests with lunar laser ranging, Class. Quantum Grav. 35, 035015 (2018).

[87] R. Lange et al., Improved limits for violations of local position invariance from atomic clock comparisons, Phys. Rev. Lett. 126, 011102 (2021).

[88] M. T. Murphy et al., Fundamental physics with ESPRESSO: precise limit on variations in the fine-structure constant towards the bright quasar HE 0515−4414, Astron. Astrophys. 658, A123 (2022).

[89] J. D. Bowman et al., An absorption profile centred at 78 megahertz in the sky-averaged spectrum, Nature 555, 67 (2018).

[90] S. Singh et al., On the detection of a cosmic dawn signal in the radio background, Nature Astron. 6, 607 (2022).

[91] S. Carniani et al., Spectroscopic confirmation of two luminous galaxies at a redshift of 14, Nature 633, 318 (2024).

[92] S. Naoz, S. Noter and R. Barkana, The first stars in the Universe, Mon. Not. R. Astron. Soc. 373, L98 (2006).

[93] R. H. Dicke, Dirac's cosmology and Mach's principle, Nature 192, 440 (1961).

[94] S. Weinberg, The cosmological constant problem, Rev. Mod. Phys. 61, 1 (1989).

[95] A. G. Cohen, D. B. Kaplan and A. E. Nelson, Effective field theory, black holes, and the cosmological constant, Phys. Rev. Lett. 82, 4971 (1999).

[96] M. Li, A model of holographic dark energy, Phys. Lett. B 603, 1 (2004).

[97] T. Padmanabhan, Equipartition of energy in the horizon degrees of freedom and the emergence of gravity, Mod. Phys. Lett. A 25, 1129 (2010).

[98] K. Gödel, An example of a new type of cosmological solutions of Einstein's field equations of gravitation, Rev. Mod. Phys. 21, 447 (1949).

[99] B. E. Szigeti, I. Szapudi, I. F. Barna and G. G. Barnaföldi, Can rotation solve the Hubble puzzle?, Mon. Not. R. Astron. Soc. 538, 3038 (2025), arXiv:2503.13525.

[100] I. Szapudi, Constraining Mach's principle with high precision astrometry, arXiv:2105.05337 (2021).

[101] V. C. Rubin, J. A. Graham and J. D. P. Kenney, Cospatial counterrotating stellar disks in the Virgo E7/S0 galaxy NGC 4550, Astrophys. J. 394, L9 (1992).

[102] S. S. McGaugh, Novel test of modified Newtonian dynamics with gas rich galaxies, Phys. Rev. Lett. 106, 121303 (2011).

[103] K.-H. Chae, Measurements of the low-acceleration gravitational anomaly from the normalized velocity profile of Gaia wide binary stars, arXiv:2402.05720 (2024).

[104] X. Hernandez et al., A recent confirmation of the wide binary gravitational anomaly, arXiv:2410.17178 (2024).

[105] I. Banik, C. Pittordis, W. Sutherland, B. Famaey, R. Ibata, S. Mieske and H. Zhao, Strong constraints on the gravitational law from Gaia DR3 wide binaries, Mon. Not. R. Astron. Soc. 527, 4573 (2024).

[106] F. Wilczek, Two applications of axion electrodynamics, Phys. Rev. Lett. 58, 1799 (1987).

[107] A. Vilenkin, Macroscopic parity-violating effects: neutrino fluxes from rotating black holes and in rotating thermal radiation, Phys. Rev. D 20, 1807 (1979).

[108] D. T. Son and P. Surówka, Hydrodynamics with triangle anomalies, Phys. Rev. Lett. 103, 191601 (2009).

[109] R. J. Scherrer, Purely kinetic k-essence as unified dark matter, Phys. Rev. Lett. 93, 011301 (2004).

[110] A. H. Chamseddine and V. Mukhanov, Mimetic dark matter, J. High Energy Phys. 1311, 135 (2013), arXiv:1308.5410.

[111] X. Fan, T. G. Myers, B. A. D. Sukra and G. Gabrielse, Measurement of the electron magnetic moment, Phys. Rev. Lett. 130, 071801 (2023).

[112] L. Morel, Z. Yao, P. Cladé and S. Guellati-Khélifa, Determination of the fine-structure constant with an accuracy of 81 parts per trillion, Nature 588, 61 (2020).

[113] R. H. Parker, C. Yu, W. Zhong, B. Estey and H. Müller, Measurement of the fine-structure constant as a test of the Standard Model, Science 360, 191 (2018).

[114] R. Penrose, *Cycles of Time: An Extraordinary New View of the Universe* (Bodley Head, London, 2010).

[115] D. Saadeh, S. M. Feeney, A. Pontzen, H. V. Peiris and J. D. McEwen, How isotropic is the Universe?, Phys. Rev. Lett. 117, 131302 (2016).

[116] E. Witten, An SU(2) anomaly, Phys. Lett. B 117, 324 (1982).

[117] W. G. Unruh, Notes on black-hole evaporation, Phys. Rev. D 14, 870 (1976).

[118] D. R. Brill, Lattice universes in 2+1-dimensional gravity, arXiv:gr-qc/0312033.

[119] T. Padmanabhan, Emergence and expansion of cosmic space as due to the quest for holographic equipartition, arXiv:1206.4916 (2012).

[120] S. Weinberg, *Gravitation and Cosmology* (Wiley, New York, 1972).

[121] G. A. Mena Marugán and S. Carneiro, Holography and the large number hypothesis, Phys. Rev. D 65, 087303 (2002).

[122] A. Alfonso-Faus, Structure of the universe: generalization of Weinberg relation, arXiv:0805.4762 (2008).

[123] S. Funkhouser, A fundamental scale for acceleration from the holographic principle, arXiv:physics/0505175 (2005).

[124] T. T. Takahashi, H. Matsufuru, Y. Nemoto and H. Suganuma, The three-quark potential in the SU(3) lattice QCD, Phys. Rev. Lett. 86, 18 (2001).

[125] D. Tong, Line operators in the Standard Model, JHEP 07 (2017) 104.

[126] J.-P. Uzan, The fundamental constants and their variation: observational and theoretical status, Rev. Mod. Phys. 75, 403 (2003).

[127] Super-Kamiokande Collaboration (A. Takenaka et al.), Search for proton decay via $p \to e^+\pi^0$ and $p \to \mu^+\pi^0$ with an enlarged fiducial volume in Super-Kamiokande I–IV, Phys. Rev. D 102, 112011 (2020).

[128] A. Antognini et al., Proton structure from the measurement of 2S–2P transition frequencies of muonic hydrogen, Science 339, 417 (2013).

[129] L. de Broglie, Recherches sur la théorie des quanta, Ann. Phys. (Paris) 3, 22 (1925).

[130] E. Schrödinger, Über die kräftefreie Bewegung in der relativistischen Quantenmechanik, Sitzungsber. Preuss. Akad. Wiss. Phys.-Math. Kl. 24, 418 (1930).

[131] D. Hestenes, The zitterbewegung interpretation of quantum mechanics, Found. Phys. 20, 1213 (1990).

[132] J. M. Bardeen, B. Carter and S. W. Hawking, The four laws of black hole mechanics, Commun. Math. Phys. 31, 161 (1973).

[133] M. Henneaux and C. Teitelboim, The cosmological constant and general covariance, Phys. Lett. B 222, 195 (1989).

[134] R. M. Wald, Black hole entropy is the Noether charge, Phys. Rev. D 48, R3427 (1993).

[135] S. Hod, Universal entropy bound for rotating systems, Phys. Rev. D 61, 024018 (2000).

[136] J. D. Bekenstein, The quantum mass spectrum of the Kerr black hole, Lett. Nuovo Cim. 11, 467 (1974).

[137] M. Maggiore, Physical interpretation of the spectrum of black hole quasinormal modes, Phys. Rev. Lett. 100, 141301 (2008).

[138] D. Kothawala, T. Padmanabhan and S. Sarkar, Is gravitational entropy quantized?, Phys. Rev. D 78, 104018 (2008).

[139] X.-X. Zeng, X.-M. Liu and W.-B. Liu, Periodicity and area spectrum of black holes, Eur. Phys. J. C 72 (2012), arXiv:1203.5947.

[140] V. Cardoso, V. F. Foit and M. Kleban, Gravitational wave echoes from black hole area quantization, J. Cosmol. Astropart. Phys. 08 (2019) 006.

[141] I. Agullo, V. Cardoso, A. del Rio, M. Maggiore and J. Pullin, Potential gravitational-wave signatures of quantum gravity, Phys. Rev. Lett. 126, 041302 (2021).

[142] D. Laghi, G. Carullo, J. Veitch and W. Del Pozzo, Quantum black hole spectroscopy: probing the quantum nature of the black hole area using LIGO–Virgo ringdown detections, Class. Quantum Grav. 38, 095005 (2021).

[143] H.-P. Nollert, Quasinormal modes of Schwarzschild black holes: the determination of quasinormal frequencies with very large imaginary parts, Phys. Rev. D 47, 5253 (1993).

[144] L. Motl, An analytical computation of asymptotic Schwarzschild quasinormal frequencies, Adv. Theor. Math. Phys. 6, 1135 (2003).

[145] M. Milgrom, The modified dynamics as a vacuum effect, Phys. Lett. A 253, 273 (1999).

[146] P. W. Anderson, Considerations on the flow of superfluid helium, Rev. Mod. Phys. 38, 298 (1966).

[147] P. A. M. Dirac, Quantised singularities in the electromagnetic field, Proc. R. Soc. Lond. A 133, 60 (1931).

[148] C. A. R. Herdeiro and E. Radu, Kerr black holes with scalar hair, Phys. Rev. Lett. 112, 221101 (2014).

[149] B. I. Ciocan, N. F. Bouché, J. Fensch et al., MUSE-DARK III: the evolution of the radial acceleration relation at intermediate redshifts, Astron. Astrophys. 709, L16 (2026).

[150] M. Milgrom, High-redshift rotation curves and MOND, arXiv:1703.06110 (2017).

[151] V. Mossa et al., The baryon density of the Universe from an improved rate of deuterium burning, Nature 587, 210 (2020).

[152] Y. Koide, U(3)-flavor nonet scalar as an origin of the flavor mass spectra, Phys. Lett. B (2008), arXiv:0802.1084.

[153] Y. Sumino, Family gauge symmetry as an origin of Koide's mass formula and charged lepton spectrum, J. High Energy Phys. 05 (2009) 075.

[154] J. Kocik, The Koide lepton mass formula and geometry of circles, arXiv:1201.2067 (2012).

[155] R. P. Naidu et al., A cosmic miracle: a remarkably luminous galaxy at $z_{\rm spec} = 14.44$ confirmed with JWST, Open J. Astrophys. (2026).

[156] F.-Y. F. Liu et al., A clear detection of proper motion confirms that the claimed $z \simeq 32$ galaxy candidate "Capotauro" is a Y-type brown dwarf, arXiv:2608.07461 (2026).

[157] K. Kadota, J. Ooba, H. Tashiro, K. Ichiki and G.-C. Liu, Cross-correlation between 21-cm radiation and CMB B modes from the cosmic birefringence in the presence of a light scalar field, Phys. Rev. D 100, 063506 (2019).

[158] D. Clowe, M. Bradač, A. H. Gonzalez, M. Markevitch, S. W. Randall, C. Jones and D. Zaritsky, A direct empirical proof of the existence of dark matter, Astrophys. J. Lett. 648, L109 (2006).

[159] M. Kunz, Degeneracy between the dark components resulting from the fact that gravity only measures the total energy-momentum tensor, Phys. Rev. D 80, 123001 (2009).

[160] E. A. Lim, I. Sawicki and A. Vikman, Dust of dark energy, J. Cosmol. Astropart. Phys. 05 (2010) 012.

[161] S. Pan, S. Paul, E. N. Saridakis and W. Yang, Interacting dark energy after DESI DR2: a challenge for the ΛCDM paradigm?, arXiv:2504.00994 (2025).

[162] Y.-H. Li and X. Zhang, Cosmic sign-reversal: non-parametric reconstruction of interacting dark energy with DESI DR2, J. Cosmol. Astropart. Phys. 12 (2025) 018.

[163] D. Eckert, S. Ettori, E. Pointecouteau, R. F. J. van der Burg and S. I. Loubser, The gravitational field of X-COP galaxy clusters, Astron. Astrophys. 662, A123 (2022).

[164] M. J. Jee, D. Park and W. Lee, A new robust constraint on the self-interaction cross-section of dark matter with double radio relic clusters, arXiv:2605.00093 (2026).

[165] S. More, B. Diemer and A. V. Kravtsov, The splashback radius as a physical halo boundary and the growth of halo mass, Astrophys. J. 810, 36 (2015).

[166] C. Chang et al. (DES), The splashback feature around DES galaxy clusters: galaxy density and weak lensing profiles, Astrophys. J. 864, 83 (2018).

[167] M. M. Brouwer et al., The weak lensing radial acceleration relation: constraining modified gravity and cold dark matter theories with KiDS-1000, Astron. Astrophys. 650, A113 (2021).

[168] T. Mistele, S. McGaugh, F. Lelli, J. Schombert and P. Li, Indefinitely flat circular velocities and the baryonic Tully–Fisher relation from weak lensing, Astrophys. J. Lett. (2024), arXiv:2406.09685.

[169] Y. Tian et al., The radial acceleration relation in CLASH galaxy clusters, Astrophys. J. 896, 70 (2020).

[170] R. C. Keenan, A. J. Barger and L. L. Cowie, Evidence for a ∼ 300 megaparsec scale under-density in the local galaxy distribution, Astrophys. J. 775, 62 (2013).

[171] W. D. Kenworthy, D. Scolnic and A. Riess, The local perspective on the Hubble tension: local structure does not impact measurement of the Hubble constant, Astrophys. J. 875, 145 (2019).

[172] D. Green and J. Meyers, The cosmological preference for negative neutrino mass, arXiv:2407.07878 (2024).

[173] DESI Collaboration (W. Elbers et al.), Constraints on neutrino physics from DESI DR2 BAO and DR1 full shape, arXiv:2503.14744 (2025).

[174] DES Collaboration (T. M. C. Abbott et al.), Dark Energy Survey Year 6 results: cosmological constraints from galaxy clustering and weak lensing, arXiv:2601.14559 (2026).

[175] A. H. Wright et al., KiDS-Legacy: cosmological constraints from cosmic shear with the complete Kilo-Degree Survey, arXiv:2503.19441 (2025).

[176] R. Watkins et al., Analysing the large-scale bulk flow using CosmicFlows4: increasing tension with the standard cosmological model, Mon. Not. R. Astron. Soc. 524, 1885 (2023), arXiv:2302.02028.

[177] A. Nusser and R. B. Tully, Cosmicflows-4 and the cosmological consistency of large-scale motions, arXiv:2608.14265 (2026).

[178] F. J. Qu et al., Unified and consistent structure growth measurements from joint ACT, SPT and Planck CMB lensing, Phys. Rev. Lett. 136, 021001 (2026), arXiv:2504.20038.

[179] Planck Collaboration, Planck 2015 results. XXI. The integrated Sachs–Wolfe effect, Astron. Astrophys. 594, A21 (2016), arXiv:1502.01595.

[180] V. Springel and G. R. Farrar, The speed of the 'bullet' in the merging galaxy cluster 1E0657−56, Mon. Not. R. Astron. Soc. 380, 911 (2007), arXiv:astro-ph/0703232.

[181] M. Milosavljević, J. Koda, D. Nagai, E. Nakar and P. R. Shapiro, The cluster-merger shock in 1E0657−56: faster than a speeding bullet?, Astrophys. J. 661, L131 (2007).

[182] R. Thompson and K. Nagamine, Pairwise velocities of dark matter halos: a test for the ΛCDM model using the Bullet Cluster, arXiv:1107.4645 (2012).

[183] M. Bradač et al., Revealing the properties of dark matter in the merging cluster MACS J0025.4−1222, arXiv:0806.2320 (2008).

[184] A. Mahdavi, H. Hoekstra, A. Babul, D. Balam and P. Capak, A dark core in Abell 520, Astrophys. J. 668, 806 (2007).

[185] M. J. Jee et al., A study of the dark core in A520 with the Hubble Space Telescope: the mystery deepens, arXiv:1202.6368 (2012).

[186] D. Clowe et al., On dark peaks and missing mass: a weak lensing mass reconstruction of the merging cluster system Abell 520, arXiv:1209.2143 (2012).

[187] A. Peel, F. Lanusse and J.-L. Starck, Sparse reconstruction of the merging A520 cluster system, arXiv:1708.00269 (2017).